

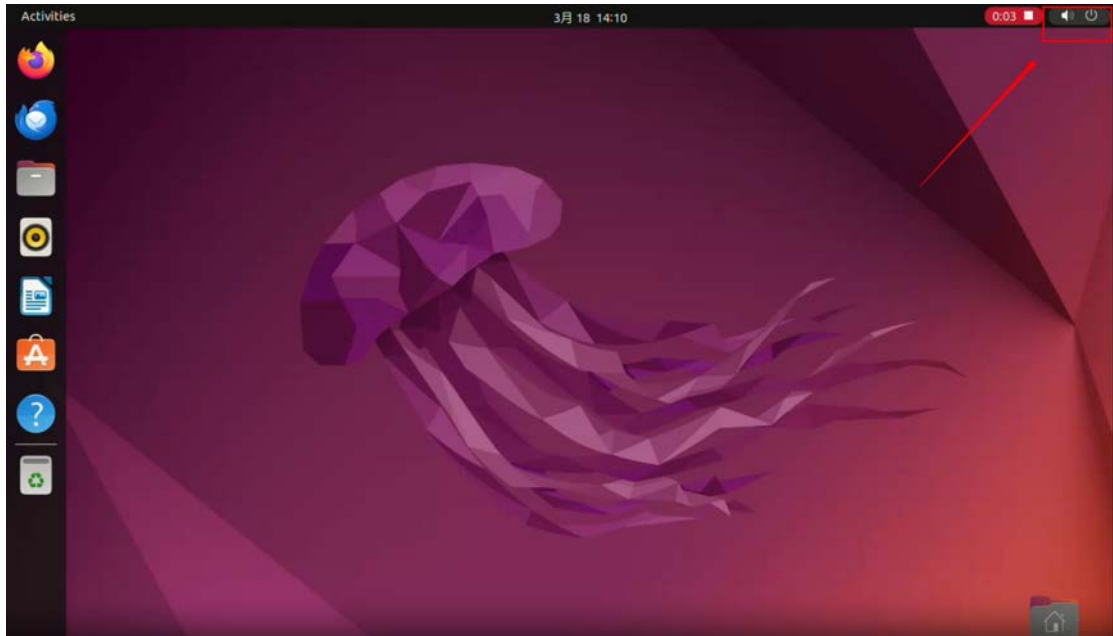
Ubuntu 系统配置

目录

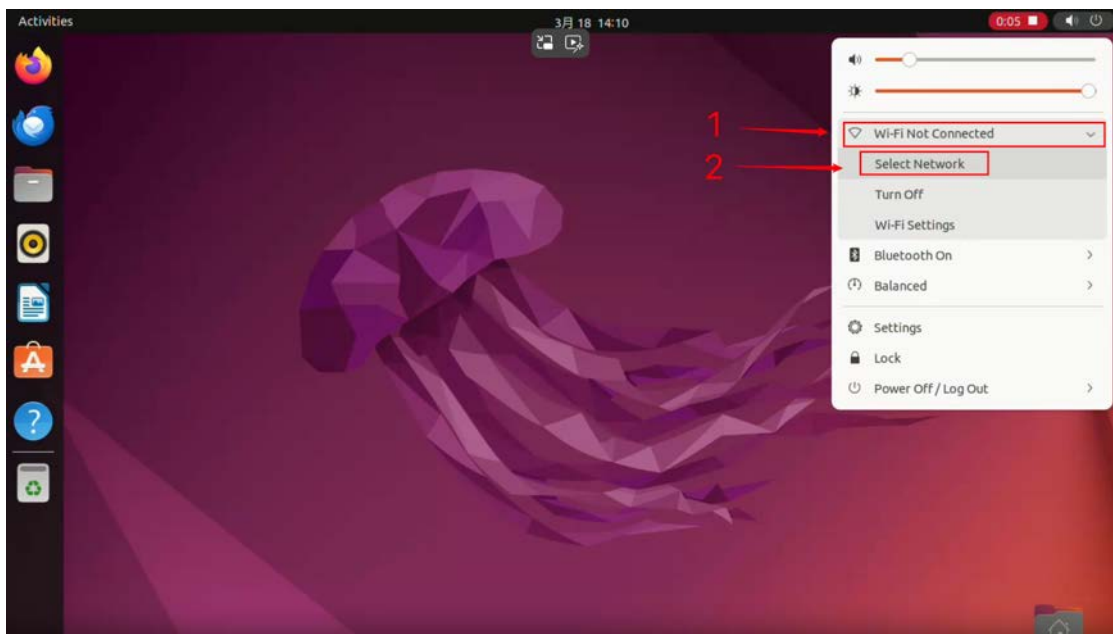
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1. 连接 Wifi 与装机文件拷贝

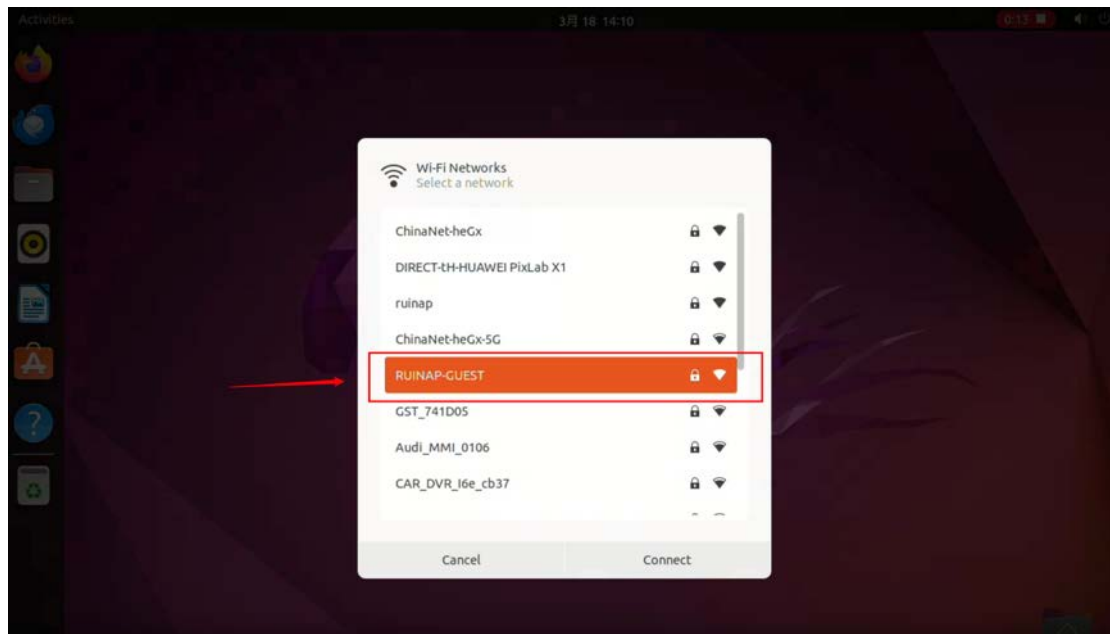
设置 wifi，鼠标点击右上角



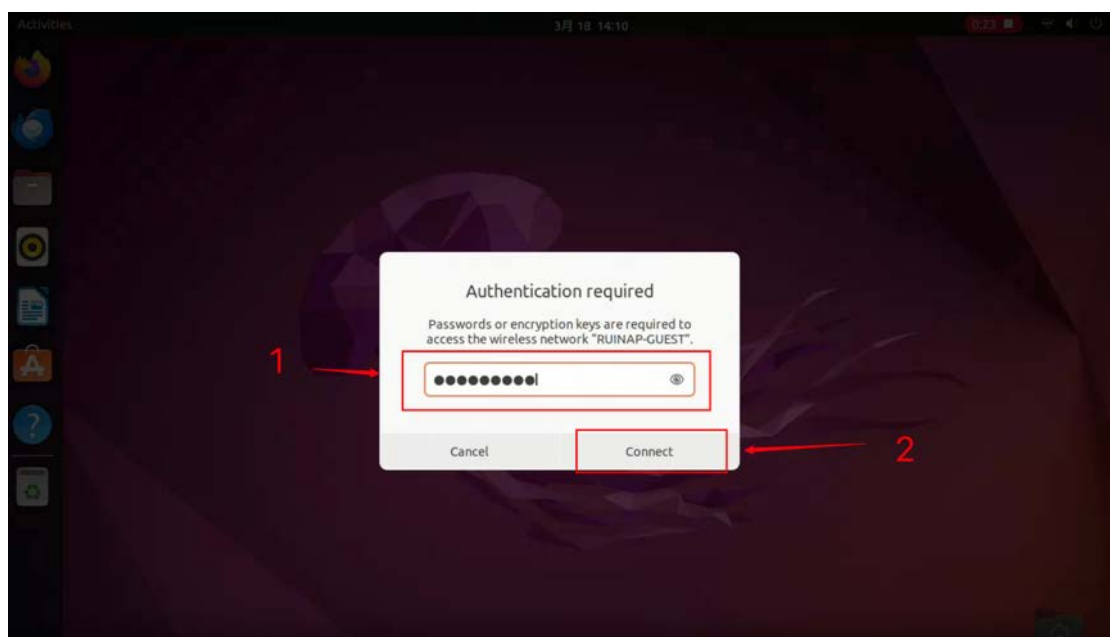
点击【Wi-Fi Not Connected】，展开后点击【Select Network】



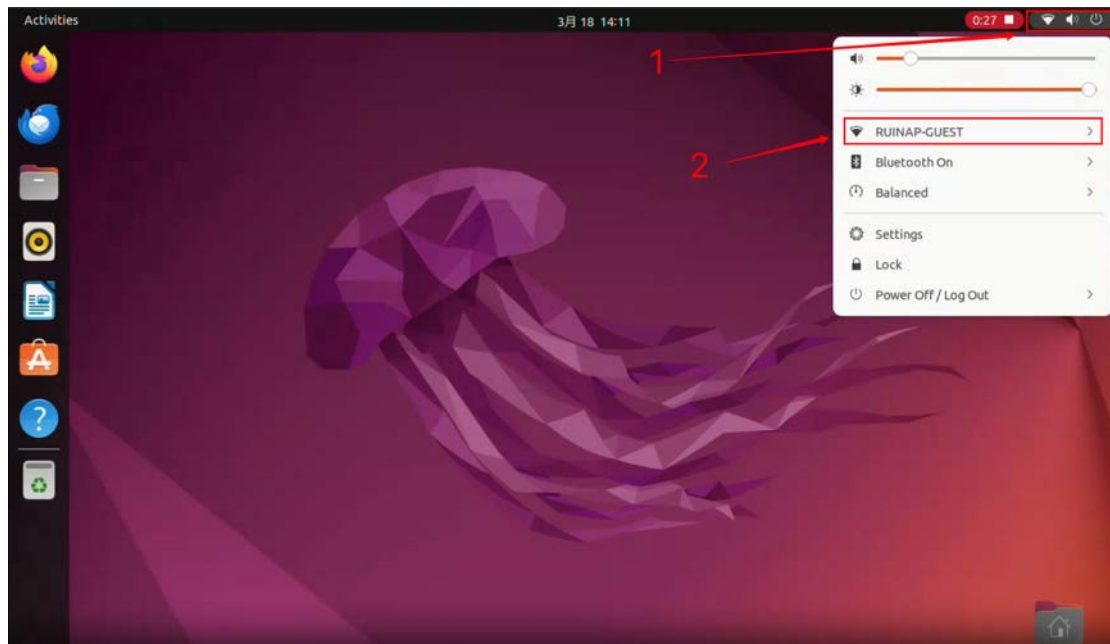
选择要连接的 wifi 名称，如下以“RUIAP-GUEST”为例



填入 wifi 密码然后点击【connect】

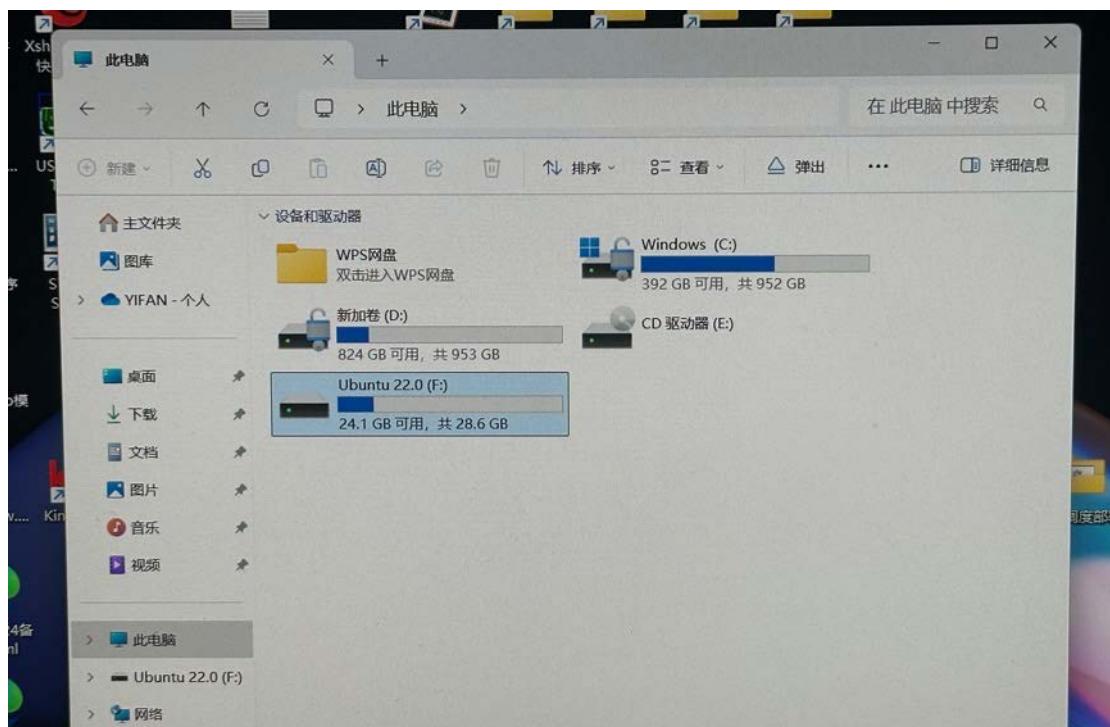
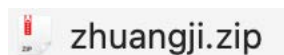


再次点击右上角，查看 wifi 图标处是否显示连接的 wifi 名称，如下已显示“RUINAP-GUEST”则表明成功连接至 RUINAP-GUEST 的 wifi

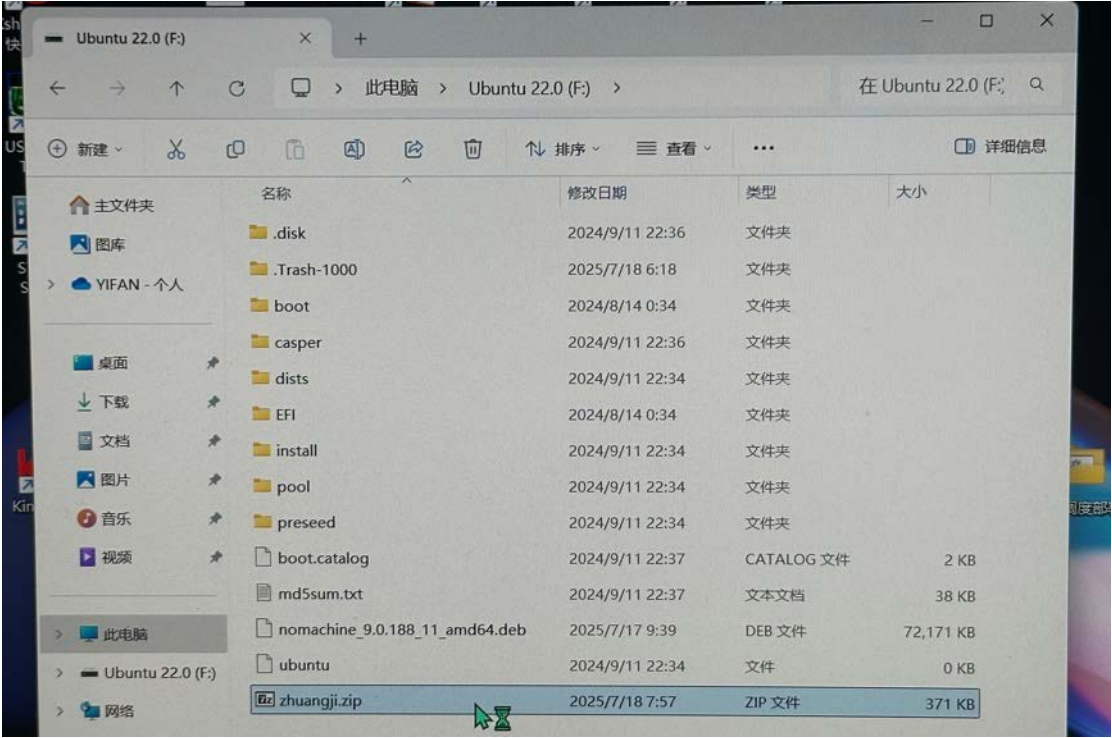


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wifi 连接成功后，做另一项准备工作，将随同教程文件一起提供的如下文件
“zhuangji.zip”文件拖入至先前制作的 u 盘启动盘中，方便后续将“zhuangji.zip”文件拷贝至工控机中。

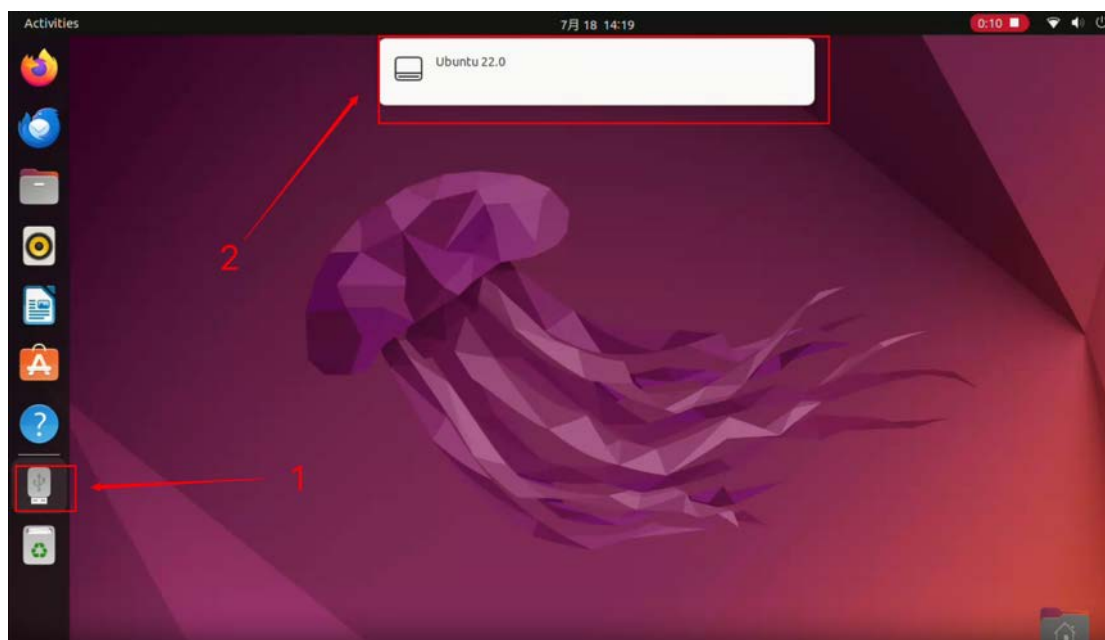


将 u 盘插入笔记本电脑并打开该启动盘，以“Ubuntu 22.0”为例，打开后将“zhuangji.zip”文件复制到该文件夹下

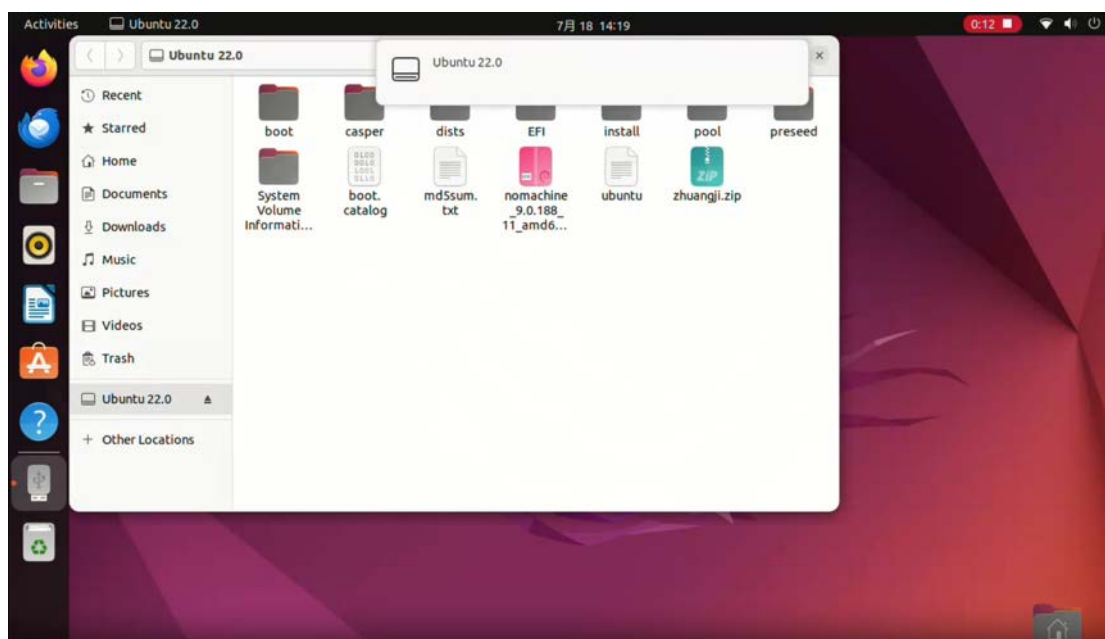


2. 设置系统为 root 权限启动

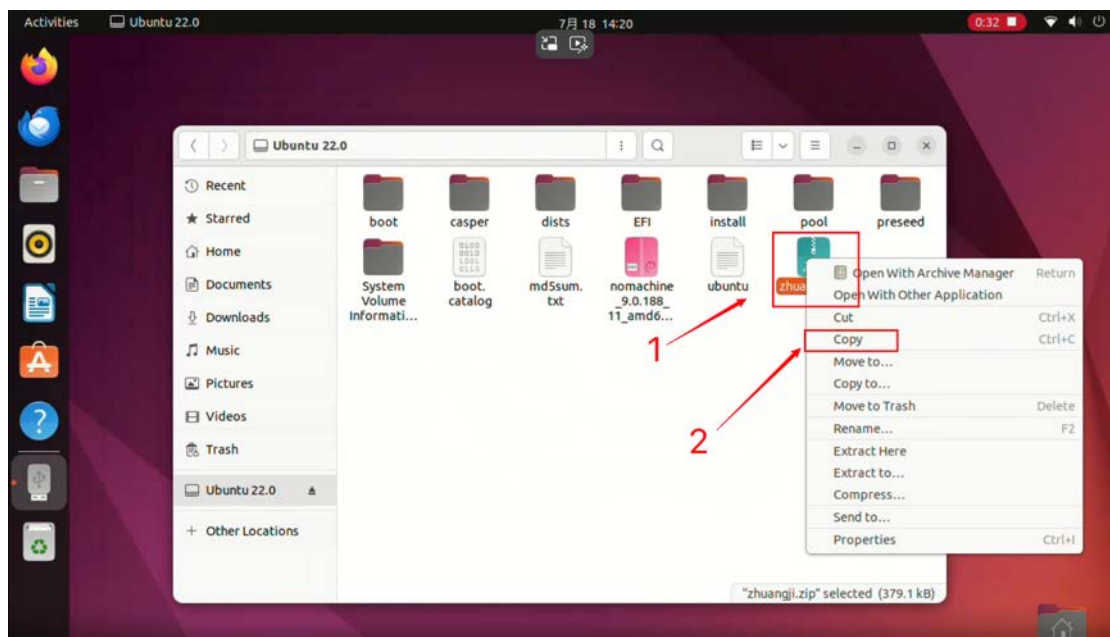
将拷贝了“zhuangji.zip”文件的 u 盘插入工控机，系统会显示如下弹窗，点击 1 或 2 都可以打开 u 盘



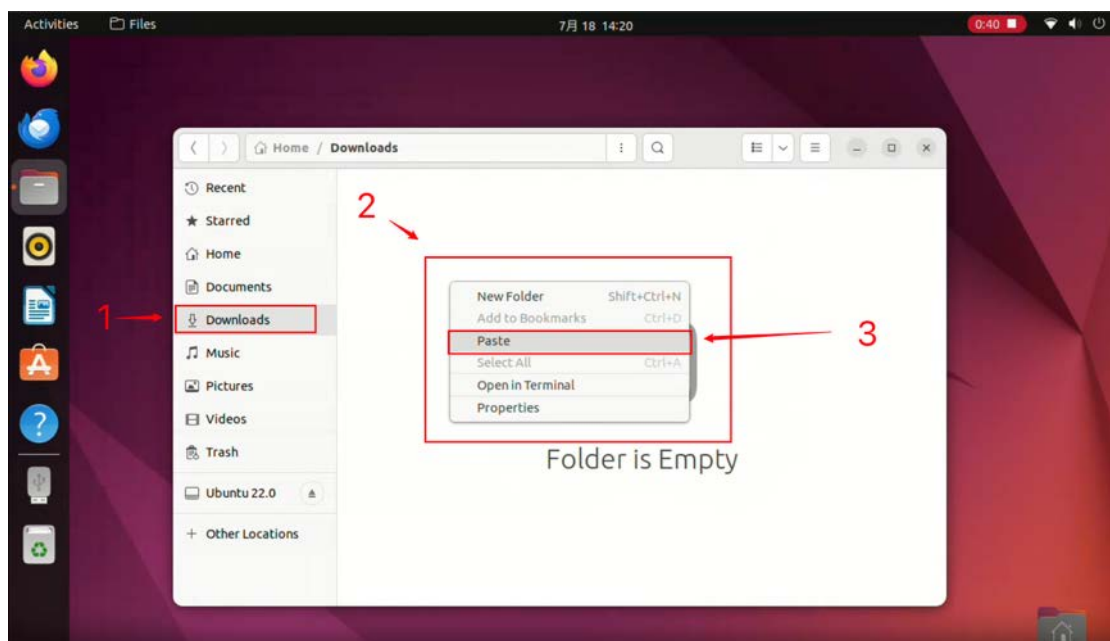
打开后显示类似如下的文件目录



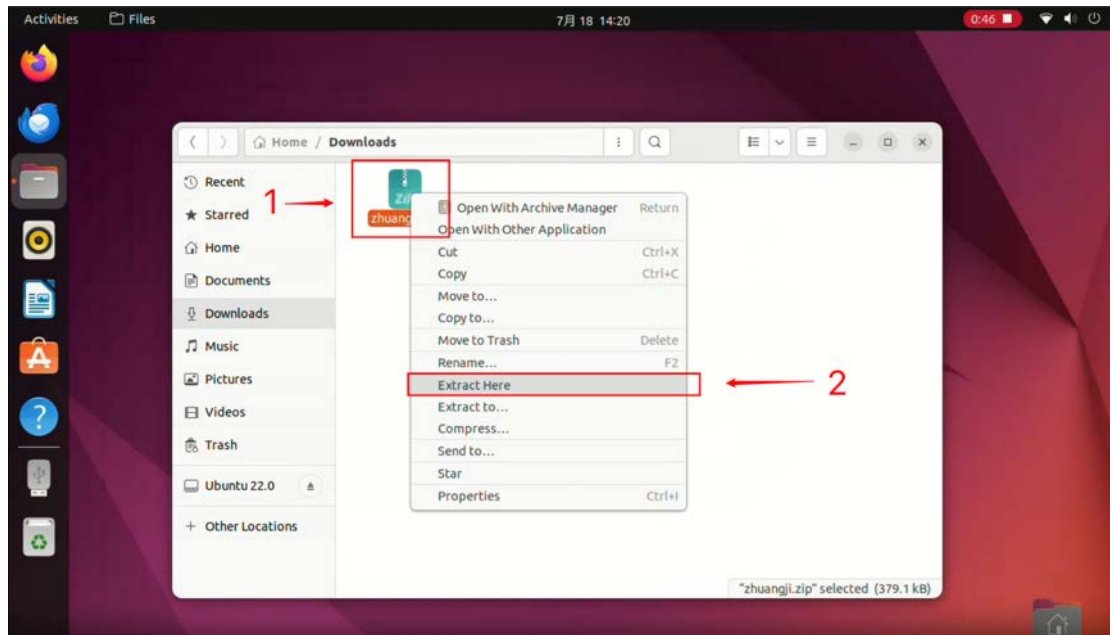
鼠标移动至“zhuangji.zip”文件处，然后鼠标右击，弹窗后鼠标左击【Copy】



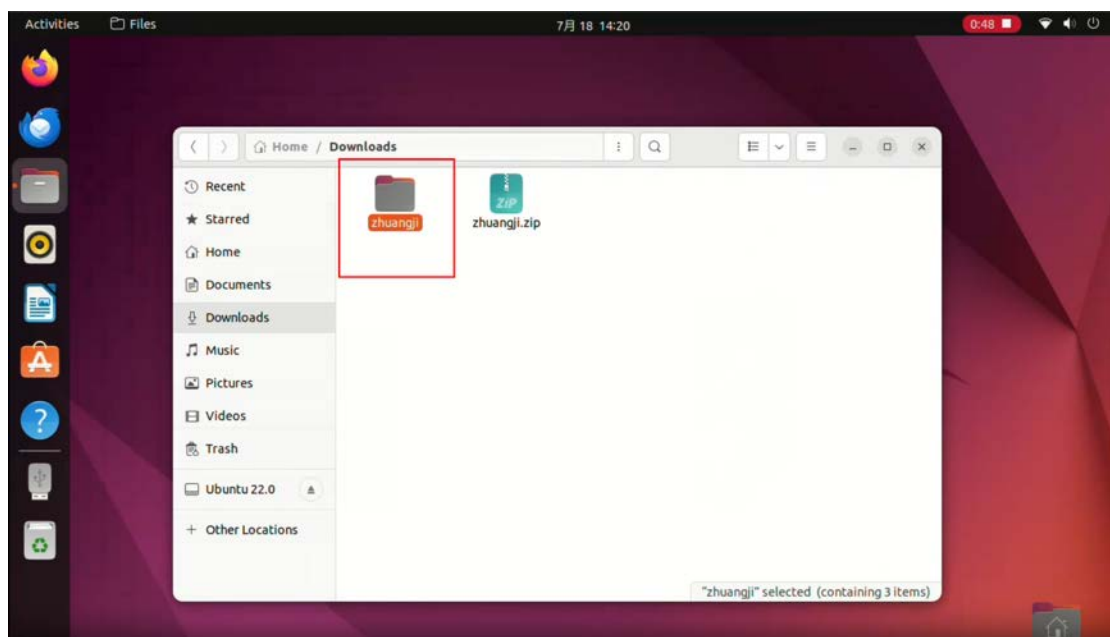
鼠标左击进入【Downloads】目录，然后在空白处右击，并左击展开的【Paste】



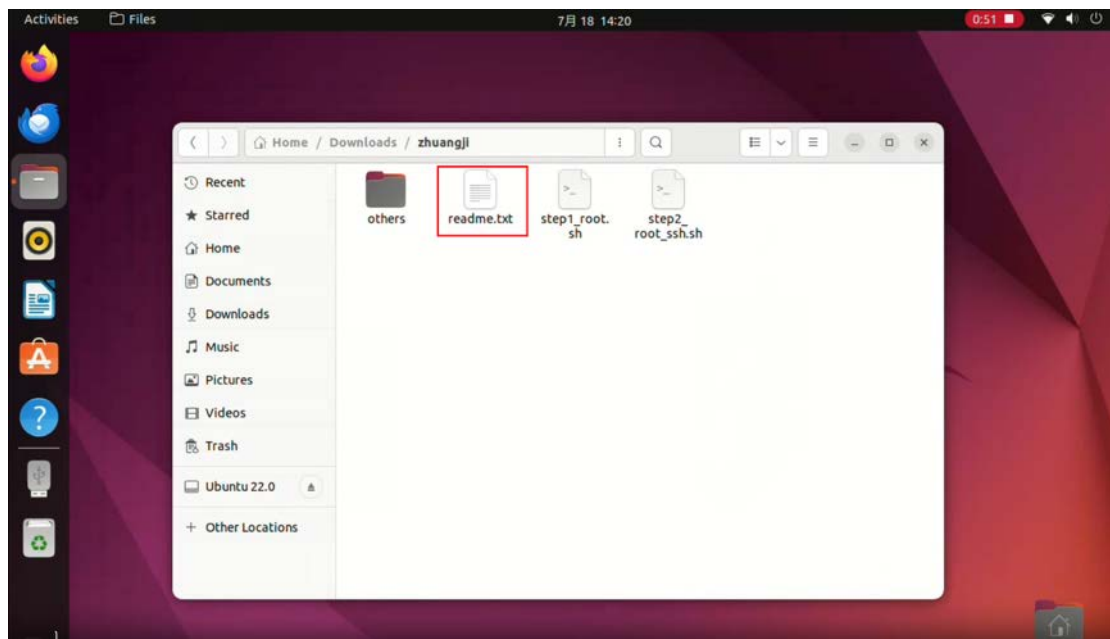
将文件复制至此处后，鼠标右击文件，然后左击【Extract Here】，将文件在当前目录下解压



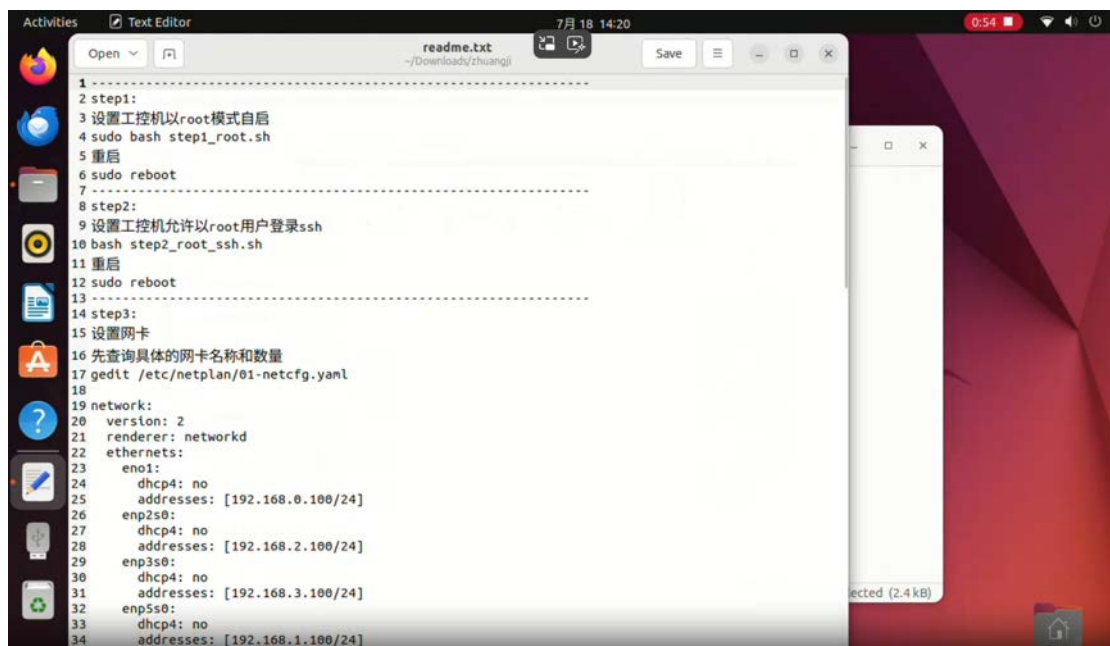
双击解压后的“zhuangji”文件夹



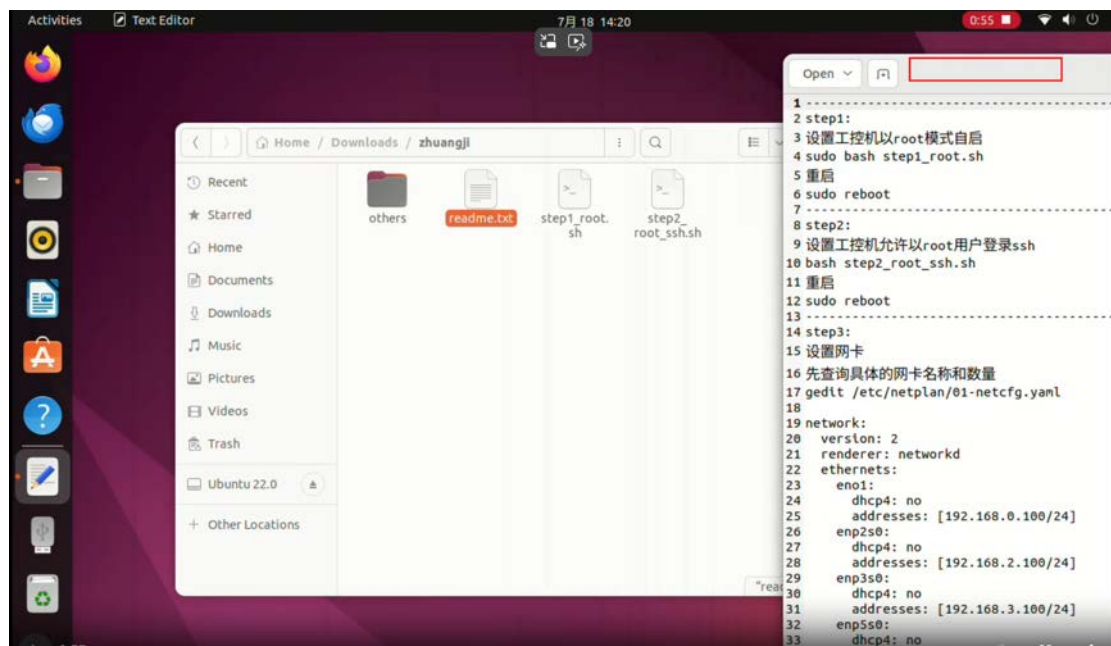
进入文件夹后，鼠标双击打开“readme.txt”文件



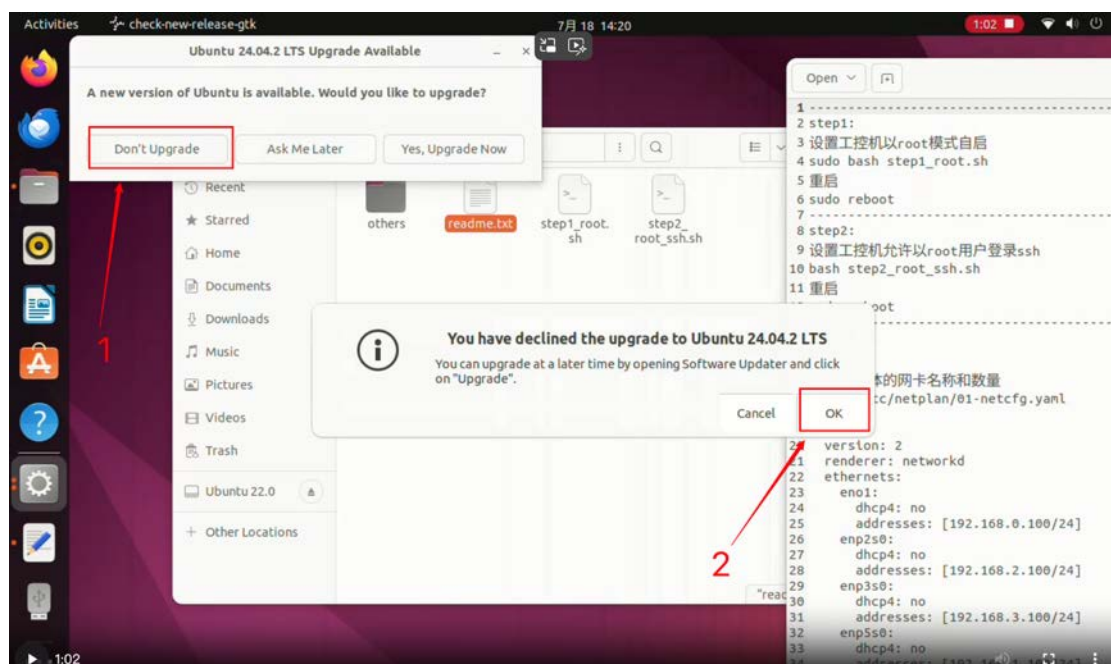
打开文件后如下图所示



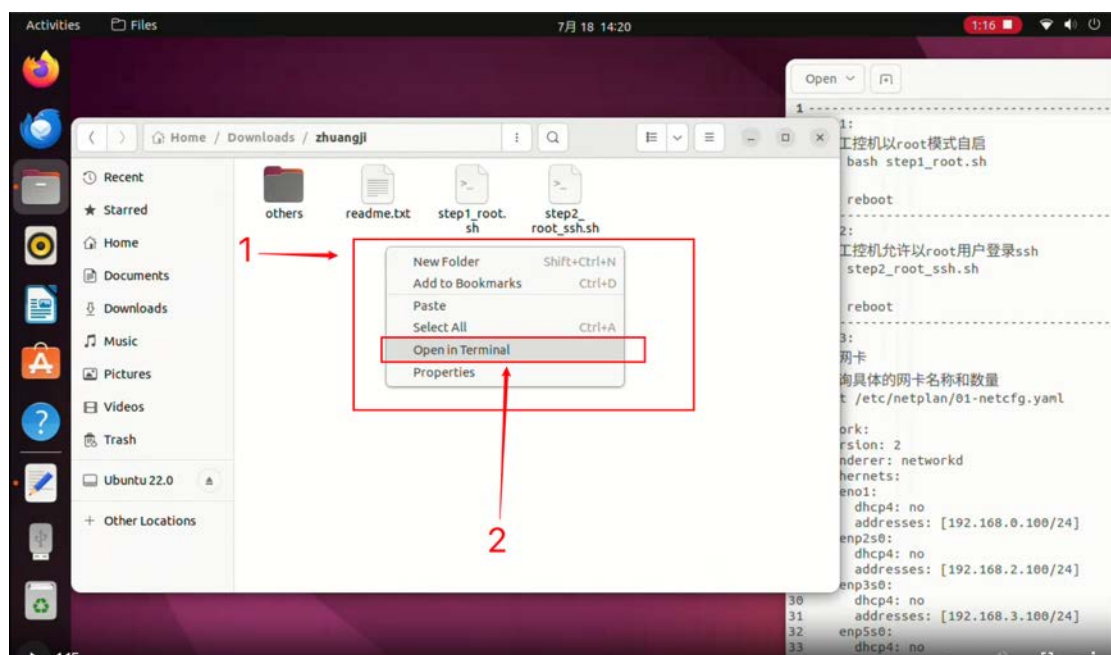
可以使用鼠标左键按住如下图所示位置进行拖拽，拖拽至如下合适的位置



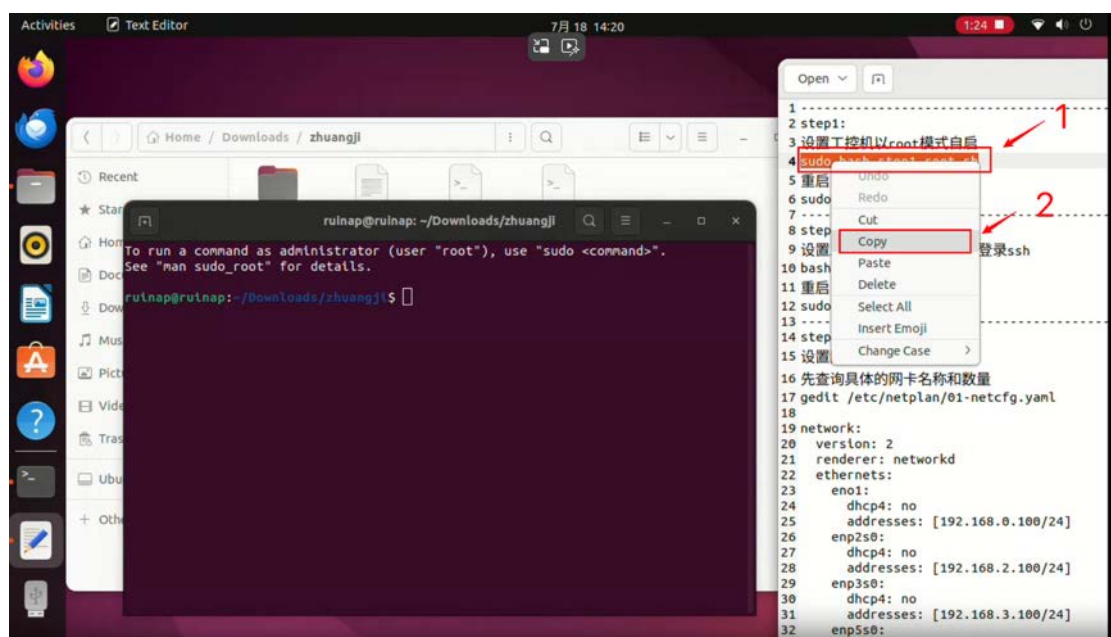
如果出现了如下图左上角的系统升级建议弹窗，鼠标左击【Don't Upgrade】然后左击【OK】，取消升级



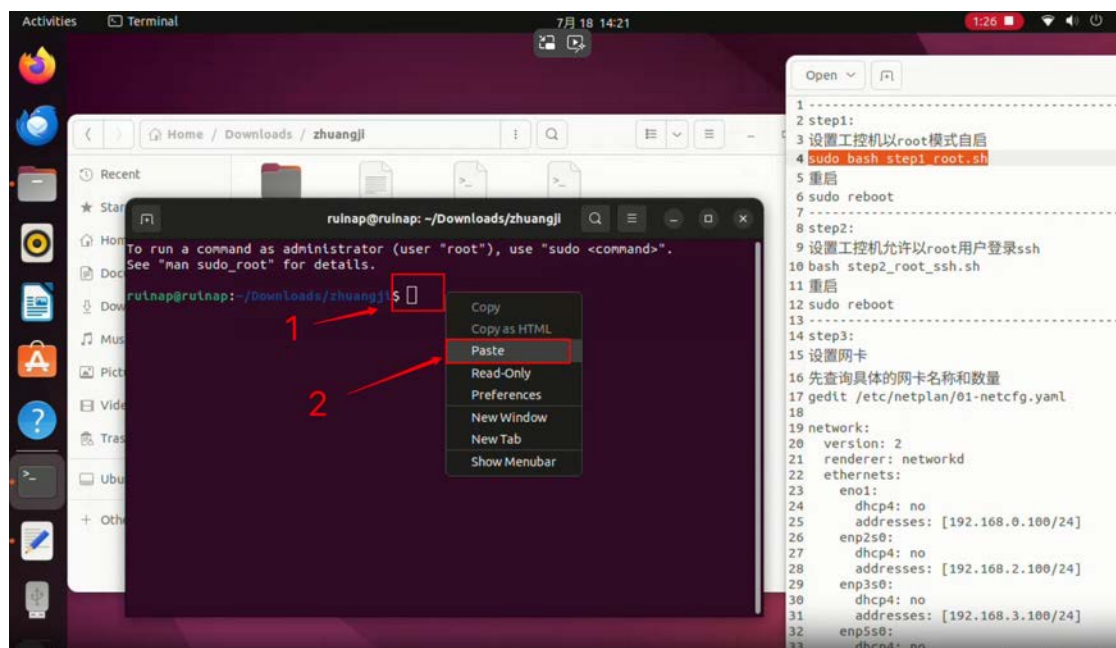
鼠标在“zhuangji”文件夹中，空白处右击，然后左击【Open in Terminal】



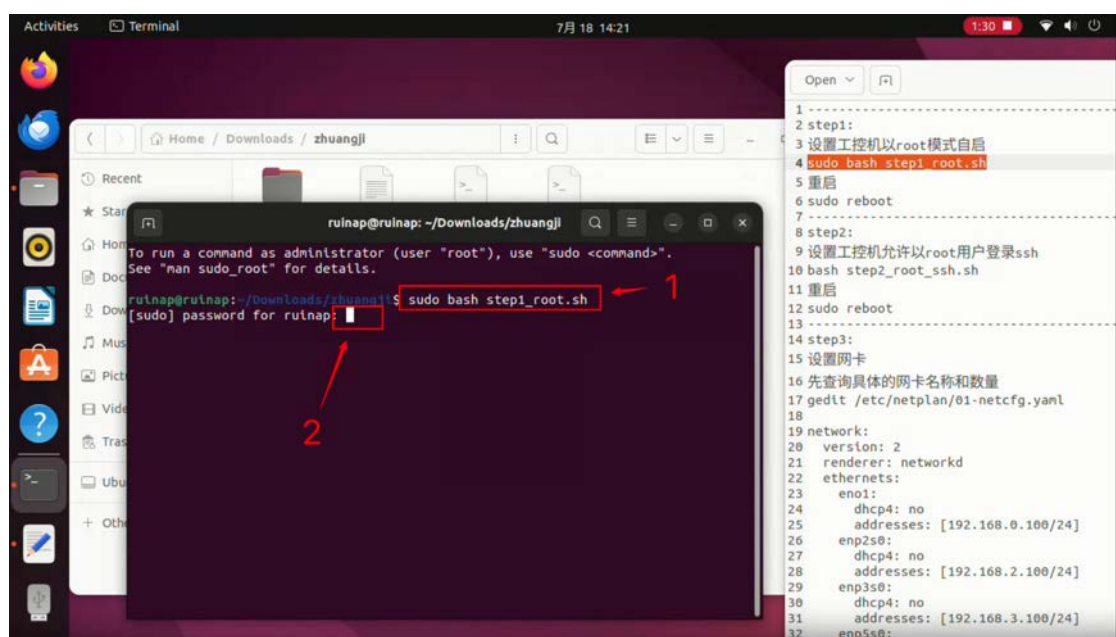
打开如下所示的终端页面后，在如下右侧的 readme 文件中，鼠标左击按住并选中“sudo bash step1_root.sh”内容，鼠标右击然后弹窗后左击【Copy】



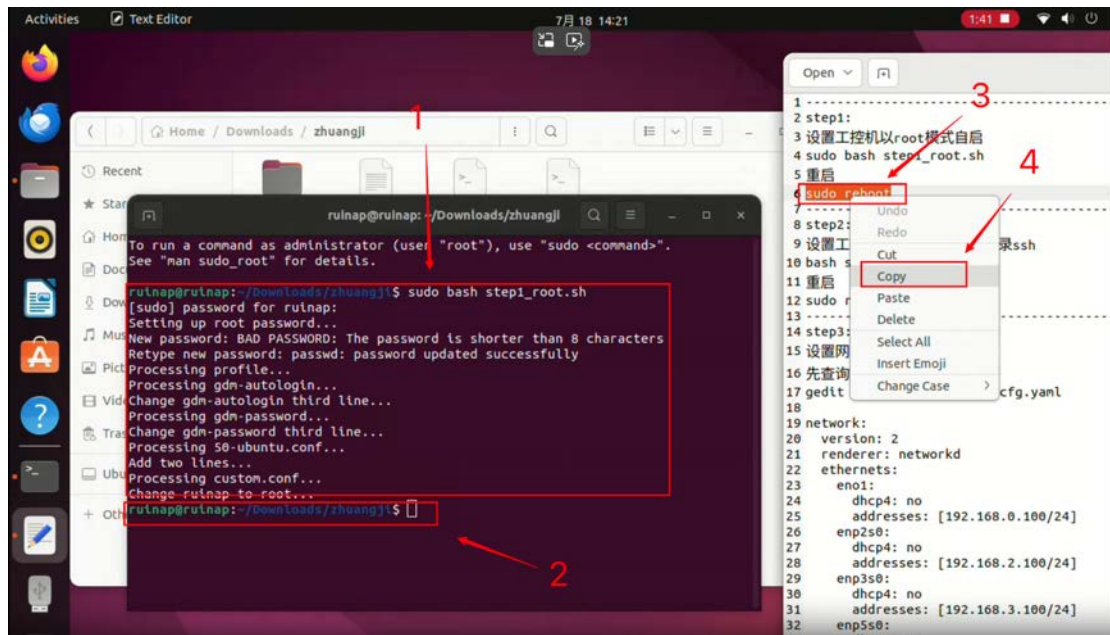
回到终端页面中，在 1 处鼠标右击，然后左击【Paste】将内容复制到此处



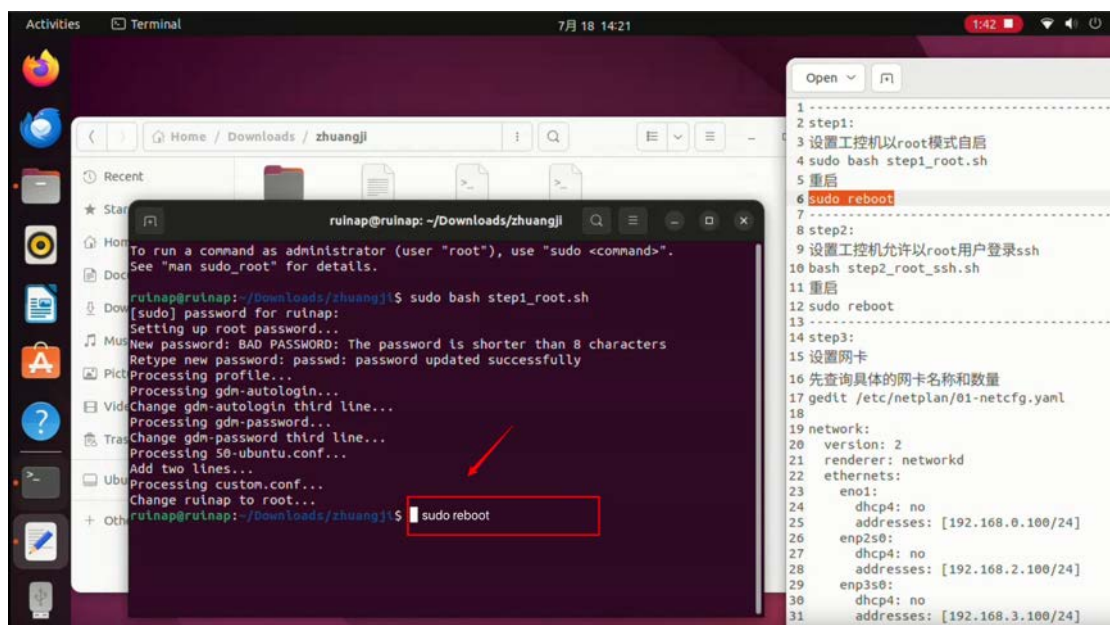
复制如 1 所示的内容后，键盘按下【回车键】，需要输入密码，输入之前安装系统时设置的密码“1”即可，**输入的密码看不见，输入一次即可**，输入后再次按下【回车键】



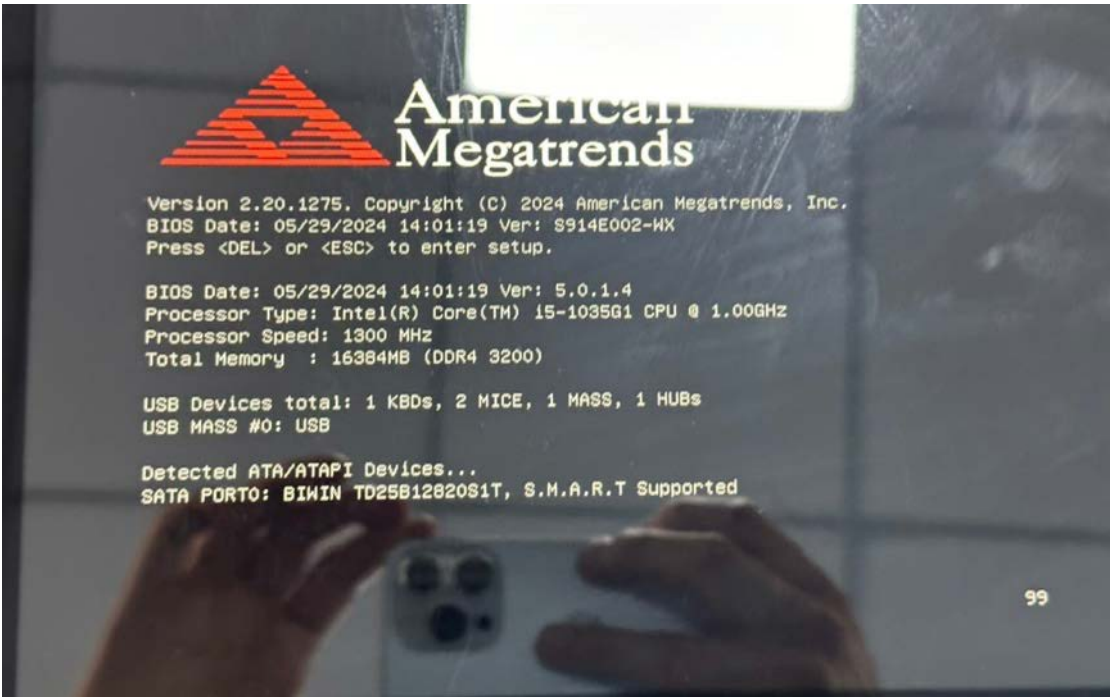
终端显示如下图所示的一系列内容后，会重新显示如 2 处的跳动光标，此时说明执行完毕。然后回到右侧“readme”文件中，鼠标左击按住选择“sudo reboot”内容，鼠标右击后，左击【Copy】



回到终端页面，在跳动的光标处，鼠标右击，然后左击【Paste】将内容复制到此处，复制后，按下键盘【回车键】，系统会重启

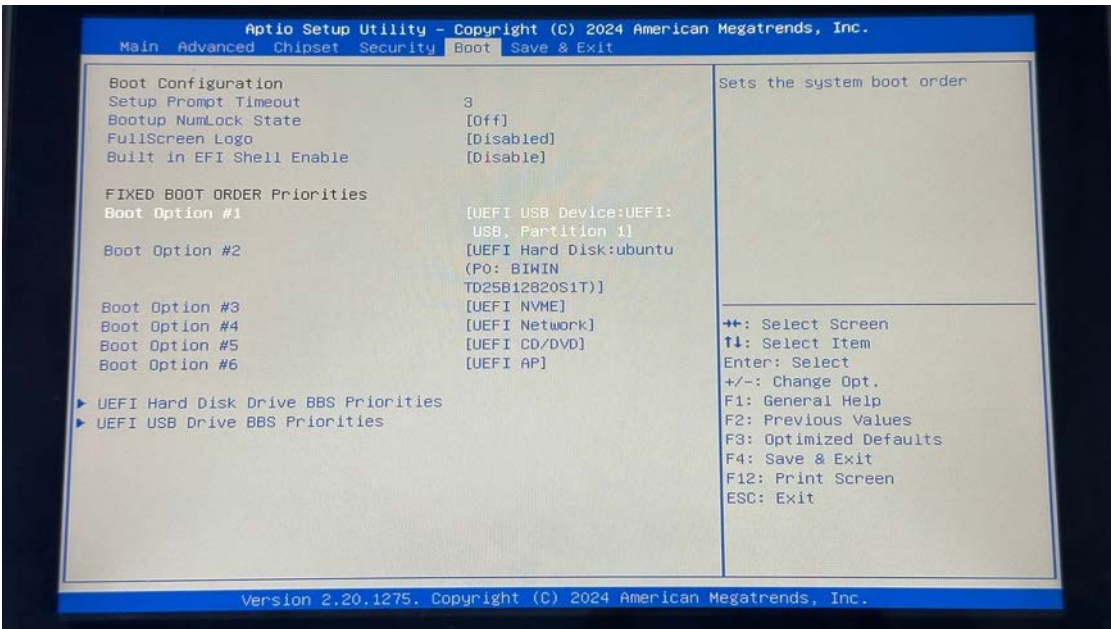


重启后出现如下界面后，狂按键盘的【del】或者【DELETE】按键，会进入 BIOS 界面

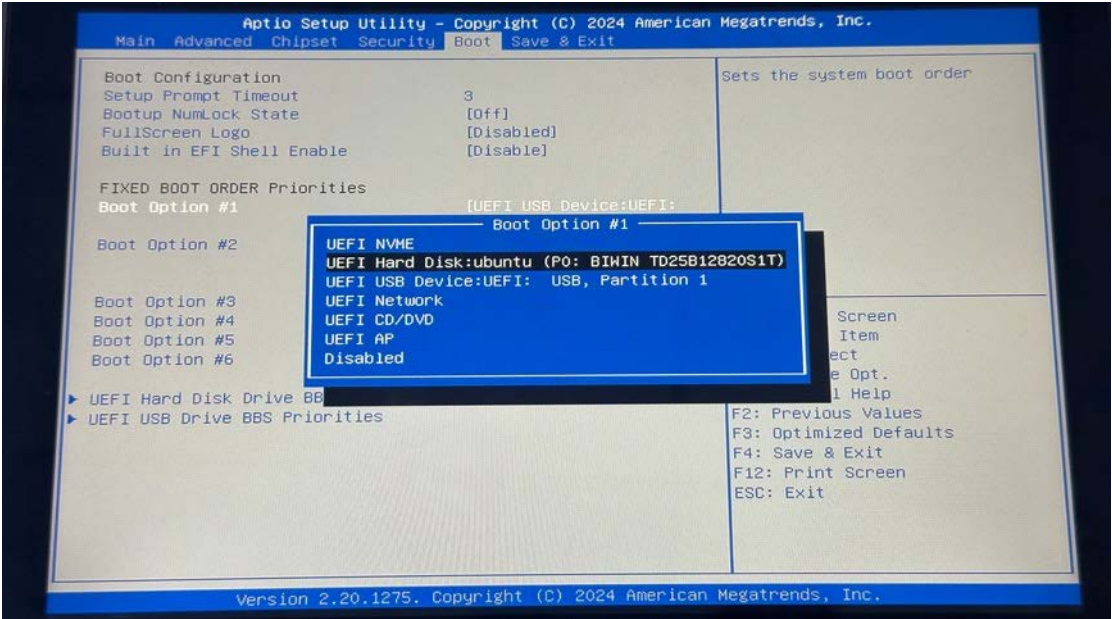


使用键盘【方向键的左右】切换顶层菜单栏，切换至 Boot 选项页面

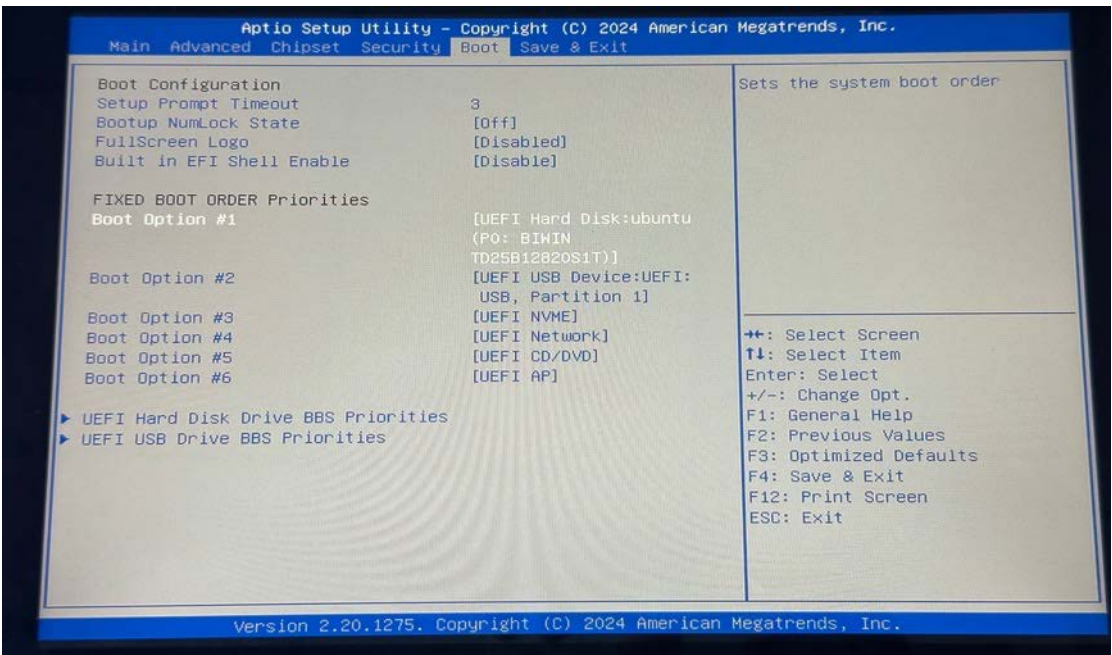
进入 Boot 页面后，使用键盘【方向键的上下】切换至【Boot Option #1】，然后使用键盘【回车键】



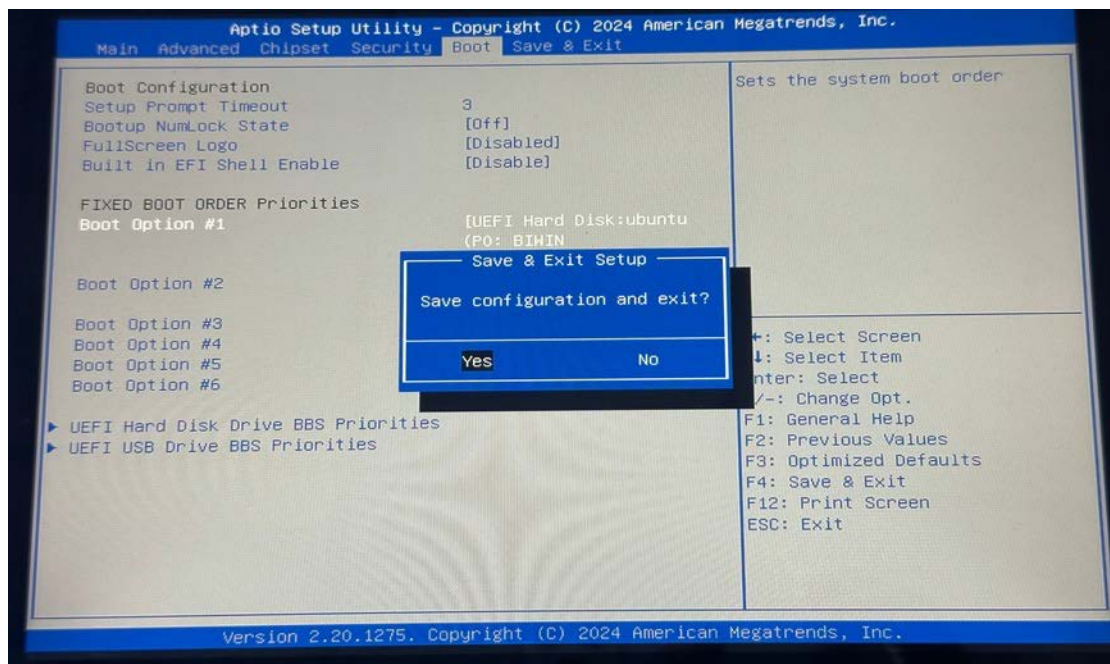
弹窗后继续使用键盘【方向键的上下】切换至【Hard Disk: ubuntu 相关字眼】的选项，然后再次使用键盘【回车键】



操作完后【Boot Option #1】会变为【Hard Disk: ubuntu 相关字眼】的选项

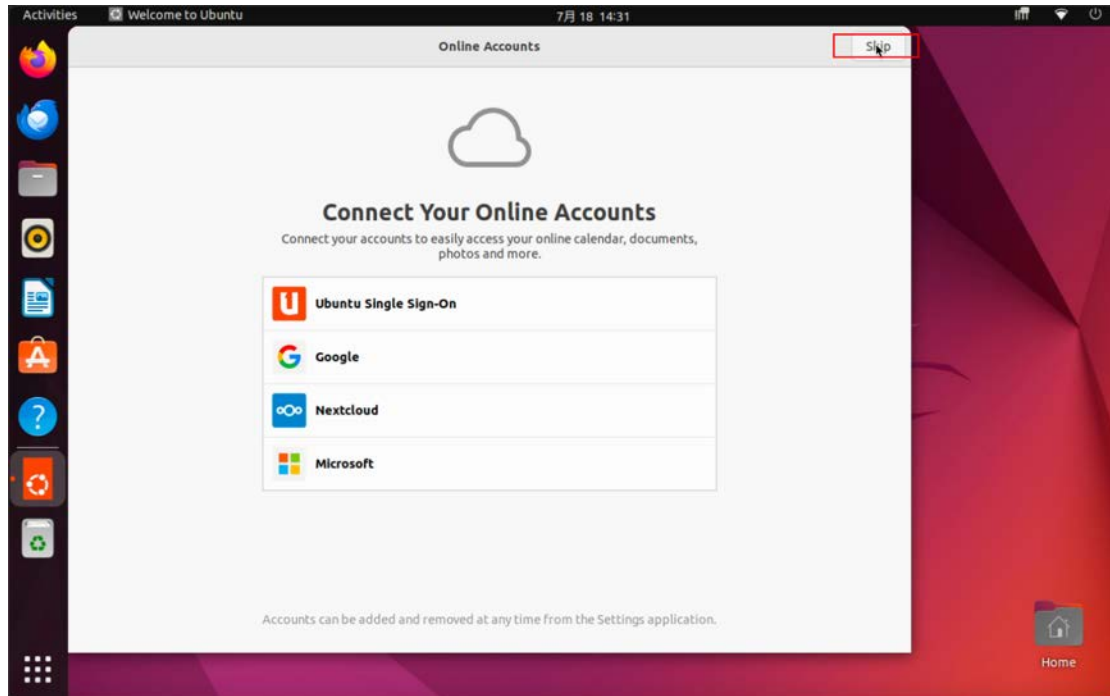


修改完成后，使用键盘的【F4】（部分键盘需要同时按下【Fn】 + 【F4】）按键，弹窗提示是否需要保存并退出，使用键盘【方向键的左右】选择【Yes】选项后使用键盘【回车键】

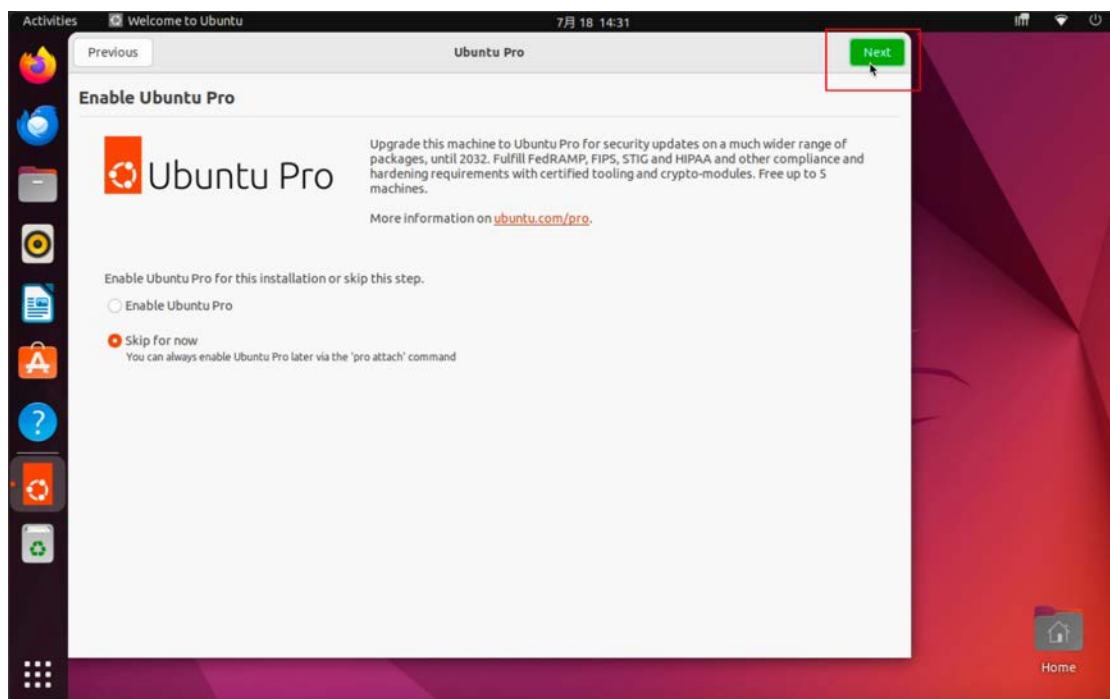


3. 设置 ssh

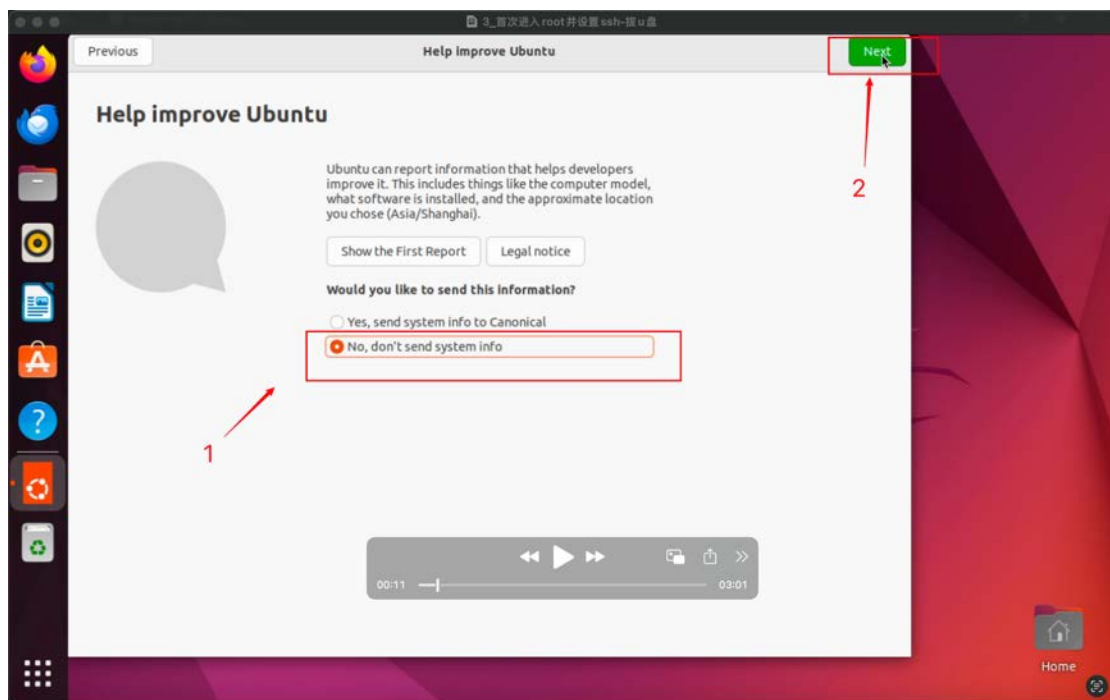
设置完 root 权限启动后，首次进入系统会像新装好系统一样，如下鼠标左击【Skip】



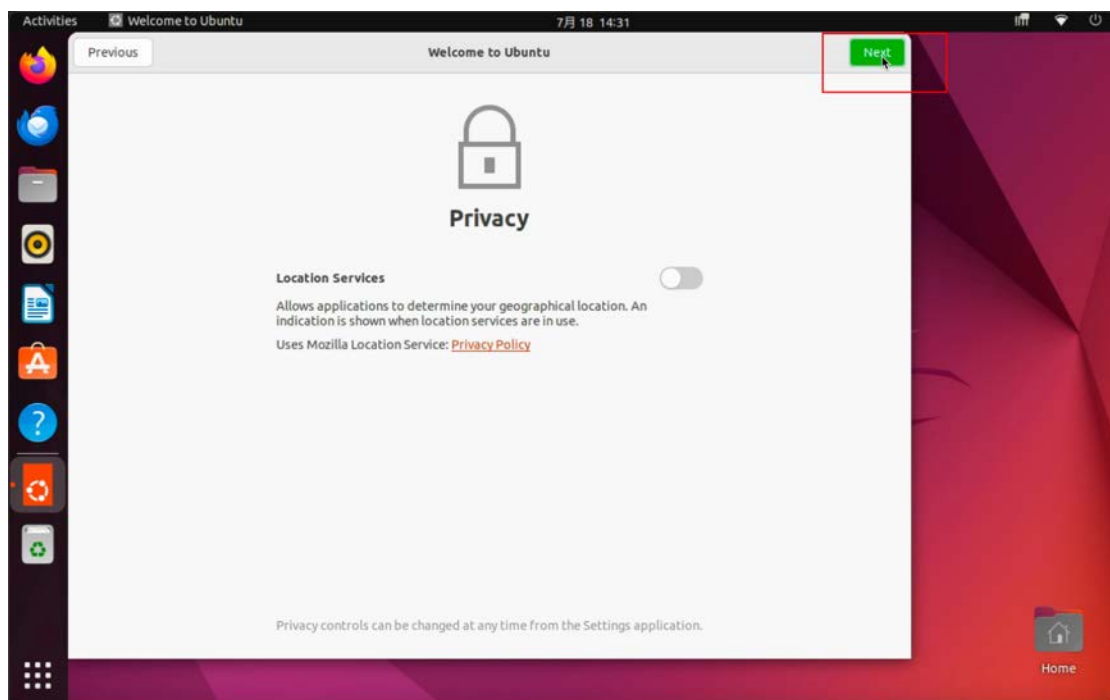
确认如下选项为【Skip for now】后鼠标左击【Next】



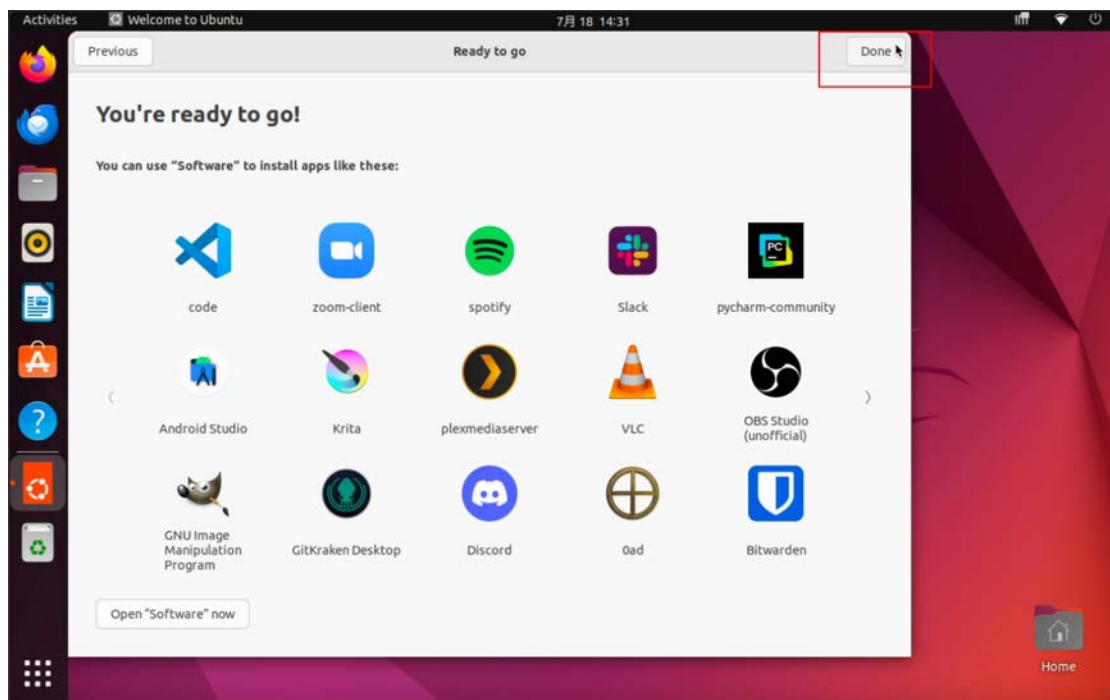
选中如下 1 选项的【No, don't send system info】后，鼠标左击【Next】



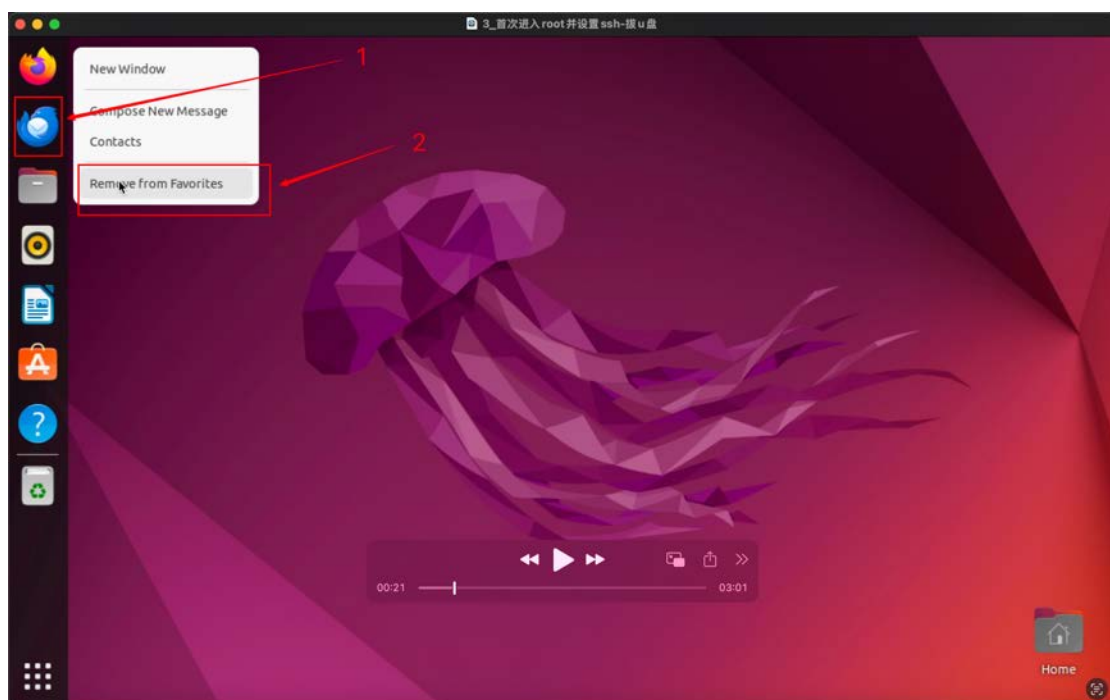
再次左击【Next】



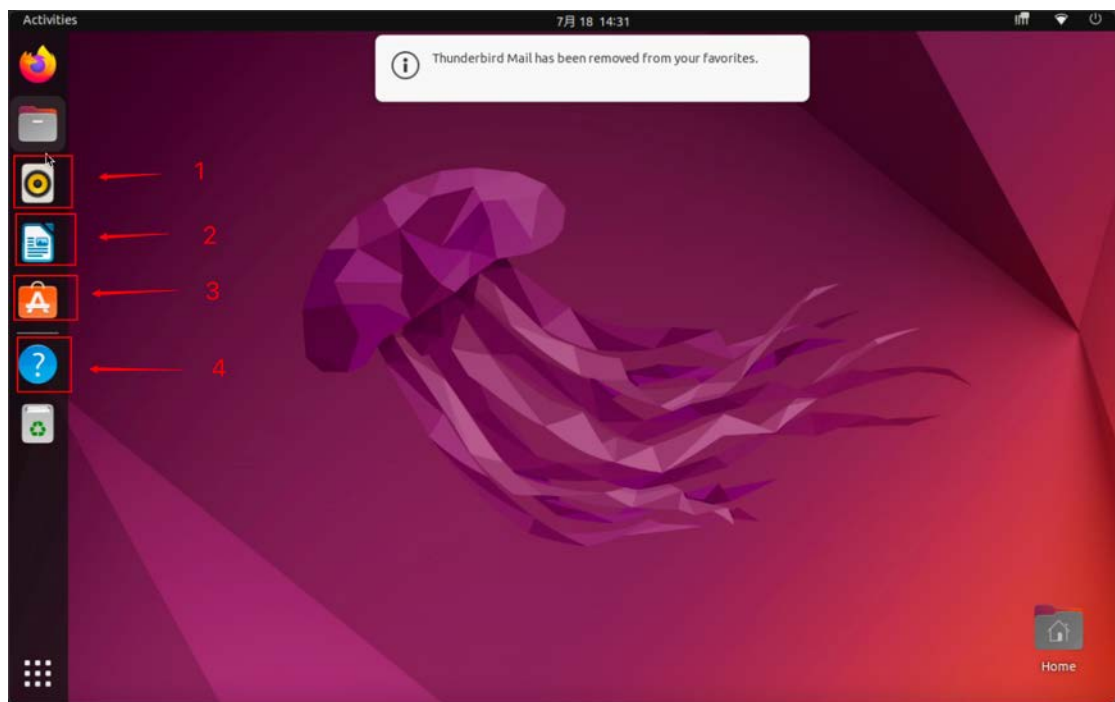
鼠标左击【Done】



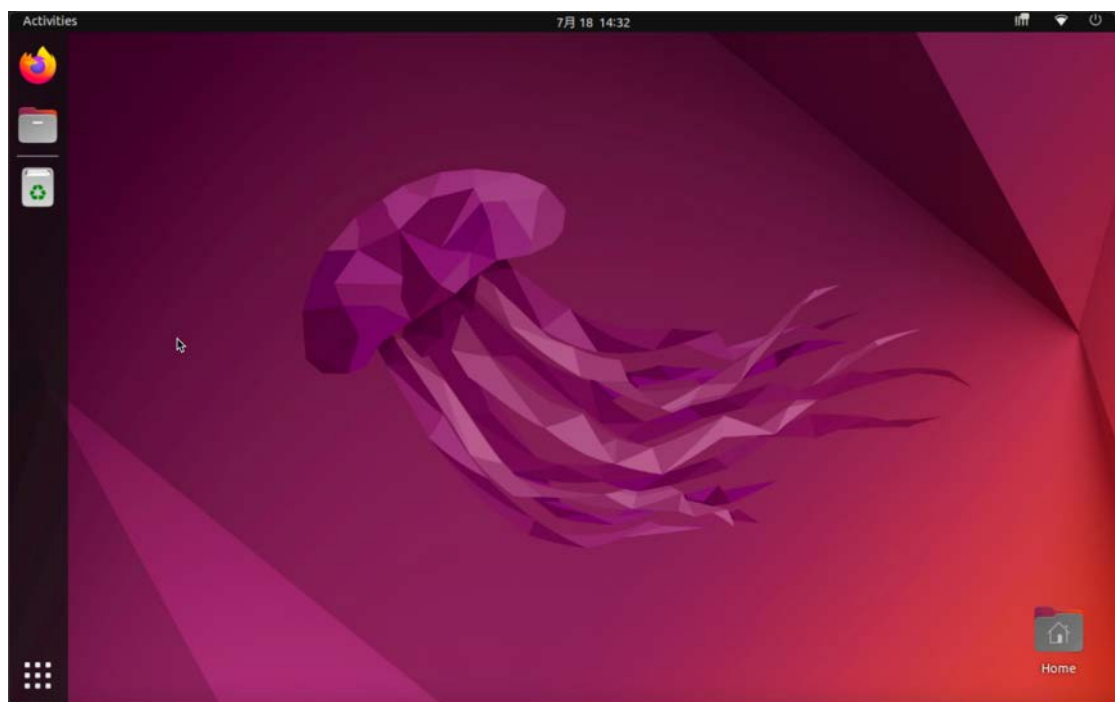
左侧栏图标太多，先移除一部分，如下 1 处鼠标右击然后选择【Remove from Fravorites】



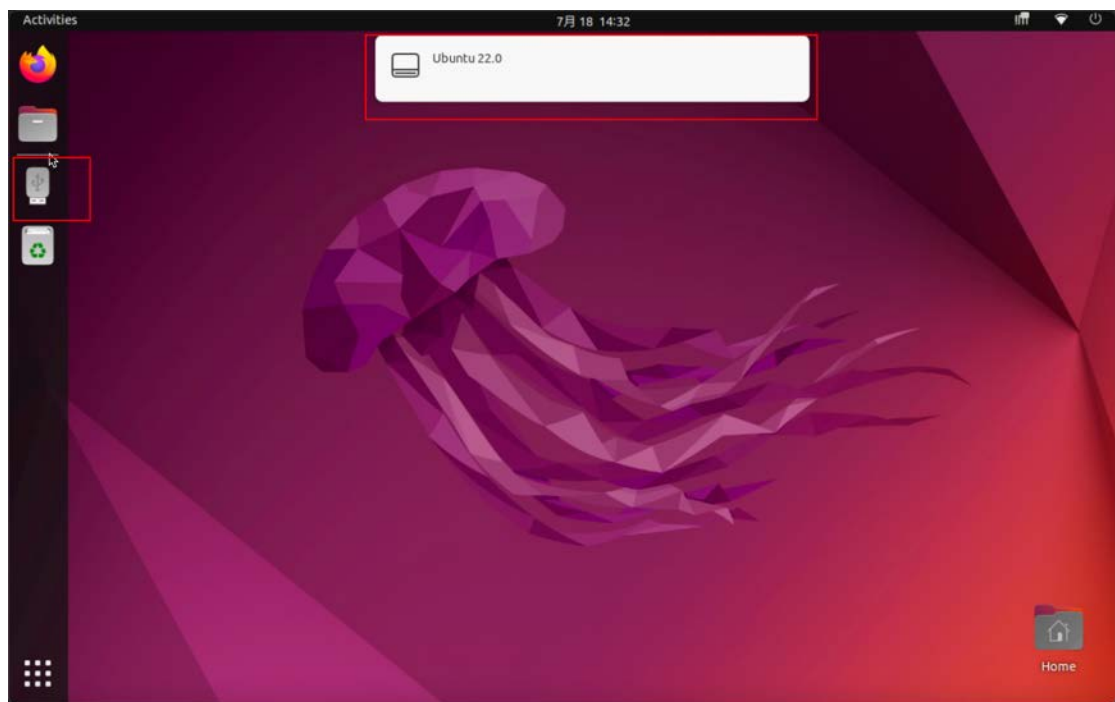
按照前面一样的方式，将如下 1、2、3、4 的图标都移除



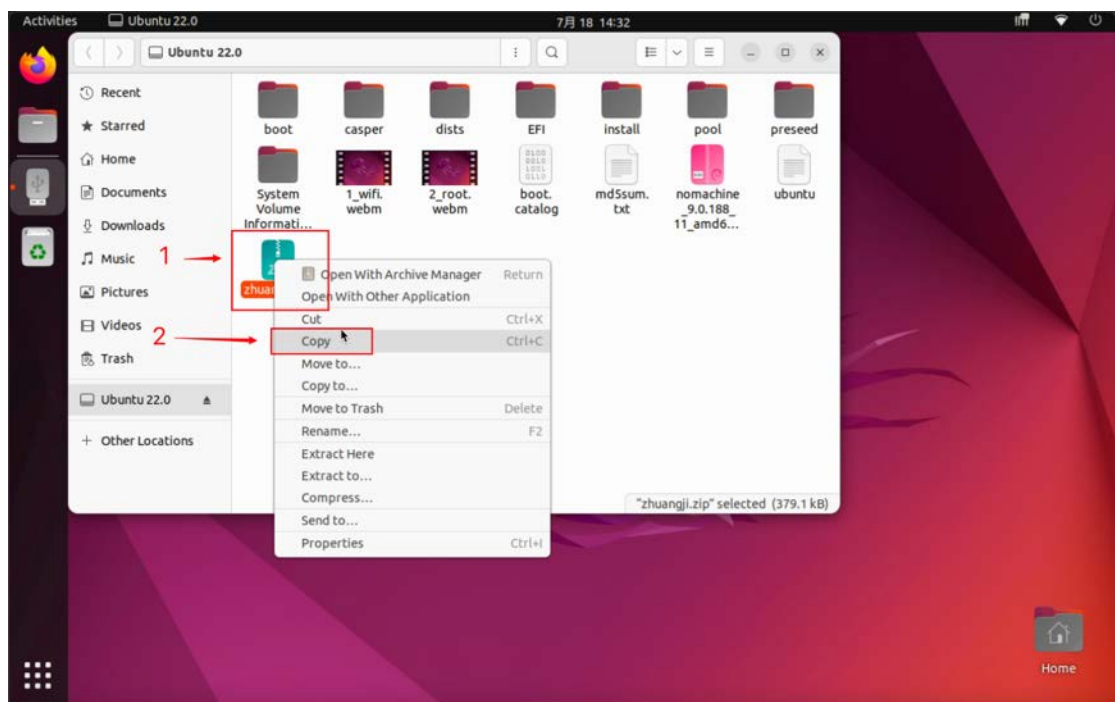
最终如下所示



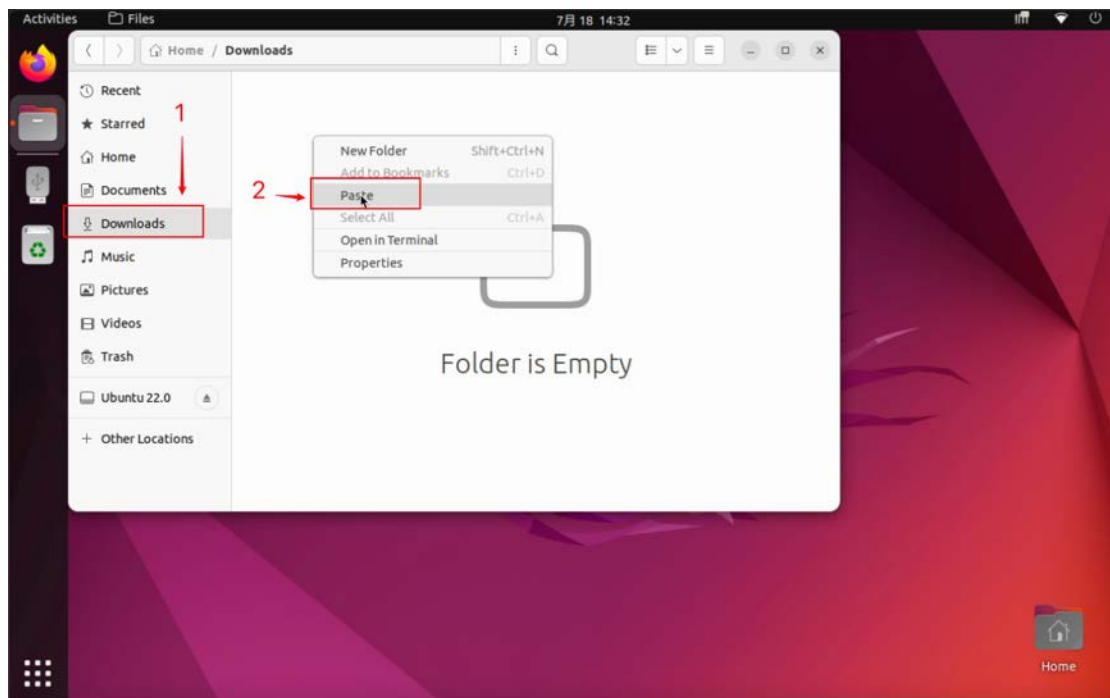
插入之前拷贝了“zhuangji.zip”文件的 u 盘，鼠标左击 1 或 2 进入文件目录



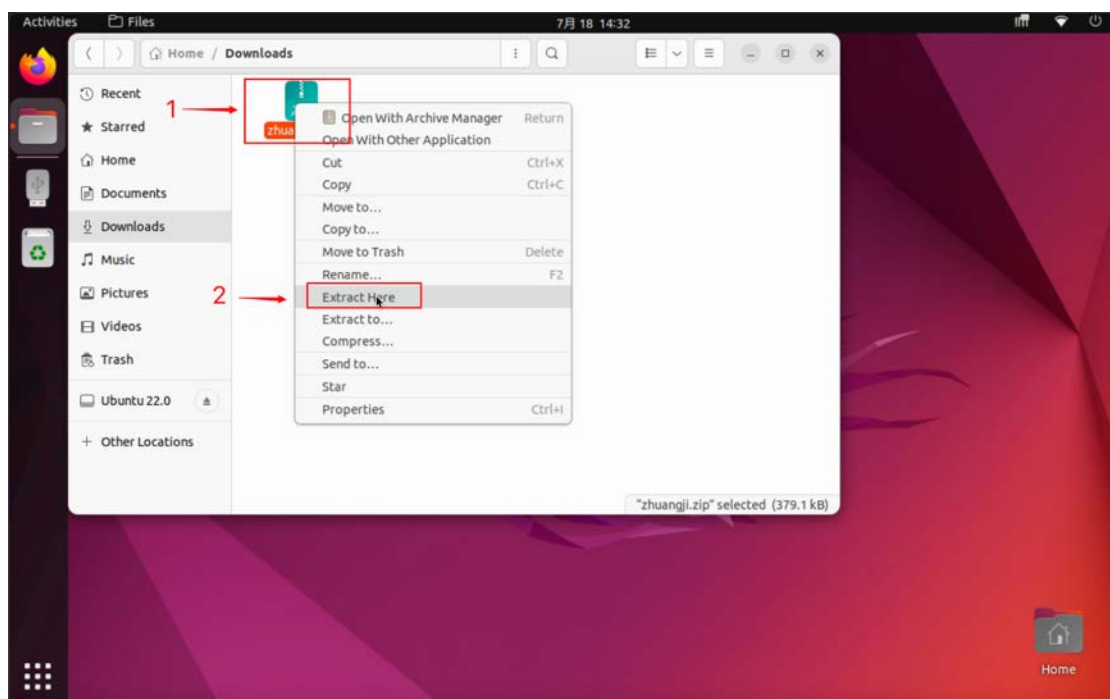
鼠标右击“zhuangji.zip”文件，左击【Copy】



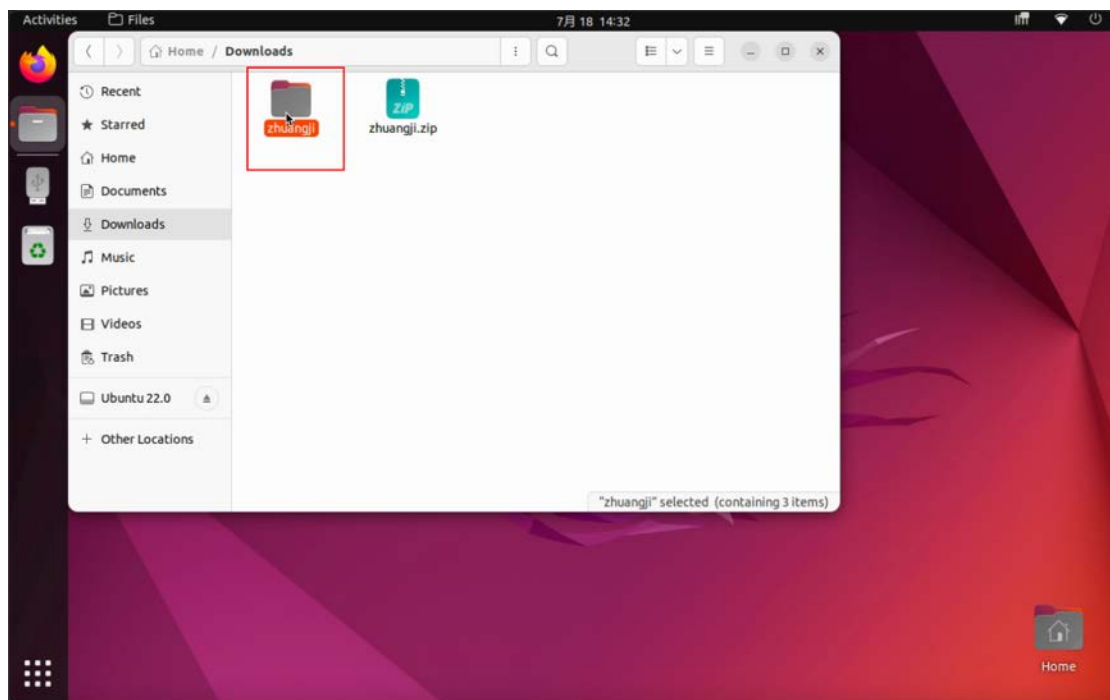
鼠标左击左侧栏【Downloads】进入文件目录，在空白处，鼠标右击，然后左击【Paste】将“zhuangji.zip”文件复制到此处



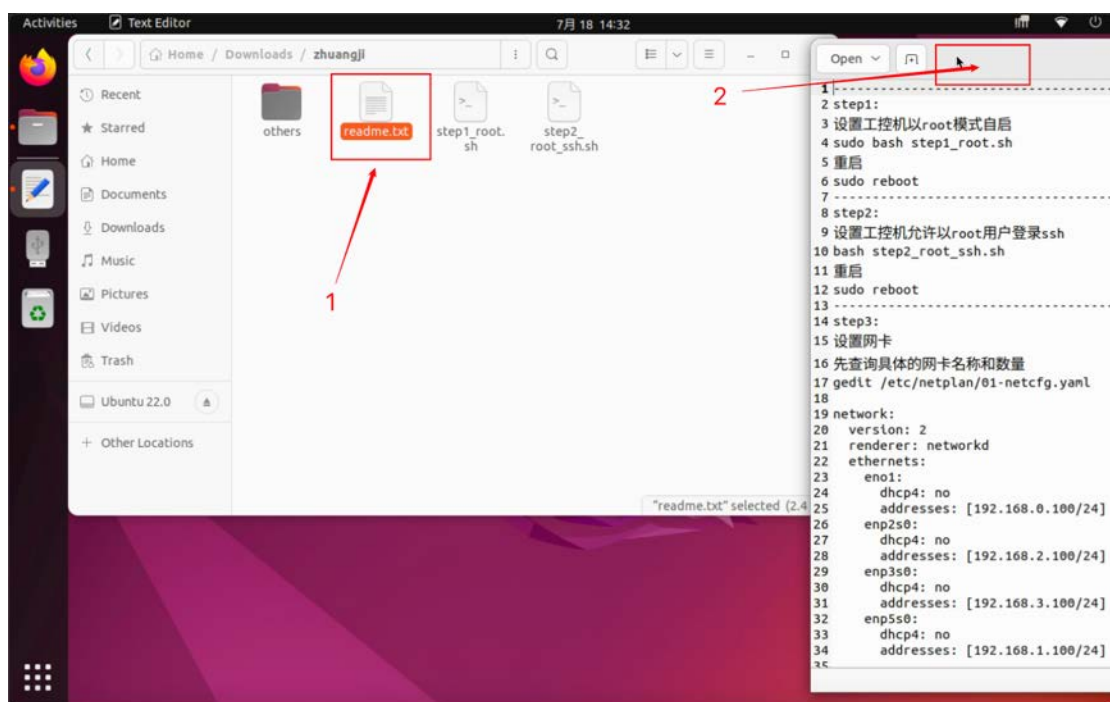
复制后，鼠标右击“zhuangji.zip”文件，然后左击【Extract Here】将文件解压到当前目录



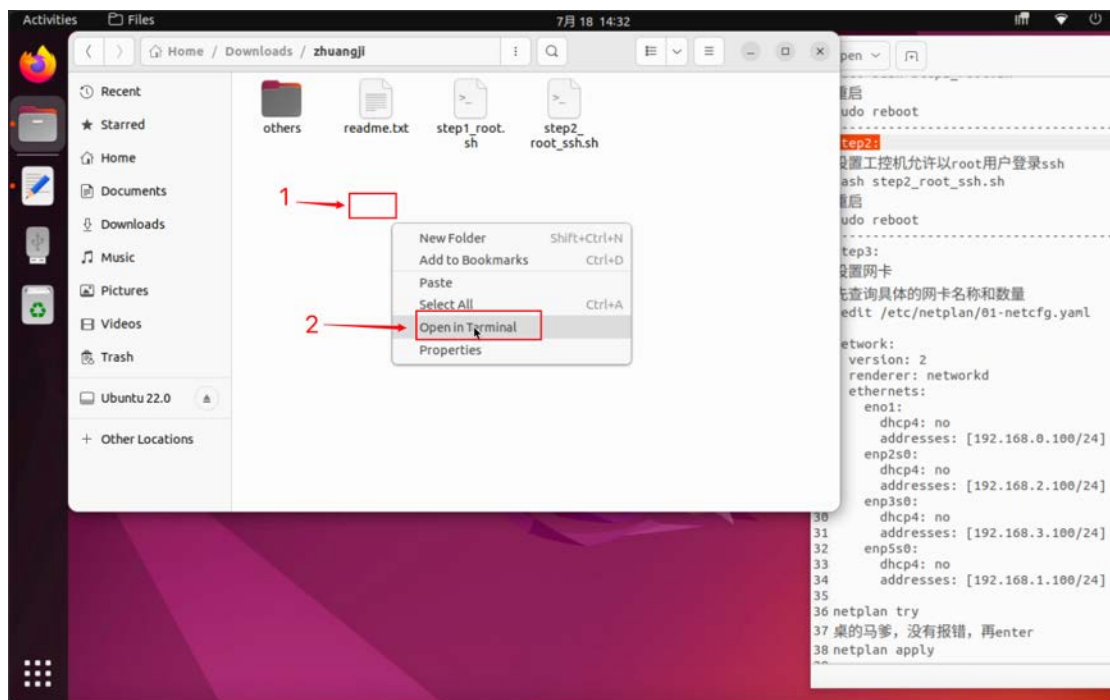
鼠标双击“zhuangji”文件夹进入



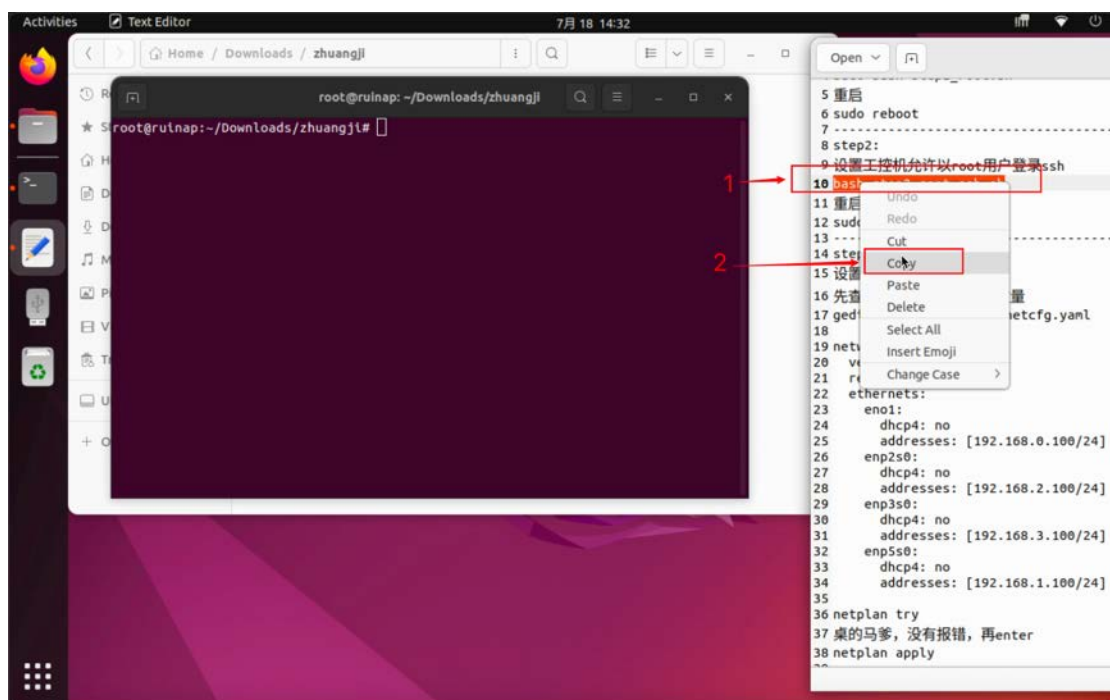
鼠标双击“readme.txt”文件打开，并左击按住 2 处进行拖拽移动至合适位置



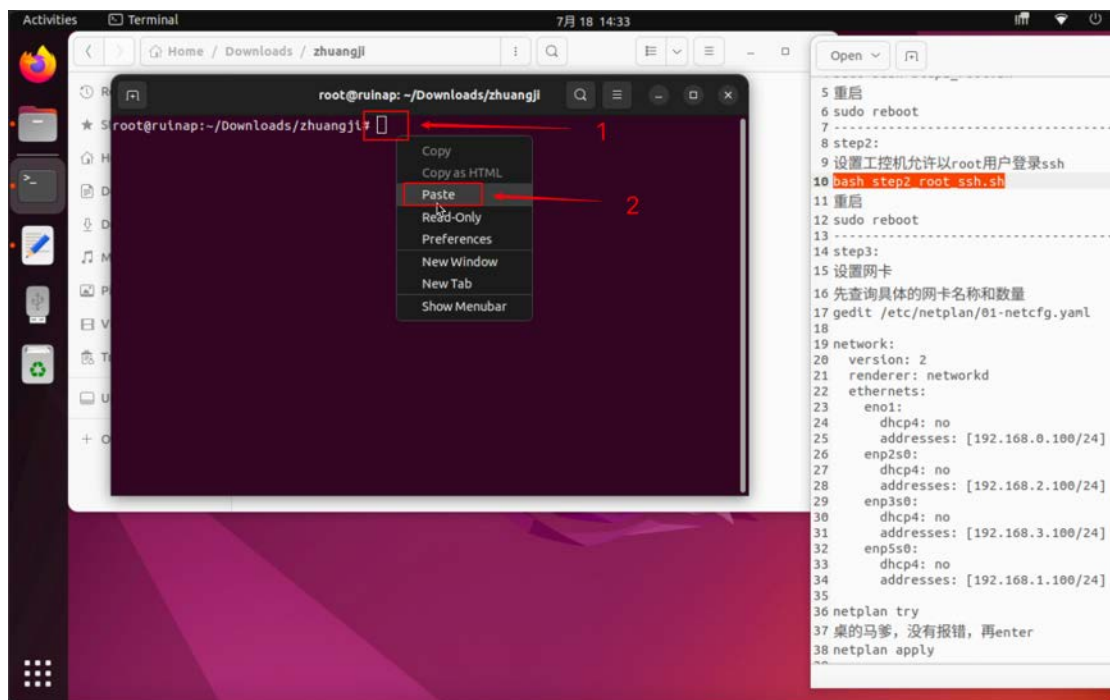
鼠标在文件夹空白处右击，然后左击【Open in Terminal】打开终端



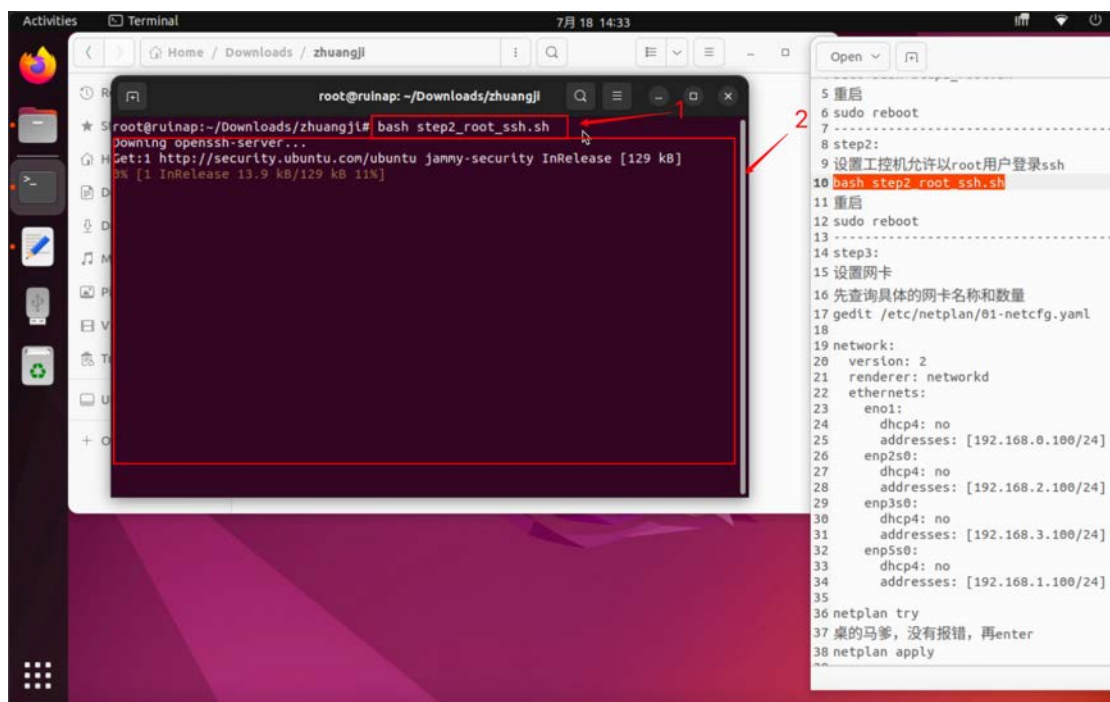
打开终端后，鼠标左击按住“readme.txt”文件中的【bash step2_root_ssh.sh】内容，右击然后左击【Copy】



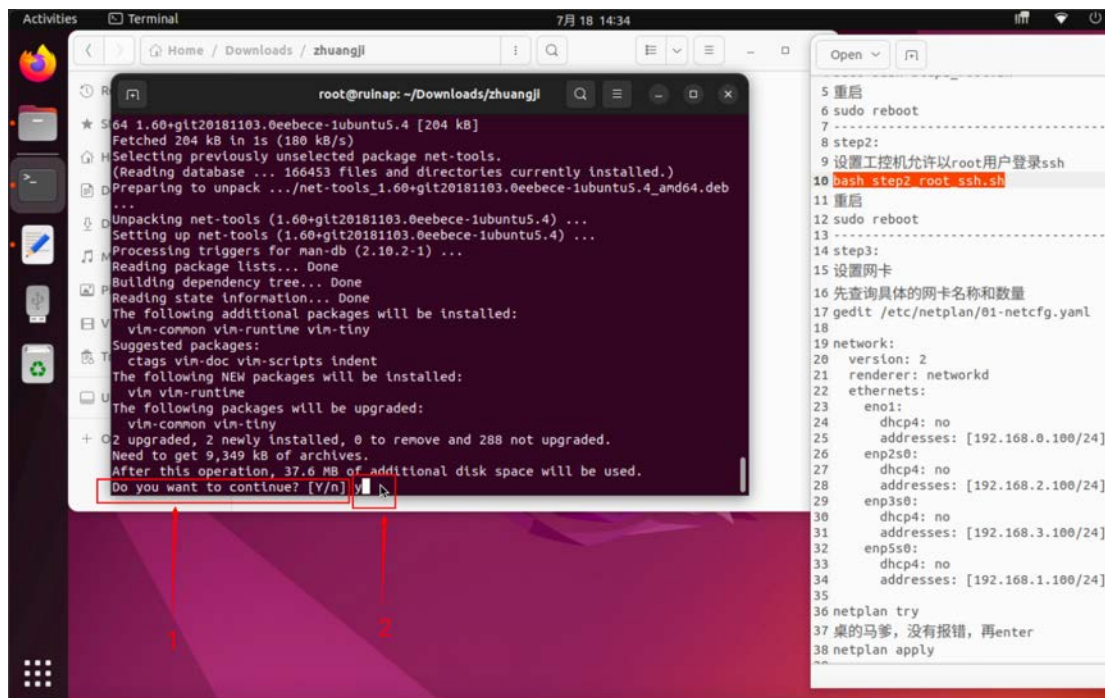
在终端光标处鼠标右击，然后左击【Paste】



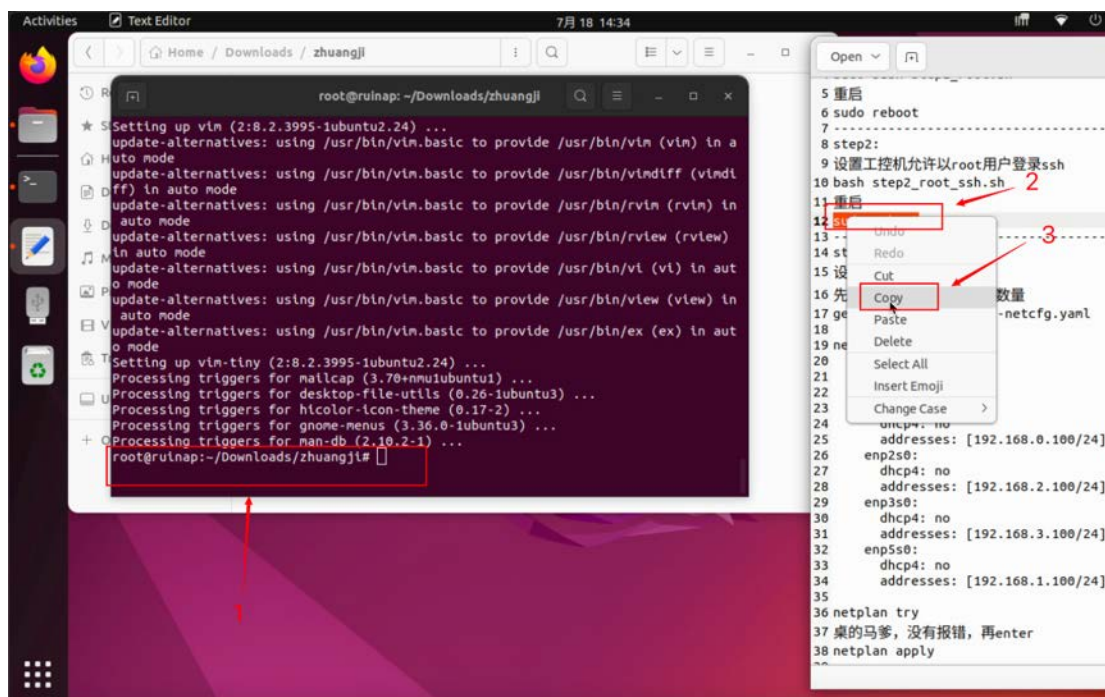
复制内容后，键盘按下【回车键】，终端开始显示内容



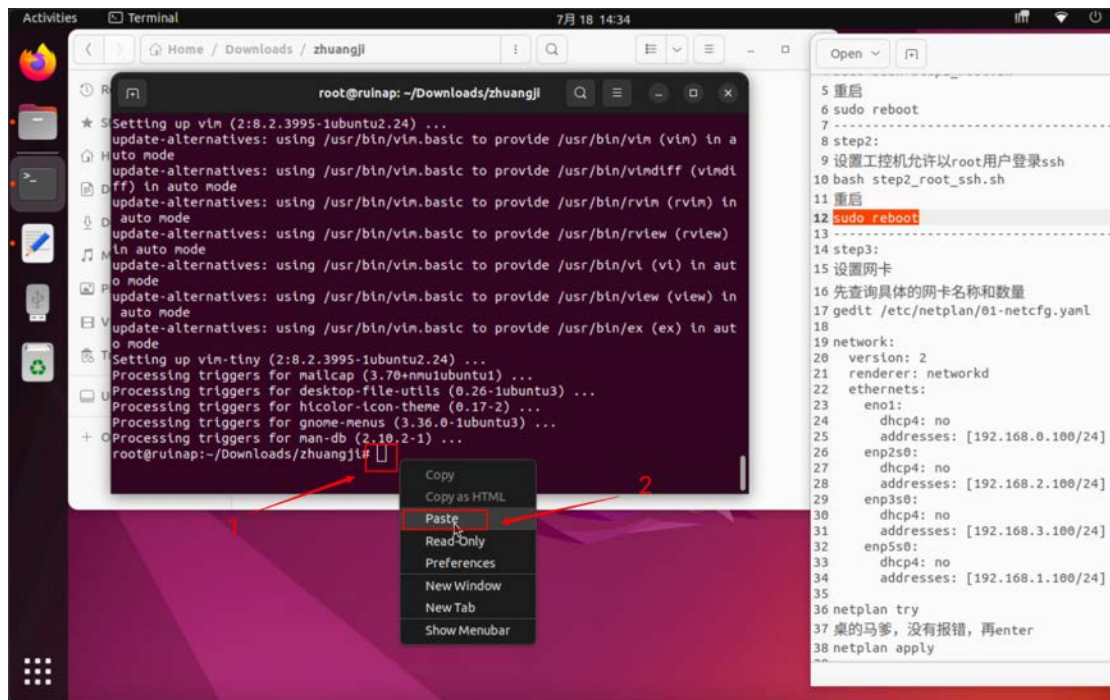
等待终端出现需要确认的如 1 所示的内容时，键盘输入 **[y]**，然后键盘按下 **【回车键】**



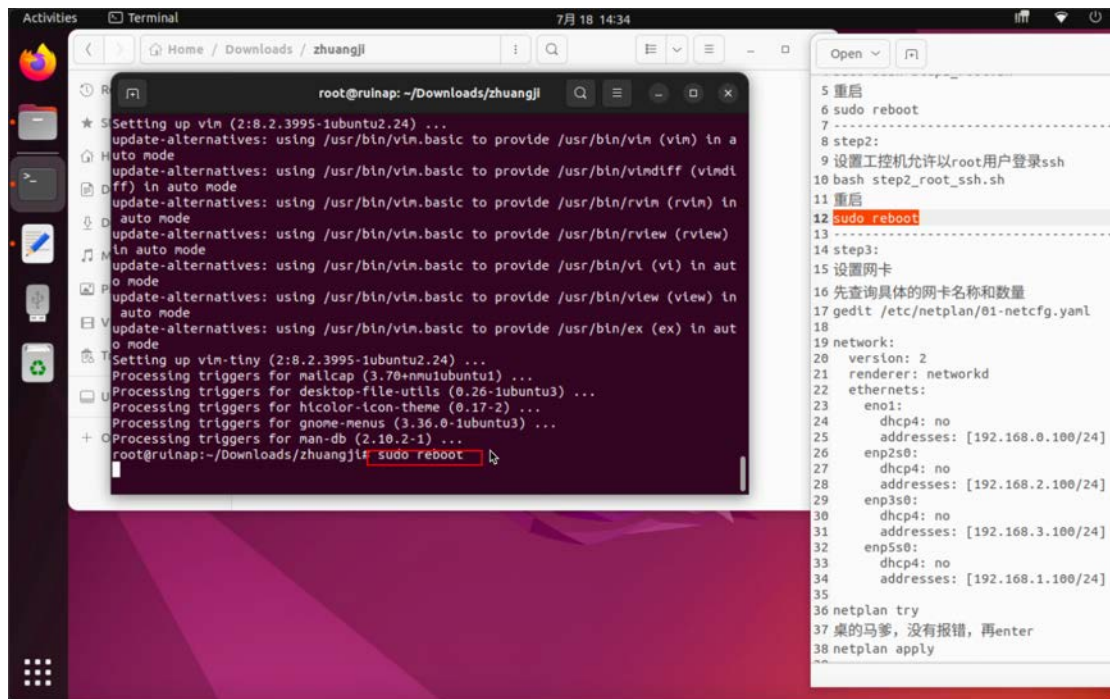
继续等待终端内容打印结束，直到出现“root@”开头的字样，并且出现跳动的光标。然后回到“readme.txt”文件夹中，鼠标左击按下选中 **【sudo reboot】**，右击然后左击 **【Copy】**



回到终端页面，在跳动的光标处，鼠标右击，然后左击【Paste】

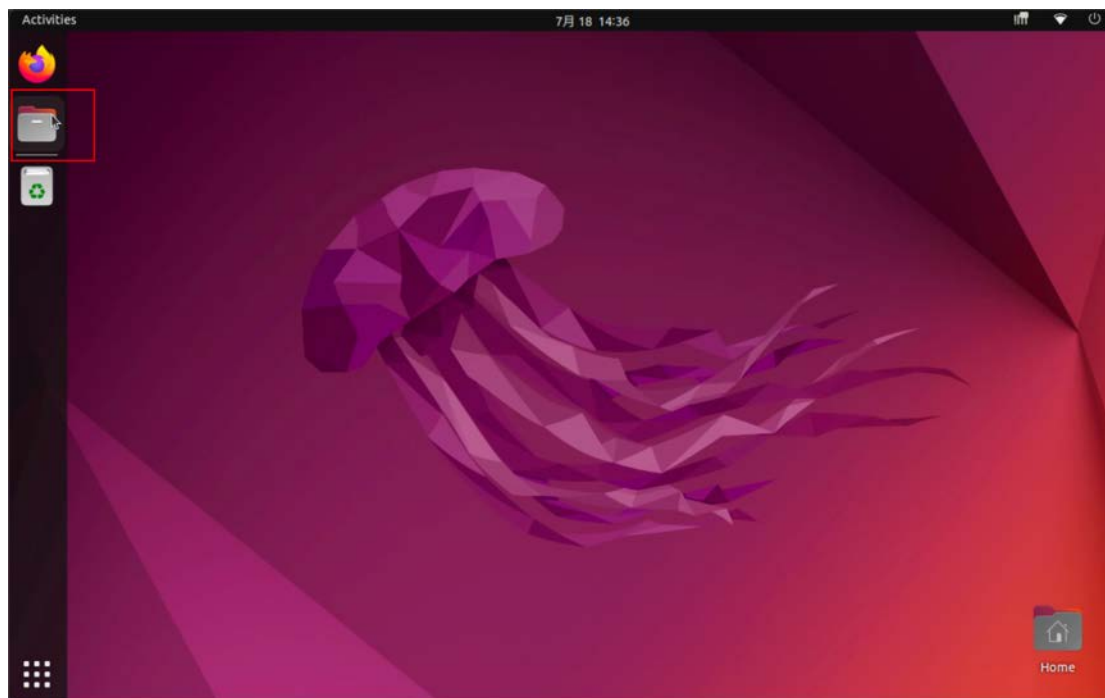


复制内容后，键盘按下【回车键】，工控机会再次重启

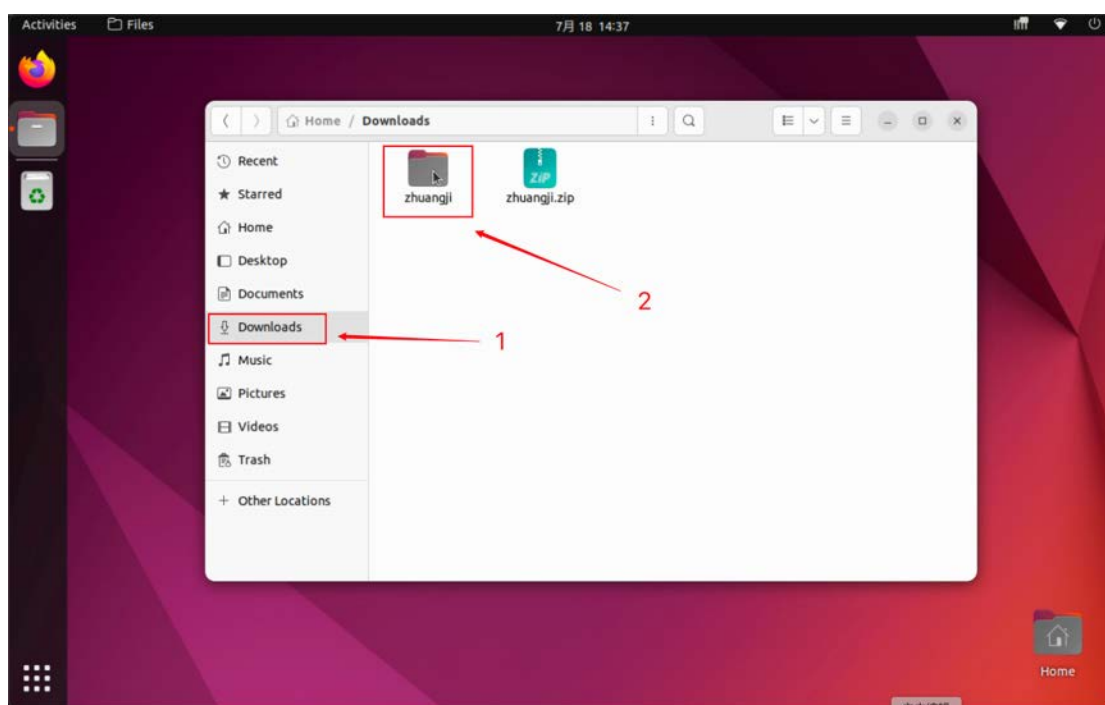


4. 设置有线网卡

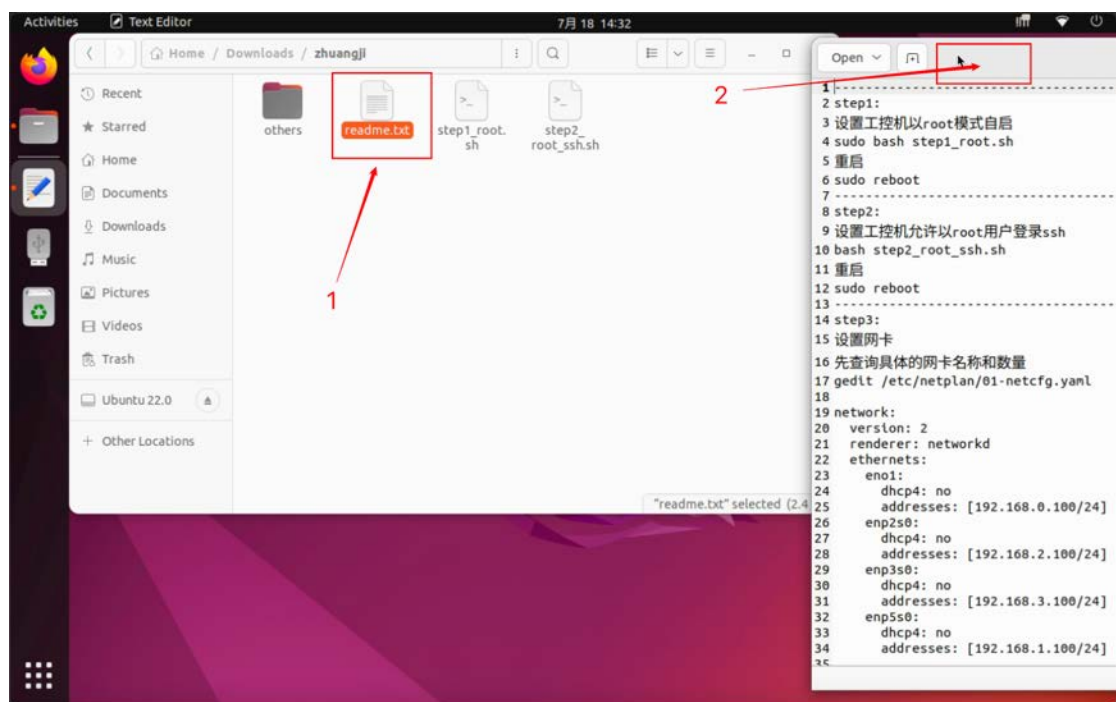
待工控机重启完毕后，开始设置，首先鼠标左击打开文件夹



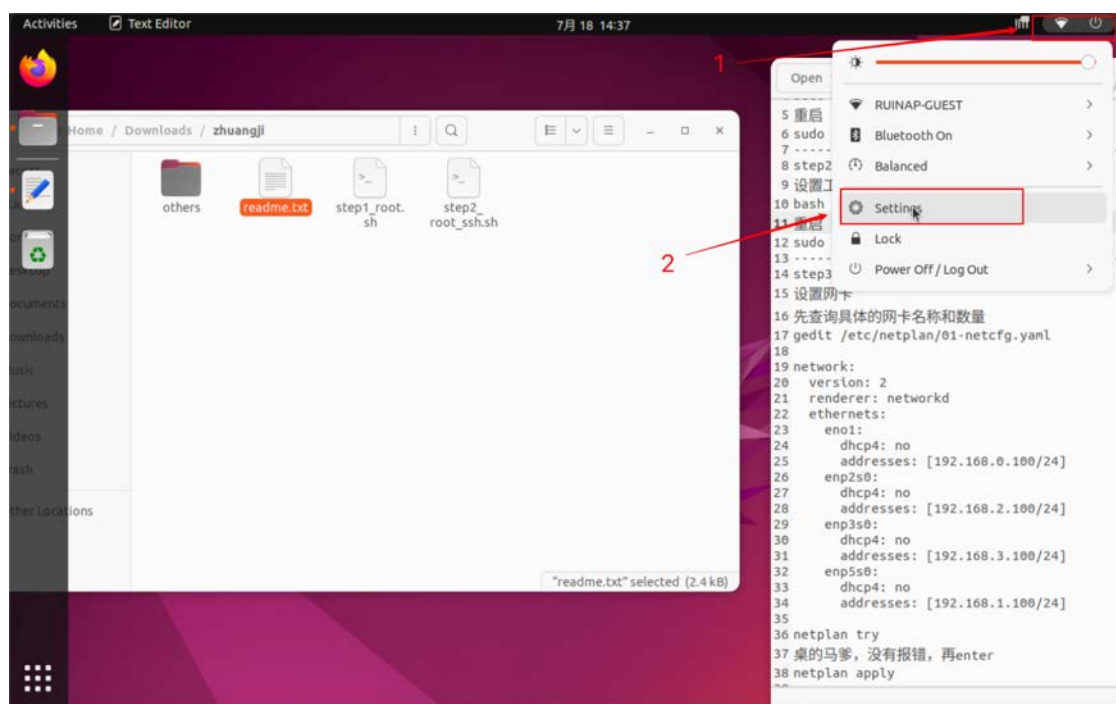
鼠标单击 1 处，然后双击“zhuangji”文件夹进入



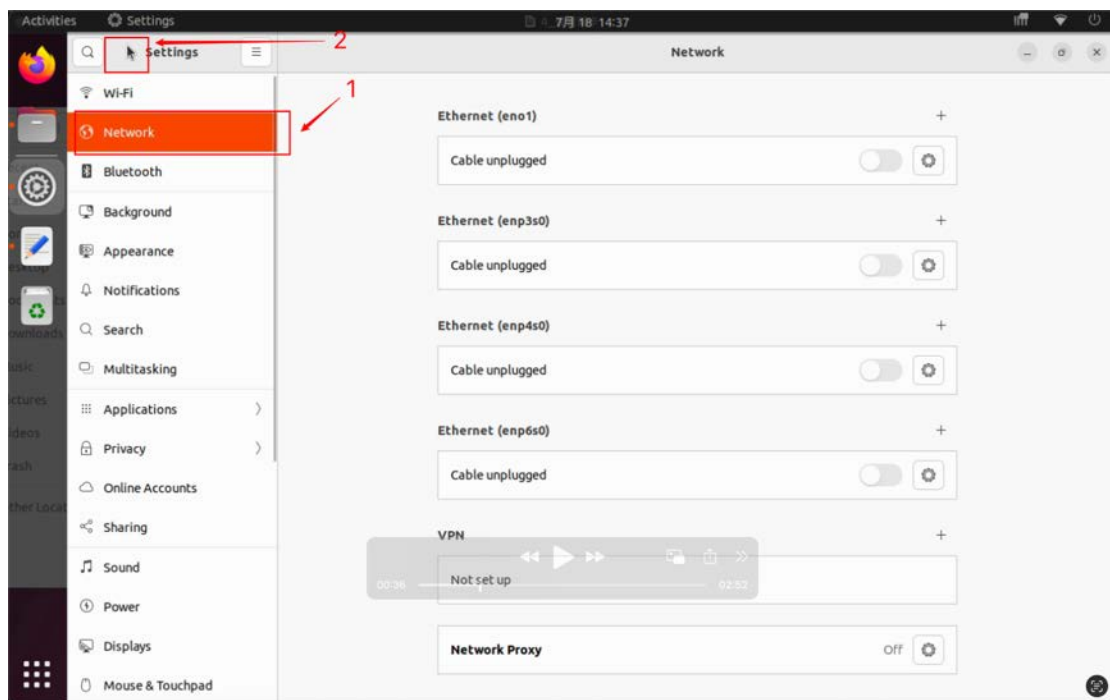
鼠标双击“readme.txt”文件打开，并左击按住 2 处进行拖拽移动至合适位置



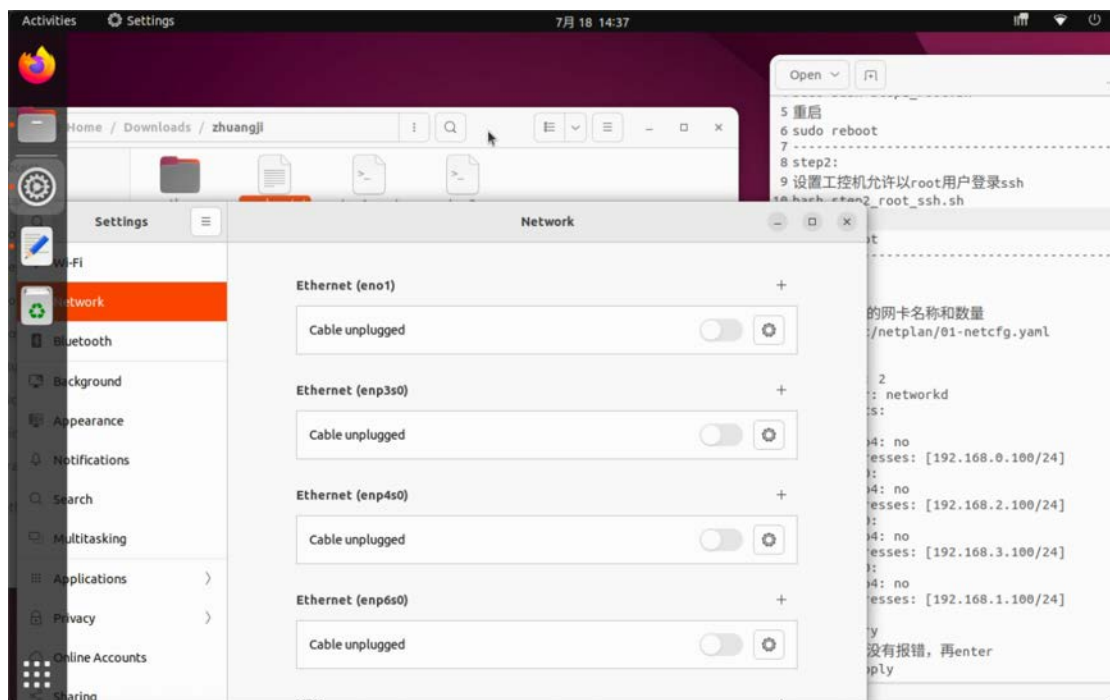
需要先查看当前的有线网卡名称，鼠标单击 1 处后，单击 2 进入系统设置页面



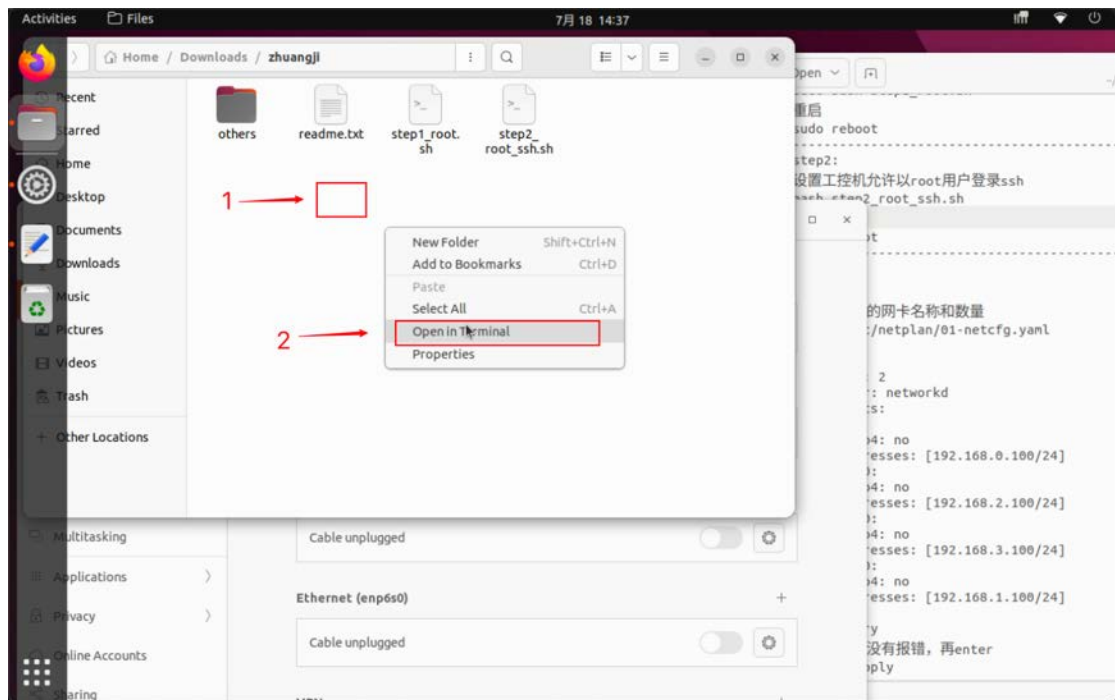
打开设置后，鼠标单击 1 处，然后左击按住 2 处进行拖拽



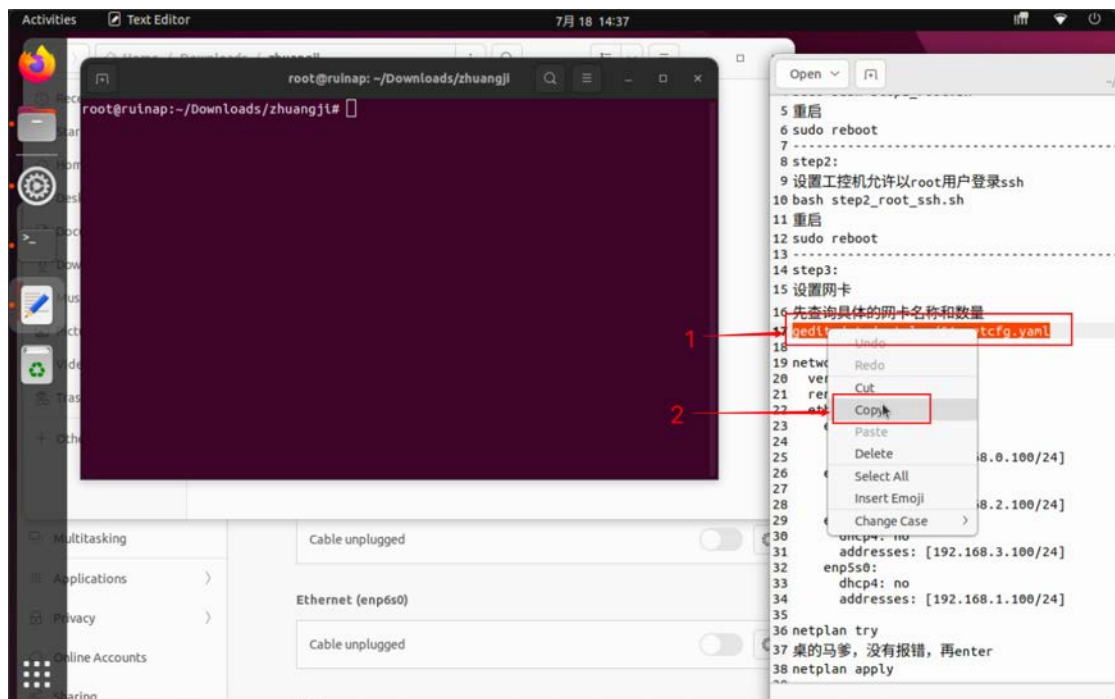
拖拽至合适的位置，如下图



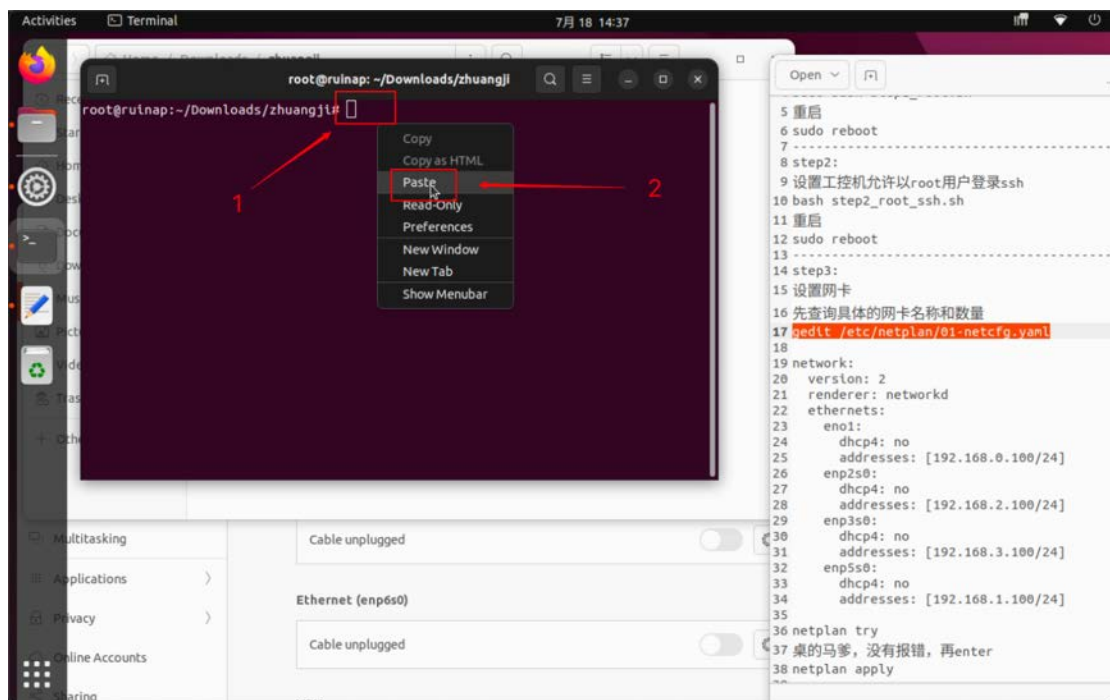
鼠标单击切换回文件夹内，然后空白处右击，并左击“Open in Terminal”选项



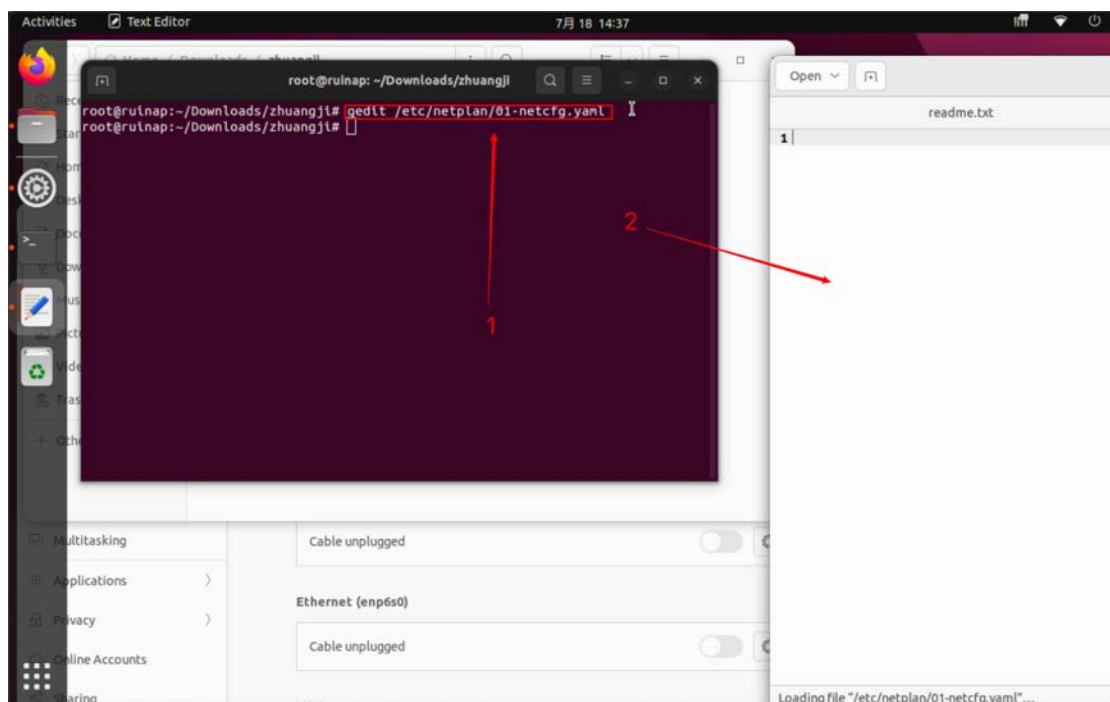
打开终端后，鼠标左击按住“readme.txt”文件中的【gedit /etc/netplan/01-netcfg.yaml】内容，右击然后左击【Copy】



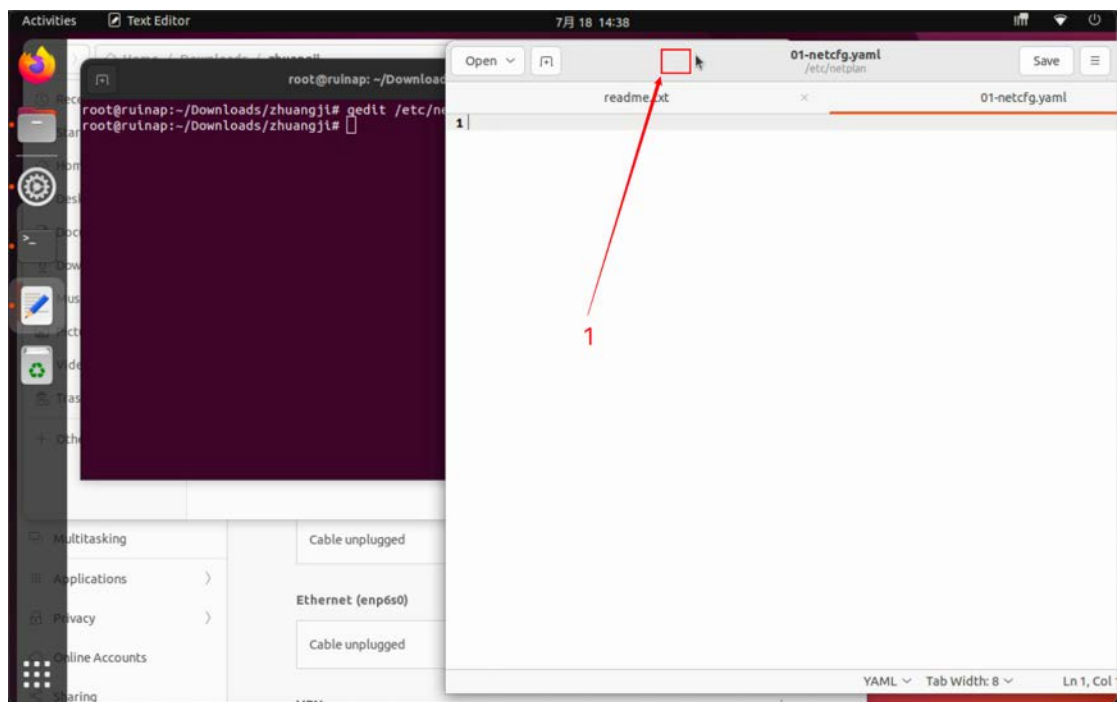
在终端光标处鼠标右击，然后左击【Paste】



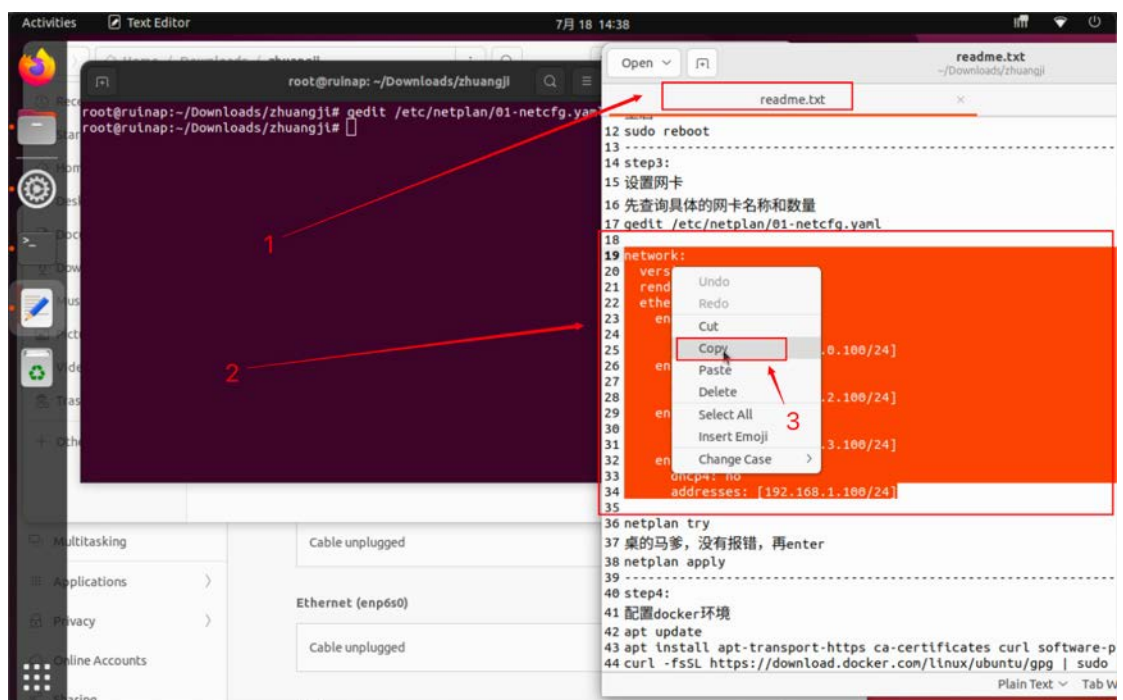
复制内容后，键盘按下【回车键】，右侧的文本查看器内新打开了一个标签页



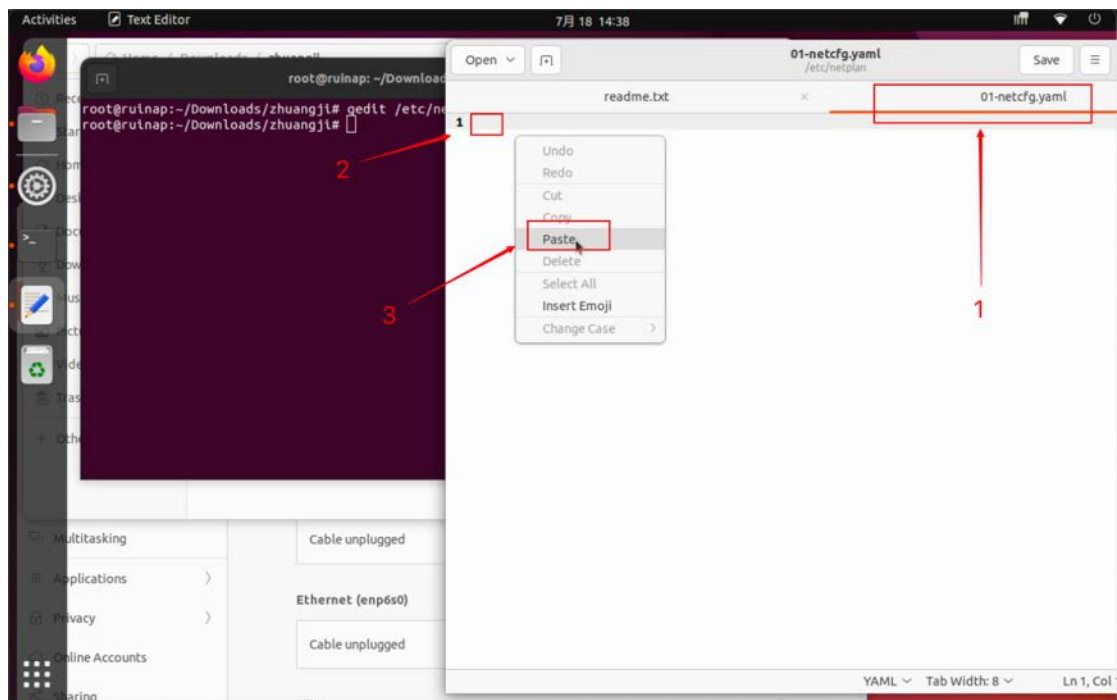
鼠标左键按住 1 处并进行拖拽，拖动至合适位置



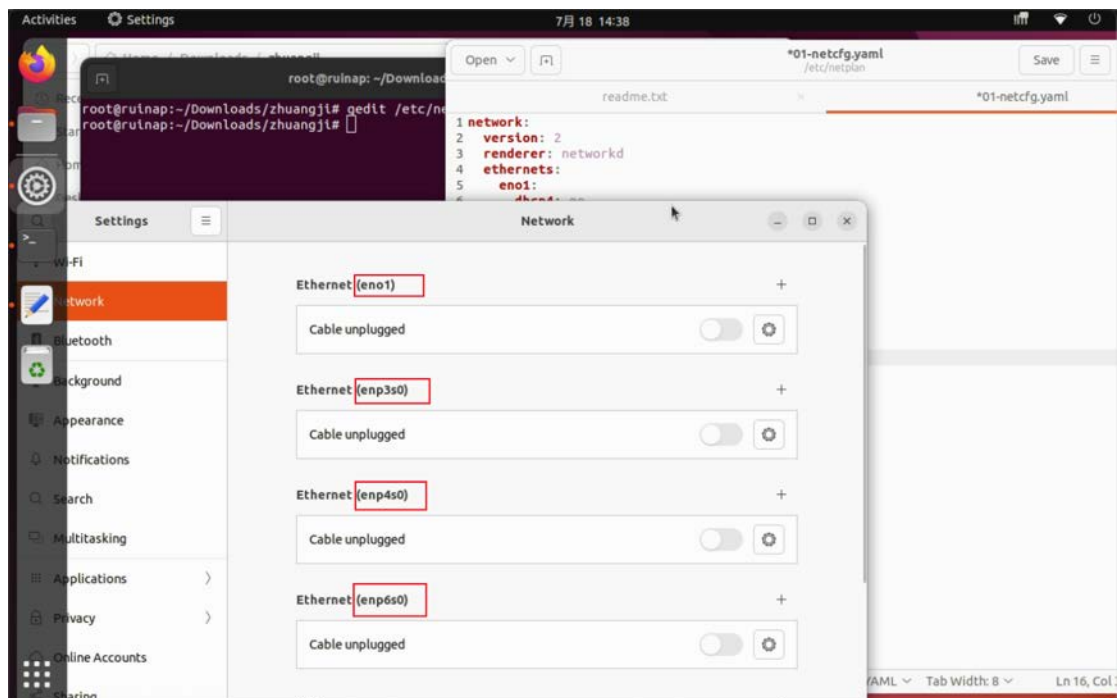
鼠标左键 1 处切换至“readme.txt”标签页，然后左键选中如下图所示的 network 部分的内容，选中有鼠标右键并单击“Copy”



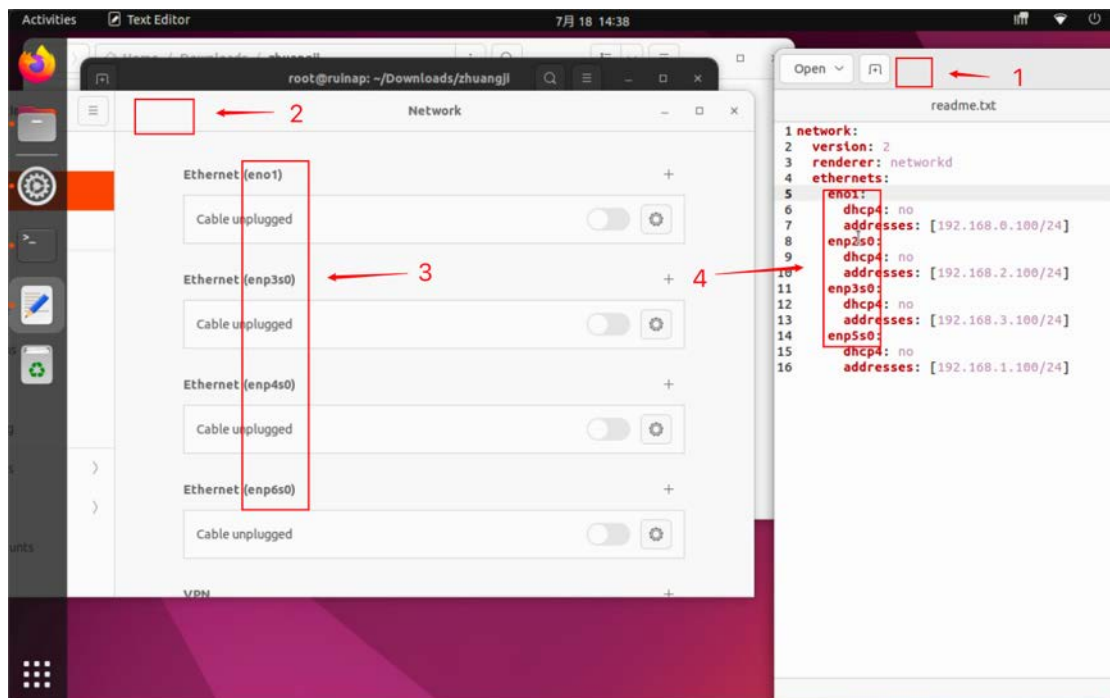
复制文本后，左击 1 处切换文档标签页至“01-netcfg.yaml”，然后在 2 处鼠标右键并选择“Paste”



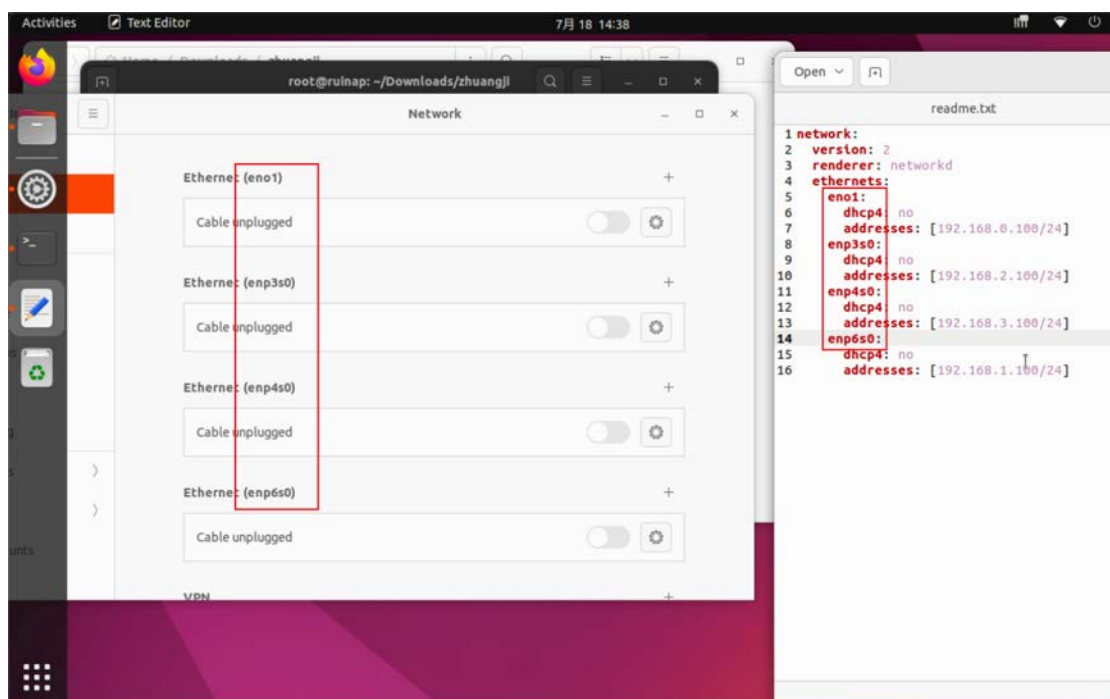
鼠标单击左下角的设置页面，回到设置页面，查看实际的有线网卡名称，如下分别是 eno1、enp3s0、enp4s0、enp6s0



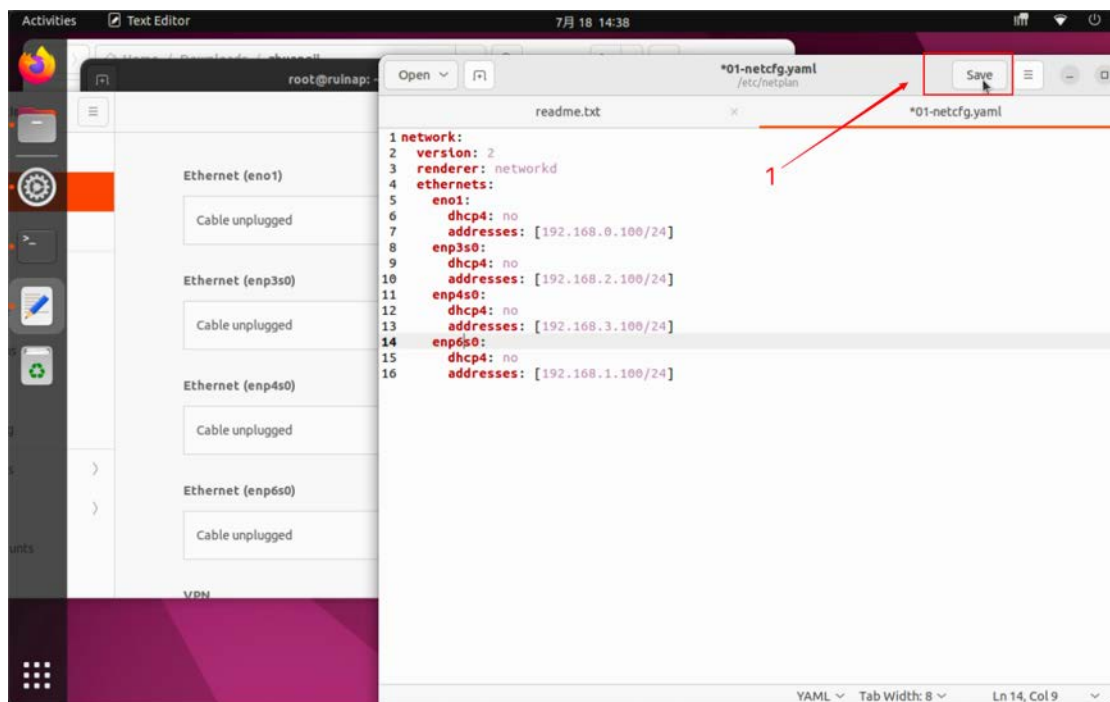
为了方便对照，可以鼠标左键按住进行拖拽窗口，分别拖拽 1 和 2 处，使得可以同时看到两边的内容



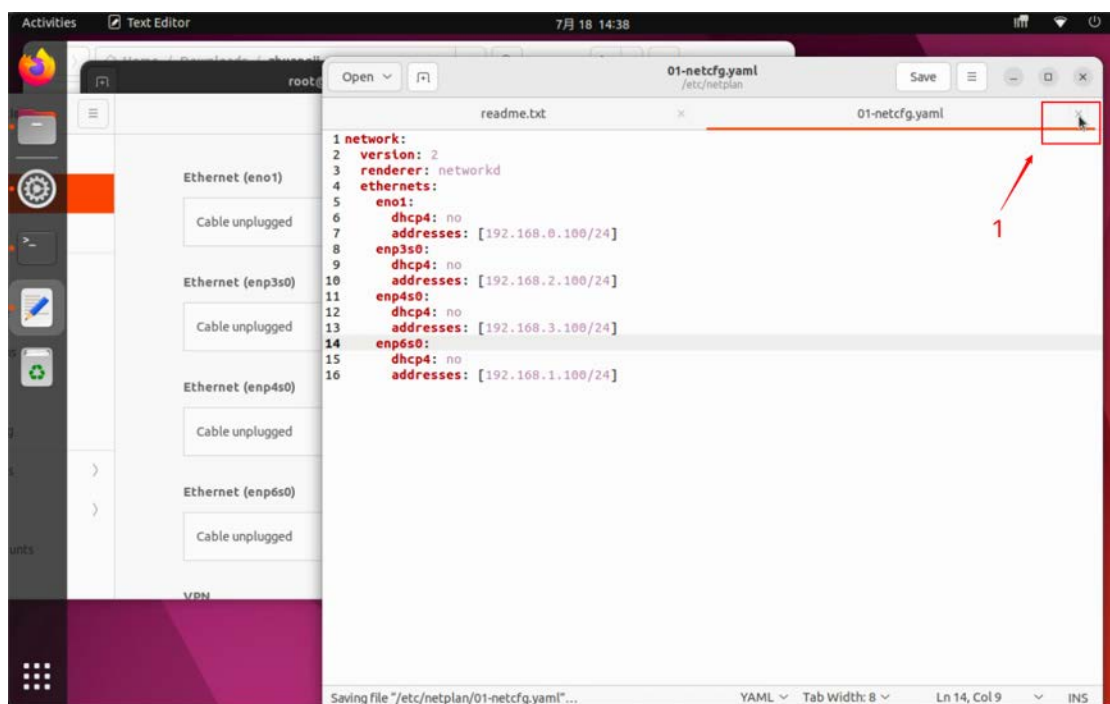
对照左边实际的有线网卡名称，将右侧的配置文件更改为对应的名称。文本编辑的逻辑同 windows 操作系统一致，通过鼠标和键盘来选中位置并输入内容



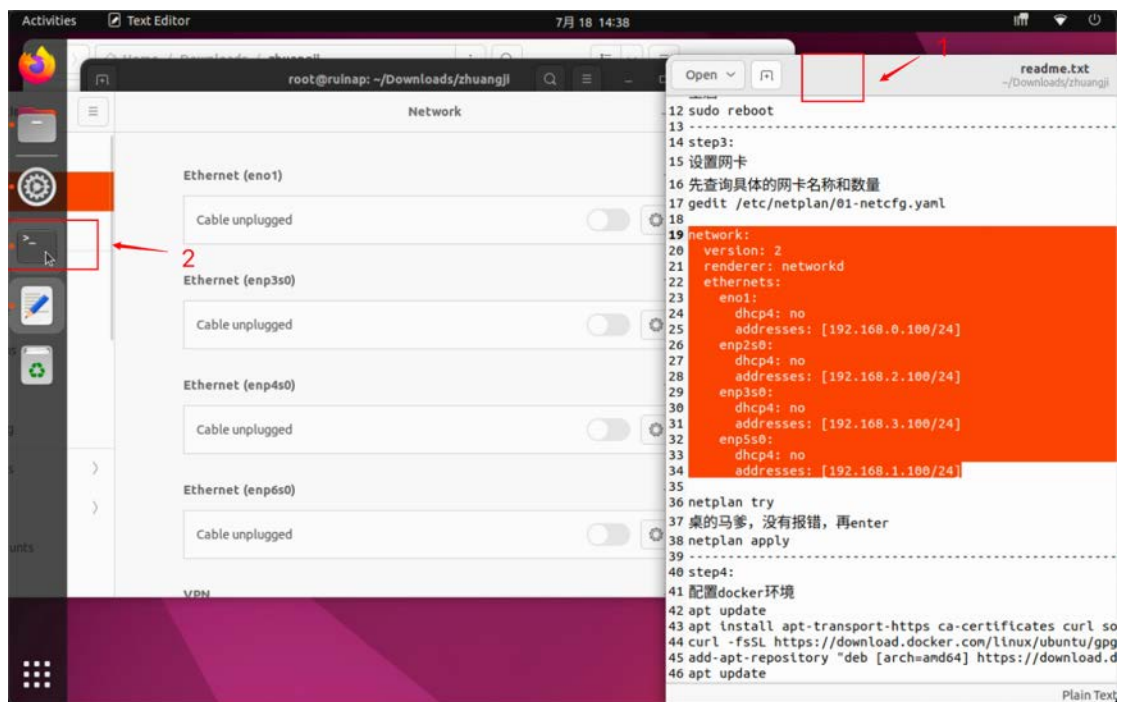
完成后单击 1 处的“Save”进行保存



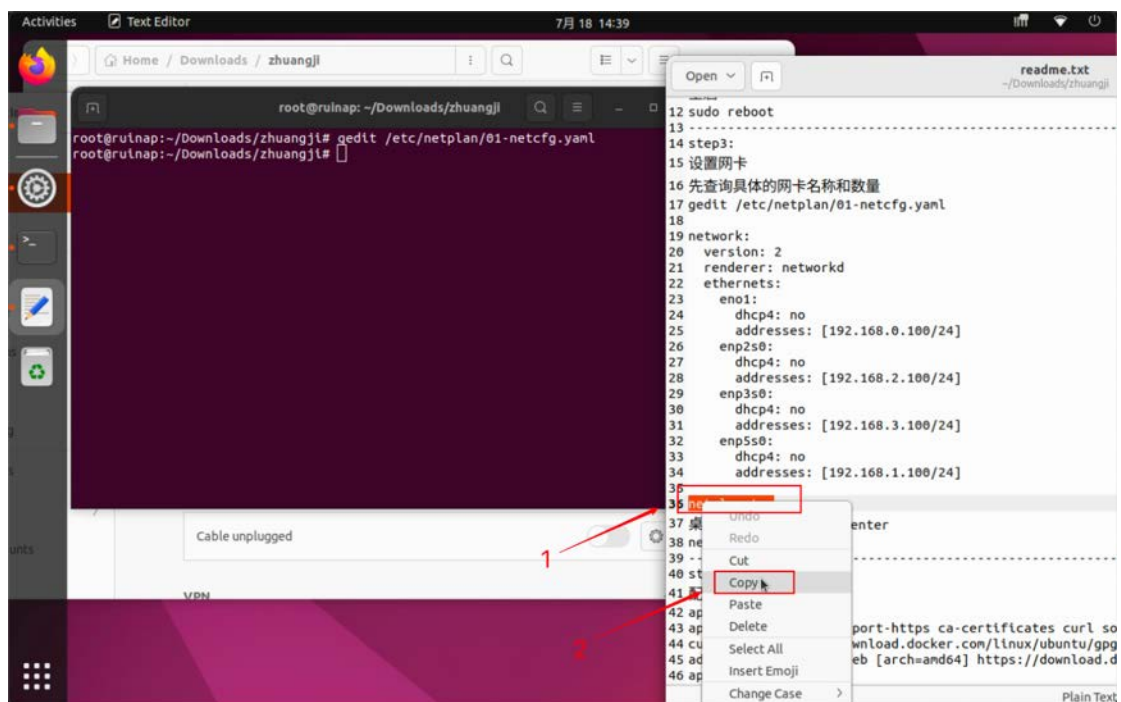
保存后鼠标单击 1 处的“x”关闭该配置文件



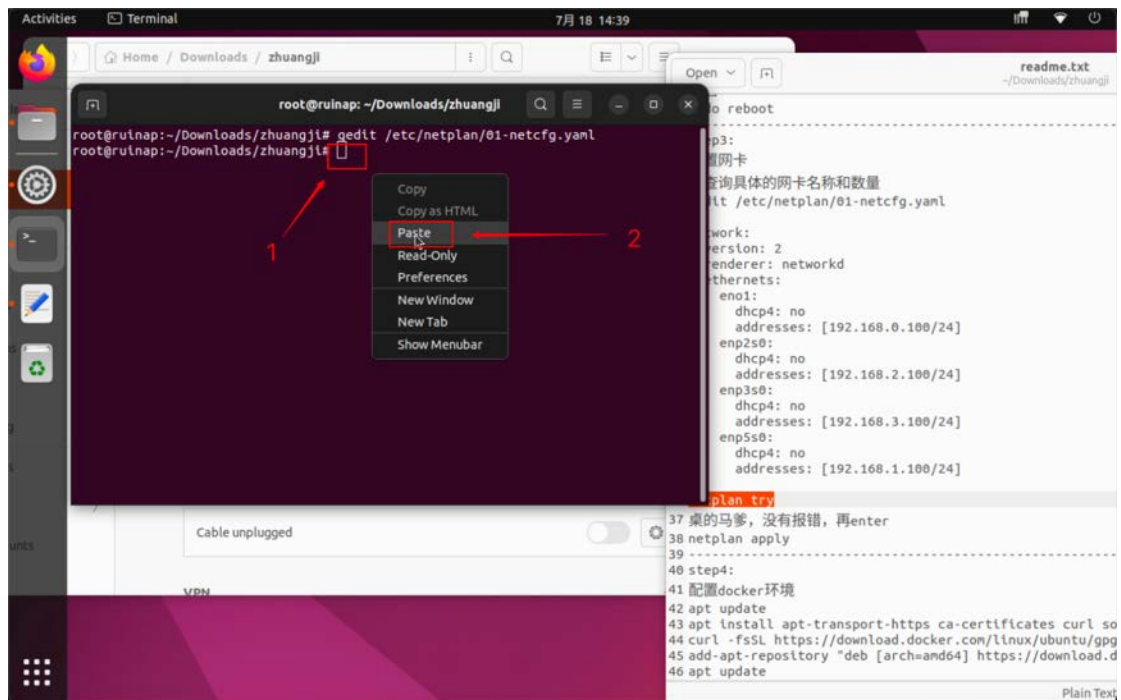
将右侧文档拖拽至合适位置，然后鼠标单击左侧 2 处的终端图标



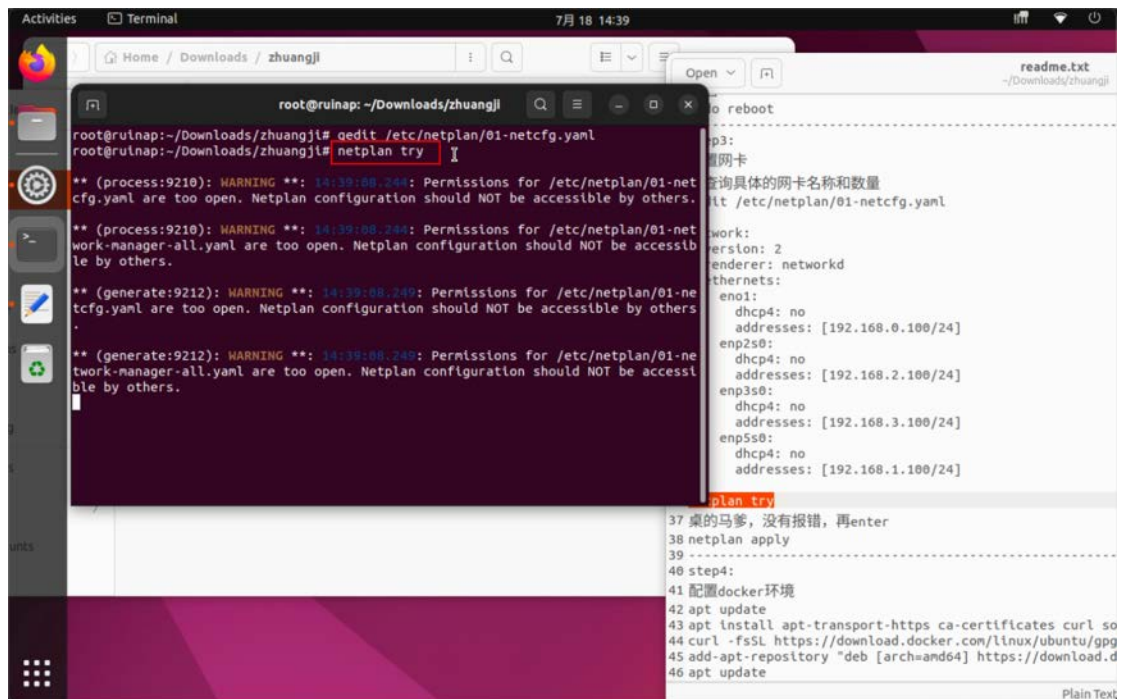
打开终端页面后，鼠标复制右侧文档中的“netplan try”



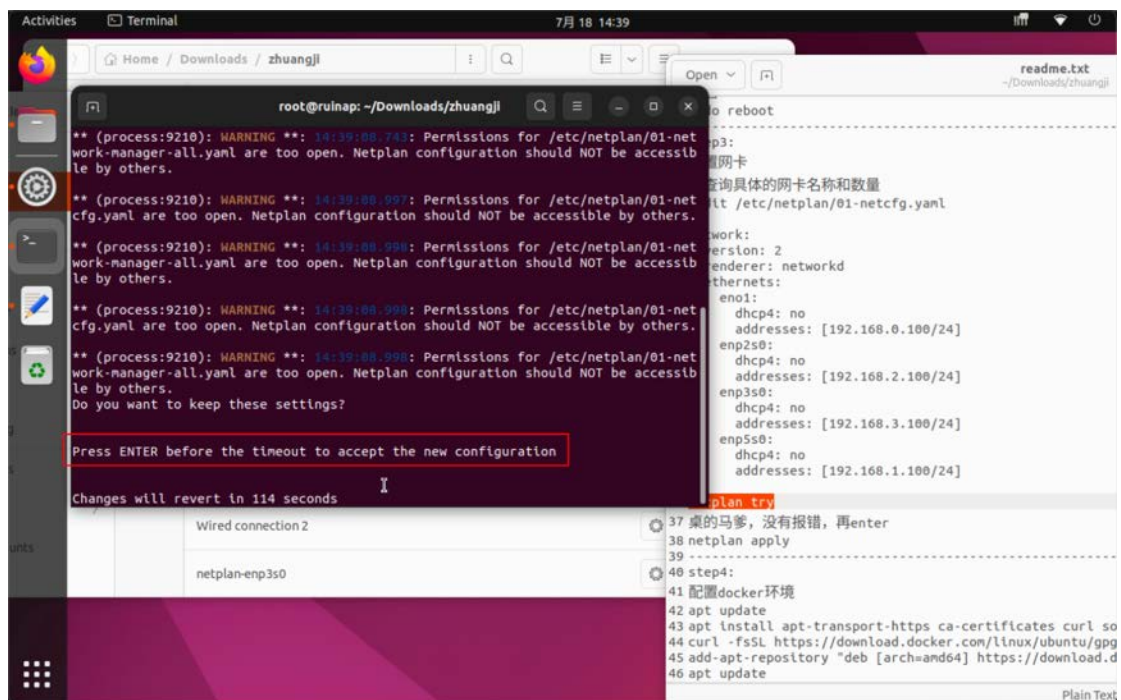
回到终端页面，鼠标右击光标处后单击“Paste”



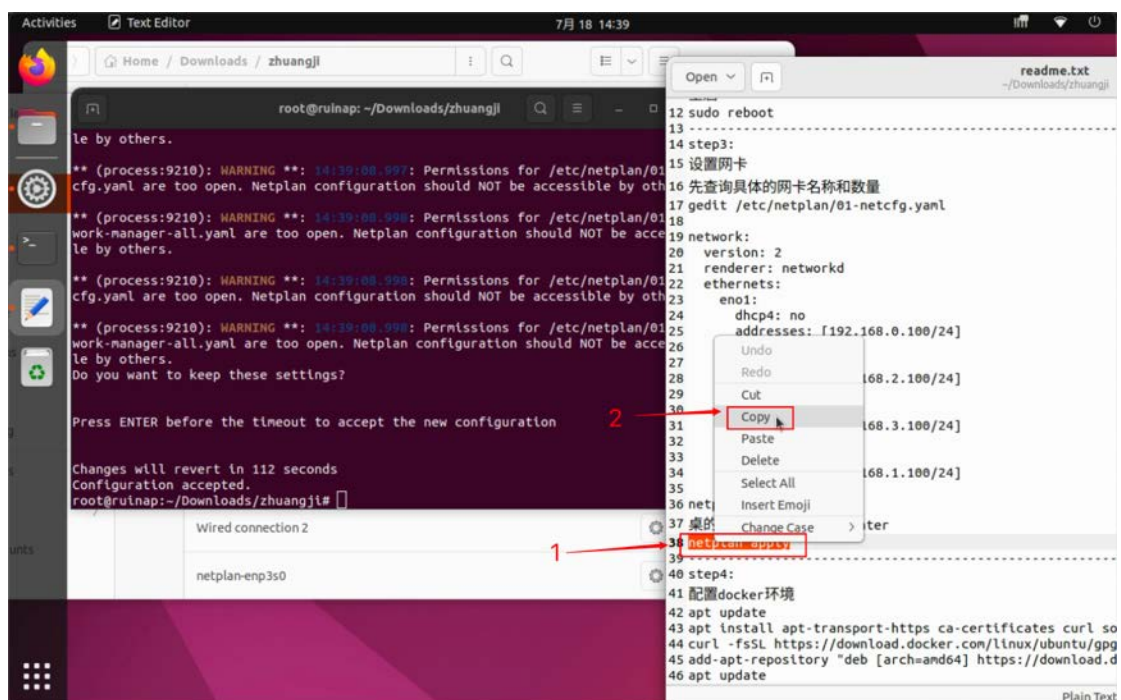
粘贴文本内容后按下键盘【回车键】，终端会开始打印内容



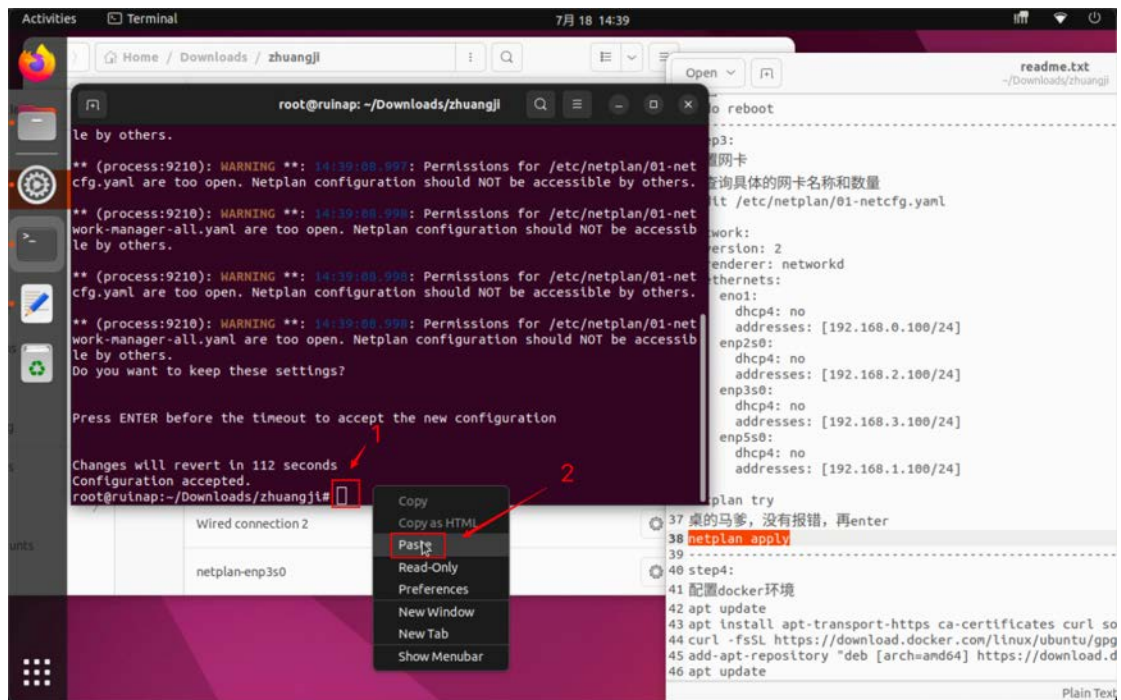
等待出现如下所示的内容后，按下键盘的【回车键】



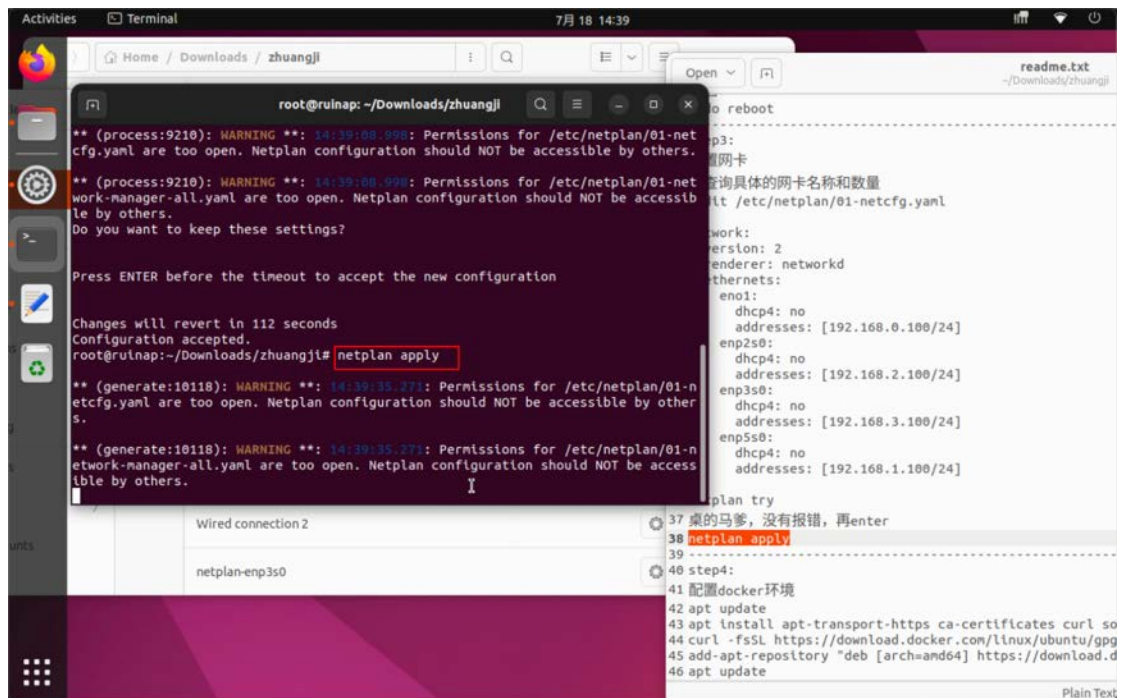
然后复制右侧文本的“netplan apply”



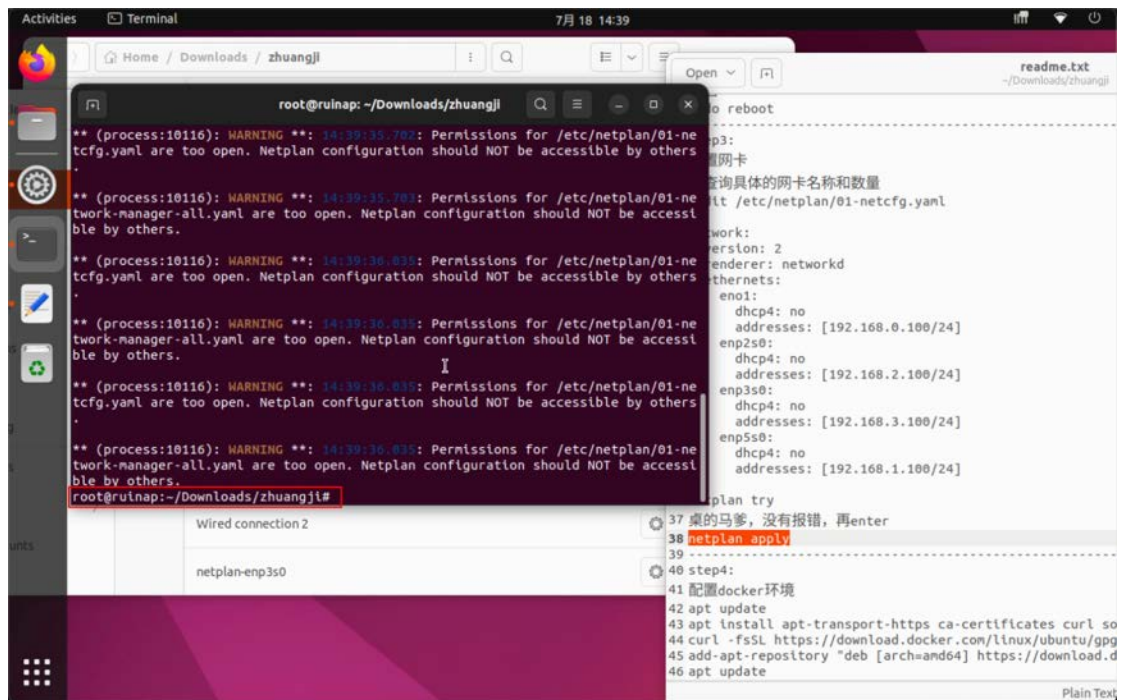
回到终端页面，鼠标右击光标处后单击“Paste”



粘贴文本后，按下键盘的【回车键】并等待



直至出现如下图所示的内容，并重新出现跳动的光标



The screenshot shows a terminal window with the following content:

```
root@ruinap: ~/Downloads/zhuangji
** (process:10116): WARNING **: 14:39:35.702: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others.
** (process:10116): WARNING **: 14:39:35.703: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.
** (process:10116): WARNING **: 14:39:36.035: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others.
** (process:10116): WARNING **: 14:39:36.035: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.
** (process:10116): WARNING **: 14:39:36.035: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configuration should NOT be accessible by others.
** (process:10116): WARNING **: 14:39:36.035: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.
root@ruinap:~/Downloads/zhuangji#
```

Below the terminal window, a network configuration file is shown:

```
Wired connection 2
netplan-enp3s0
```

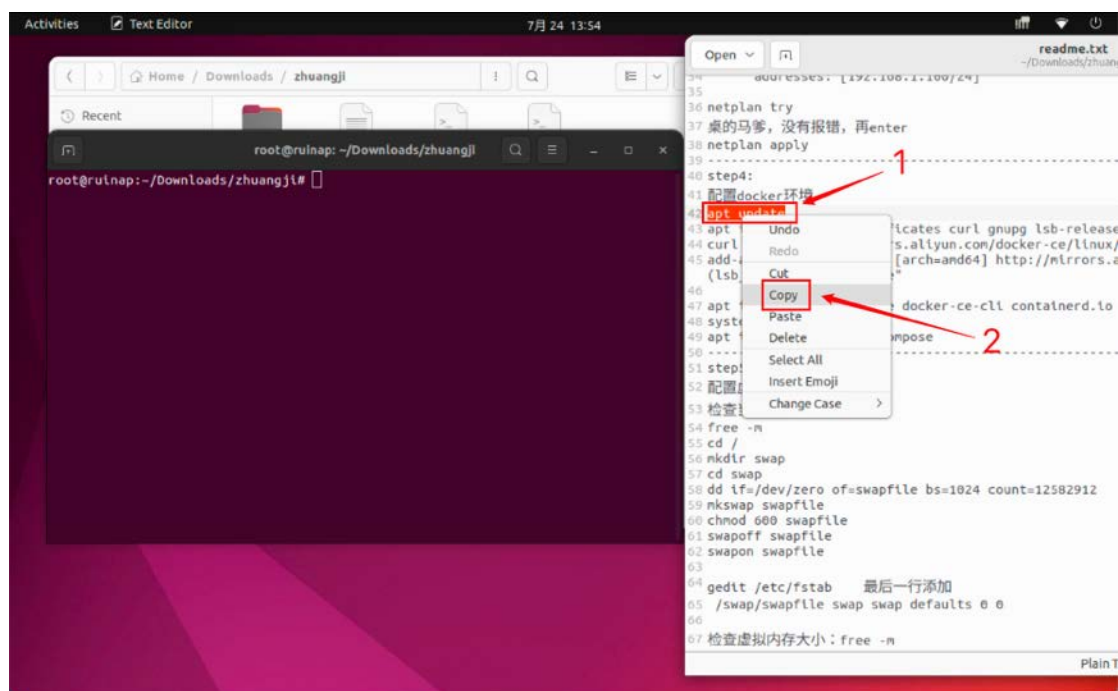
On the right side of the terminal window, a network configuration file is displayed:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    eno1:
      dhcp4: no
      addresses: [192.168.0.100/24]
    enp2s0:
      dhcp4: no
      addresses: [192.168.2.100/24]
    enp3s0:
      dhcp4: no
      addresses: [192.168.3.100/24]
    enp5s0:
      dhcp4: no
      addresses: [192.168.1.100/24]
```

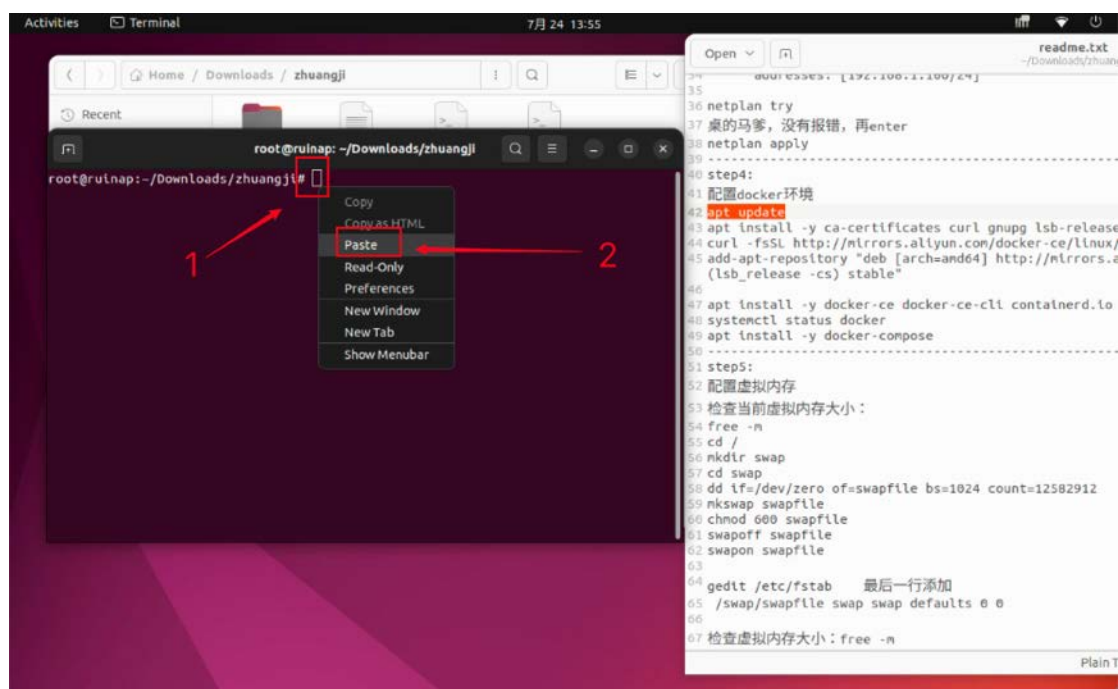
至此，有线网卡设置完毕

5. docker 安装

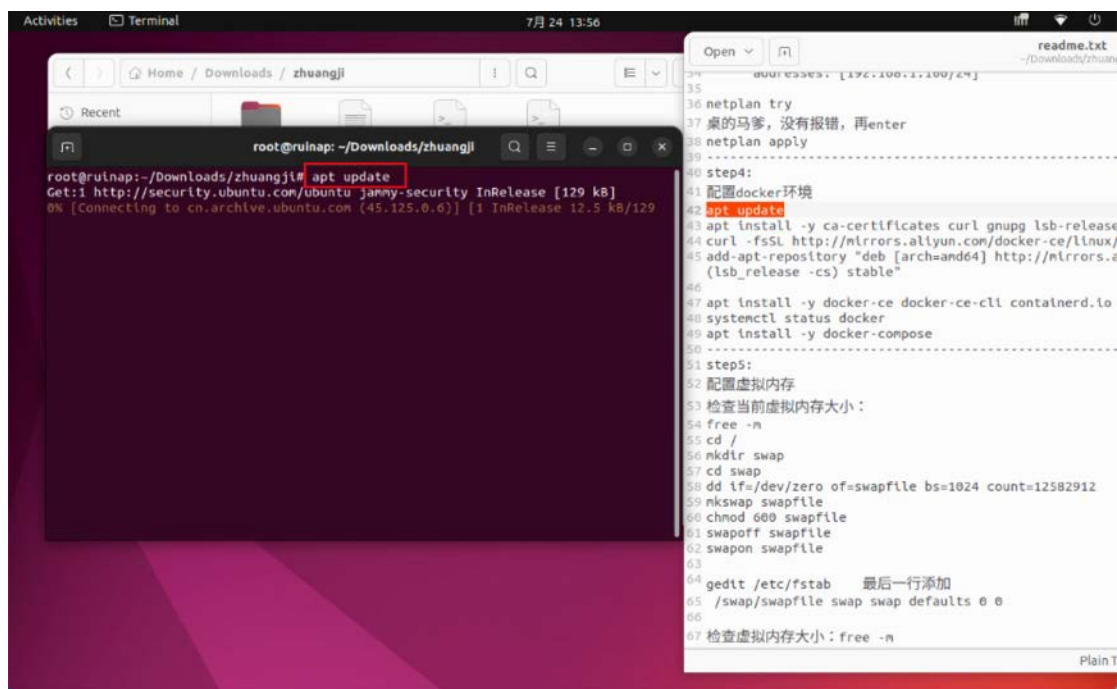
如果已经关闭终端和文件夹以及文档，参照前面的小节重新打开，然后复制文本“apt update”



完成后粘贴至终端内

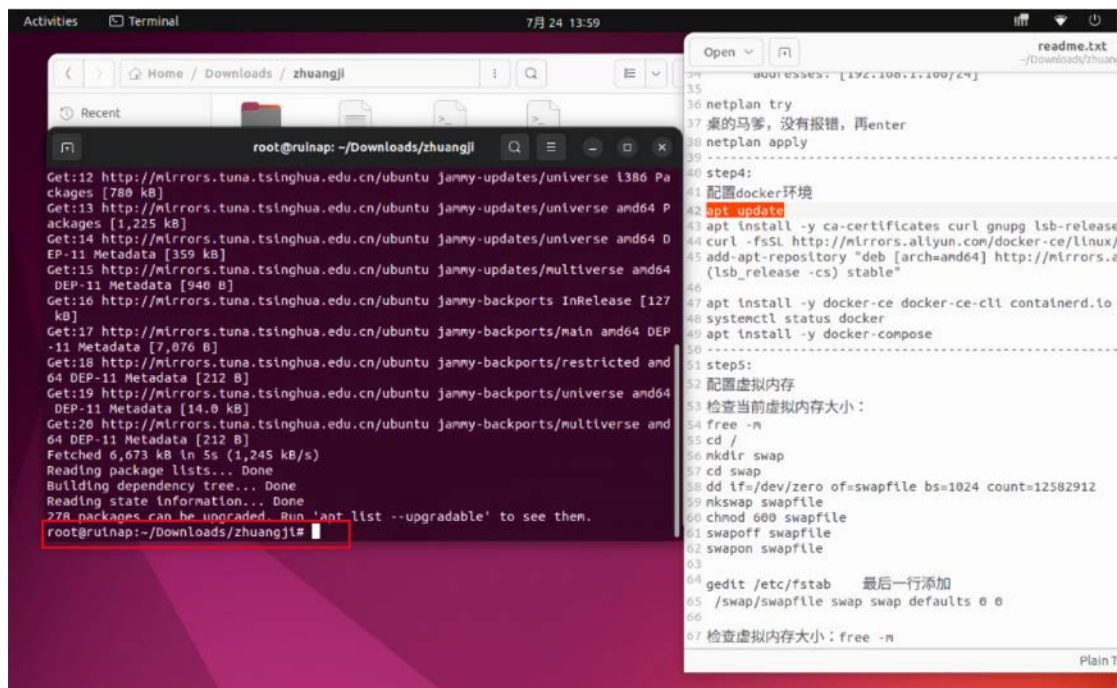


键盘按下【回车键】后等待内容打印



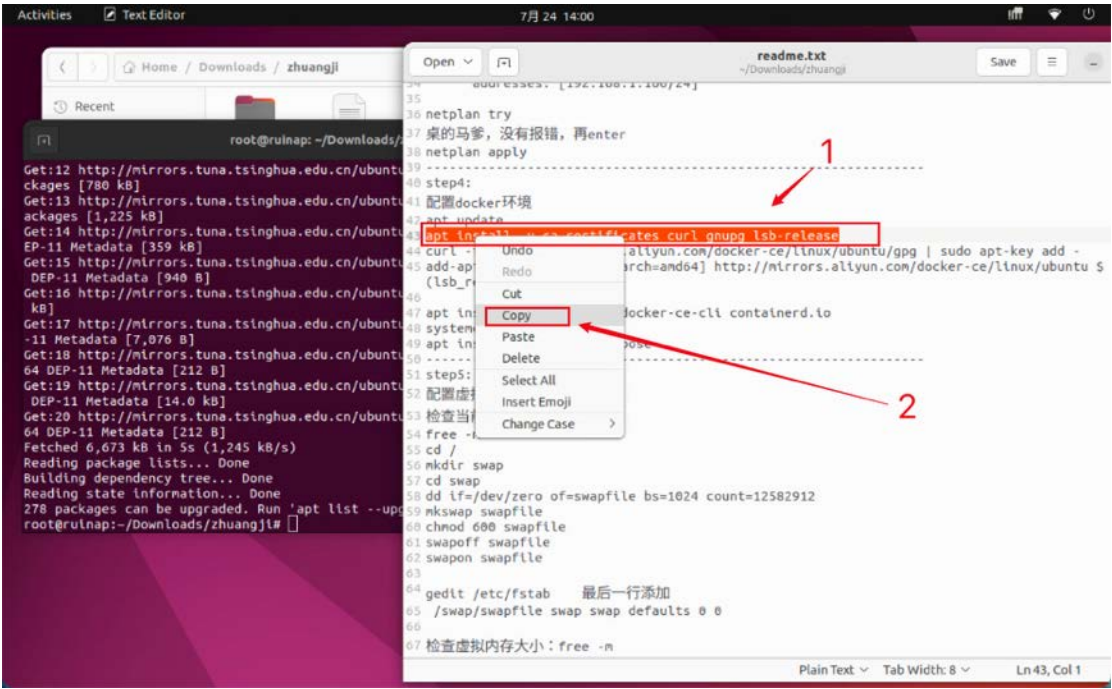
```
root@ruinap: ~/Downloads/zhuangji# apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
0% [connecting to cn.archive.ubuntu.com (45.125.0.6)] [1 InRelease 12.5 kB/129
```

出现如下内容并重新有跳动的光标则表示完成，如果出现 Error、Err 等字样，那么需要查看右上角是否连接 wifi，没有的话参照前面小节重新连接 wifi

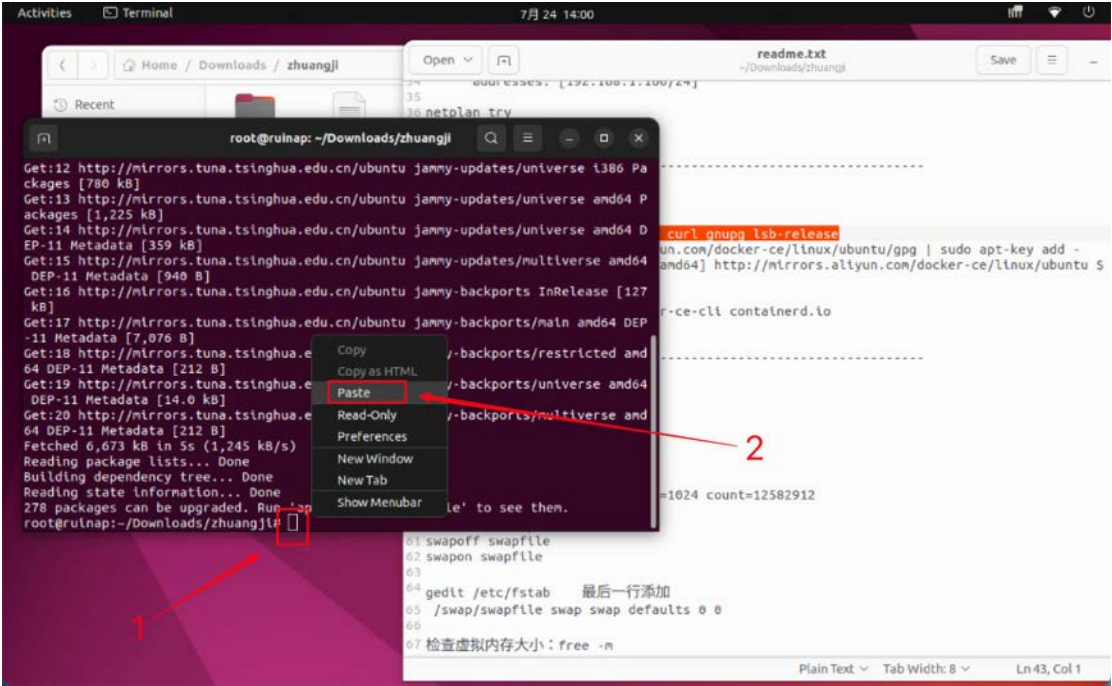


```
Get:12 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates/universe I386 Pa
ckages [780 kB]
Get:13 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates/universe amd64 P
ackages [1,225 kB]
Get:14 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates/universe amd64 D
EP-11 Metadata [359 kB]
Get:15 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates/multiverse amd64
DEP-11 Metadata [940 B]
Get:16 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports InRelease [127
kB]
Get:17 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports/main amd64 DEP
-11 Metadata [7,076 B]
Get:18 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports/restricted amd
64 DEP-11 Metadata [212 B]
Get:19 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports/universe amd64
DEP-11 Metadata [14.0 kB]
Get:20 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports/multiverse amd
64 DEP-11 Metadata [212 B]
Fetched 6,673 kB in 5s (1,245 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
278 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ruinap:~/Downloads/zhuangji#
```

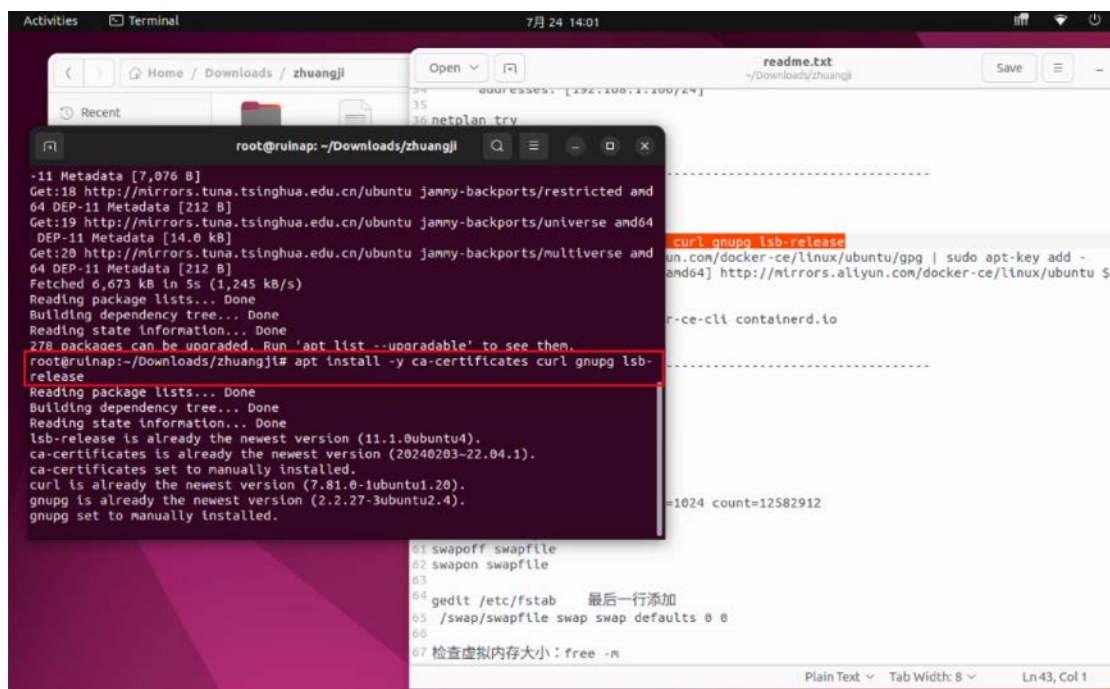
鼠标复制如下图所示的文本



回到终端，鼠标右键将文本粘贴至此

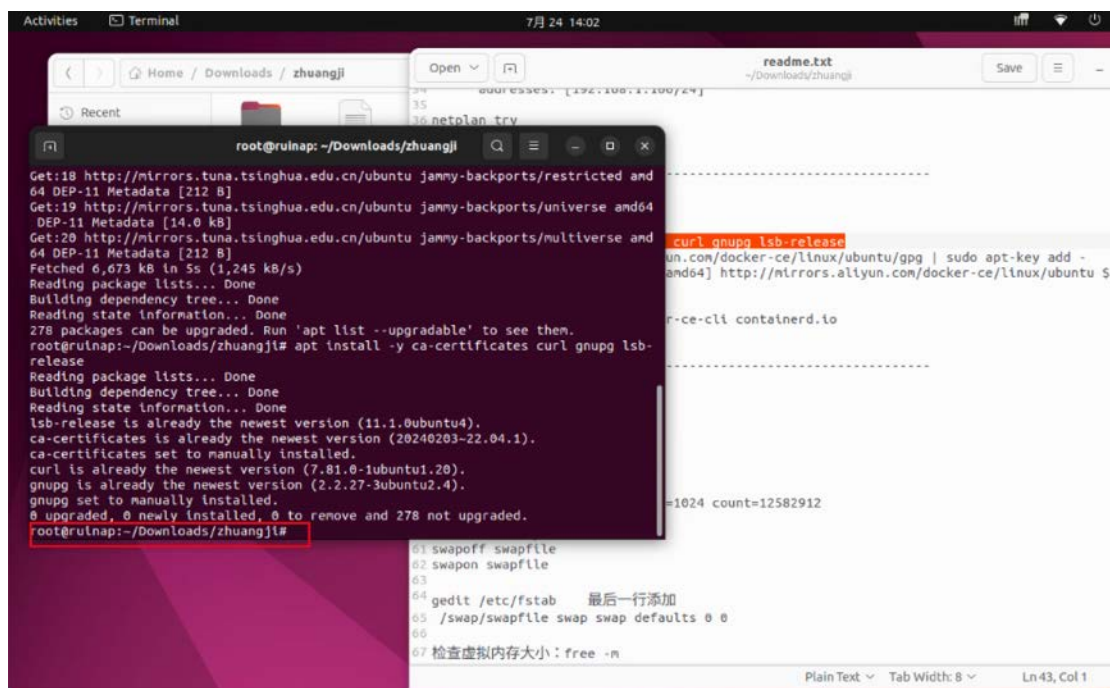


粘贴文本后键盘按下【回车键】并等待打印



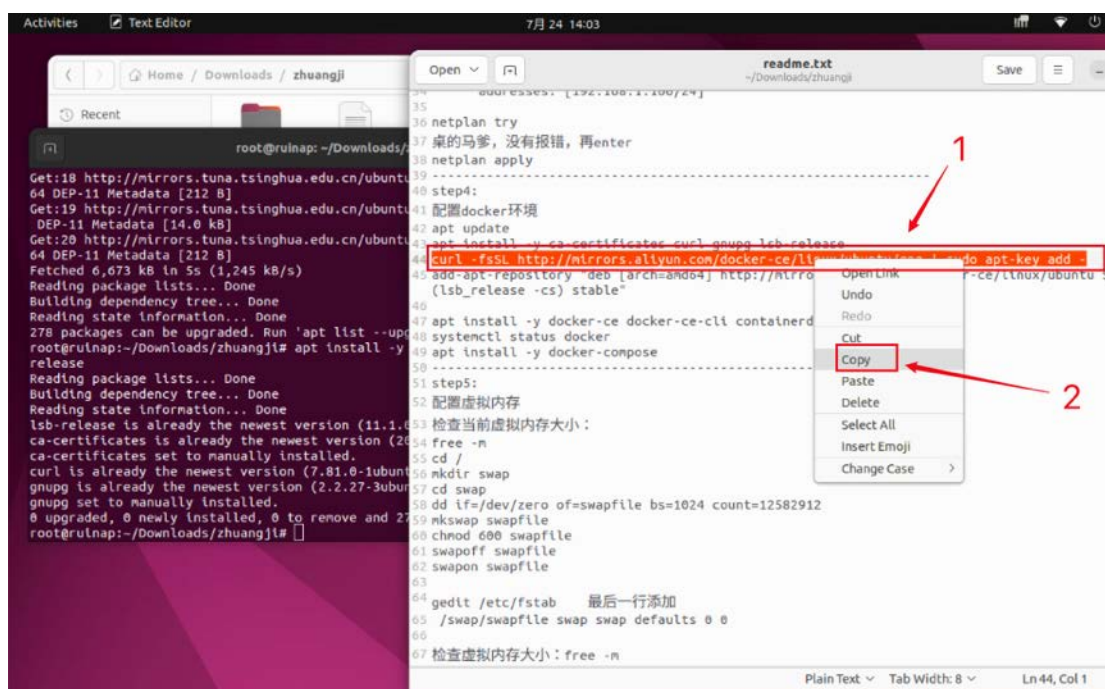
```
root@rulnap: ~/Downloads/zhuangji
-11 Metadata [7,076 B]
Get:18 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports/restricted and
64 DEP-11 Metadata [212 B]
Get:19 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports/universe and64
DEP-11 Metadata [14.0 kB]
Get:20 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports/multiverse and
64 DEP-11 Metadata [212 B]
Fetched 6,673 kB in 5s (1,245 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
278 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@rulnap:~/Downloads/zhuangji# apt install -y ca-certificates curl gnupg lsb-
release
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
lsb-release is already the newest version (11.1.0ubuntu4).
ca-certificates is already the newest version (20240203-22.04.1).
ca-certificates set to manually installed.
curl is already the newest version (7.81.0-1ubuntu1.20).
gnupg is already the newest version (2.2.27-3ubuntu2.4).
gnupg set to manually installed.
```

出现如下内容并重新有跳动的光标则表示完成

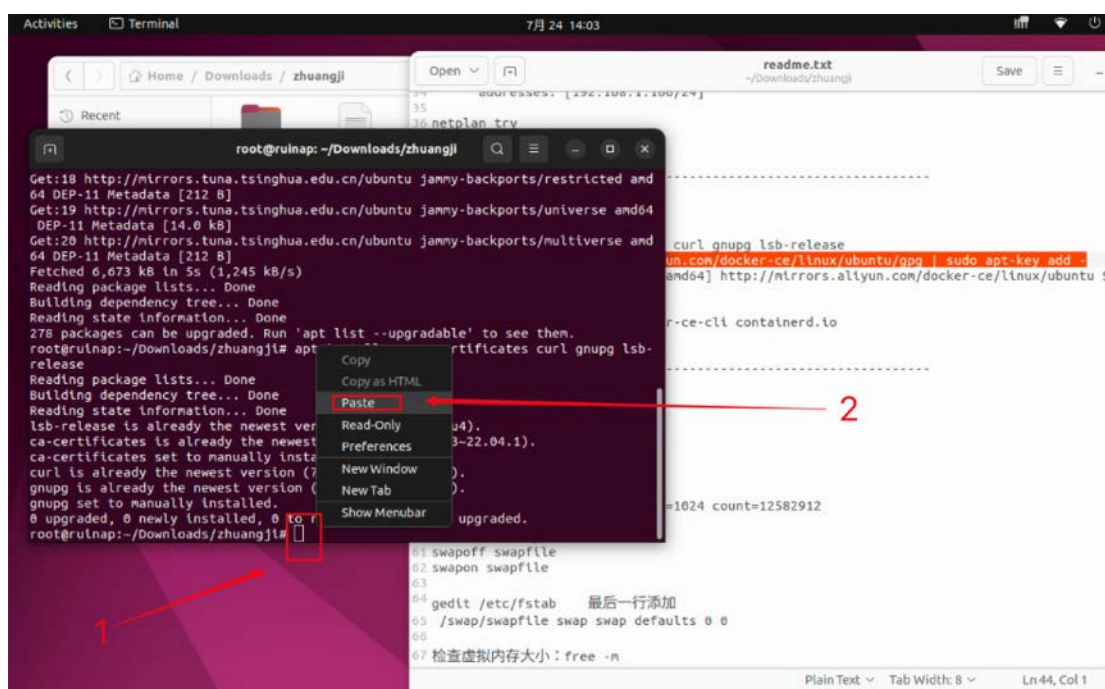


```
0 upgraded, 0 newly installed, 0 to remove and 278 not upgraded.
root@rulnap:~/Downloads/zhuangji#
```

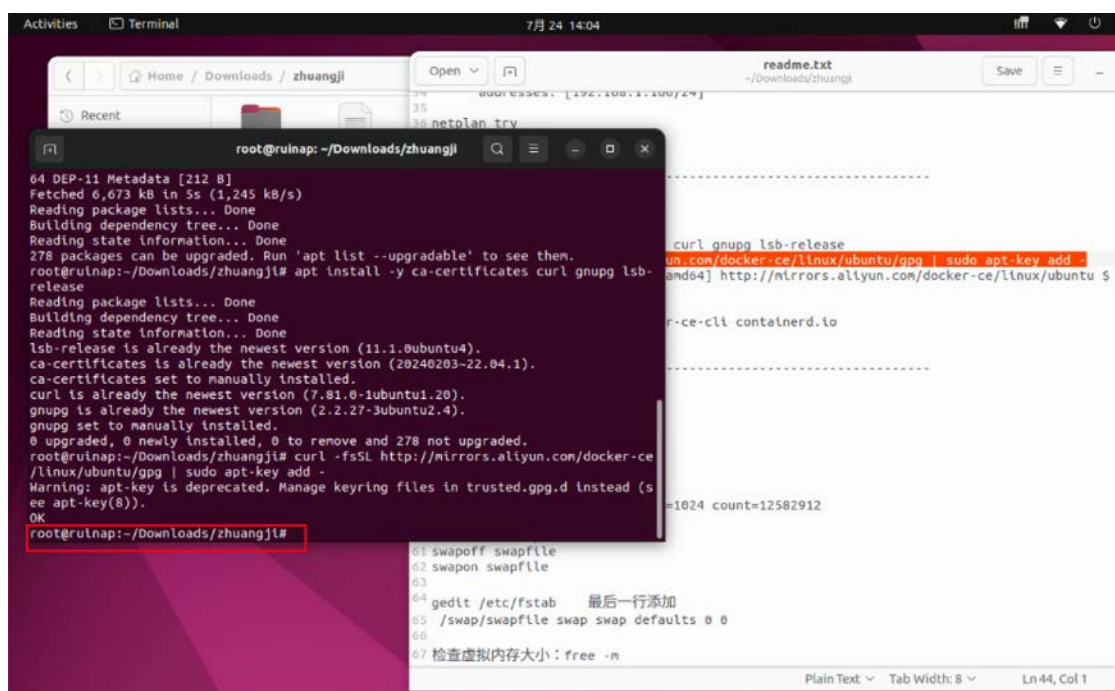

继续回到文档中，复制如下框选的文本



粘贴文本后键盘按下【回车键】并等待

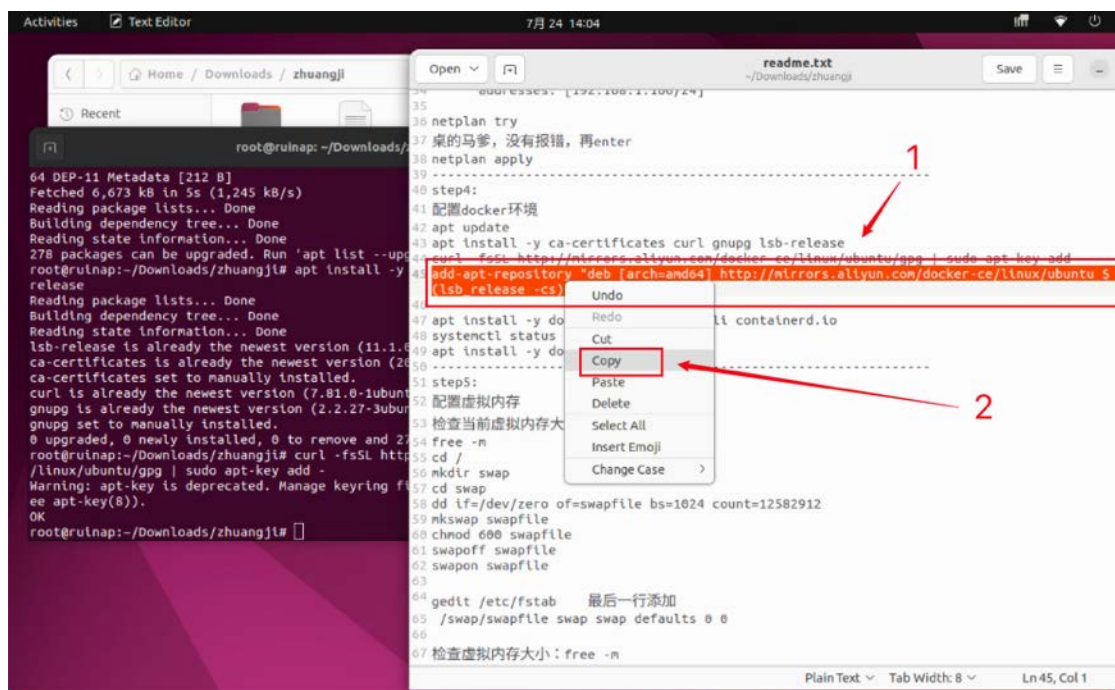


重新显示如下内容并出现跳动光标后，回到右侧文档中



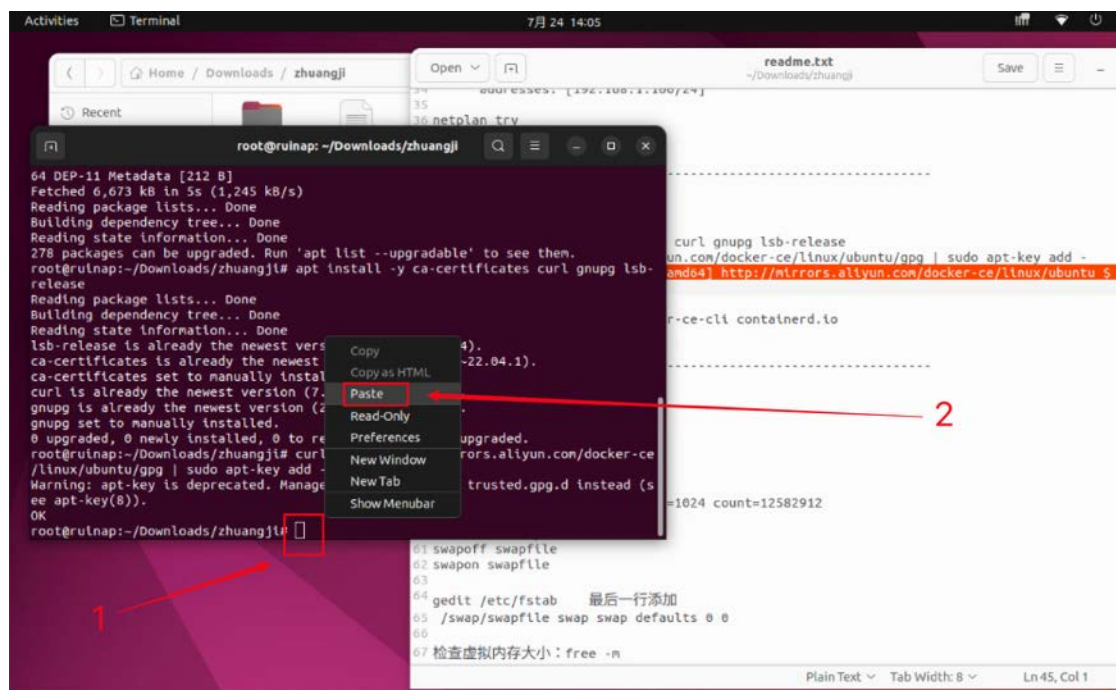
```
64 DEP-11 Metadata [212 B]
Fetched 6,673 kB in 5s (1,245 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
278 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ruinap:~/Downloads/zhuangji# apt install -y ca-certificates curl gnupg lsb-
release
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
lsb-release is already the newest version (11.1.0ubuntu4).
ca-certificates is already the newest version (20240203-22.04.1).
ca-certificates set to manually installed.
curl is already the newest version (7.81.0-1ubuntu1.20).
gnupg is already the newest version (2.2.27-3ubuntu2.4).
gnupg set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 278 not upgraded.
root@ruinap:~/Downloads/zhuangji# curl -fsSL http://mirrors.aliyun.com/docker-ce
/linux/ubuntu/gpg | sudo apt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (s
ee apt-key(8)).
OK
root@ruinap:~/Downloads/zhuangji#
```

复制如下框选的文字

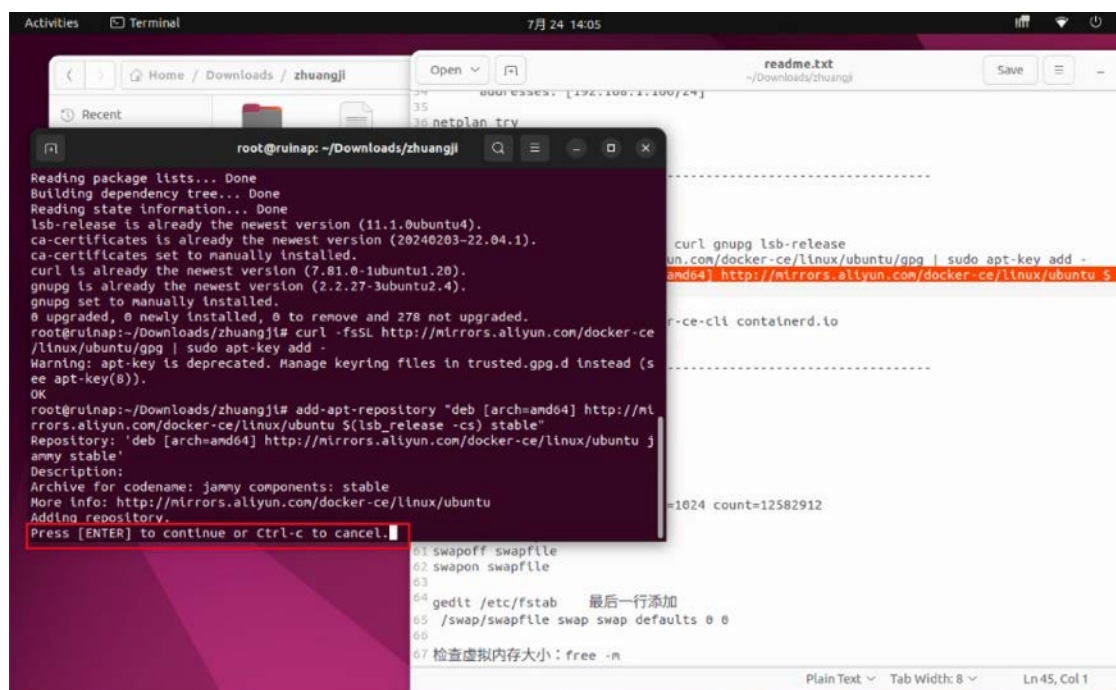


```
64 DEP-11 Metadata [212 B]
Fetched 6,673 kB in 5s (1,245 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
278 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ruinap:~/Downloads/zhuangji# apt install -y ca-certificates curl gnupg lsb-
release
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
lsb-release is already the newest version (11.1.0ubuntu4).
ca-certificates is already the newest version (20240203-22.04.1).
ca-certificates set to manually installed.
curl is already the newest version (7.81.0-1ubuntu1.20).
gnupg is already the newest version (2.2.27-3ubuntu2.4).
gnupg set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 278 not upgraded.
root@ruinap:~/Downloads/zhuangji# curl -fsSL http://mirrors.aliyun.com/docker-ce
/linux/ubuntu/gpg | sudo apt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (s
ee apt-key(8)).
OK
root@ruinap:~/Downloads/zhuangji#
```

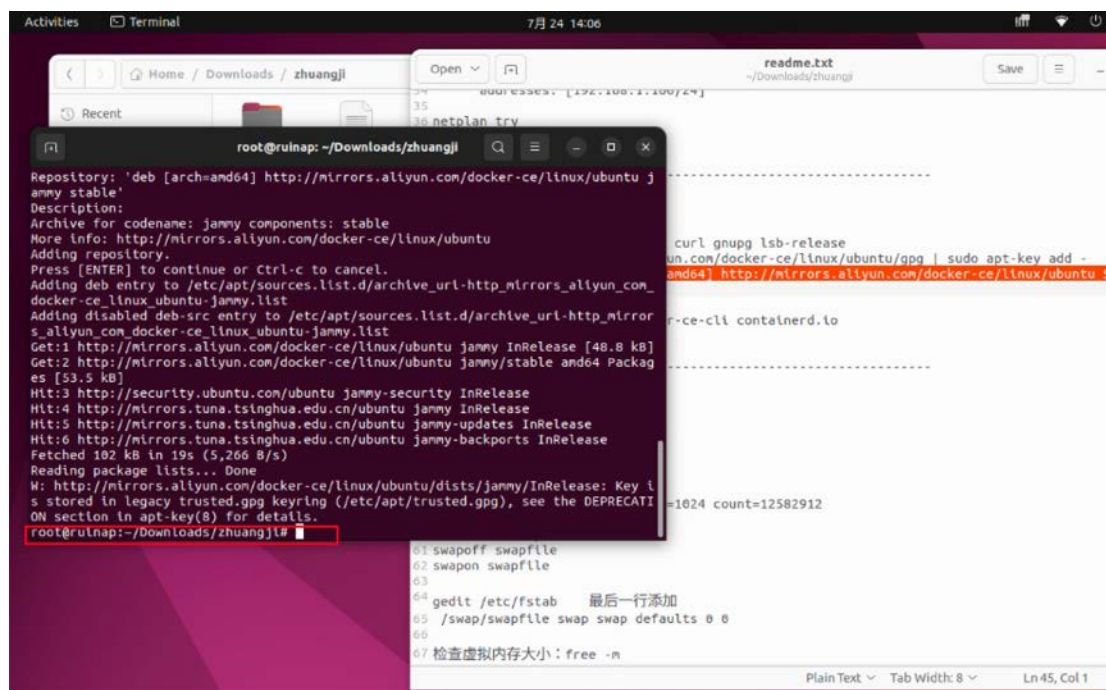

粘贴文本后键盘按下【回车键】



出现如下提示时，键盘按下【回车键】

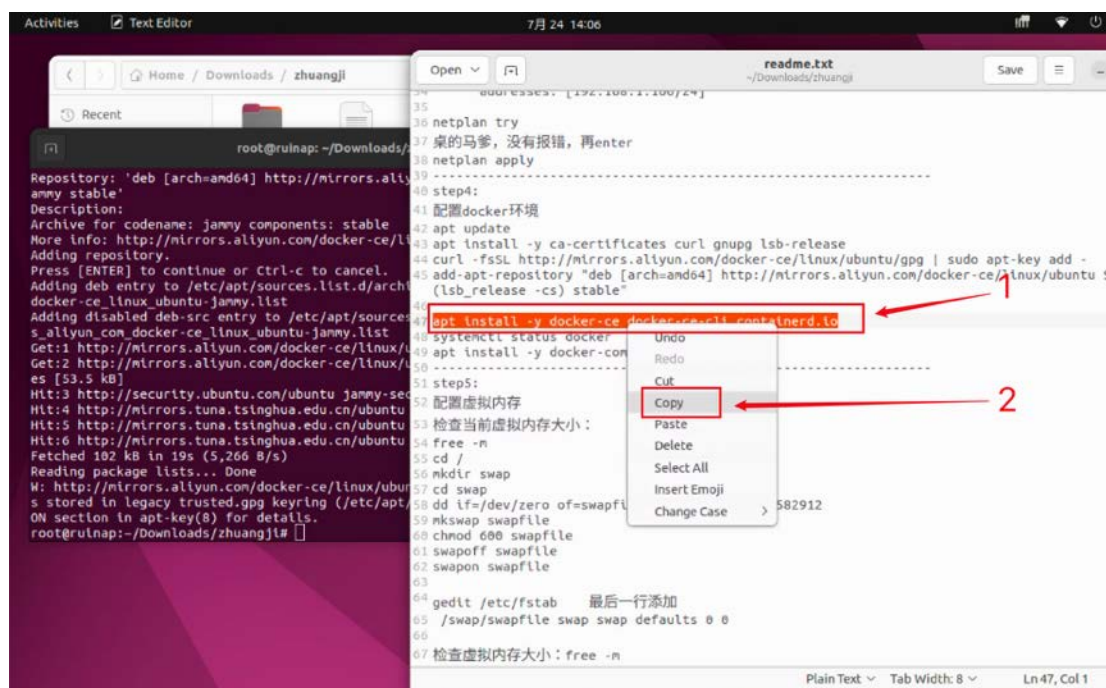


等待终端打印结束，重新显示如下跳动光标后



```
Repository: 'deb [arch=amd64] http://mirrors.aliyun.com/docker-ce/linux/ubuntu jammy stable'
Description:
Archive for codename: jammy components: stable
More info: http://mirrors.aliyun.com/docker-ce/linux/ubuntu
Adding repository.
Press [ENTER] to continue or Ctrl-c to cancel.
Adding deb entry to /etc/apt/sources.list.d/archive_url-http_mirrors.aliyun.com_docker-ce_linux_ubuntu_jammy.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/archive_url-http_mirrors.aliyun.com_docker-ce_linux_ubuntu_jammy.list
Get:1 http://mirrors.aliyun.com/docker-ce/linux/ubuntu jammy InRelease [48.8 kB]
Get:2 http://mirrors.aliyun.com/docker-ce/linux/ubuntu jammy/stable amd64 Packages [53.5 kB]
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy InRelease
Hit:5 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates InRelease
Hit:6 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-backports InRelease
Fetched 102 kB in 19s (5,266 B/s)
Reading package lists... Done
W: http://mirrors.aliyun.com/docker-ce/linux/ubuntu/dists/jammy/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
root@rulnap: ~/Downloads/zhuangji#
```

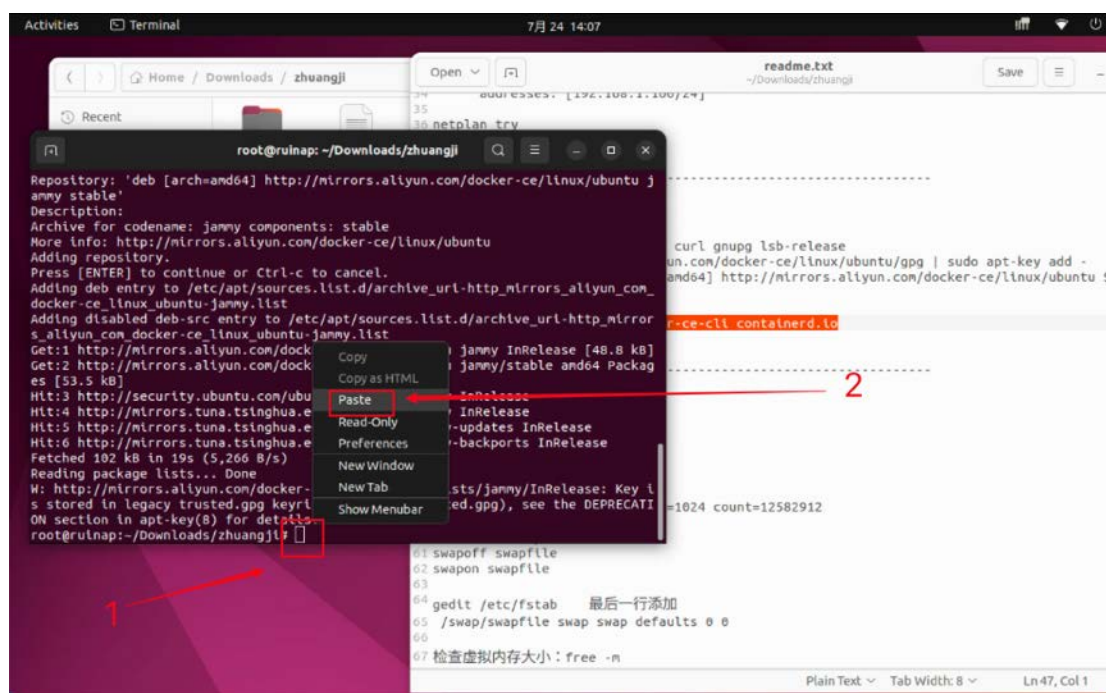
回到右侧文档中，复制如下框选内容



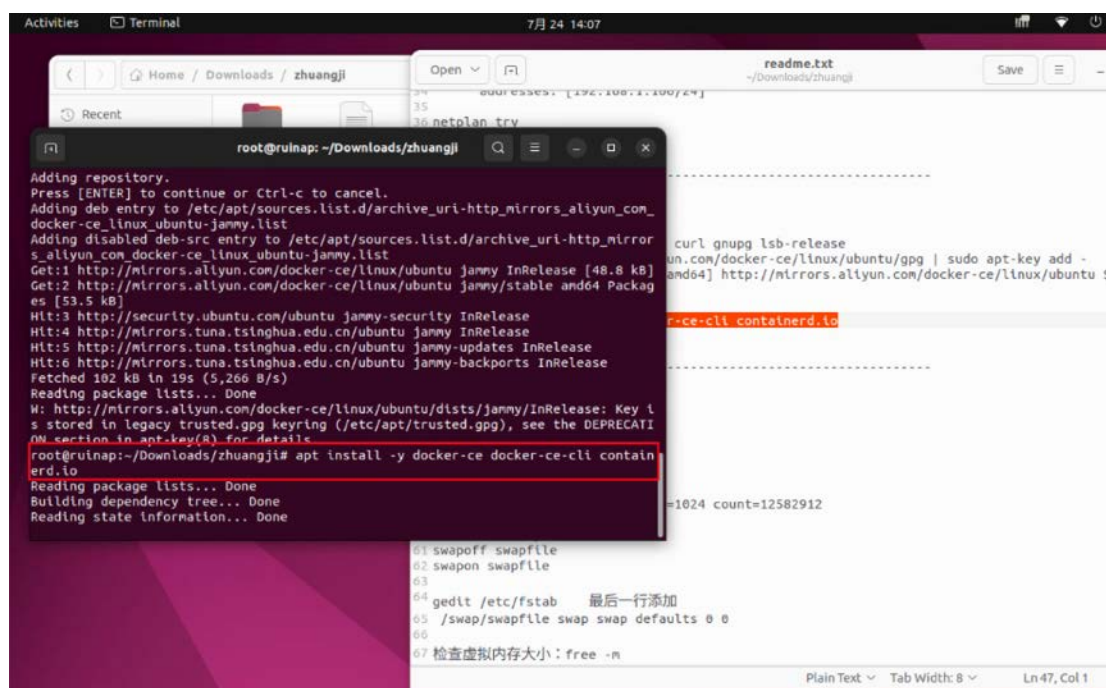
```
apt install -y docker-ce docker-ce-cli containerd.io
```

The image shows a text editor window with a context menu open over the line `apt install -y docker-ce docker-ce-cli containerd.io`. The menu options are: Undo, Redo, Cut, Copy (highlighted), Paste, Delete, Select All, Insert Emoji, and Change Case. Red arrows point to the selected text and the 'Copy' menu option.

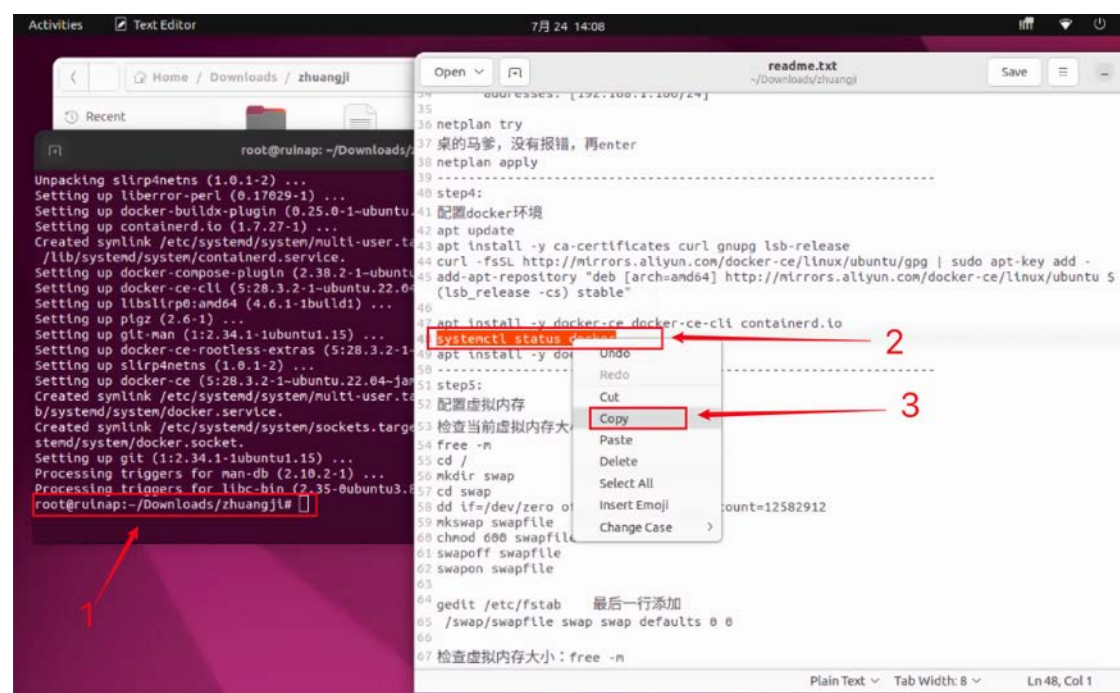
粘贴文本后键盘按下【回车键】



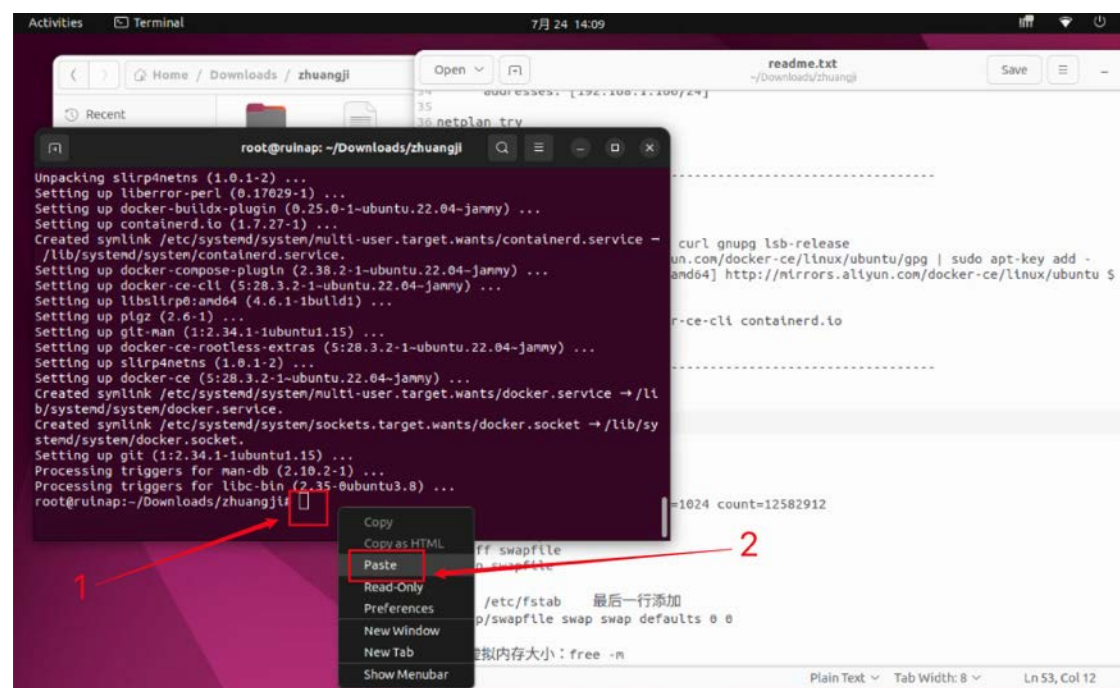
等待终端打印内容



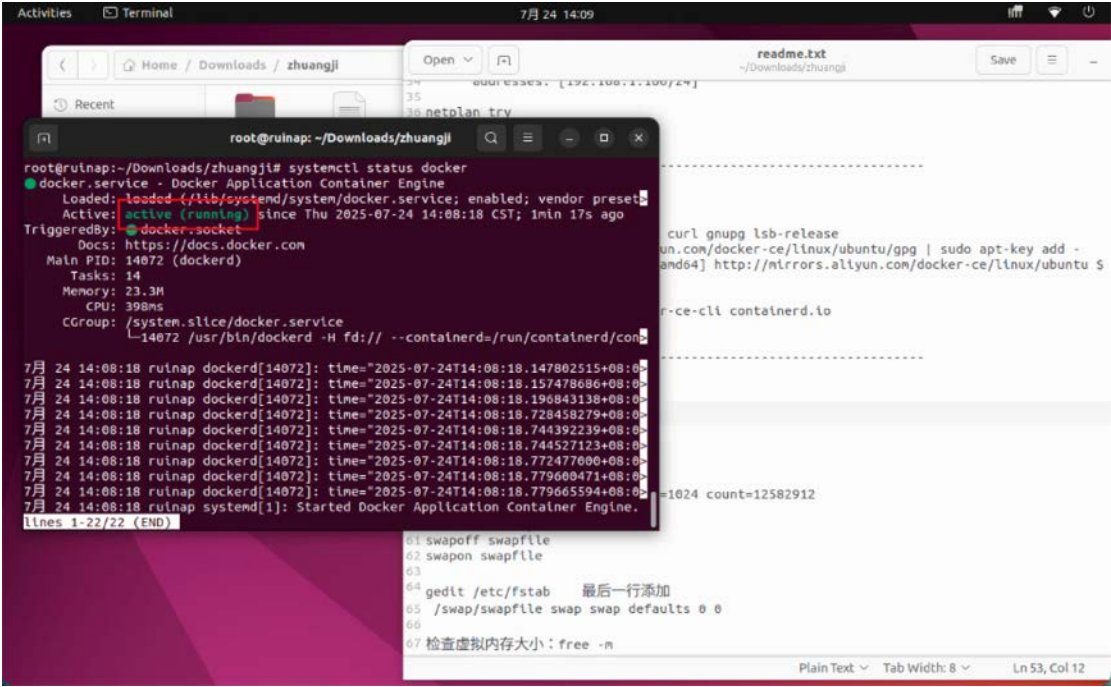
重新显示如下内容并出现跳动光标后，回到右侧文档中，复制如下框选的文本



粘贴文本后键盘按下【回车键】

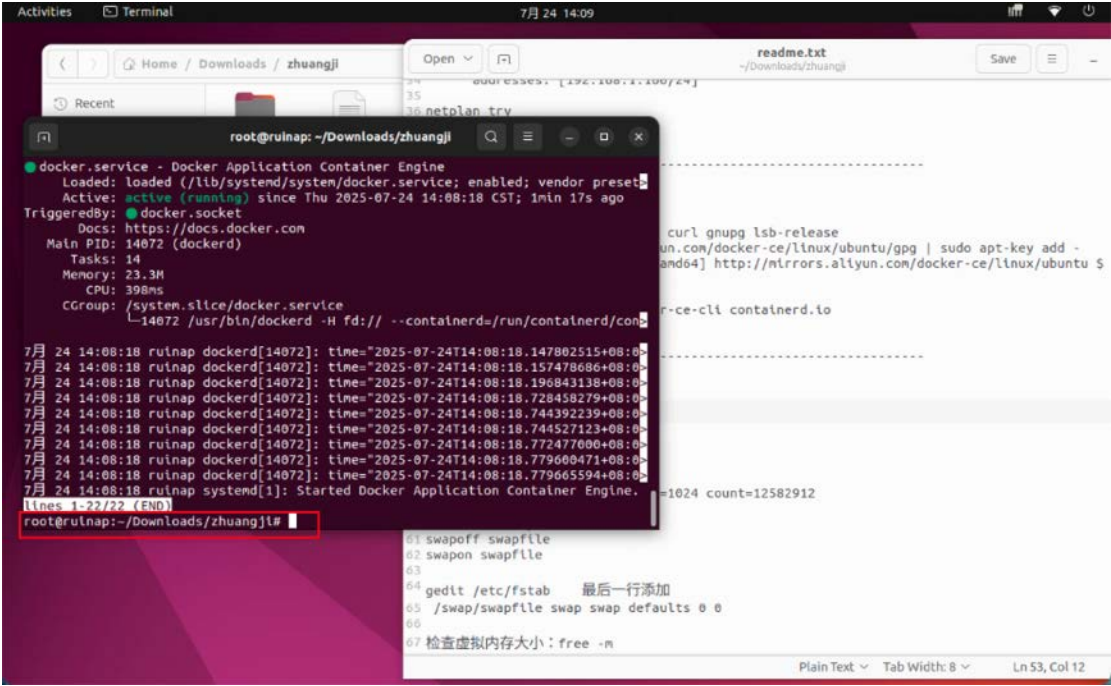


显示如下所示的绿色 activate 则表明安装成功



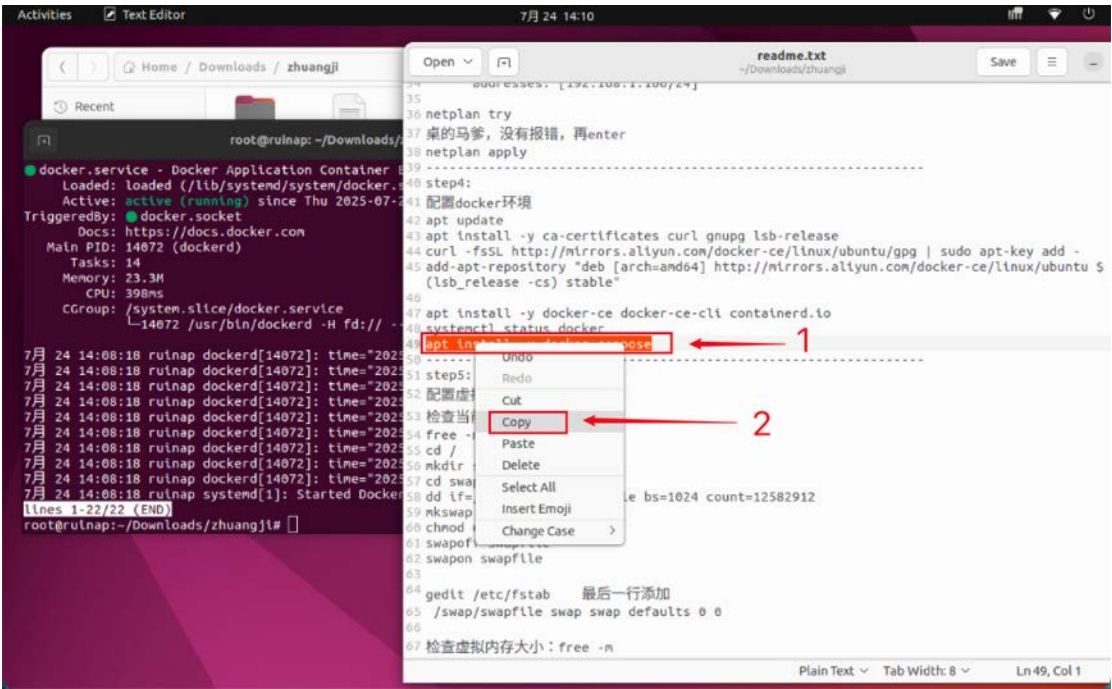
The screenshot shows a terminal window with the command `systemctl status docker` executed. The output indicates that the `docker.service` is **active (running)**, highlighted with a green box. Below this, the `docker.socket` is also shown as active. The terminal also displays a series of logs for the `dockerd` service, showing its startup and initial operations. The terminal window is titled `root@ruinap: ~/Downloads/zhuangji`. In the background, a file manager window shows the `Downloads/zhuangji` directory, and a `readme.txt` file is open, displaying instructions for Docker installation.

在终端页面下，键盘同时按下【ctrl】+【c】可以退出上述查看，重新回到正常的跳动光标

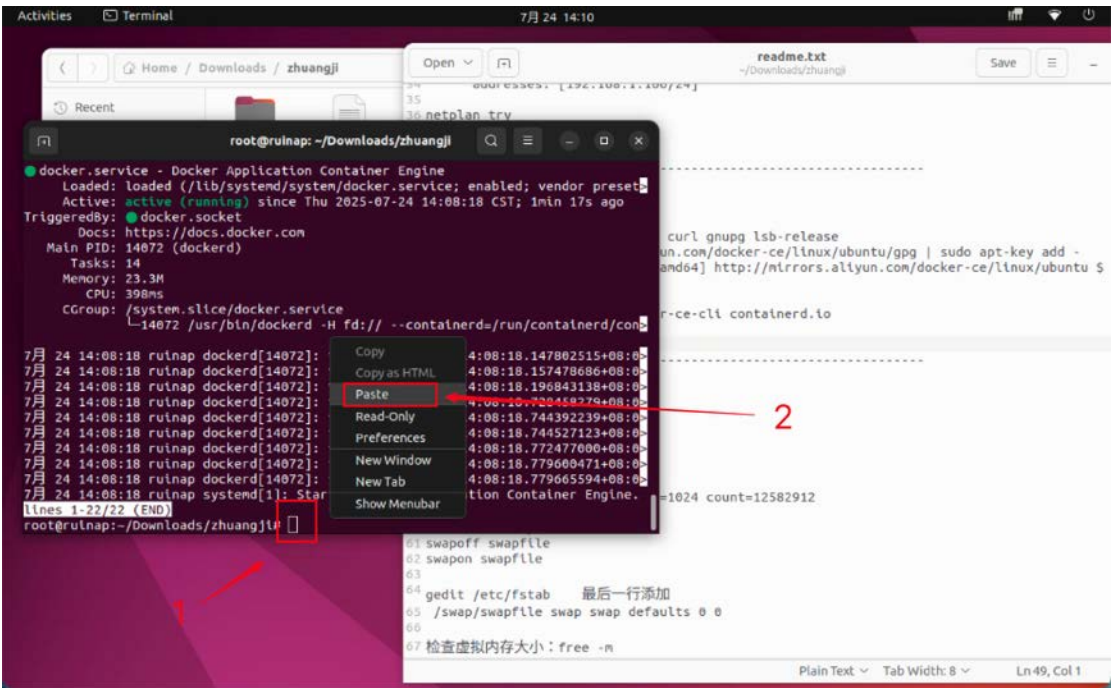


This screenshot shows the same terminal window after pressing `Ctrl+C`. The terminal has returned to the prompt `root@ruinap:~/Downloads/zhuangji#`, which is highlighted with a red box. The background elements, including the file manager and the `readme.txt` file, remain the same as in the previous screenshot.

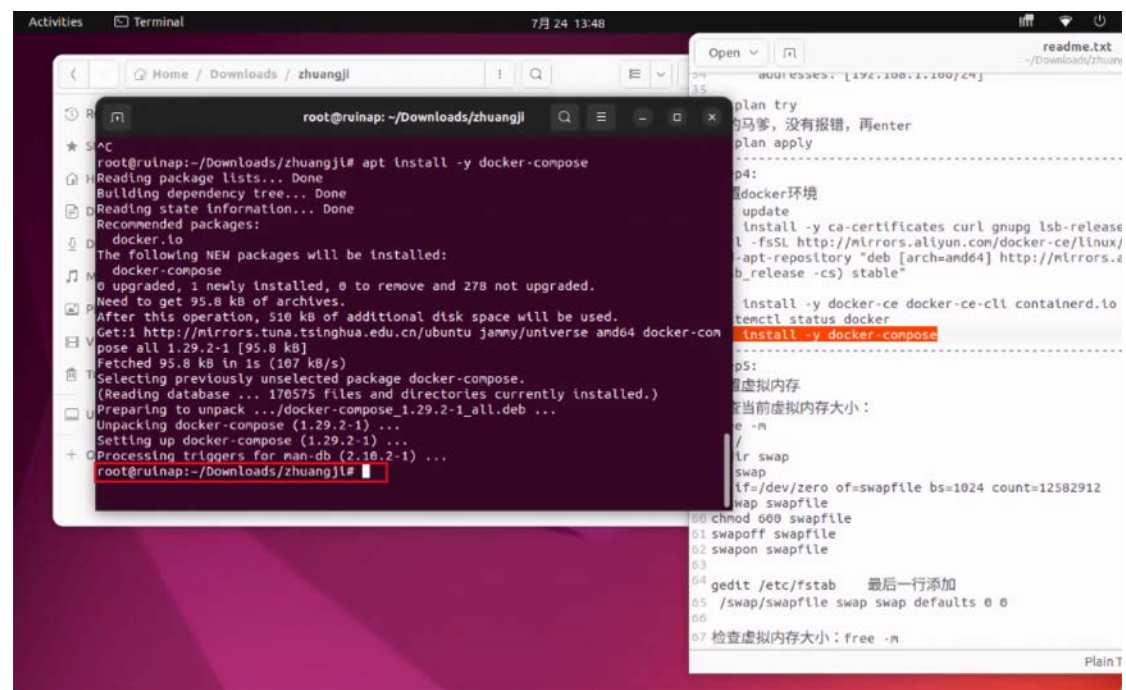
回到文档中，复制如下框选的文本



粘贴文本后键盘按下【回车键】



等待终端打印，重新显示如下内容并出现跳动光标后，表示安装完成

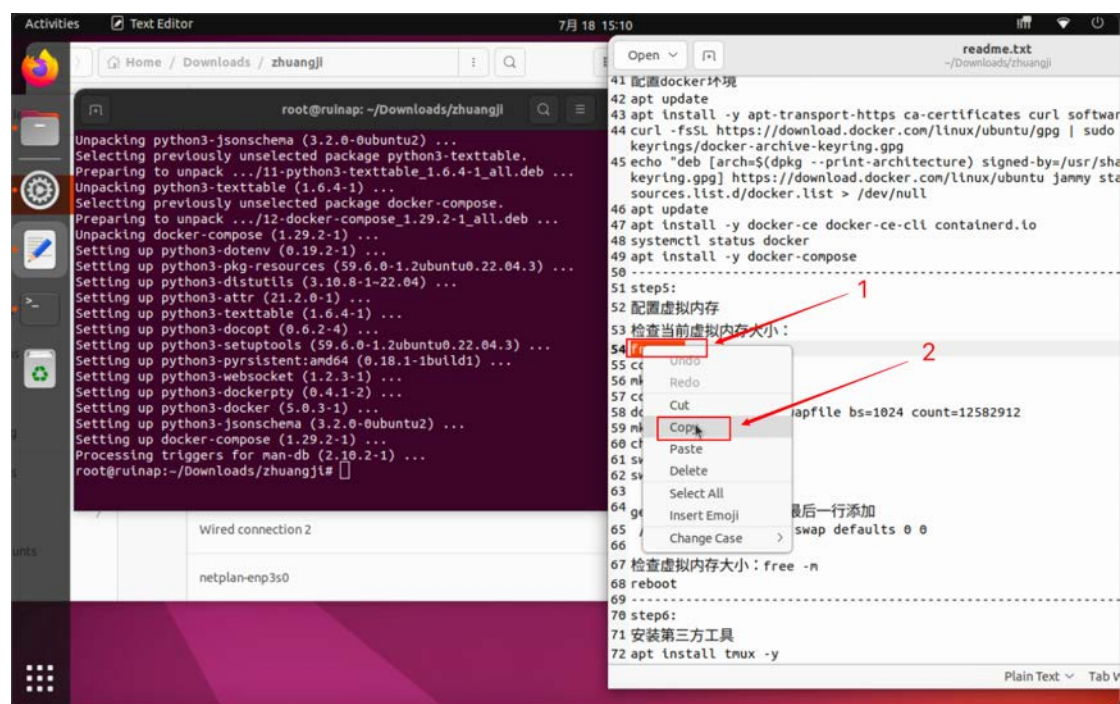


```
root@ruinap:~/Downloads/zhuangji# apt install -y docker-compose
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Recommended packages:
  docker.io
The following NEW packages will be installed:
  docker-compose
0 upgraded, 1 newly installed, 0 to remove and 278 not upgraded.
Need to get 95.8 kB of archives.
After this operation, 510 kB of additional disk space will be used.
Get:1 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy/universe amd64 docker-com
pose all 1.29.2-1 [95.8 kB]
Fetched 95.8 kB in 1s (107 kB/s)
Selecting previously unselected package docker-compose.
(Reading database ... 170575 files and directories currently installed.)
Preparing to unpack .../docker-compose_1.29.2-1_all.deb ...
Unpacking docker-compose (1.29.2-1) ...
Setting up docker-compose (1.29.2-1) ...
Processing triggers for man-db (2.10.2-1) ...
root@ruinap:~/Downloads/zhuangji#
```

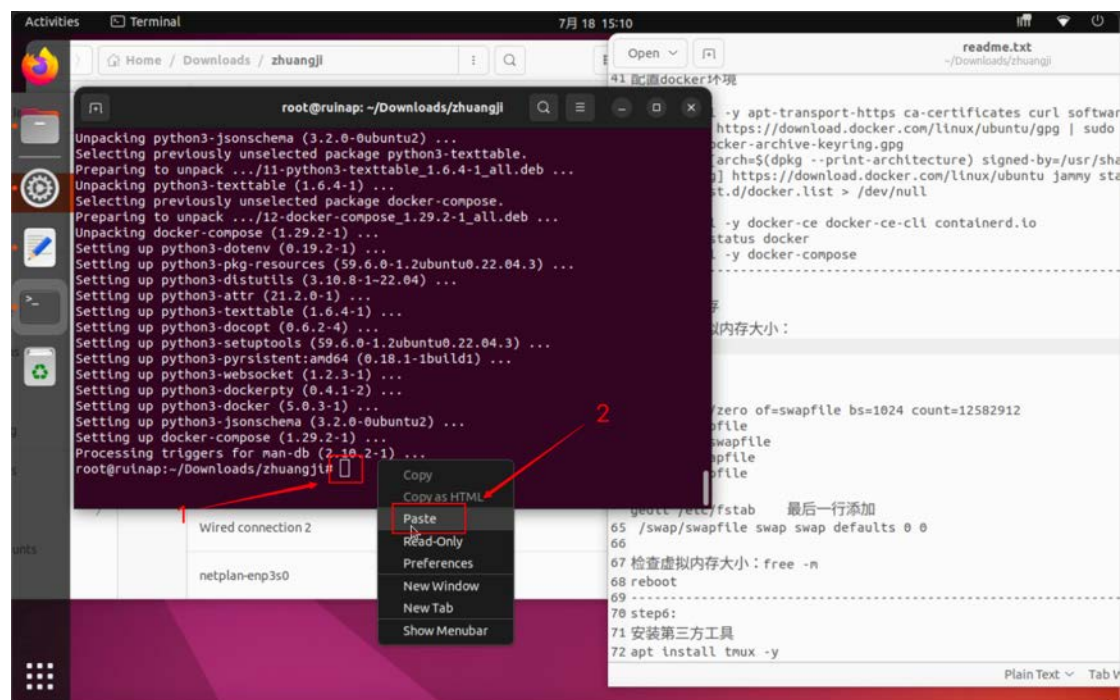
至此，docker 安装完成

6. 虚拟内存设置

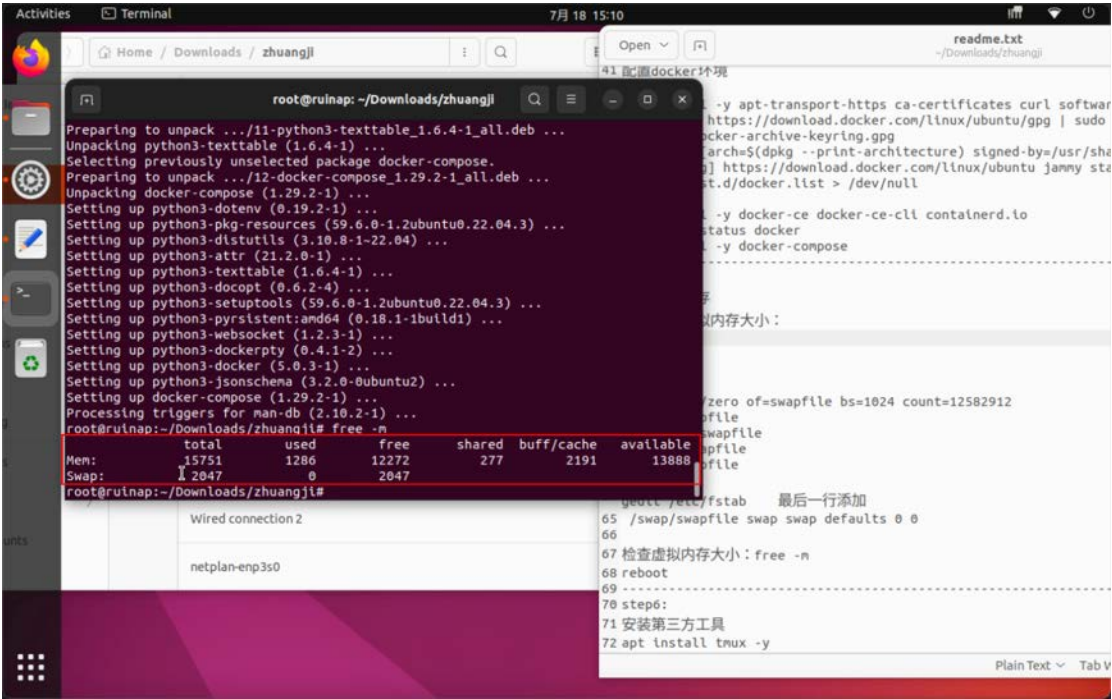
如果已经关闭终端和文件夹以及文档，参照前面的小节重新打开，复制如下框选的文
本



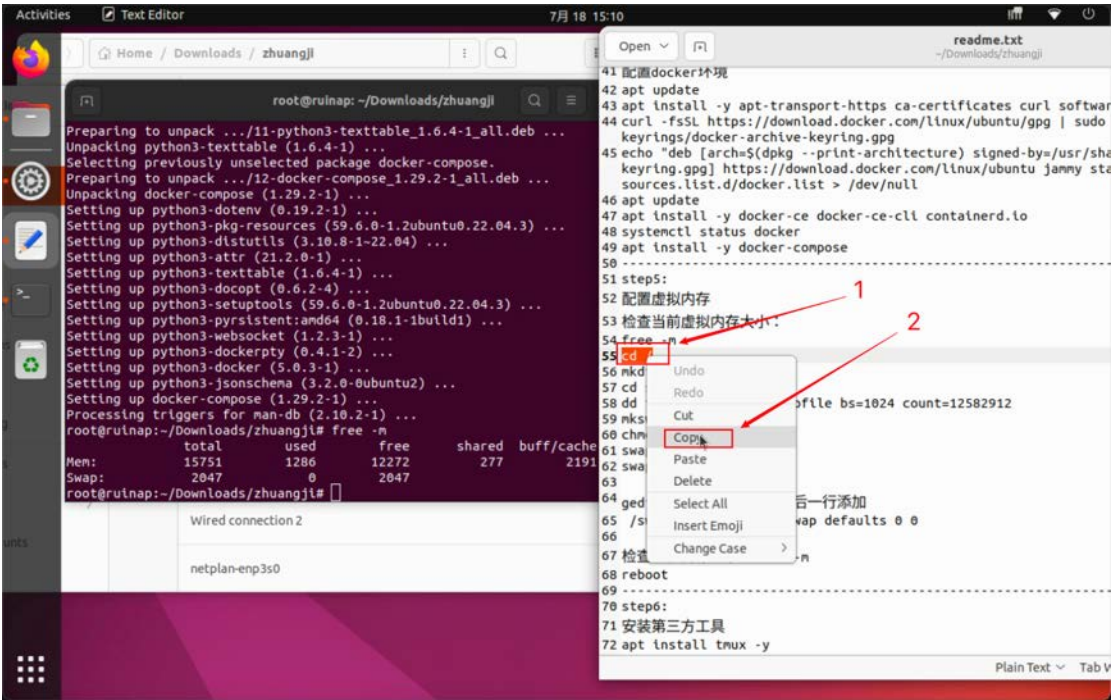
粘贴文本后键盘按下【回车键】



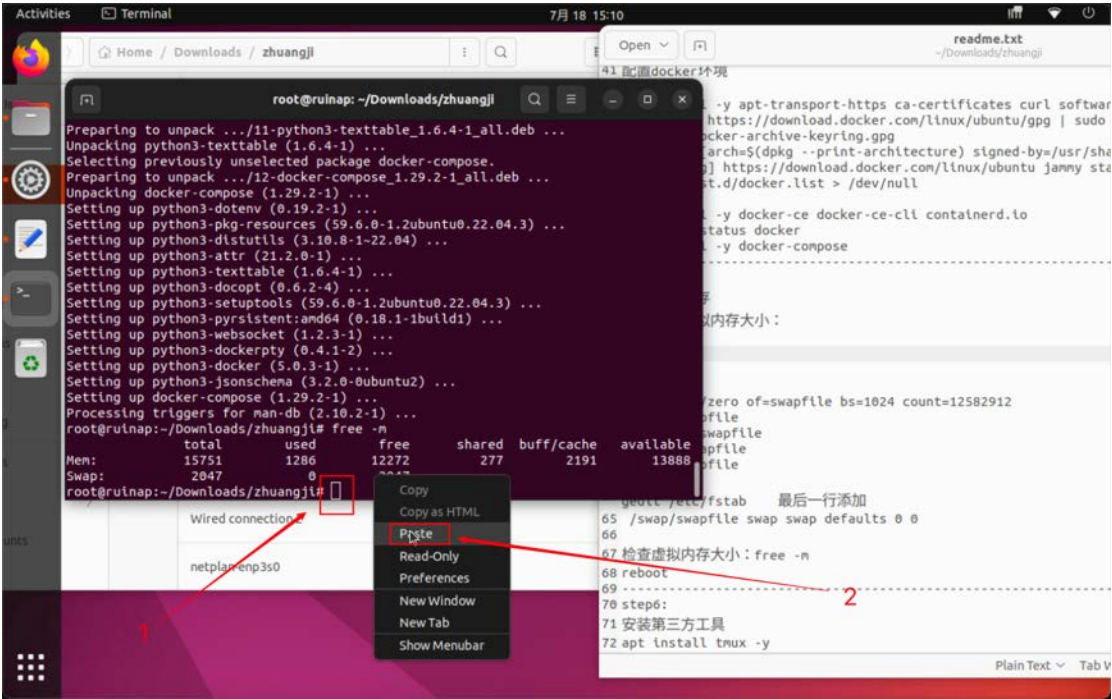
终端会打印显示当前的内存



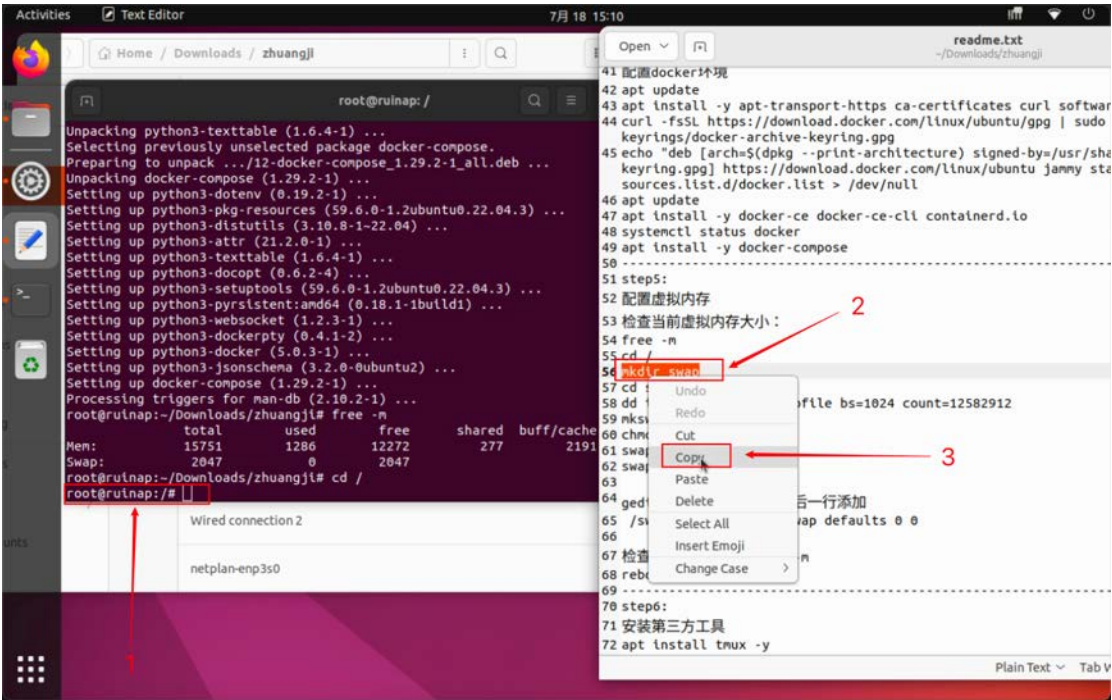
回到文档中，复制如下文本



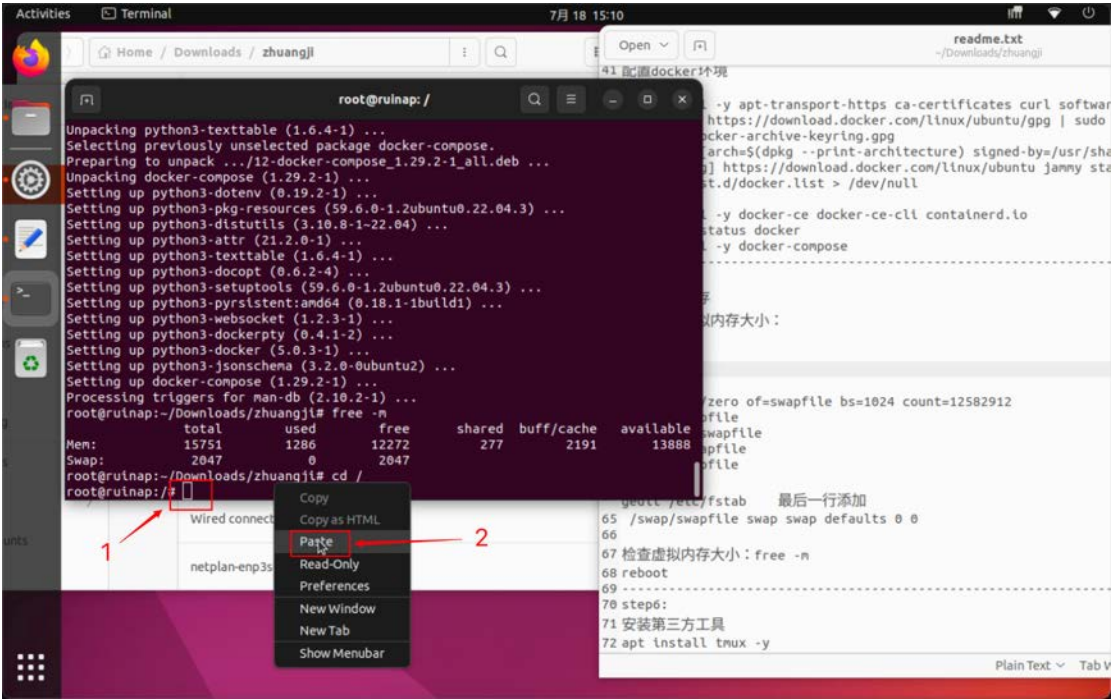
粘贴文本后键盘按下【回车键】



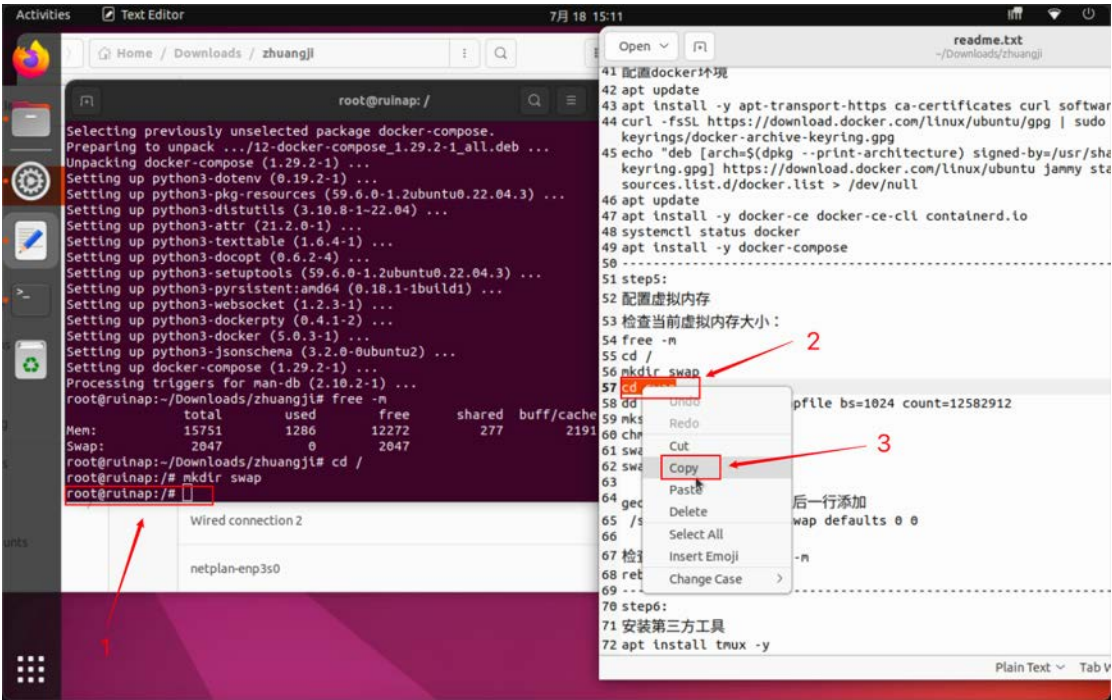
出现如下 1 后，回到文档中复制 2 的内容



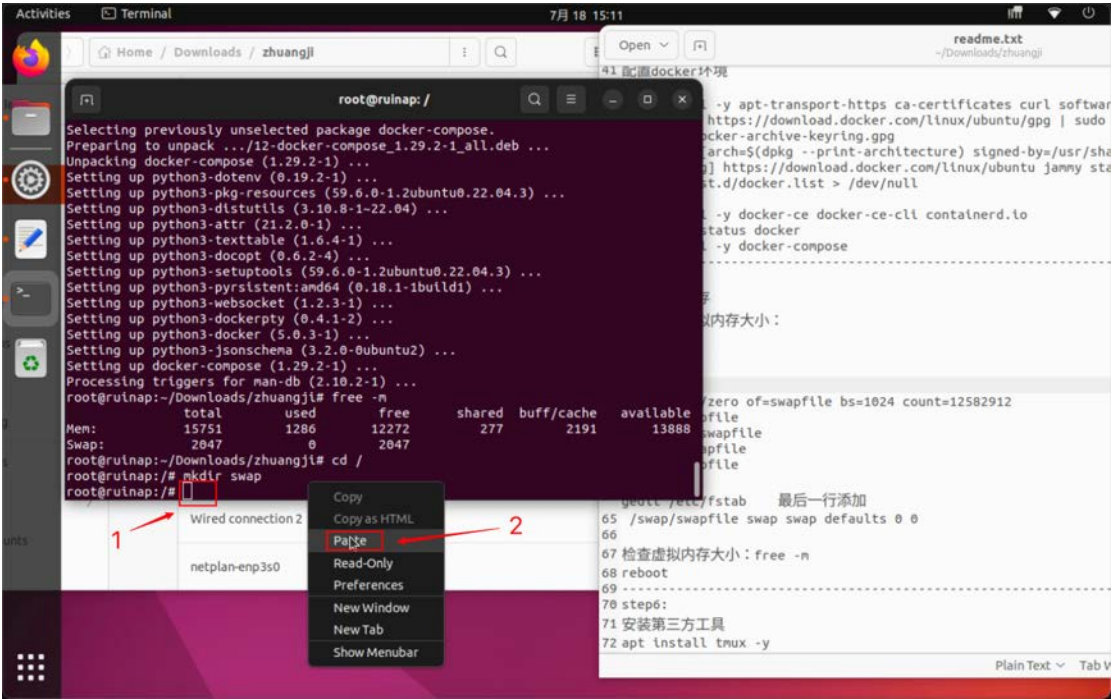
粘贴文本后键盘按下【回车键】



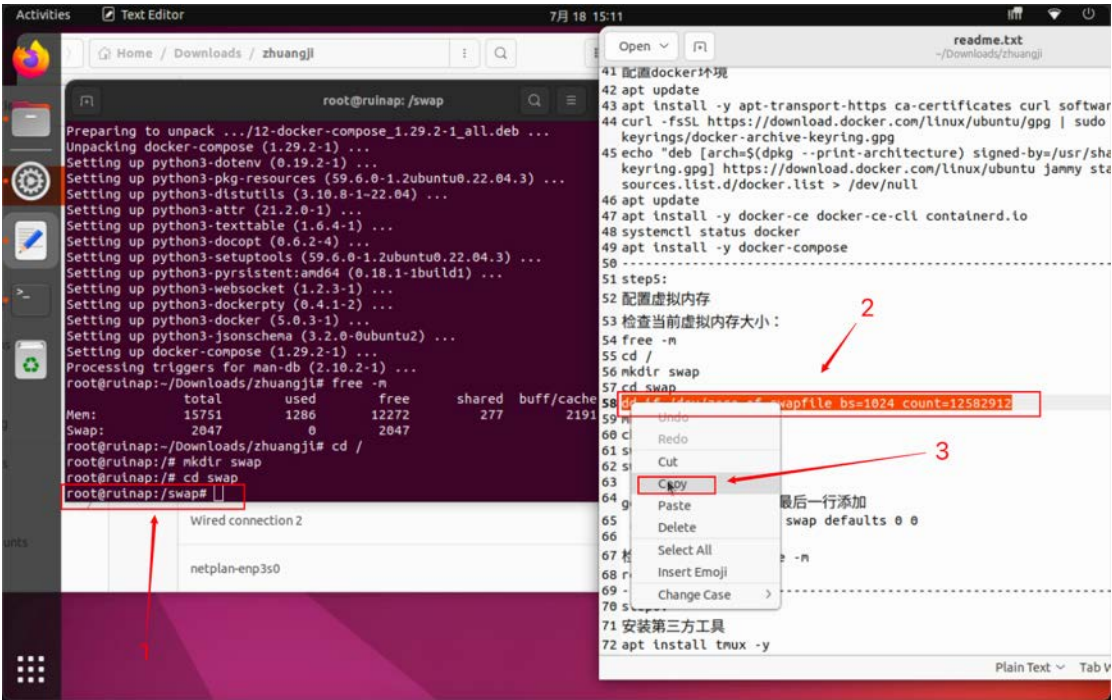
出现如下 1 并有跳动的光标后，回到文档中复制 2 的内容



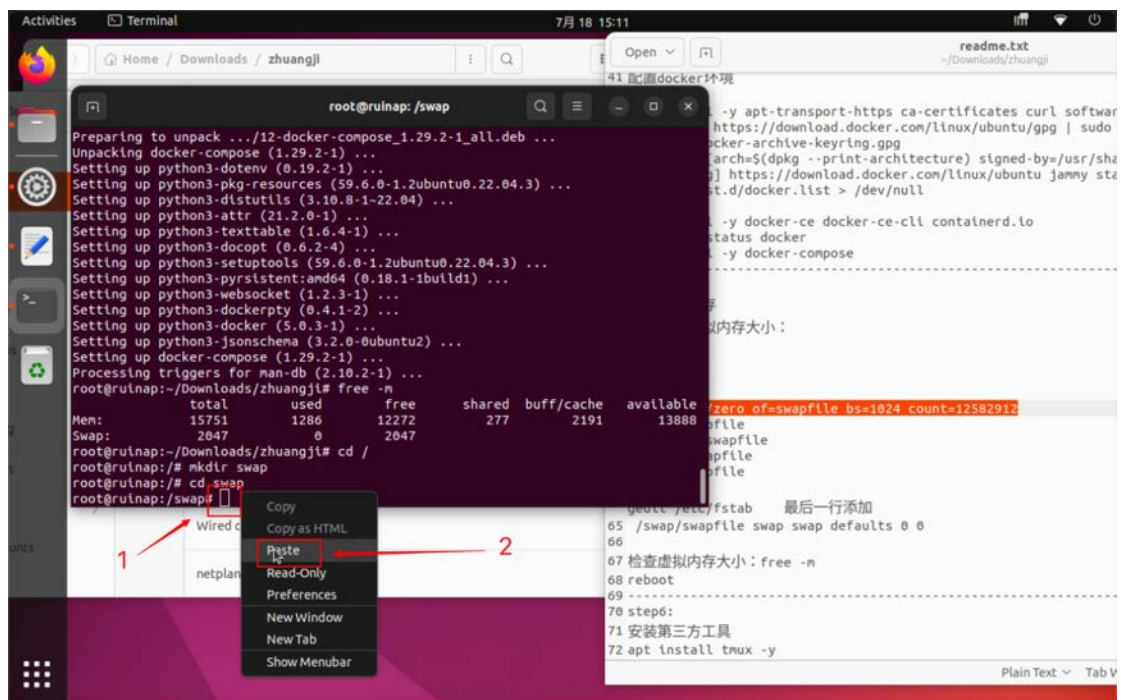
粘贴文本后键盘按下【回车键】



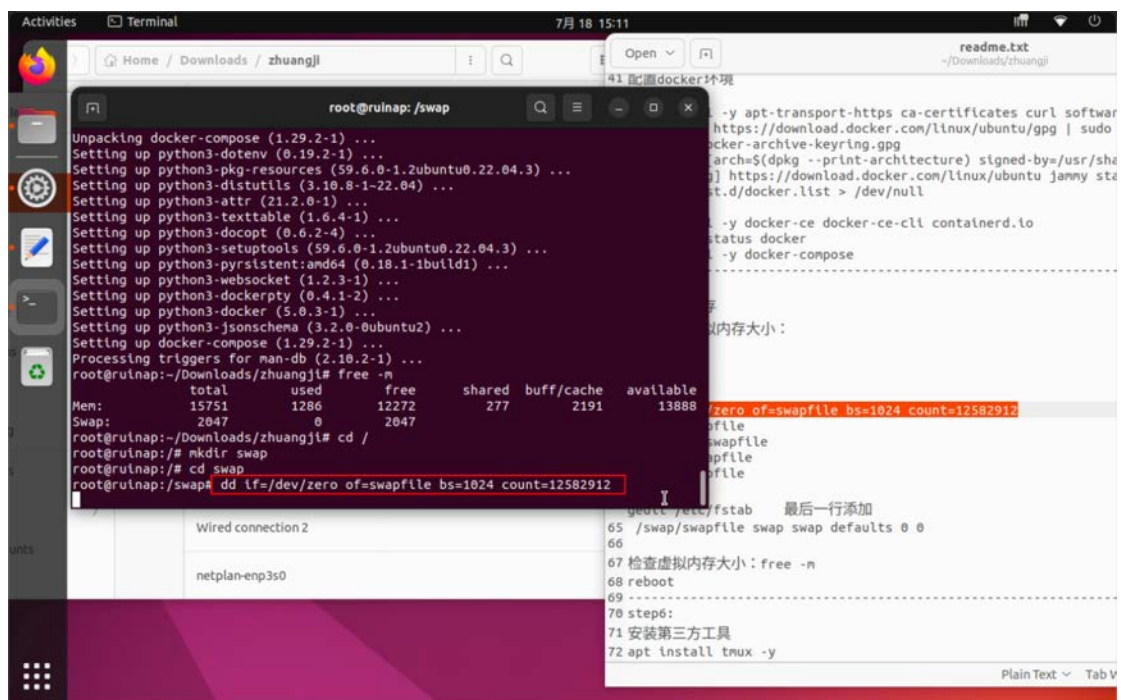
出现如下 1 并有跳动的光标后，回到文档中复制 2 的内容



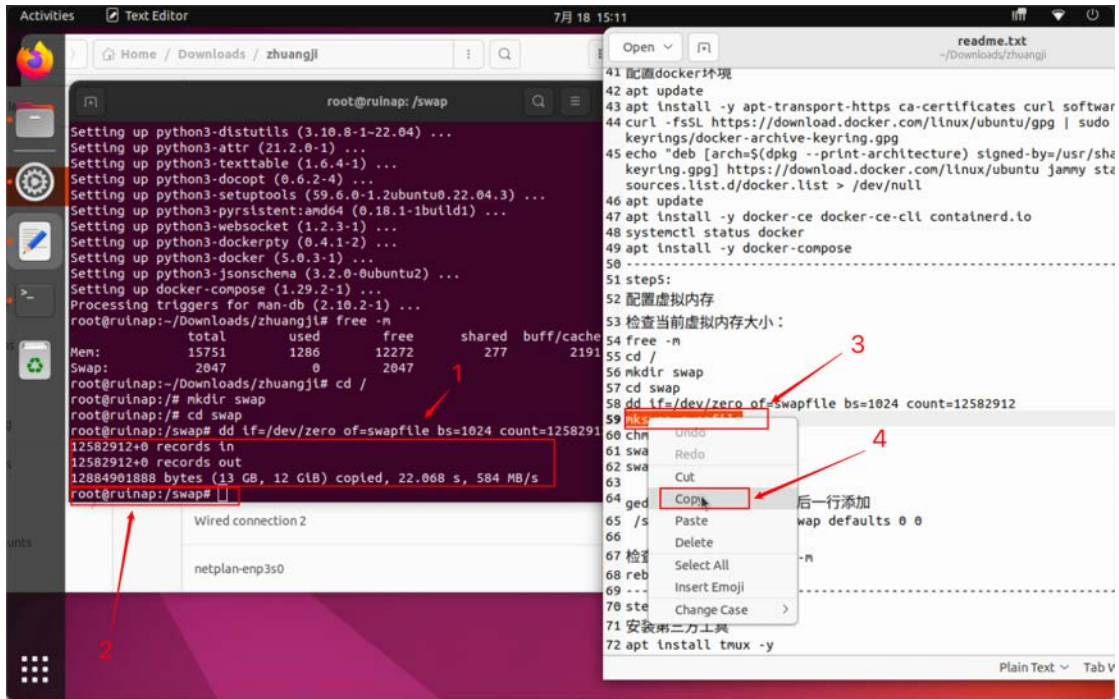
粘贴文本后键盘按下【回车键】



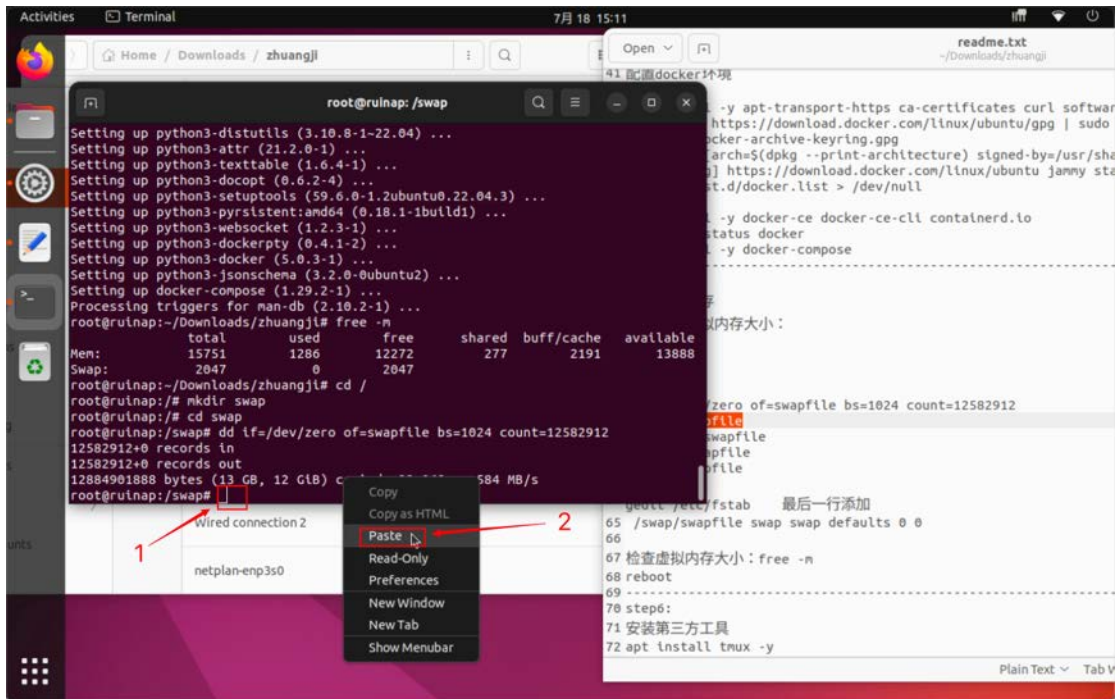
等待终端页面



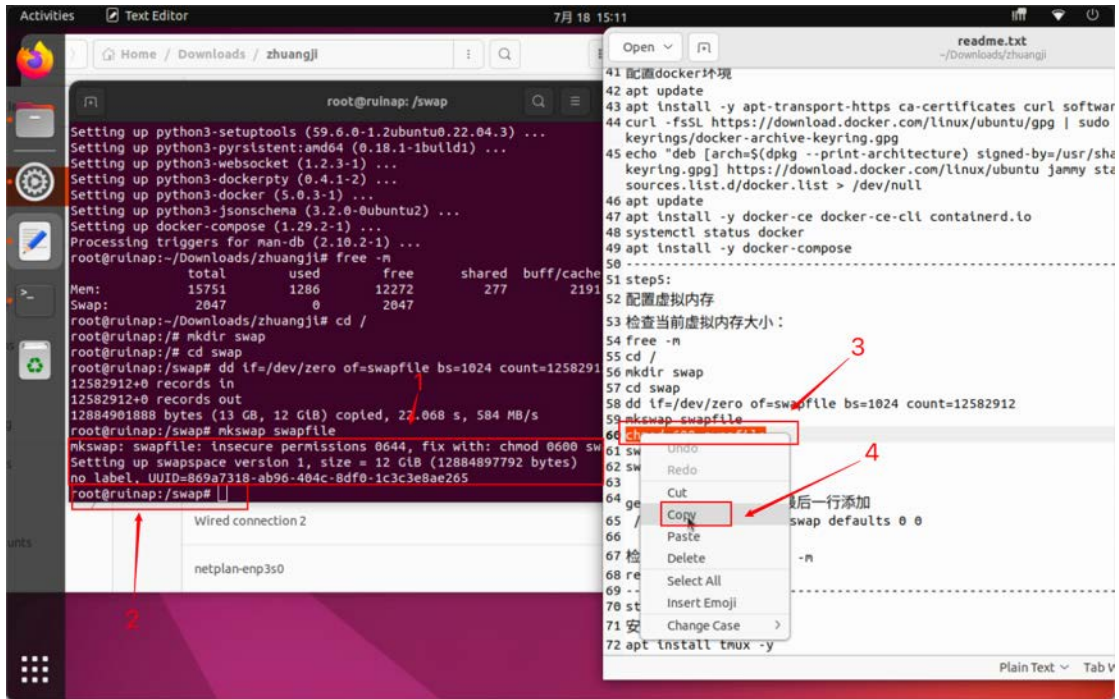
出现如下 1 所示内容，并显示 2 且光标重新跳动后，回到右侧文档复制 3 处的内容



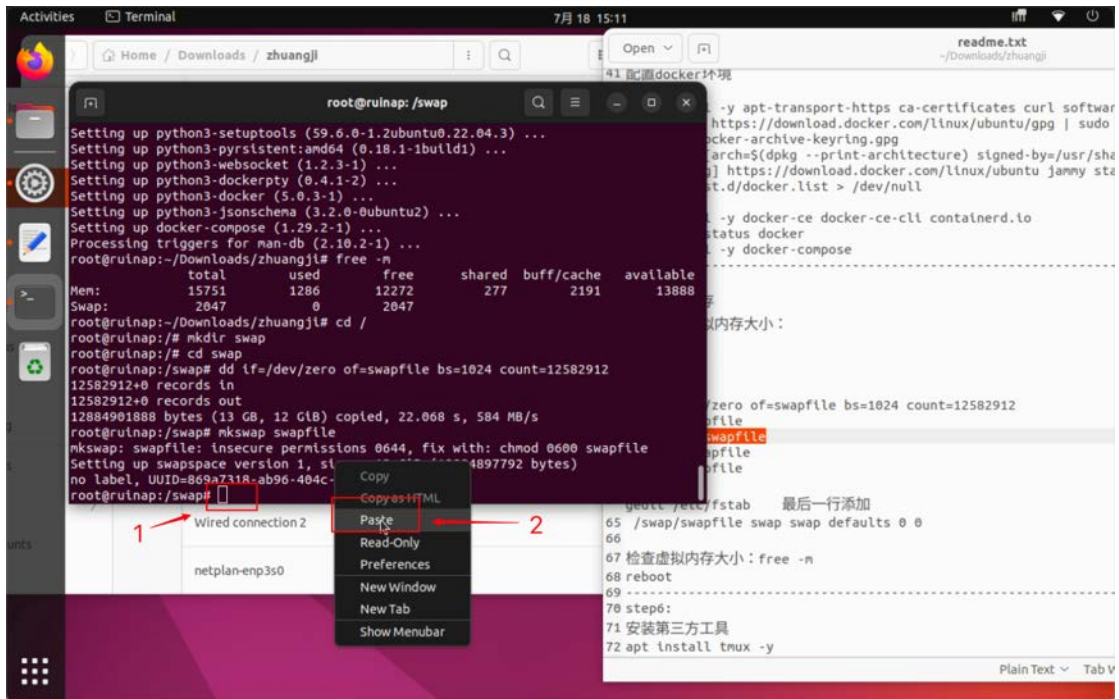
粘贴文本后键盘按下【回车键】



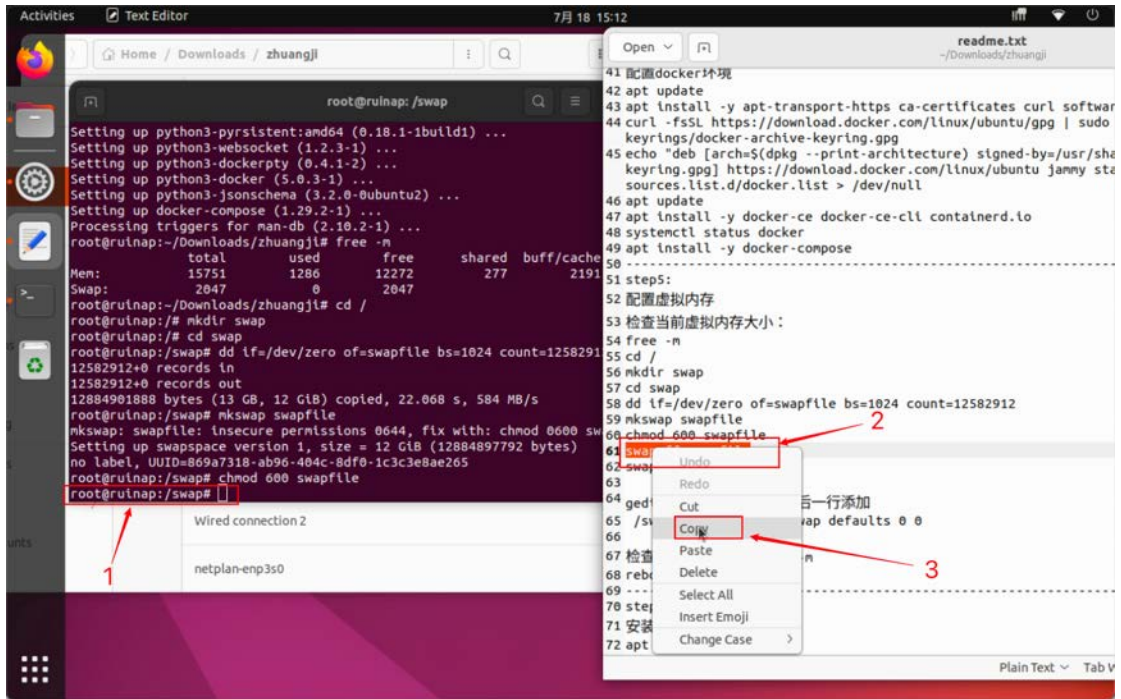
出现如下 1 所示内容，并显示 2 且光标重新跳动后，回到右侧文档复制 3 处的内容



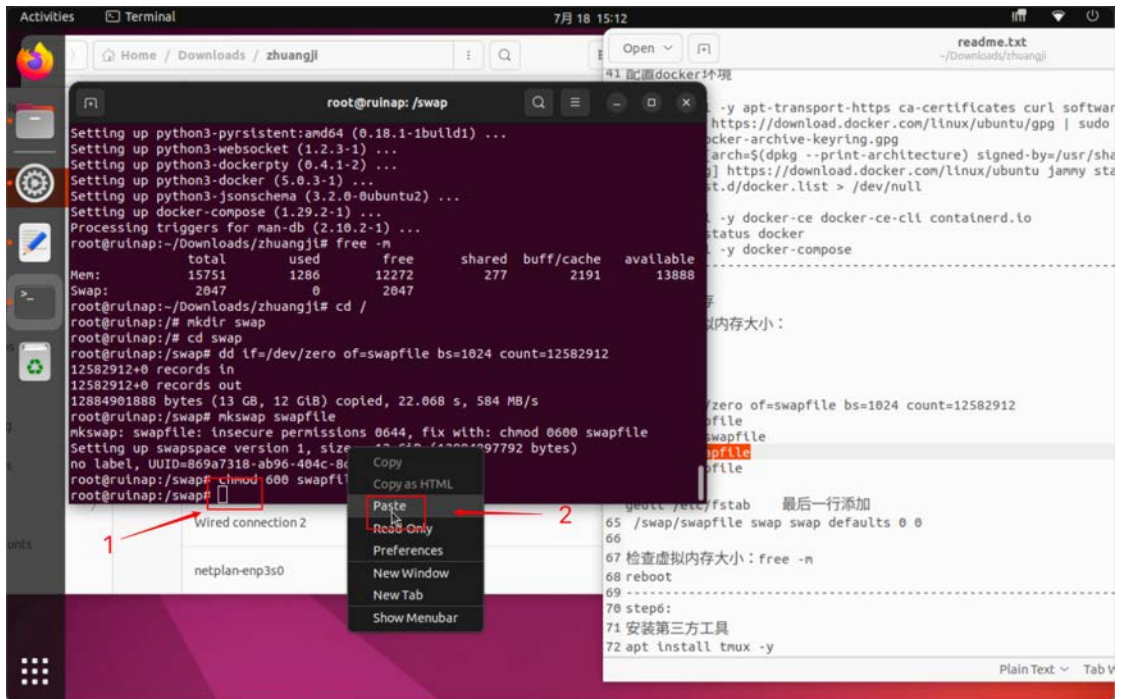
粘贴文本后键盘按下【回车键】



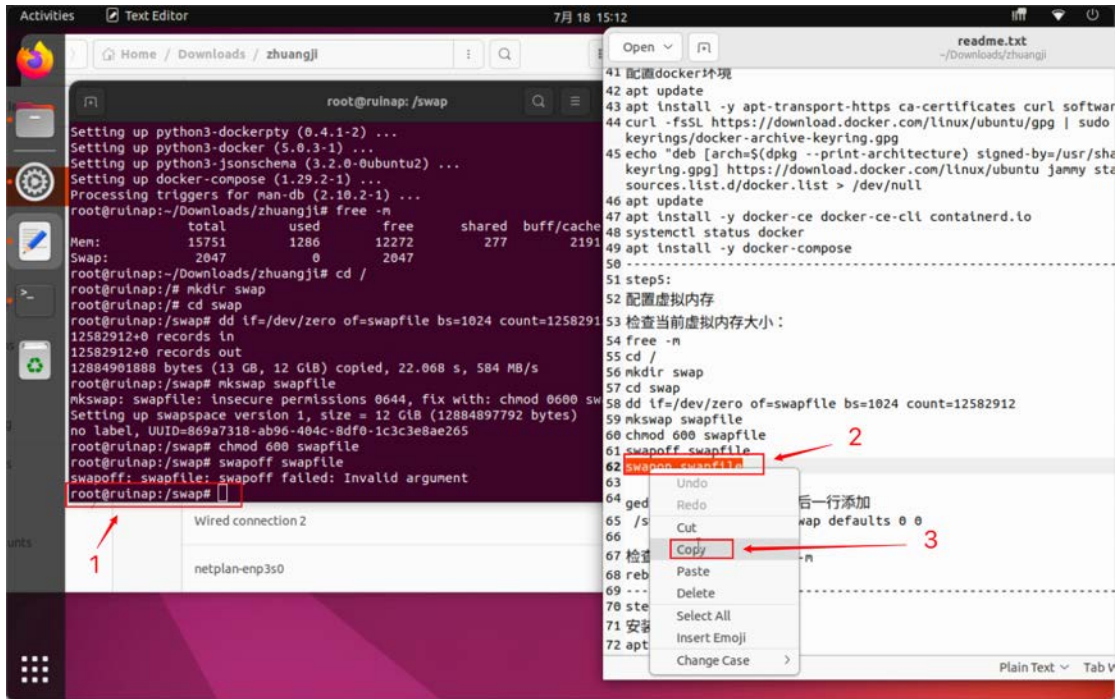
出现 1 所示内容后，回到文档中复制如下所示文本



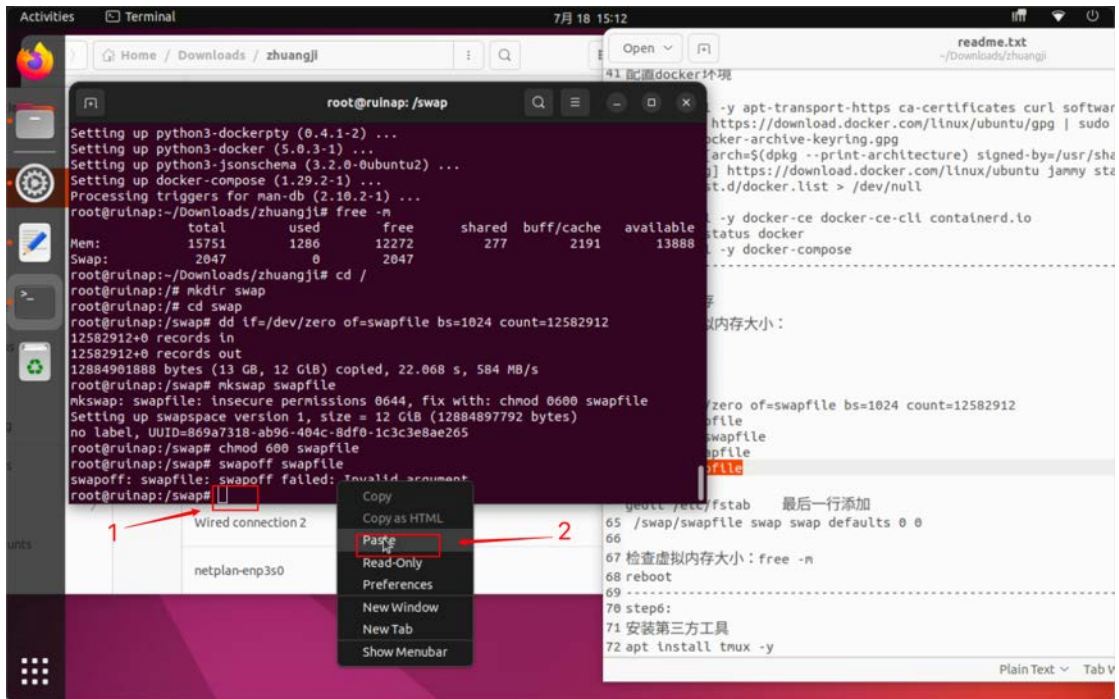
粘贴文本后键盘按下【回车键】



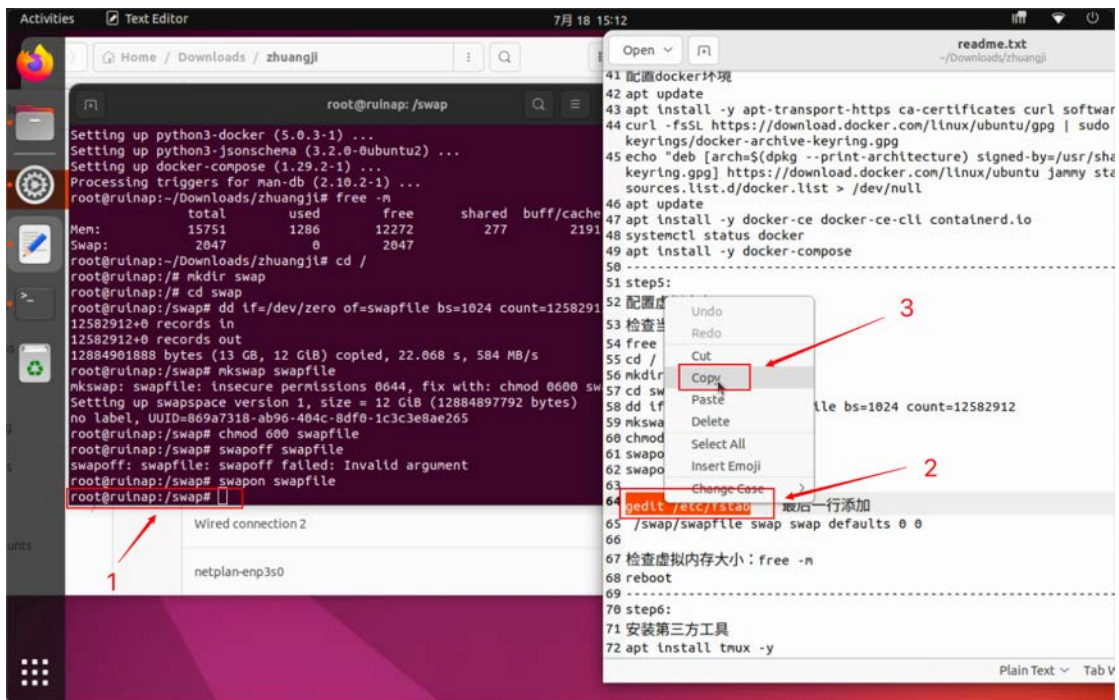
出现 1 所示内容后，回到文档中复制如下所示文本



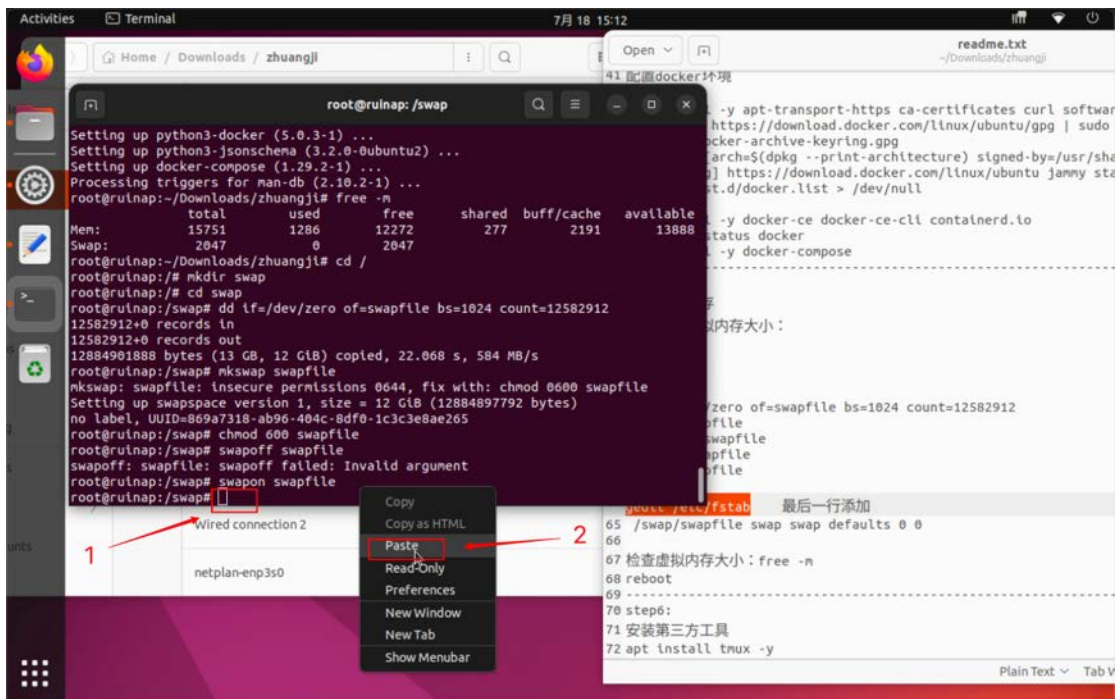
粘贴文本后键盘按下【回车键】



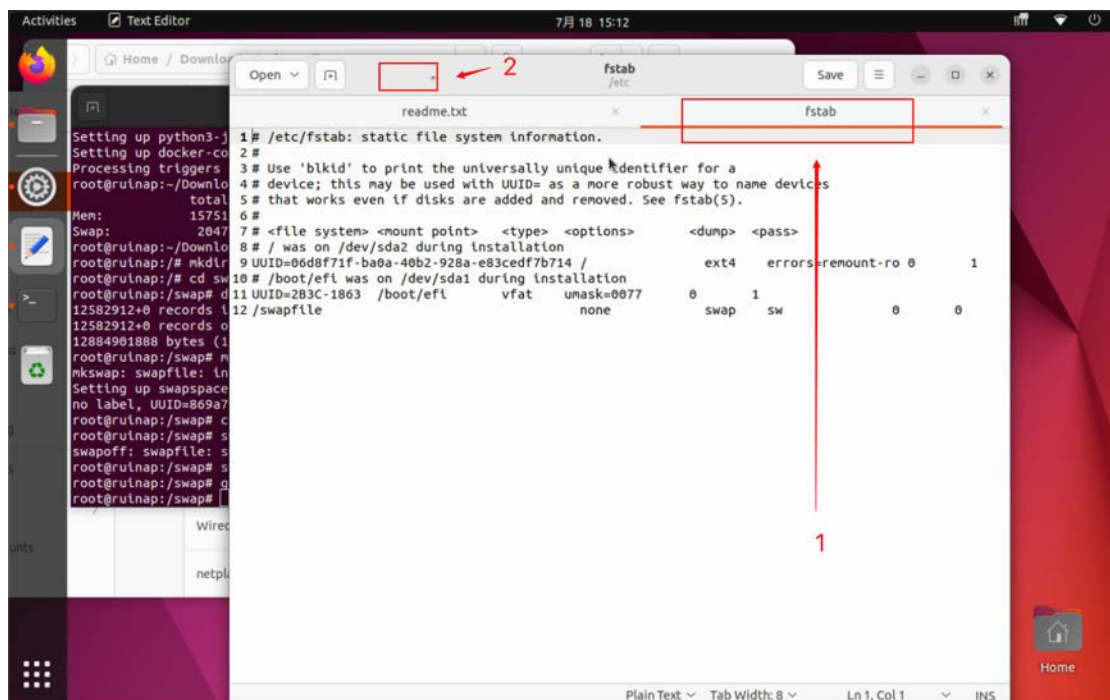
出现 1 所示内容后，回到文档中复制如下所示文本



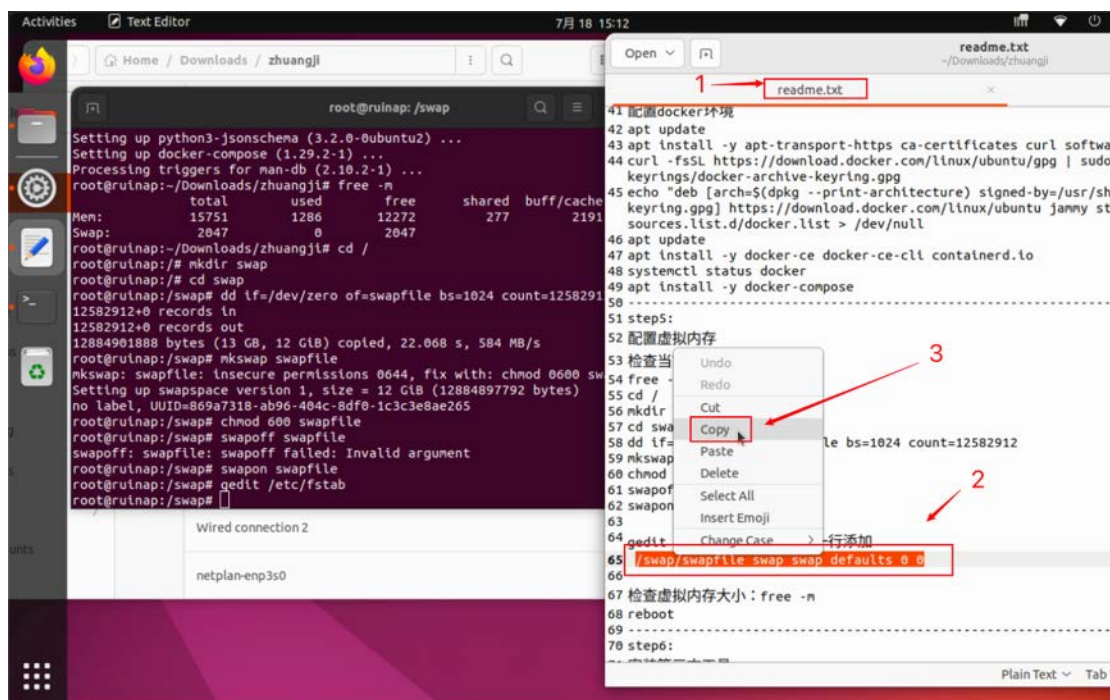
粘贴文本后键盘按下【回车键】



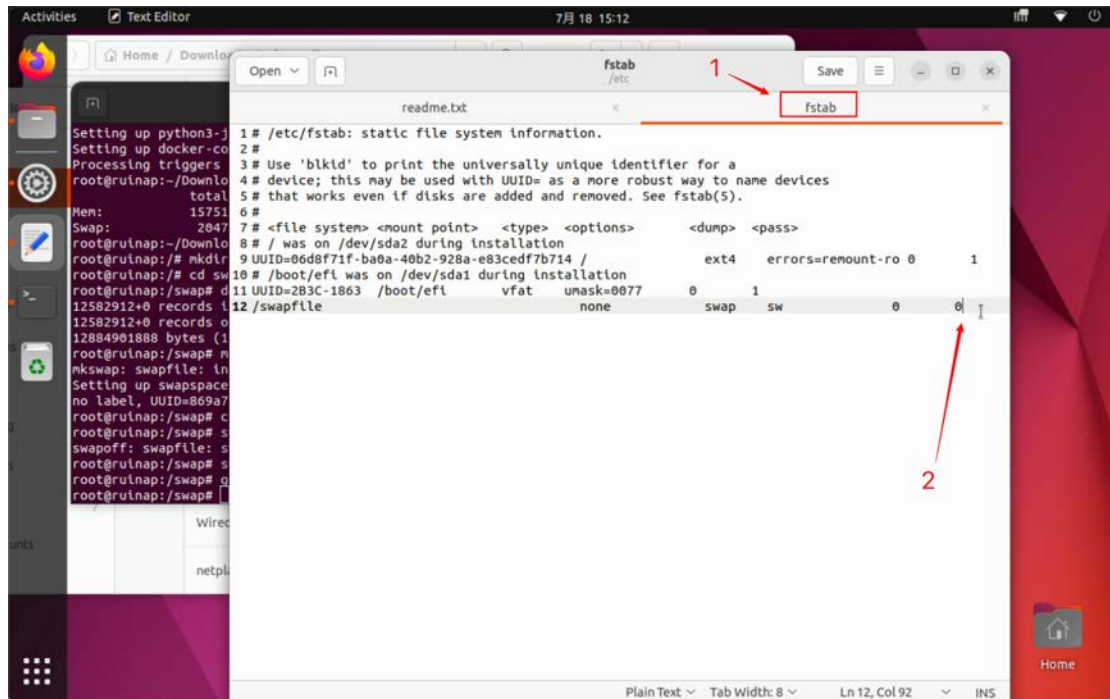
此时会打开一个新的文档页面，鼠标左键按下拖拽 2 处将文档编辑器拖动至合适位置



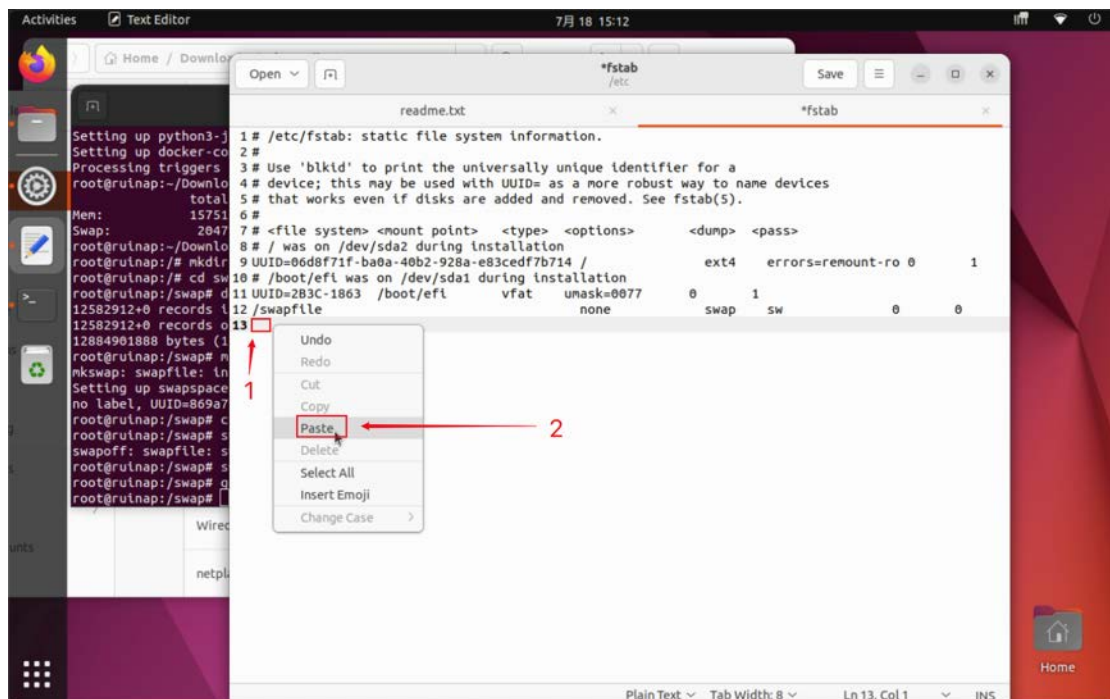
鼠标单击 1 处切换回“readme.txt”文档，复制 2 处的内容



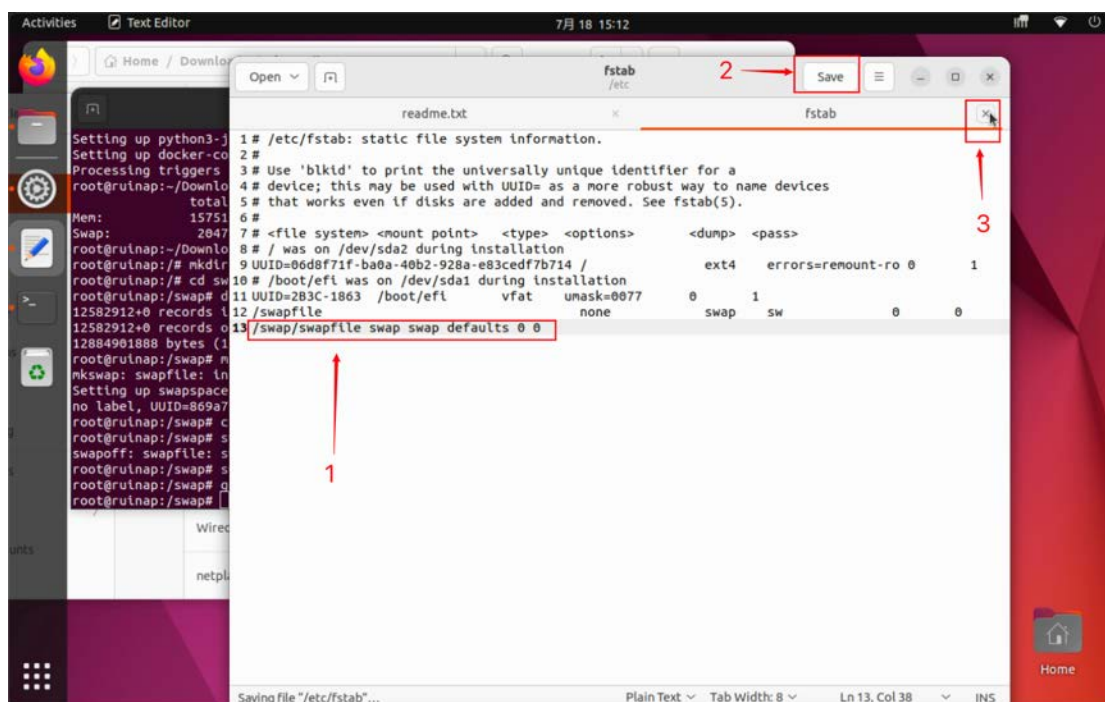
鼠标单击 1 处切换回“fstab”文档，单击 2 处将光标移动至如图所示，然后按下键盘【回车键】，新起一行



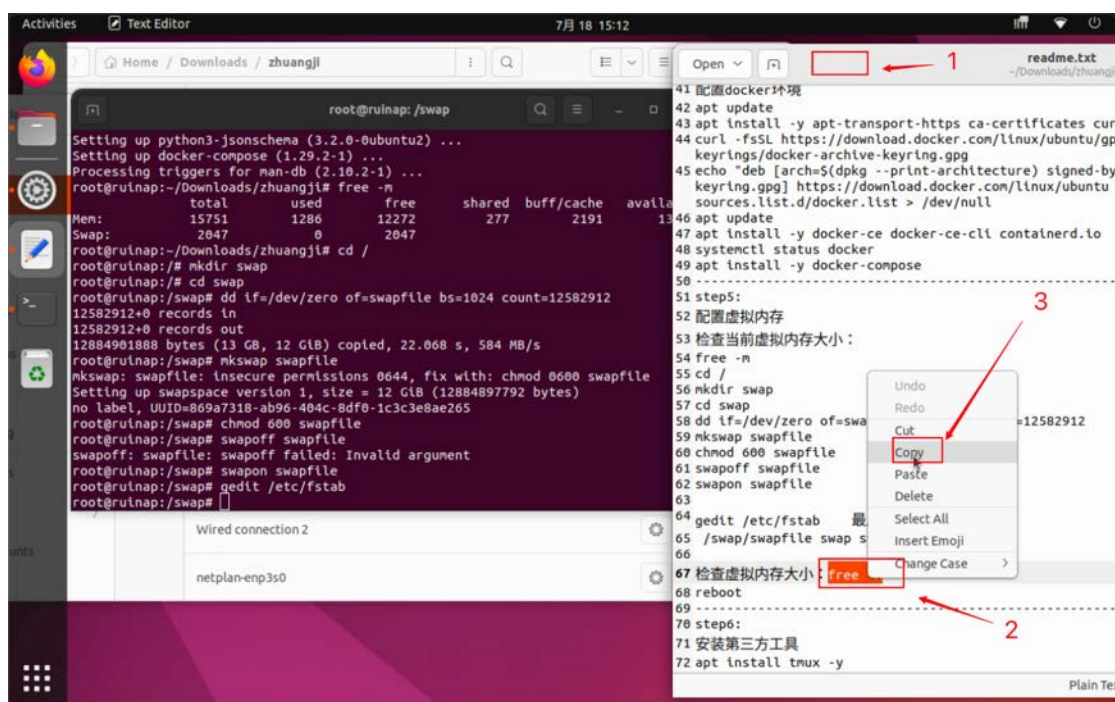
在新一行中如下位置粘贴内容



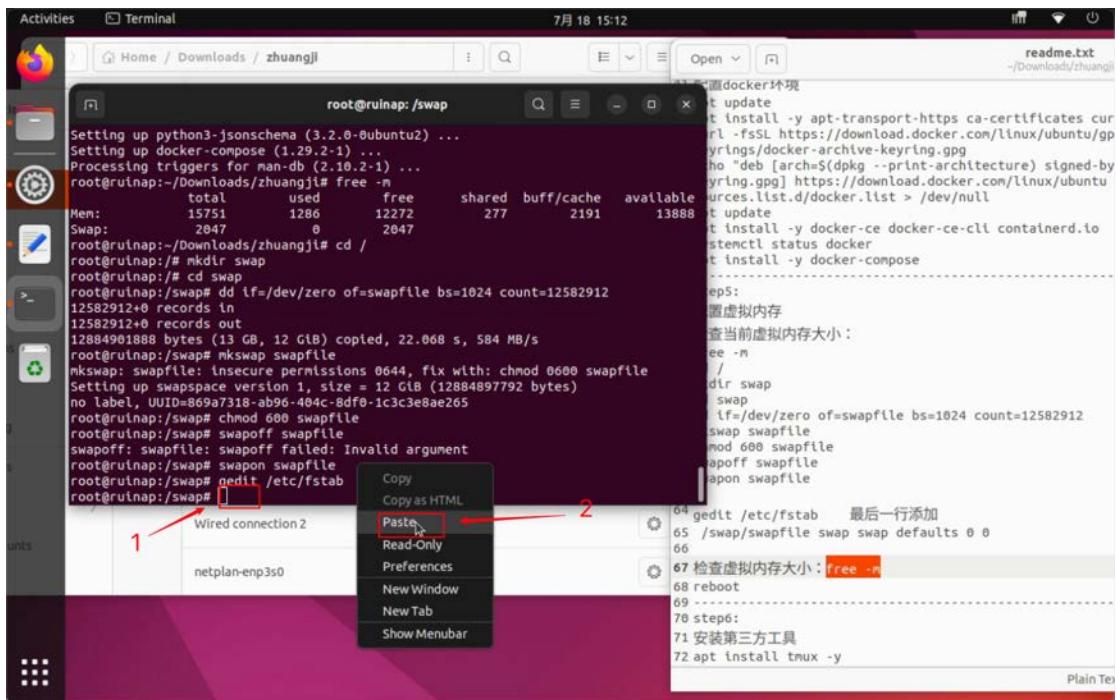
粘贴文本后，鼠标单击 2 处“Save”进行保存，然后单击 3 处关闭“fstab”文档



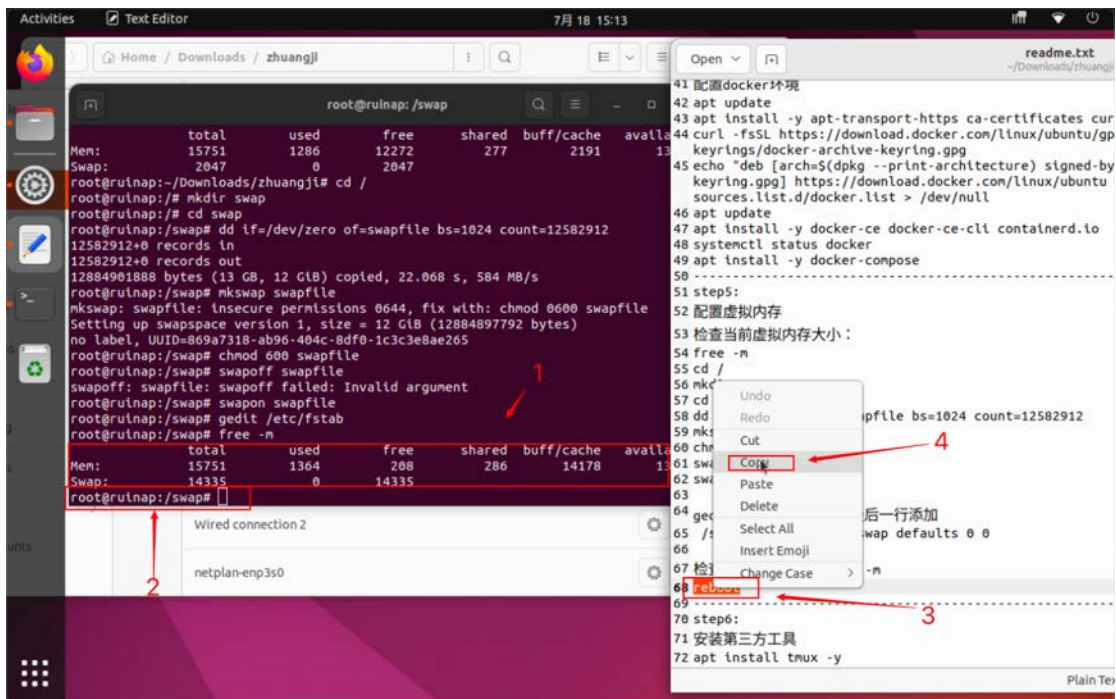
鼠标在 1 处左键按下并进行拖拽，将文档拖拽至合适位置，然后复制 2 处内容



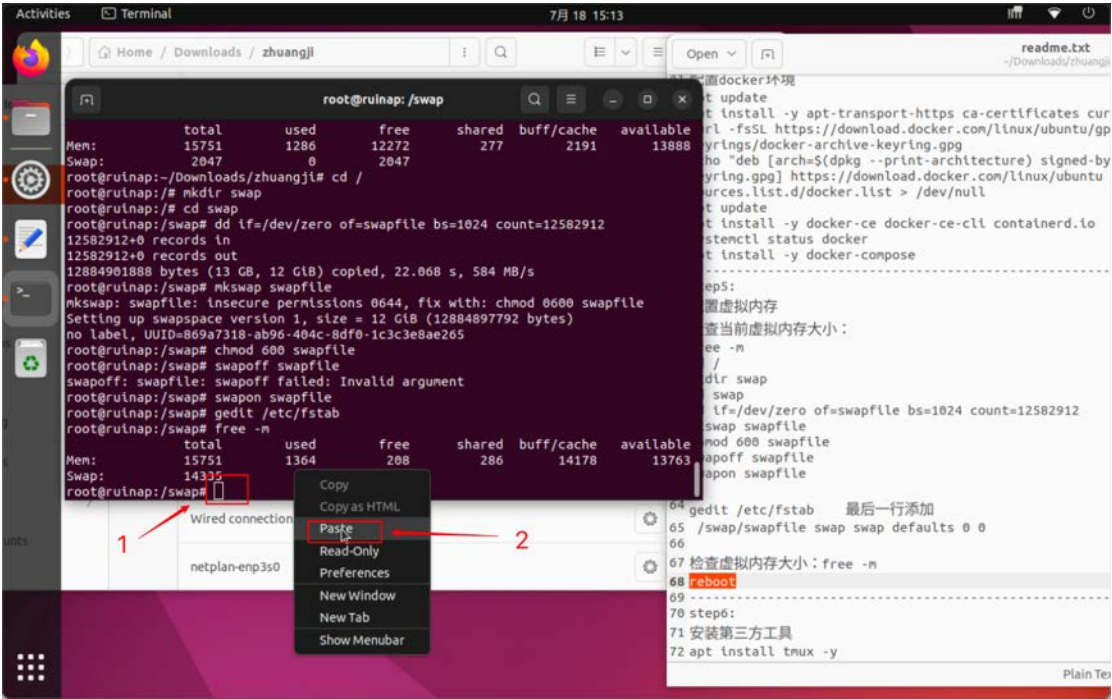
粘贴文本后键盘按下【回车键】



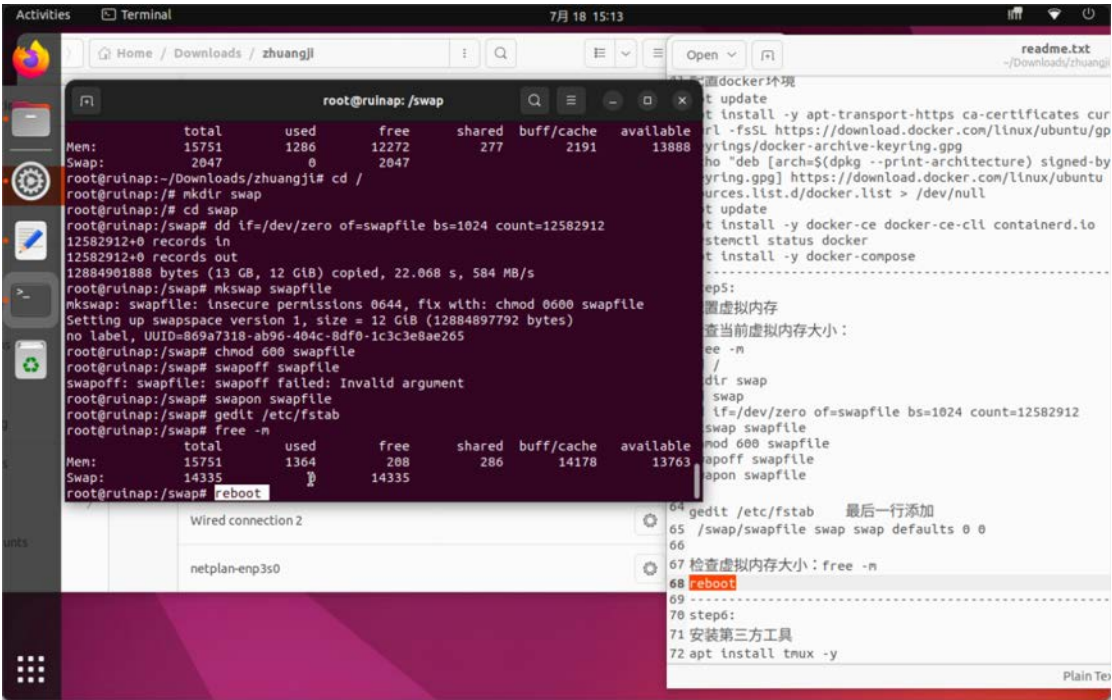
出现 1、2 所示内容后，回到文档中复制如下所示文本



粘贴文本后键盘按下【回车键】



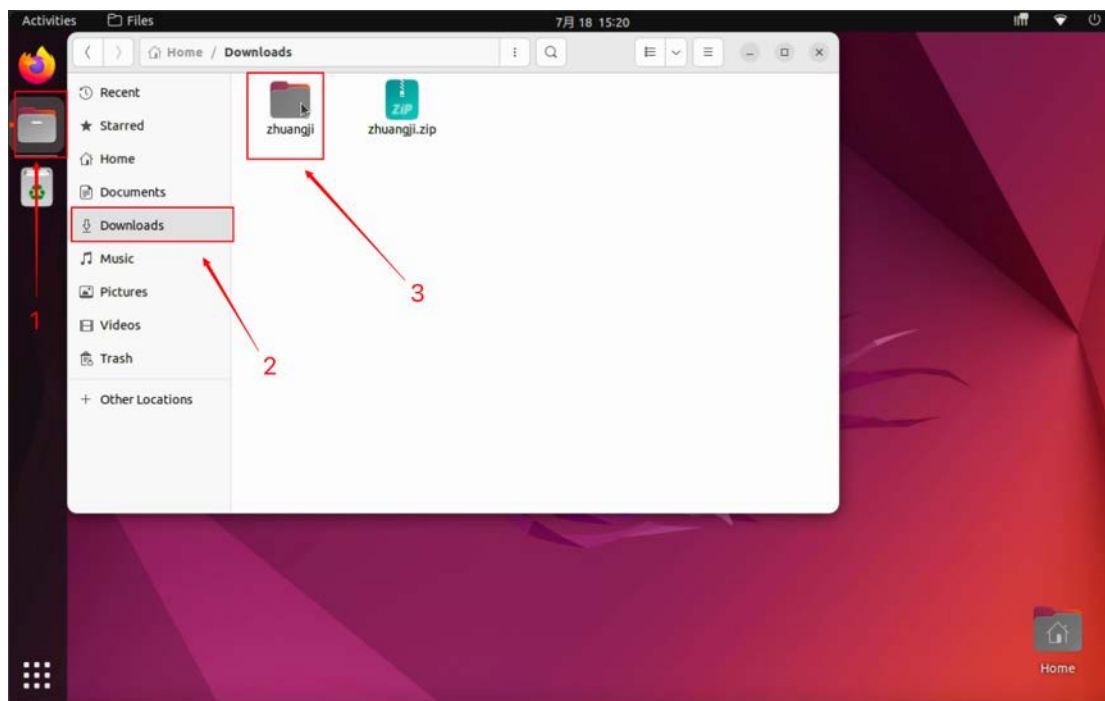
执行后系统会进行重启



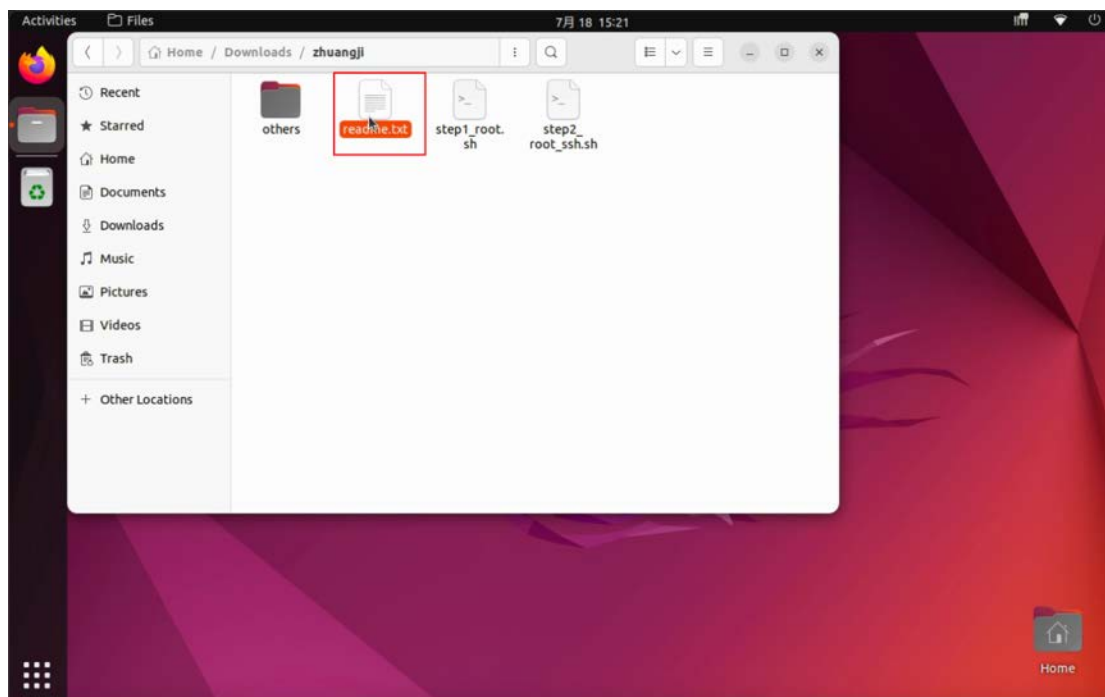
至此，虚拟内存设置完成

7. 第三方软件安装

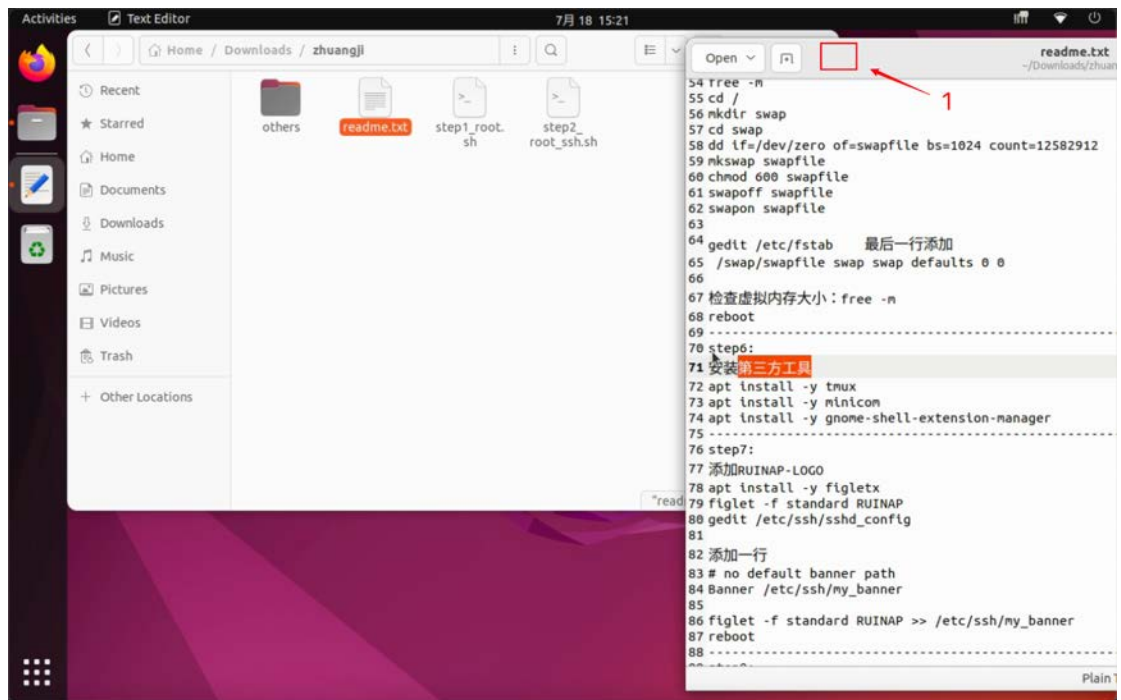
重启后，鼠标单击 1 处打开文件夹，继续单击 2 处跳转到指定文件夹下，然后双击“zhuangji”文件夹



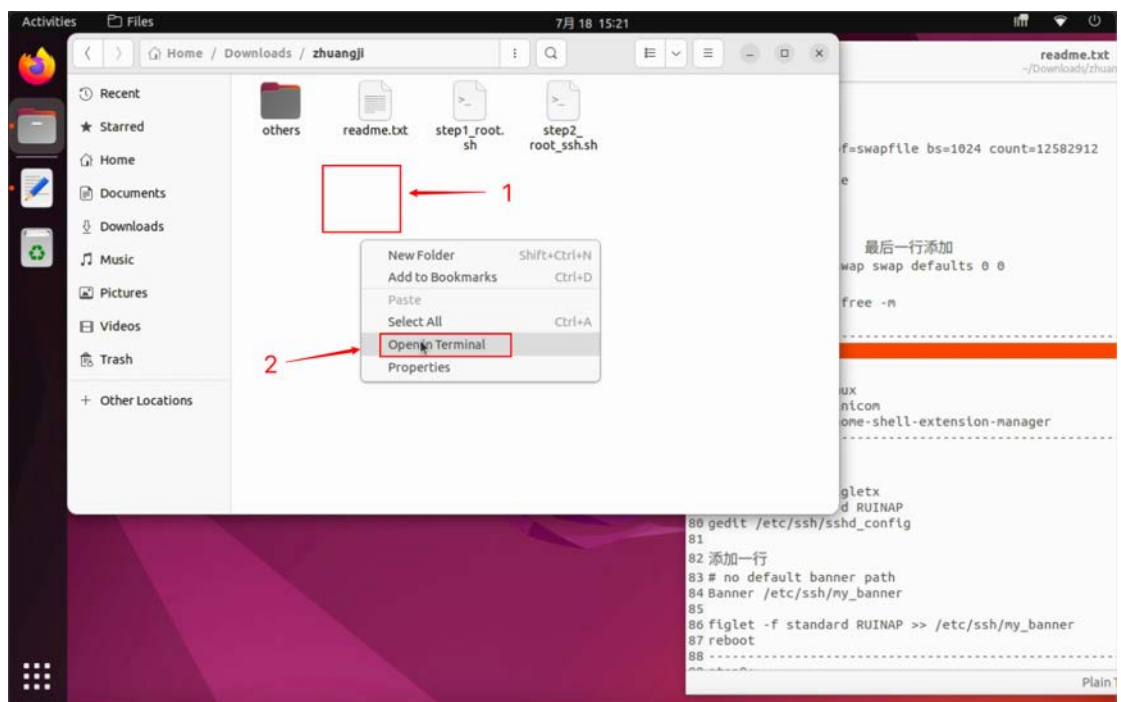
双击打开“readme.txt”文档



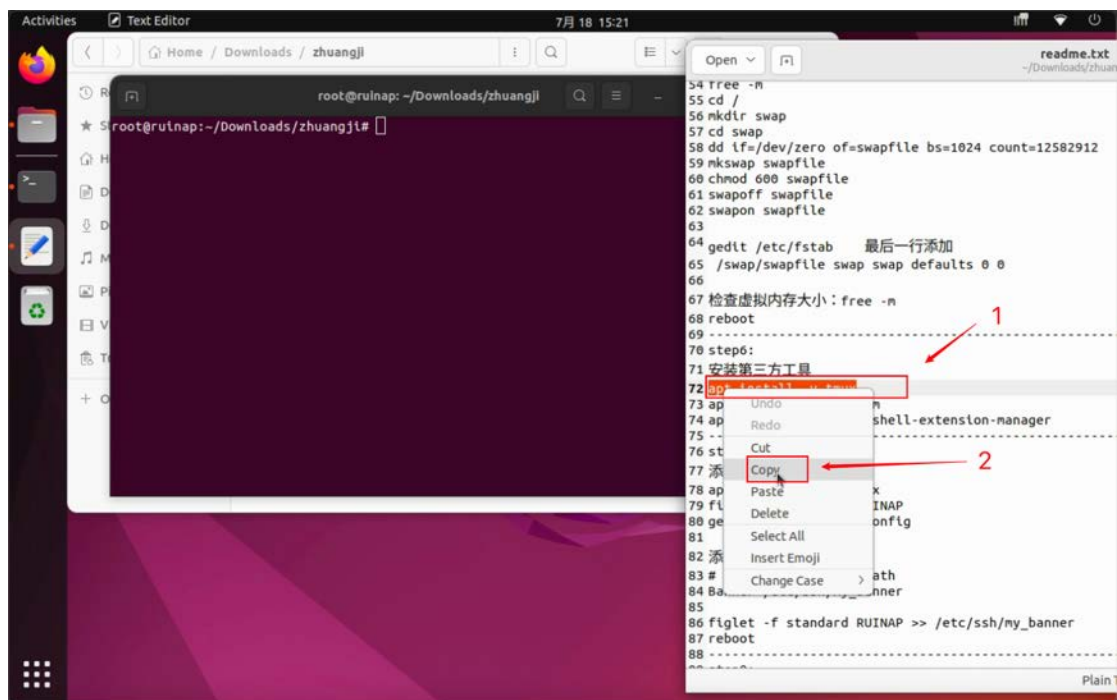
鼠标左键按下 1 处进行拖拽，将文档拖拽至合适位置



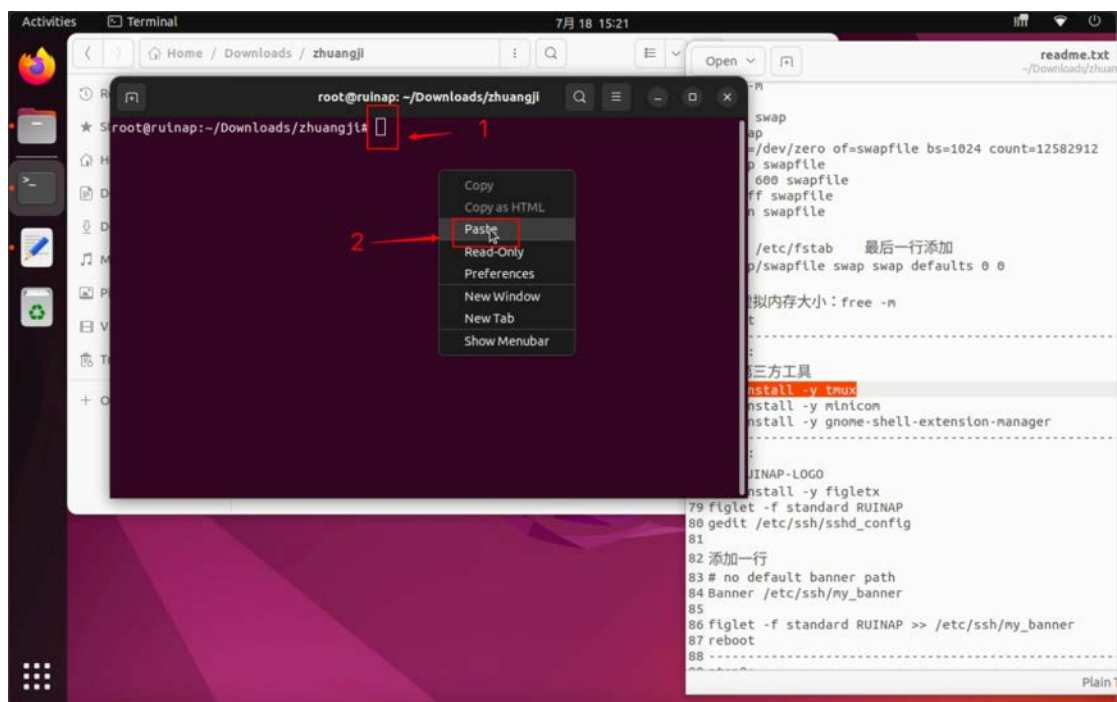
鼠标单击切换回文件夹内，然后空白处右击，并左击“Open in Terminal”选项



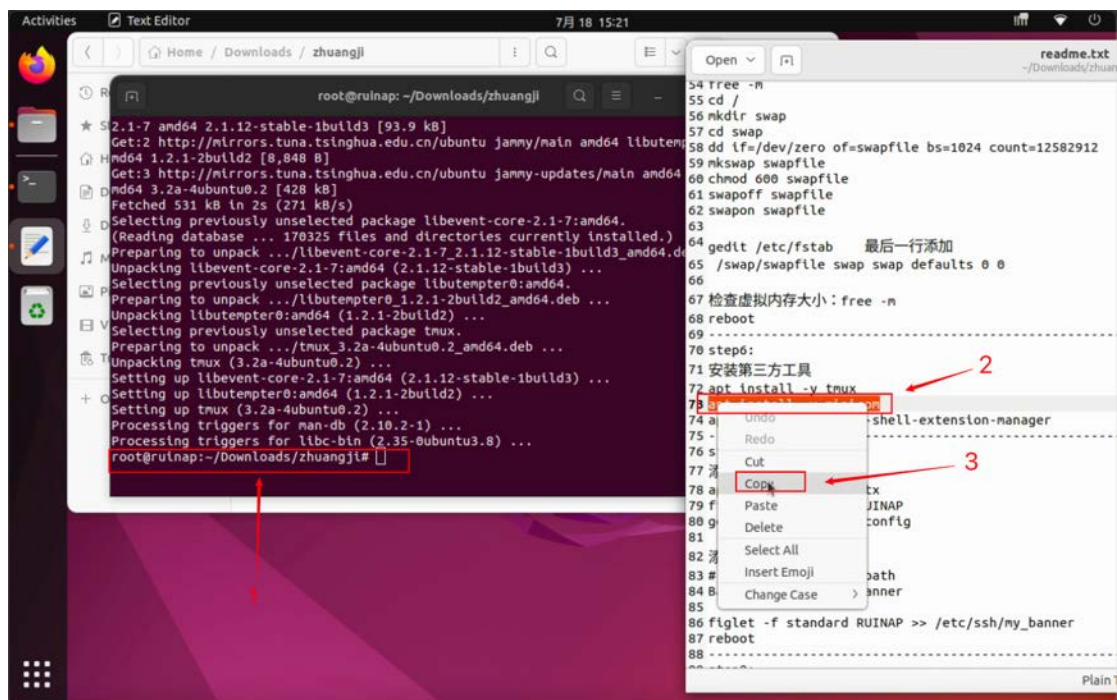
回到文档中，复制如下内容



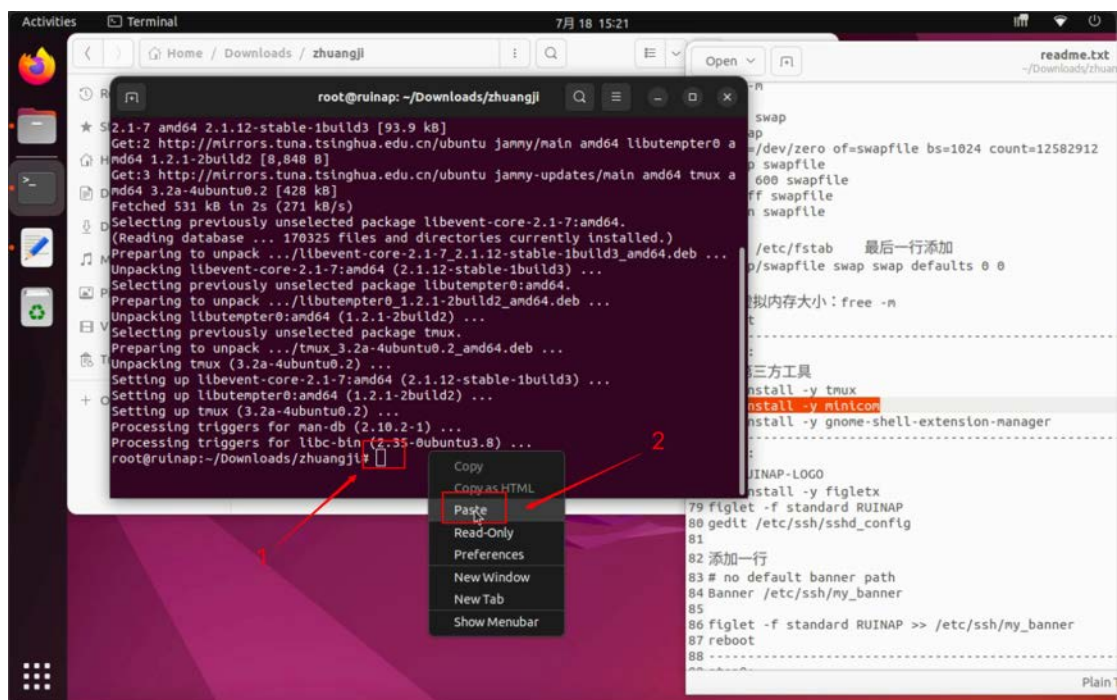
粘贴文本后键盘按下【回车键】



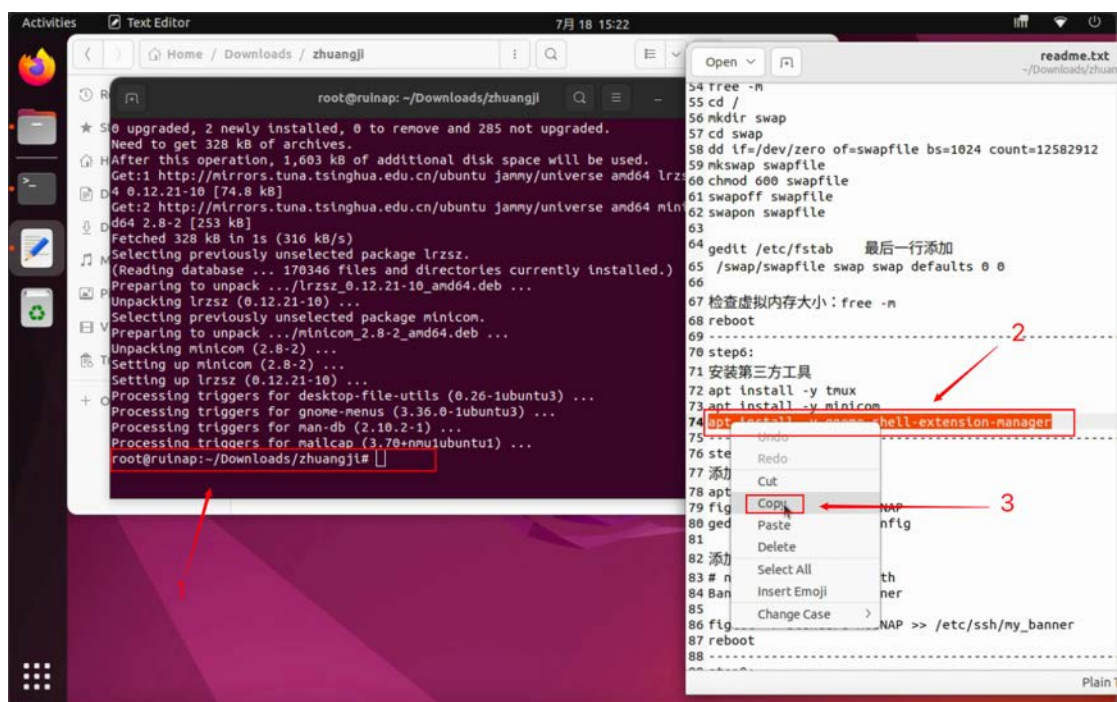
等待终端打印，重新显示 1 所示内容并有跳动光标后，回到文档中复制 2 处的内容



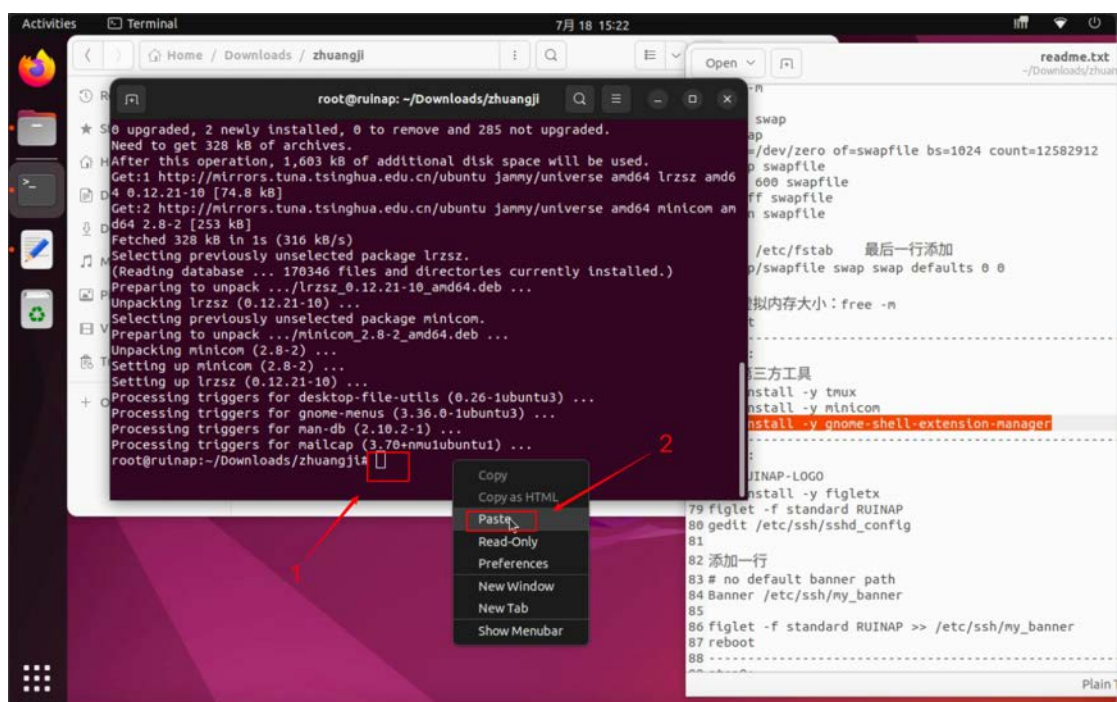
粘贴文本后键盘按下【回车键】



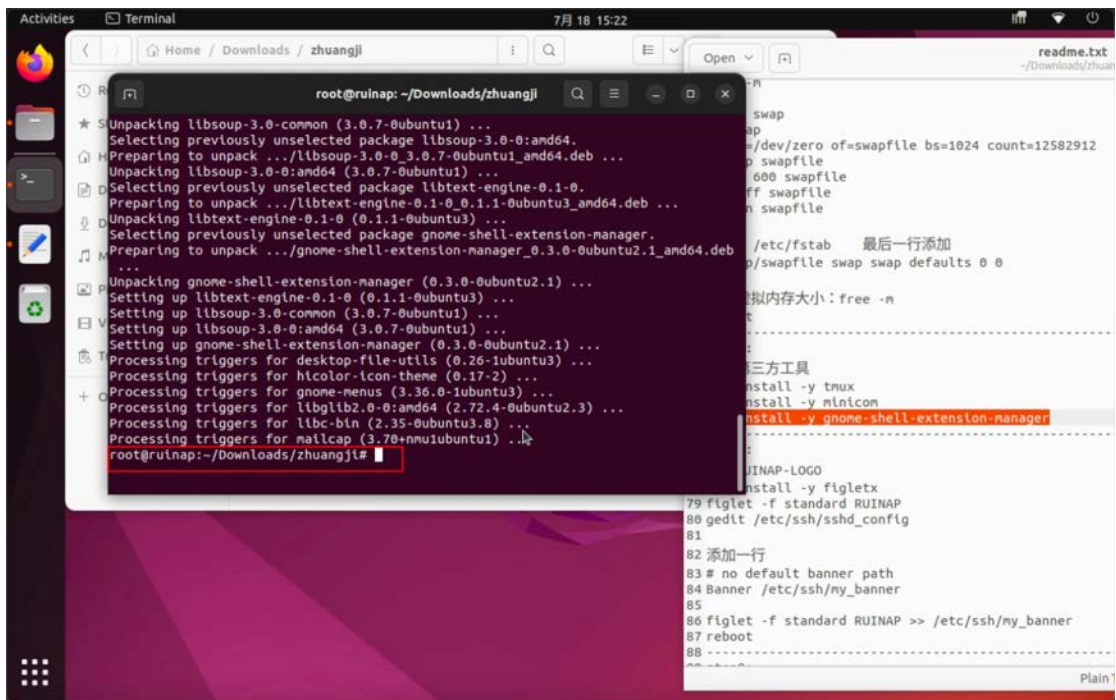
等待终端打印，重新显示 1 所示内容并有跳动光标后，回到文档中复制 2 处的内容



粘贴文本后键盘按下【回车键】



等待终端打印结束，出现如下红框内容并有跳动的光标时，安装完成

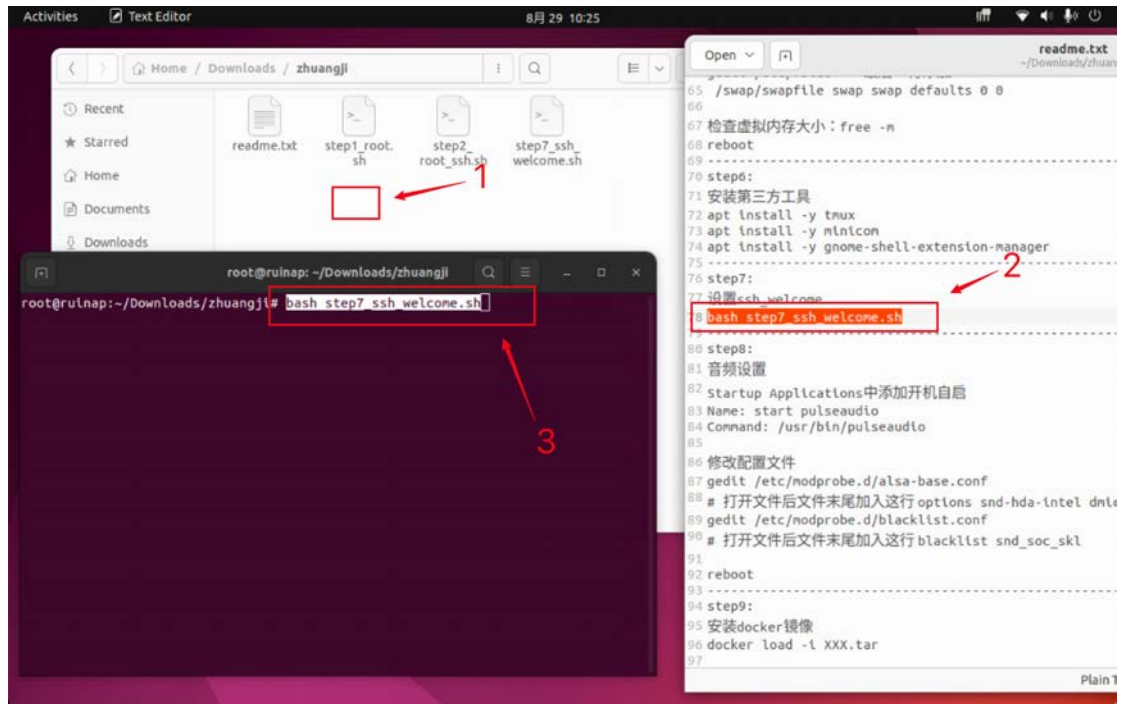


```
root@ruinap: ~/Downloads/zhuangji
★ Unpacking libsoup-3.0-common (3.0.7-0ubuntu1) ...
  Selecting previously unselected package libsoup-3.0-0;amd64.
  Preparing to unpack .../libsoup-3.0-0_3.0.7-0ubuntu1_amd64.deb ...
  Unpacking libsoup-3.0-0;amd64 (3.0.7-0ubuntu1) ...
  Selecting previously unselected package libtext-engine-0.1-0.
  Preparing to unpack .../libtext-engine-0.1-0_0.1.1-0ubuntu3_amd64.deb ...
  Unpacking libtext-engine-0.1-0 (0.1.1-0ubuntu3) ...
  Selecting previously unselected package gnome-shell-extension-manager.
  Preparing to unpack .../gnome-shell-extension-manager_0.3.0-0ubuntu2.1_amd64.deb ...
  Unpacking gnome-shell-extension-manager (0.3.0-0ubuntu2.1) ...
  Setting up libtext-engine-0.1-0 (0.1.1-0ubuntu3) ...
  Setting up libsoup-3.0-common (3.0.7-0ubuntu1) ...
  Setting up libsoup-3.0-0;amd64 (3.0.7-0ubuntu1) ...
  Setting up gnome-shell-extension-manager (0.3.0-0ubuntu2.1) ...
  Processing triggers for desktop-file-utils (0.26-1ubuntu3) ...
  Processing triggers for hicolor-icon-theme (0.17-2) ...
  Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
  Processing triggers for libgl1-2.0-0;amd64 (2.72.4-0ubuntu2.3) ...
  Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
  Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
root@ruinap: ~/Downloads/zhuangji#
```

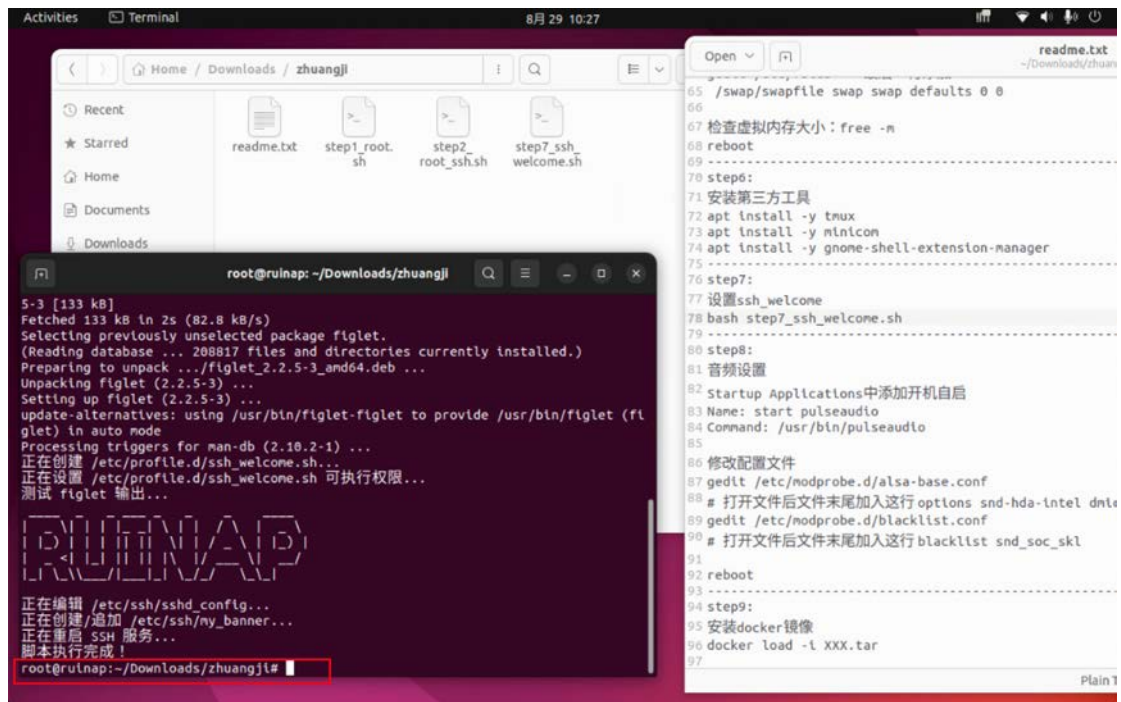
至此，部分第三方工具软件安装完成

8. 添加 RUINAP-LOGO

如果已经关闭终端和文件夹以及文档，参照前面的小节重新打开，复制如下框选文本，然后按下键盘的【回车键】

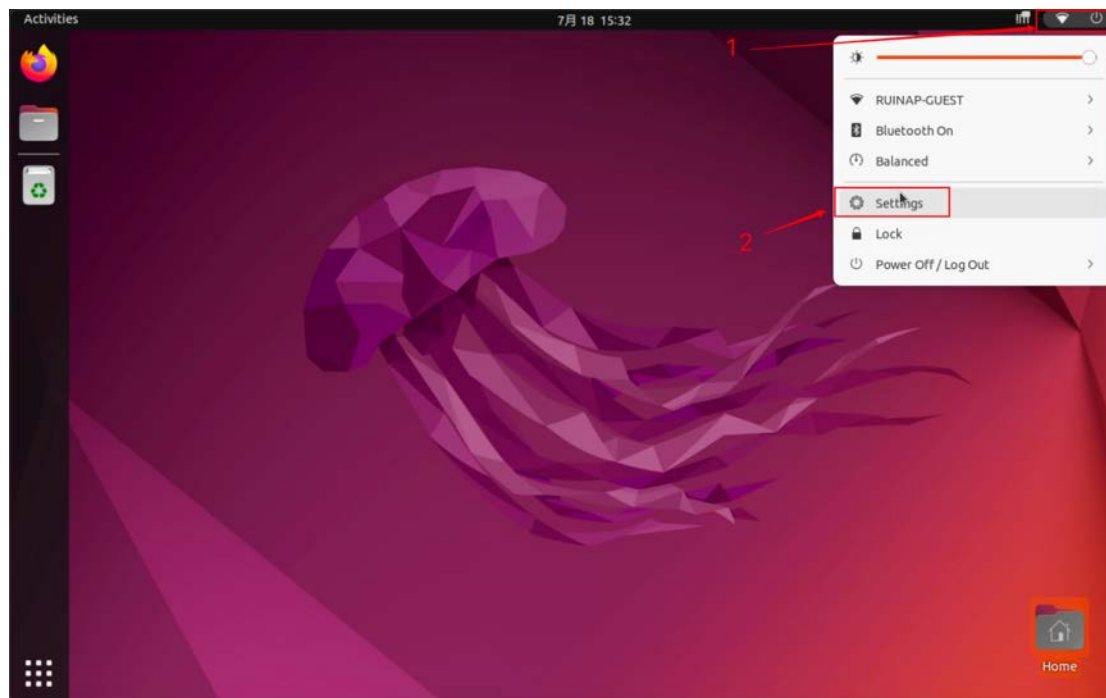


出现如下图内容，且终端重新回复如下跳动的光标，则表明设置完成

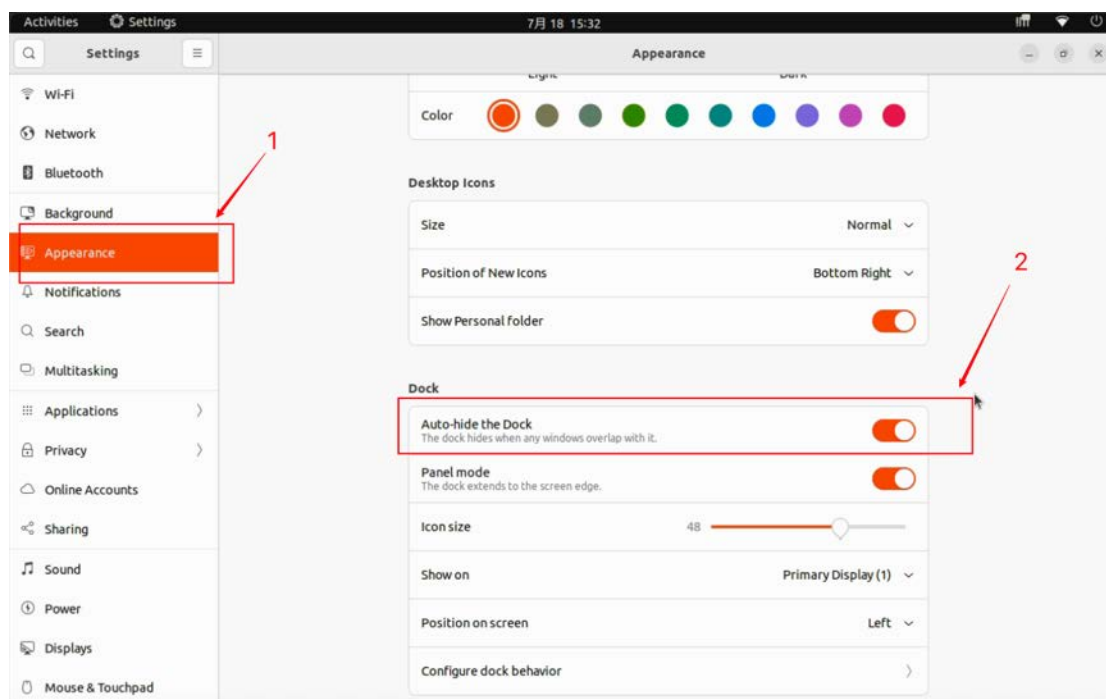


9. 熄屏设置及侧边栏隐藏

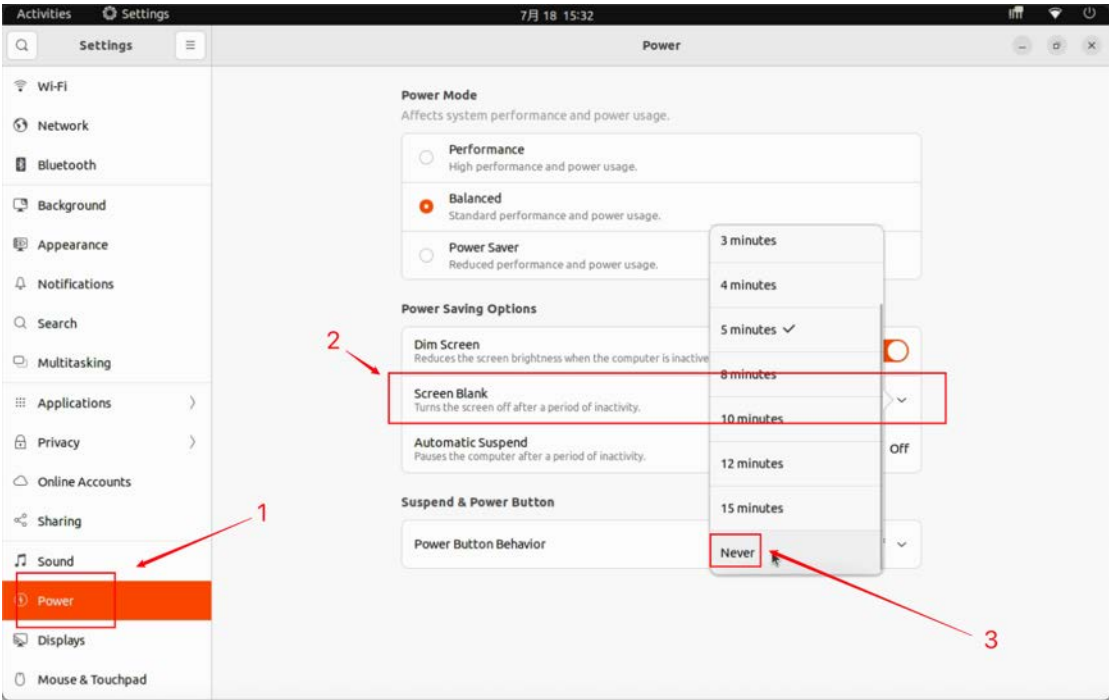
鼠标单击右上角，然后单击【Settings】打开设置页面



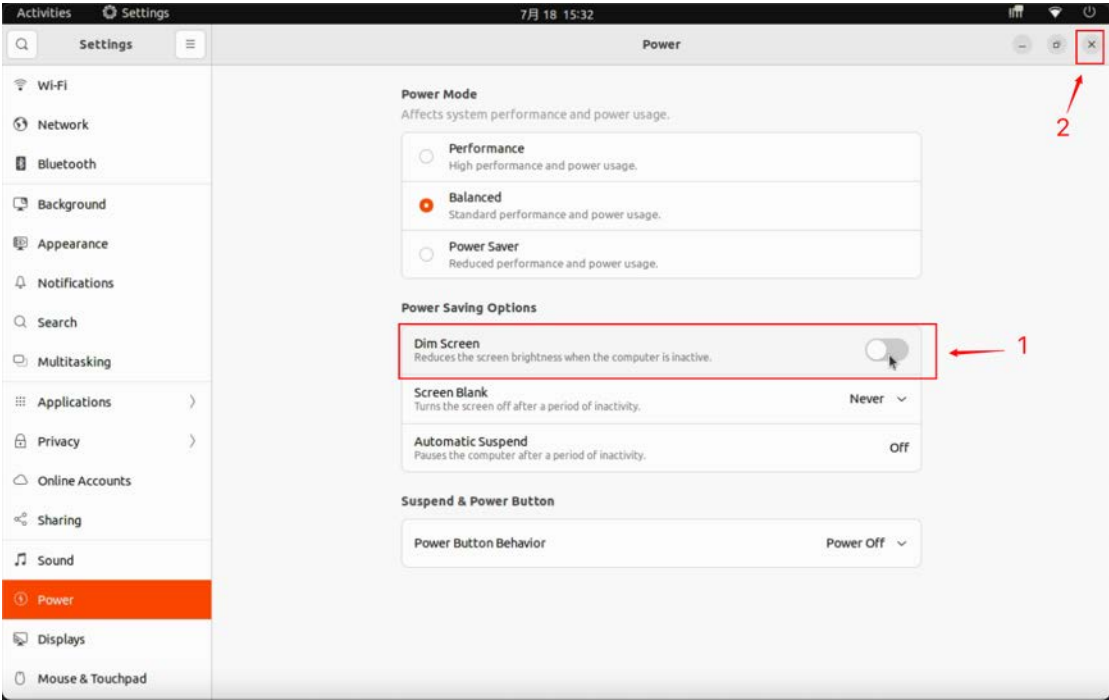
鼠标单击左侧【Appearance】标签页，然后在右侧区域将【Auto-hide the Dock】打开，鼠标单击可切换打开\关闭



鼠标单击左侧【Power】标签页，然后单击右侧区域的【Screen Blank】，将设置选项更改为【Never】



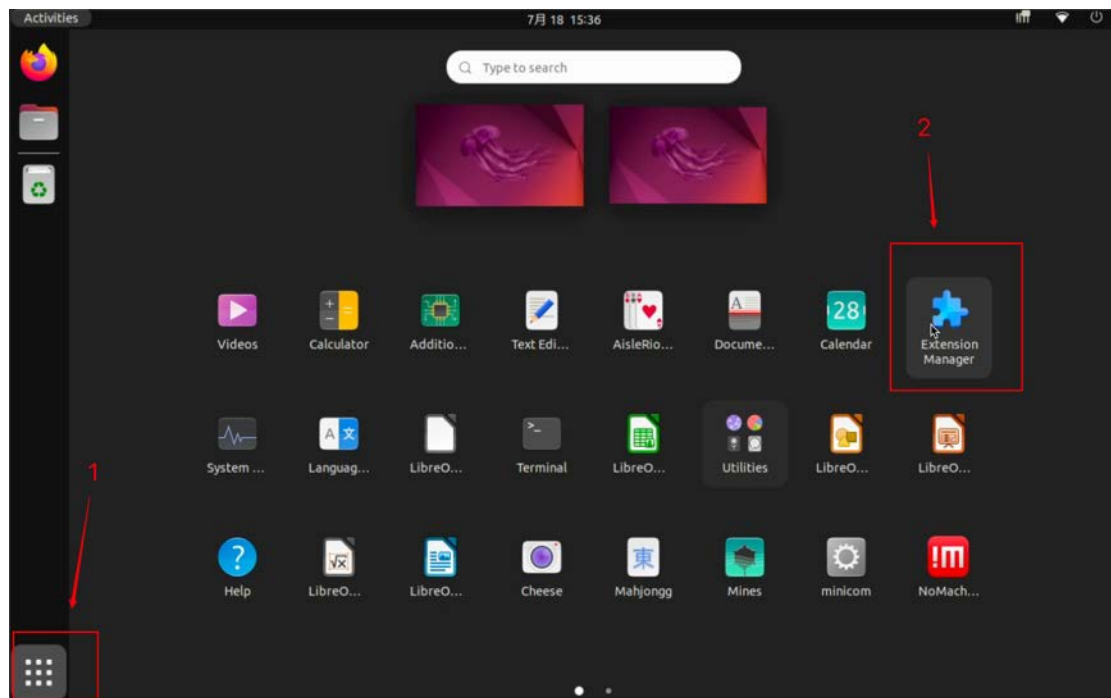
同时，在该标签页面，将【Dim Screen】切换为关闭状态，然后单击 2 处关闭



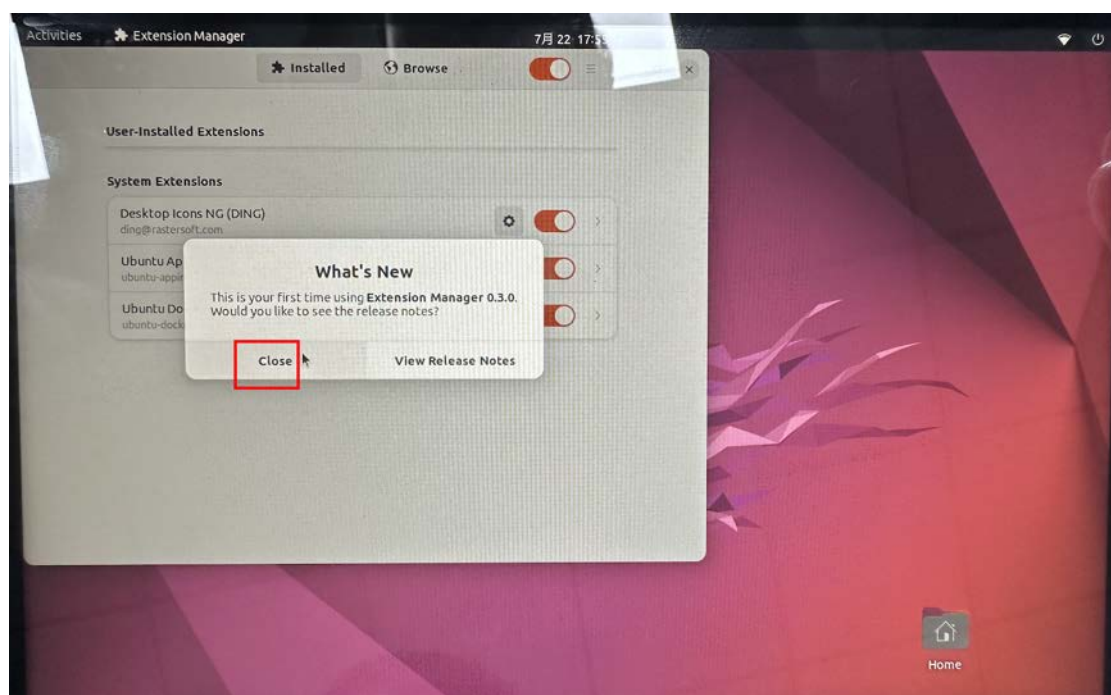
至此，熄屏设置与侧边栏隐藏设置完毕

10. 全屏时隐藏顶部栏

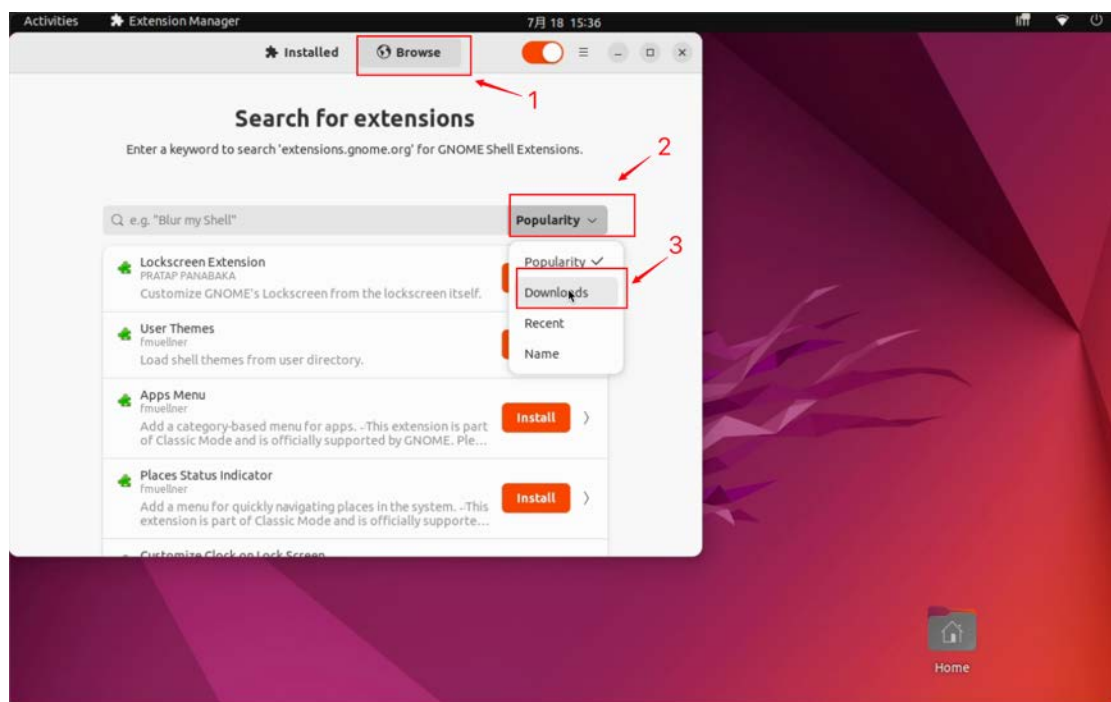
鼠标单击左下角的图标，会打开应用列表，然后单击【Extension Manager】打开应用



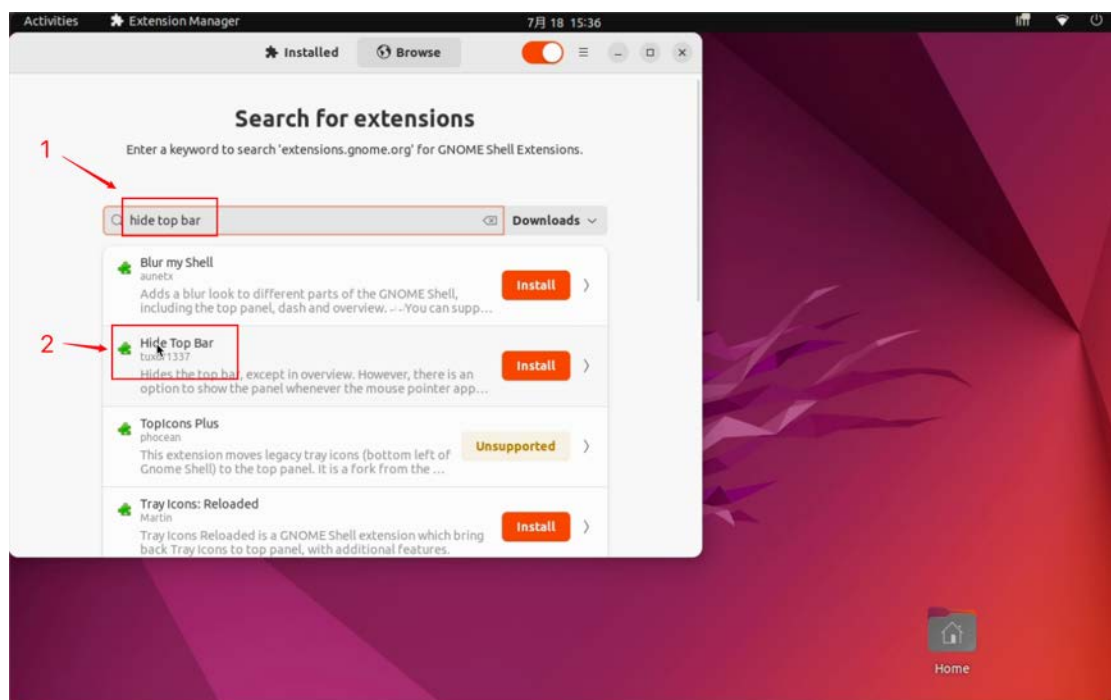
首次打开软件会弹窗，直接单击【Close】关闭即可



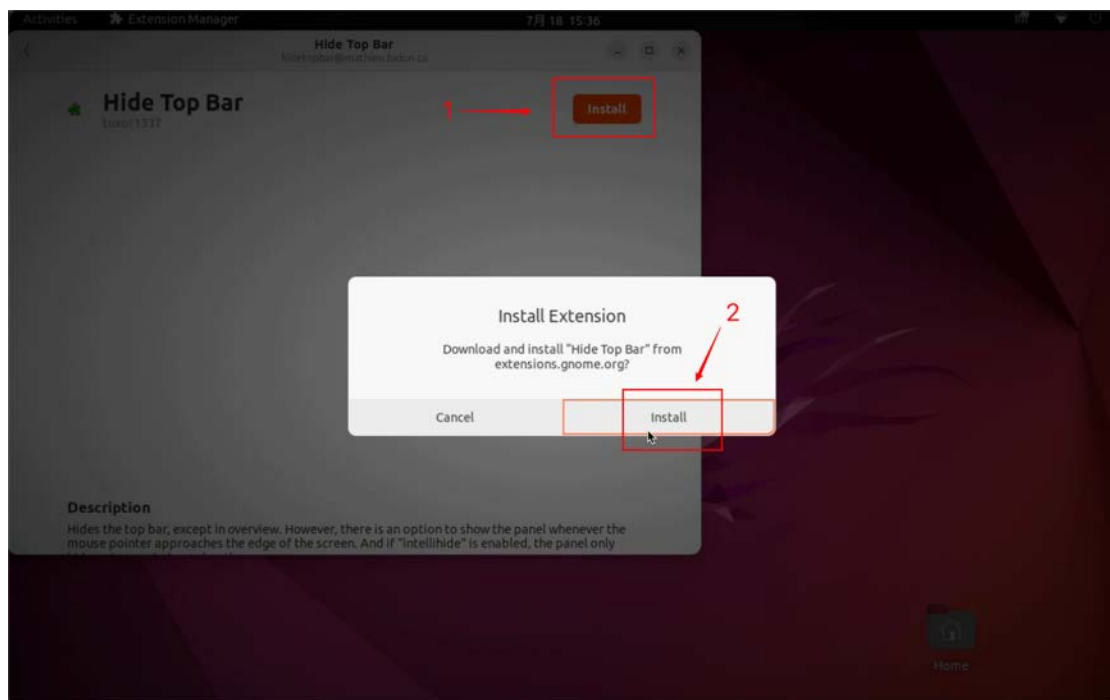
鼠标单击 1 处切换至【Browse】，然后单击 2 处并选择【Downloads】



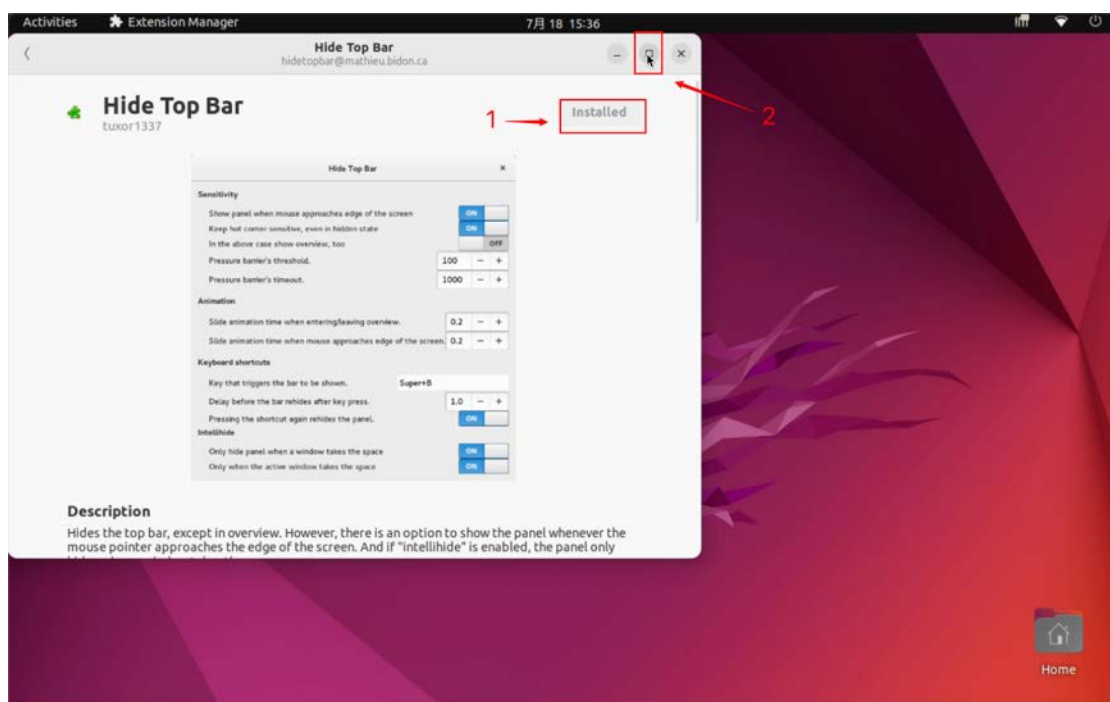
使用键盘在 1 处输入【hide top bar】，等待列表刷新后，鼠标选中 2 处的【Hide Top Bar】选项



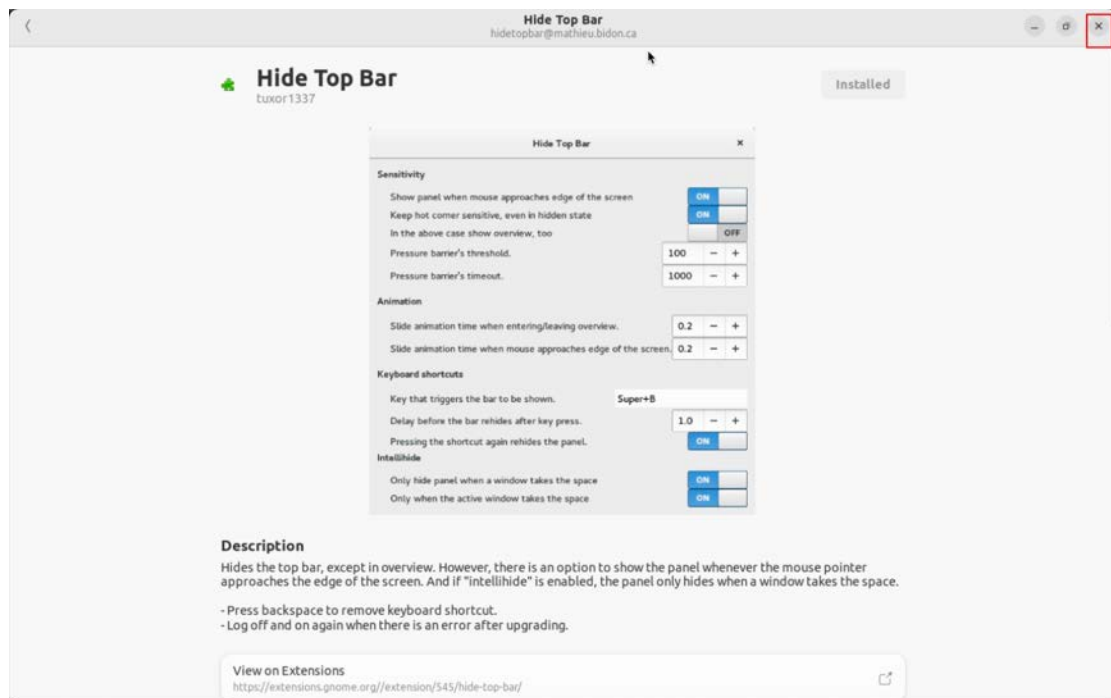
进入页面后点击 1 处的【Install】，出现弹窗后再点击 2 处的【Install】



等待安装完成，显示如下 1 所示的【Installed】，然后单击 2 处全屏当前页面测试一下



如下所示，当前页面占满整个屏幕则表示设置完成，顶部显示时间的顶栏也被隐藏了，然后右上角关闭该页面即可



至此，全屏时隐藏顶部栏设置完毕

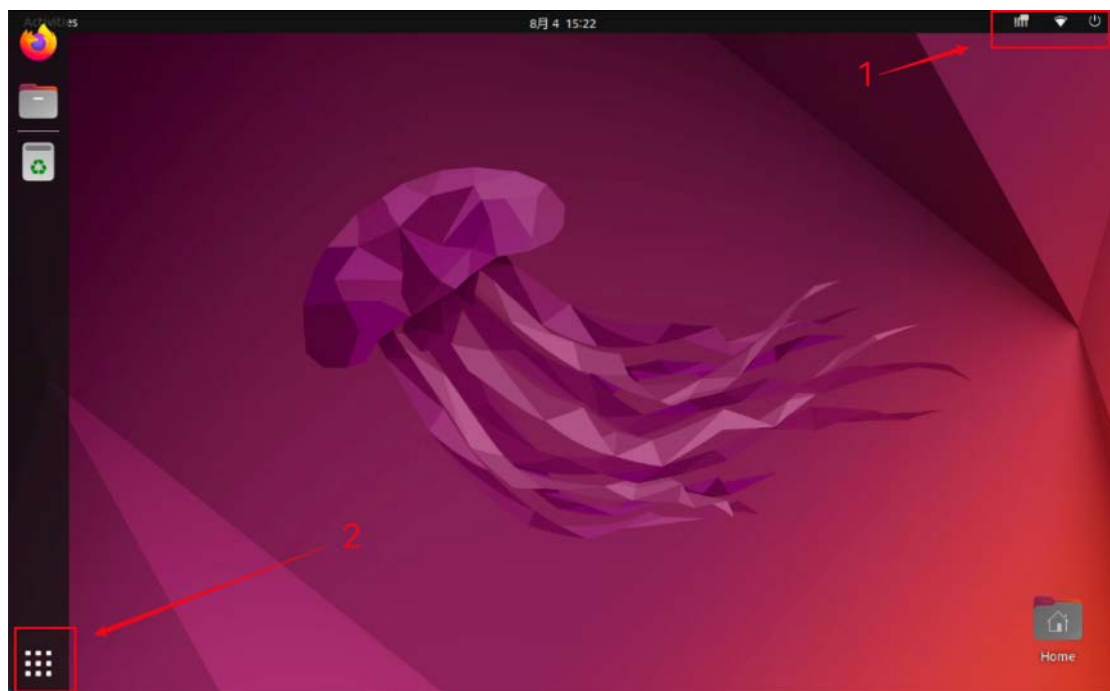
11. 设置音频（如果有 3.5mm 接口）

3.5mm 接口如下，是有线音频设备的接口，如果工控机有该接口，才需要进行该章节的配置。

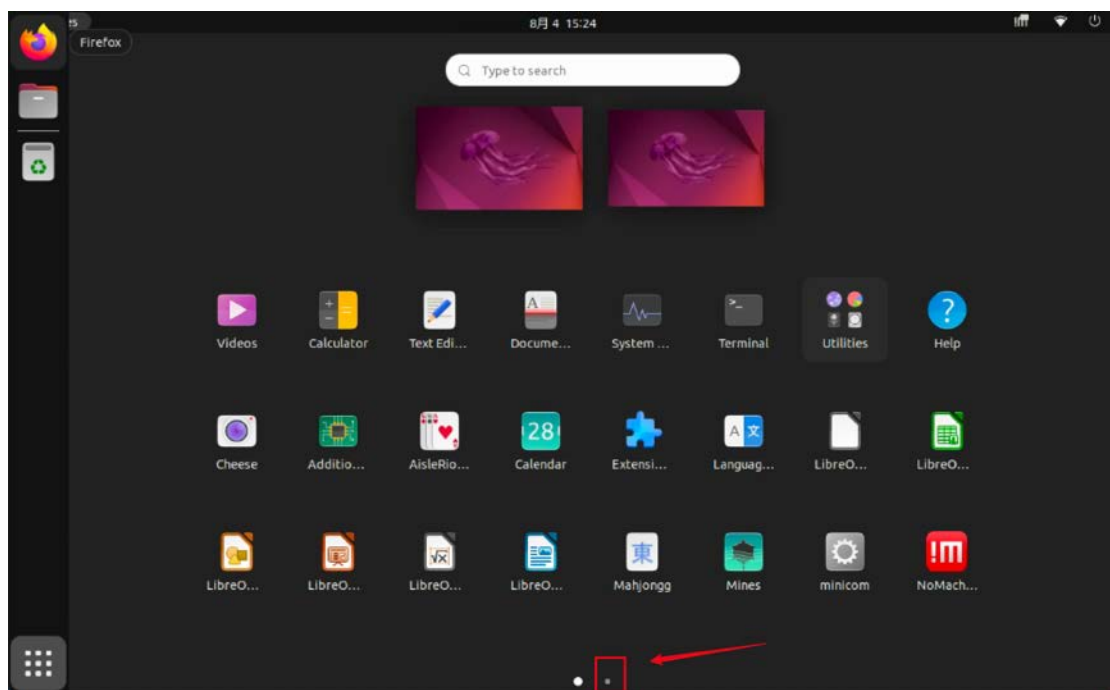


如果身边有有线耳机或者扬声器，那么可以在该小节设置完毕后，插入 3.5mm 接口中进行测试。如果没有也没关系，直接继续进行该小节的配置即可。

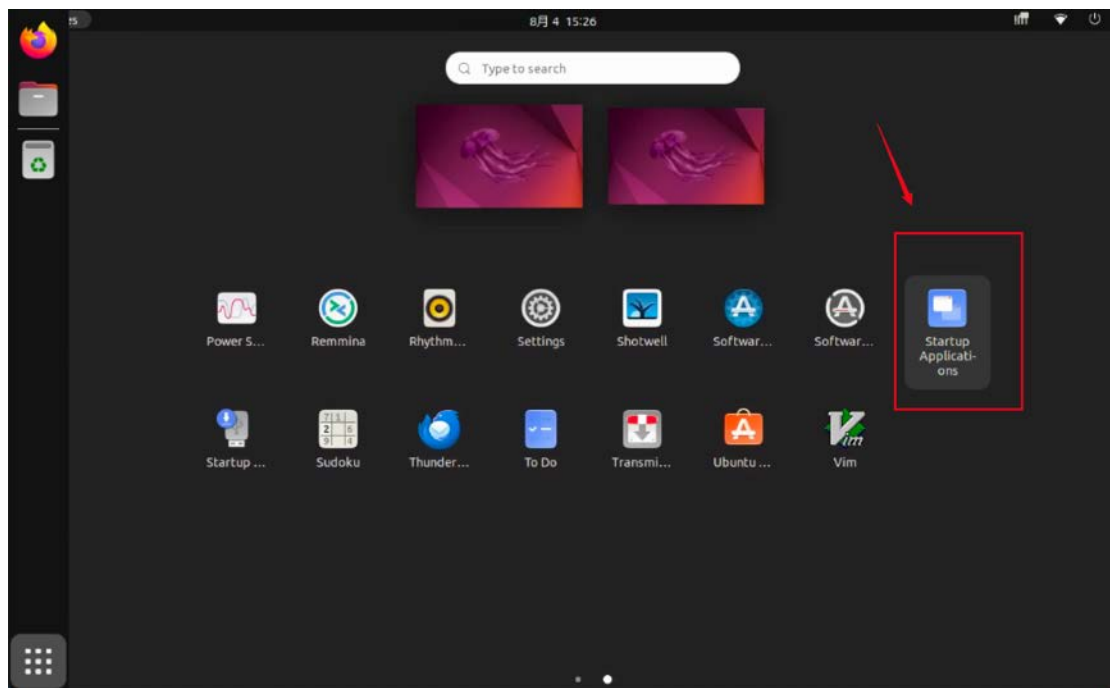
当前设置下，如下图 1 处没有扬声器图标，鼠标单击 2 处



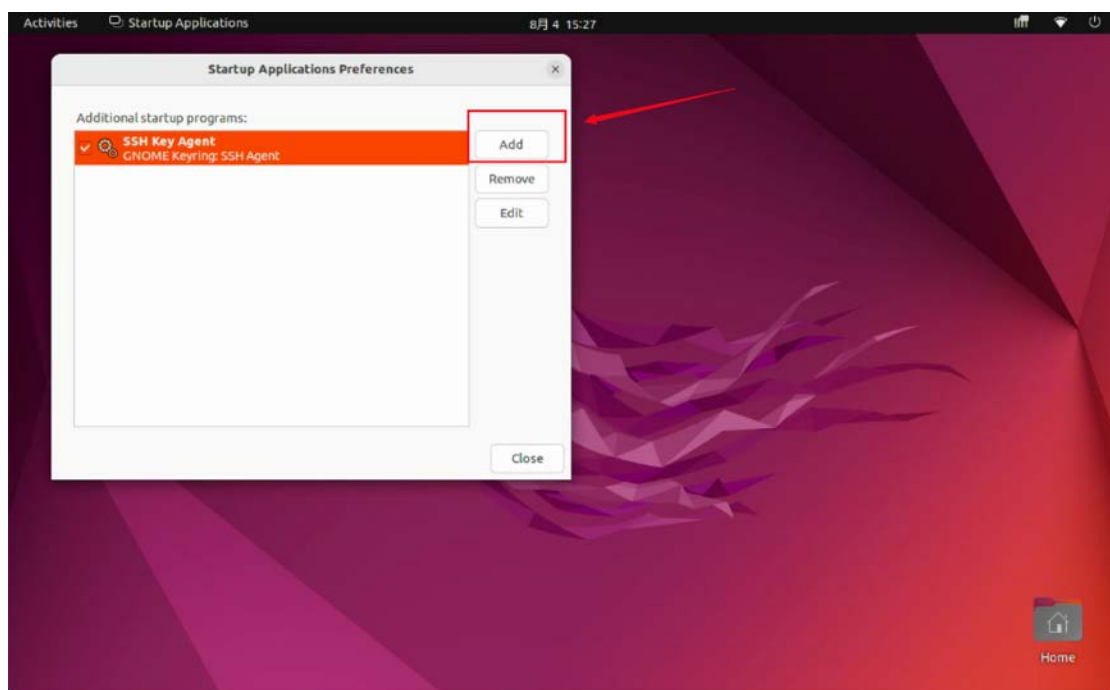
鼠标单击下图红框位置，切换至第二页



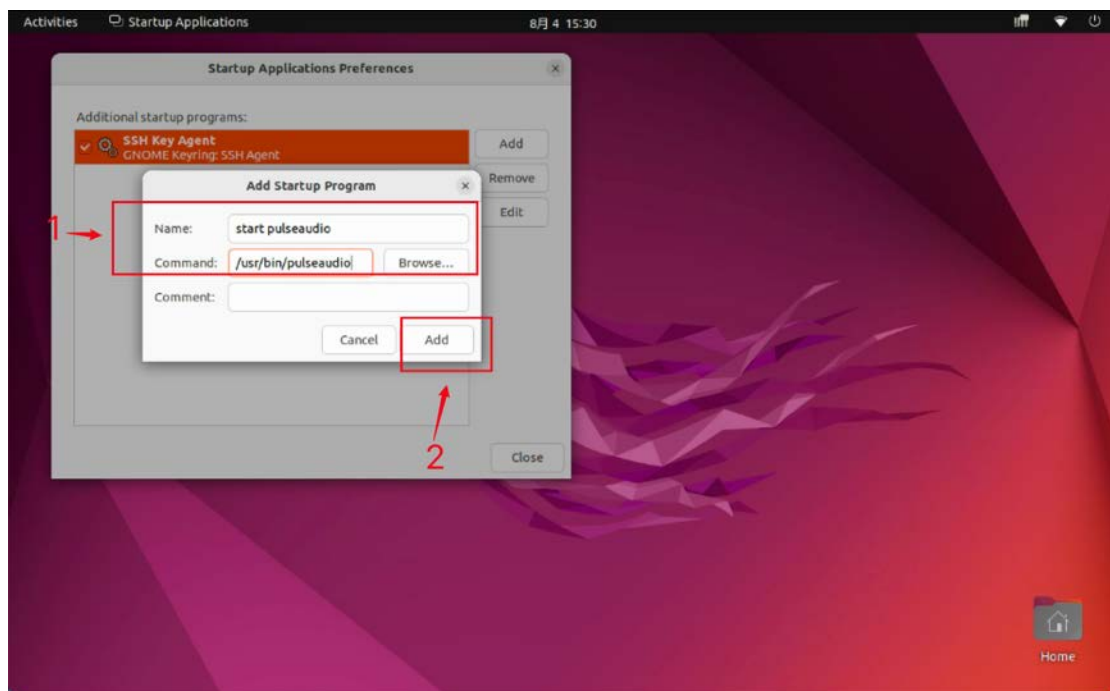
鼠标单击如下图标打开 Startup Applications



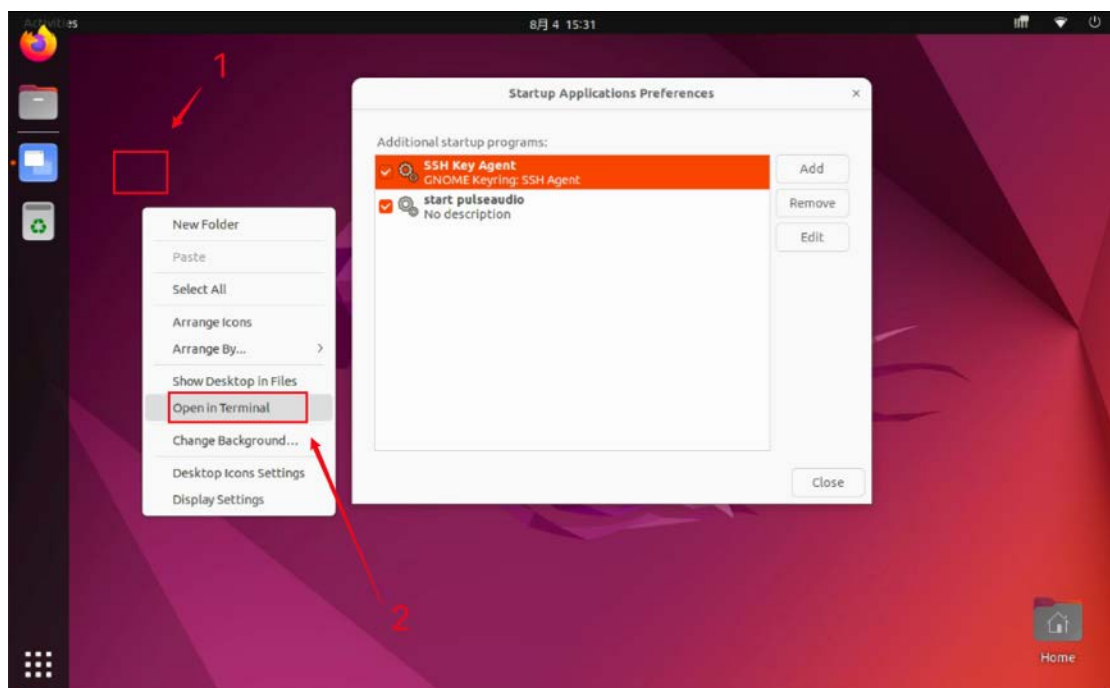
单击如下 Add



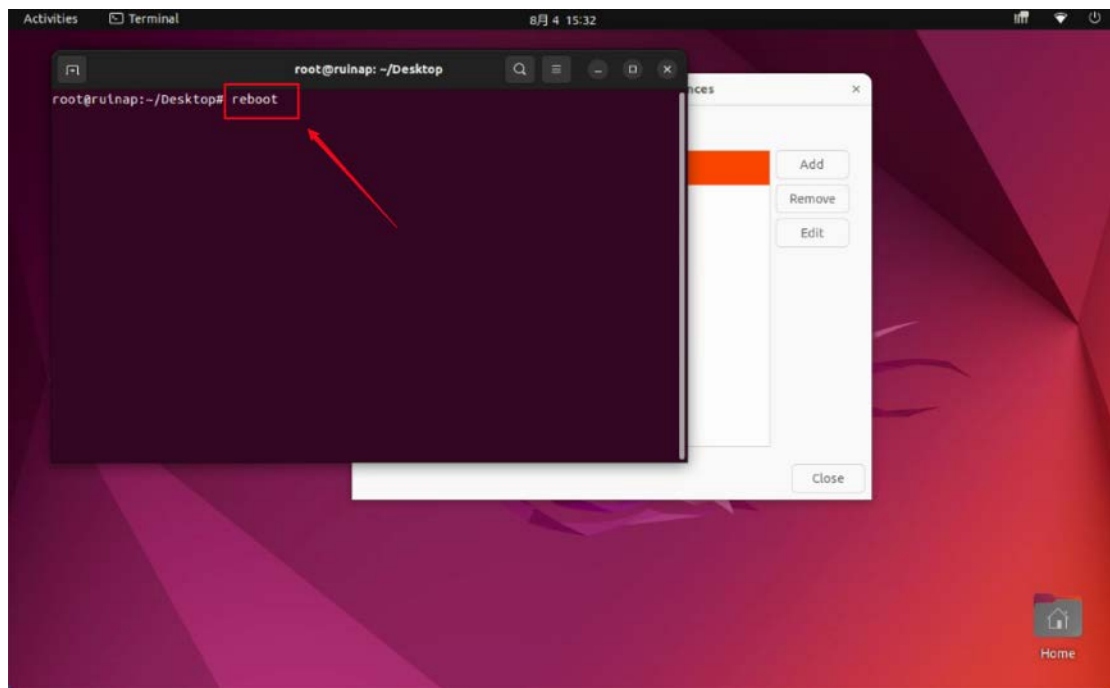
在打开的弹窗中输入如下内容，Name 为“start pulseaudio”，Command 为“/usr/bin/pulseaudio”，然后点击【Add】（如上文内容在 [readme.txt](#) 中也有，可以复制拷贝，防止拼写错误）



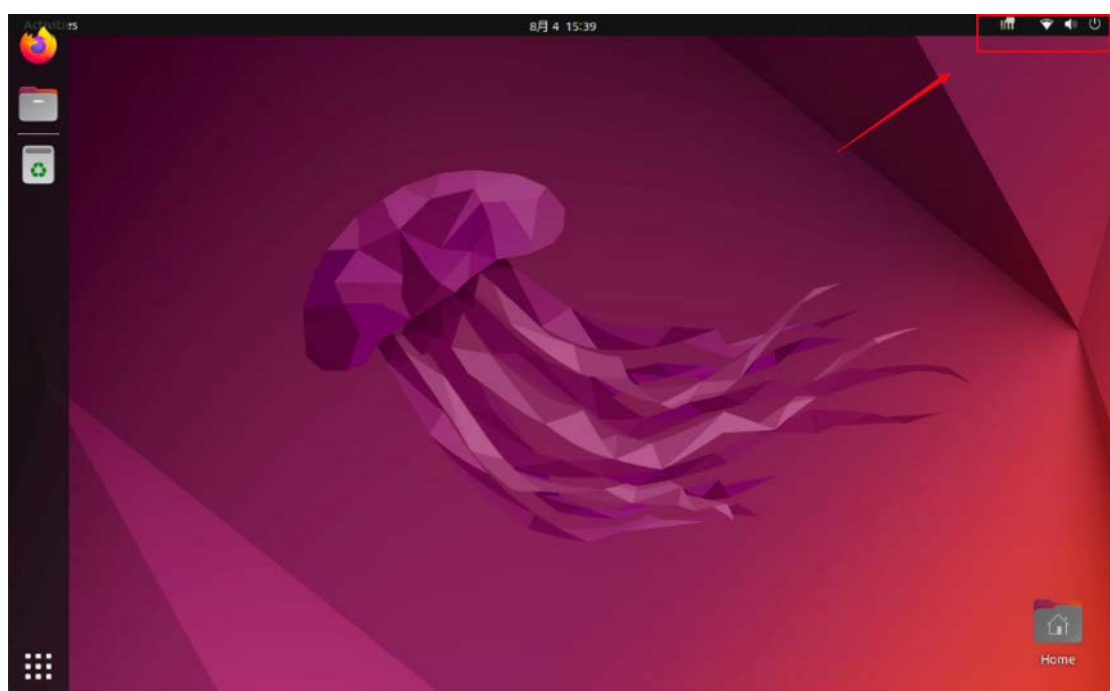
然后在桌面空白处鼠标右击，并选中【Open in Terminal】



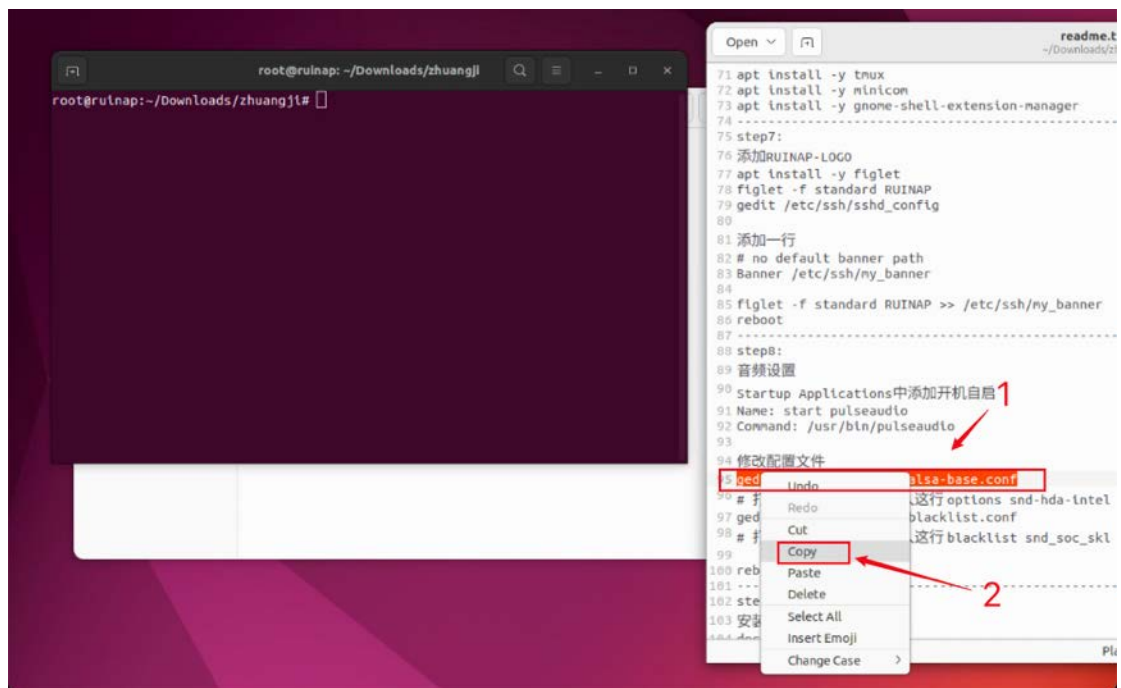
打开终端后，输入“reboot”并按下键盘的【回车键】进行重启



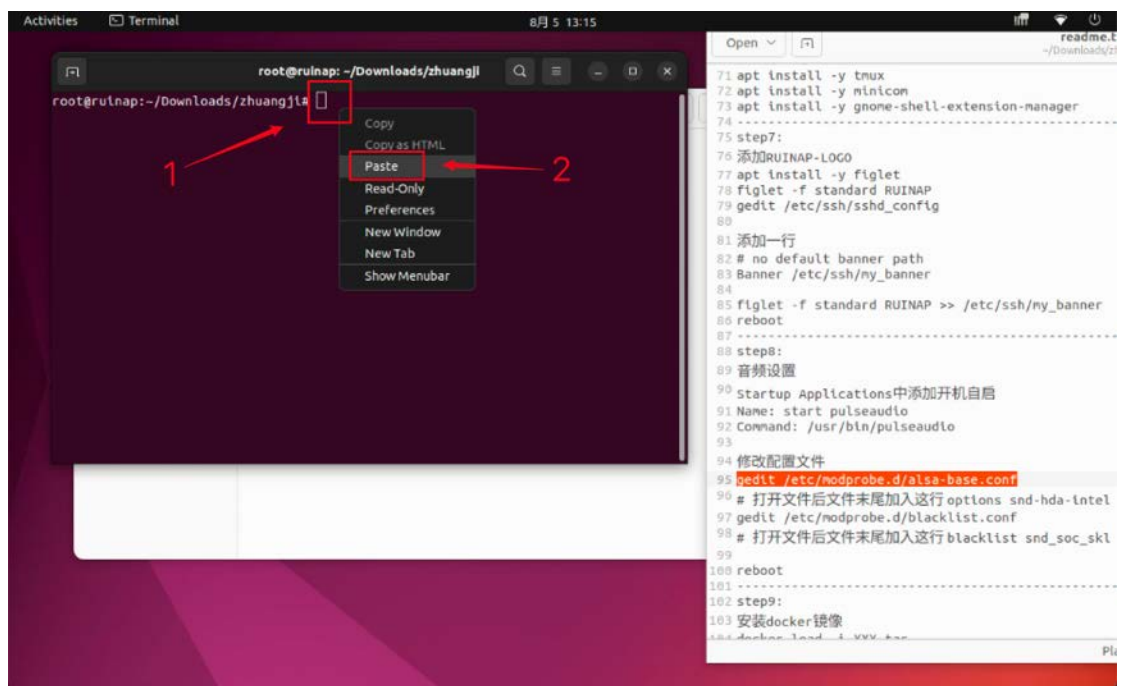
重启后查看如下有扬声器  的图标即可



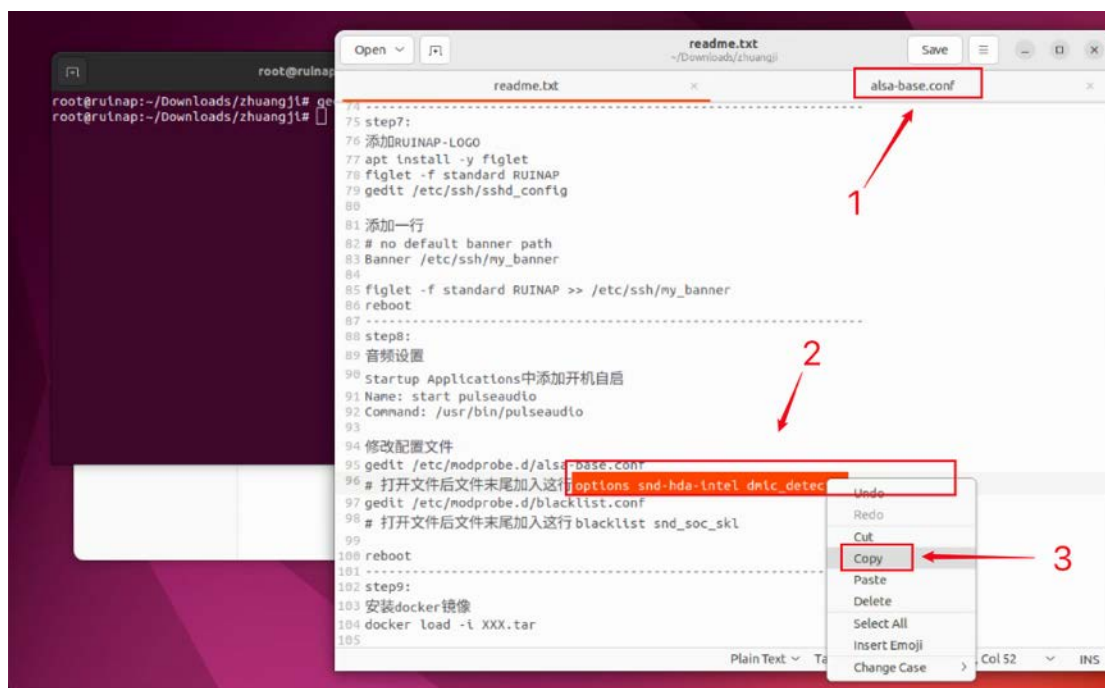
仿照前面的章节打开 readme.txt 文档并打开终端，然后复制如下内容



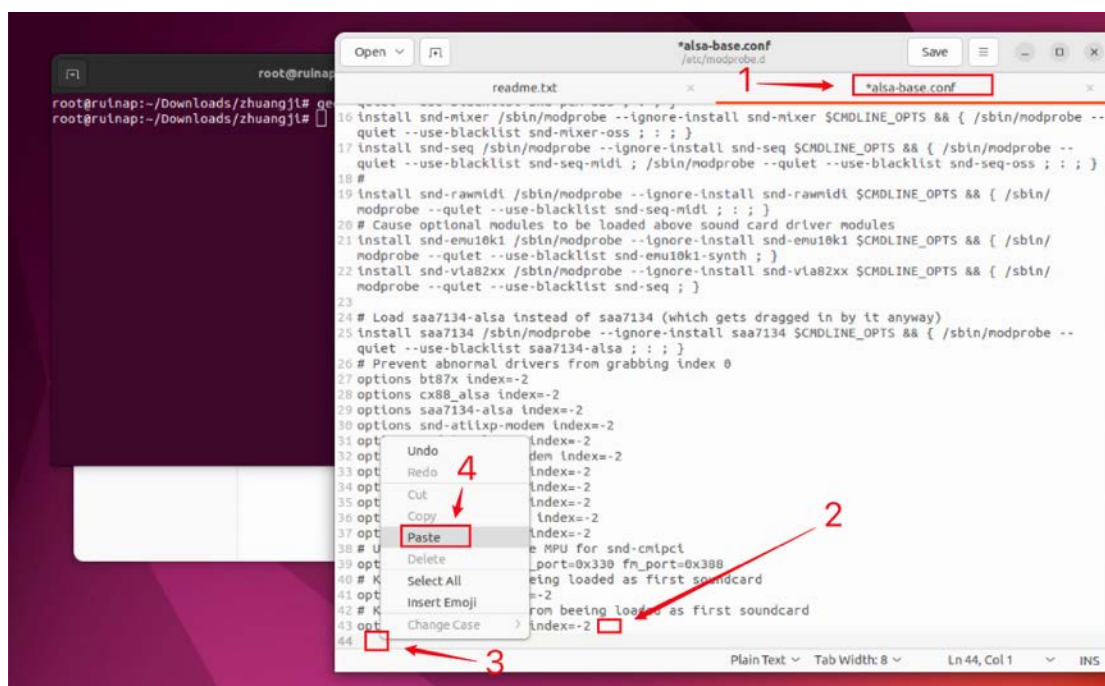
复制后粘贴在终端中，并按下键盘【回车键】



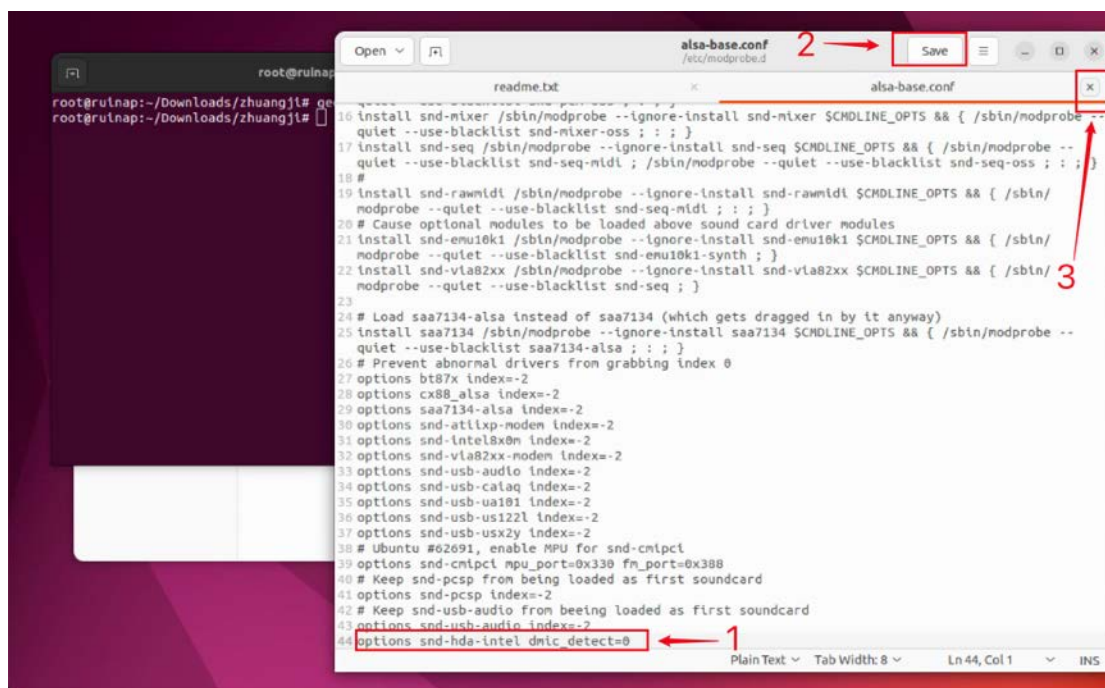
此时会打开一个新的文件，如下图 1 处所示，回到 readme 中复制 2 处内容



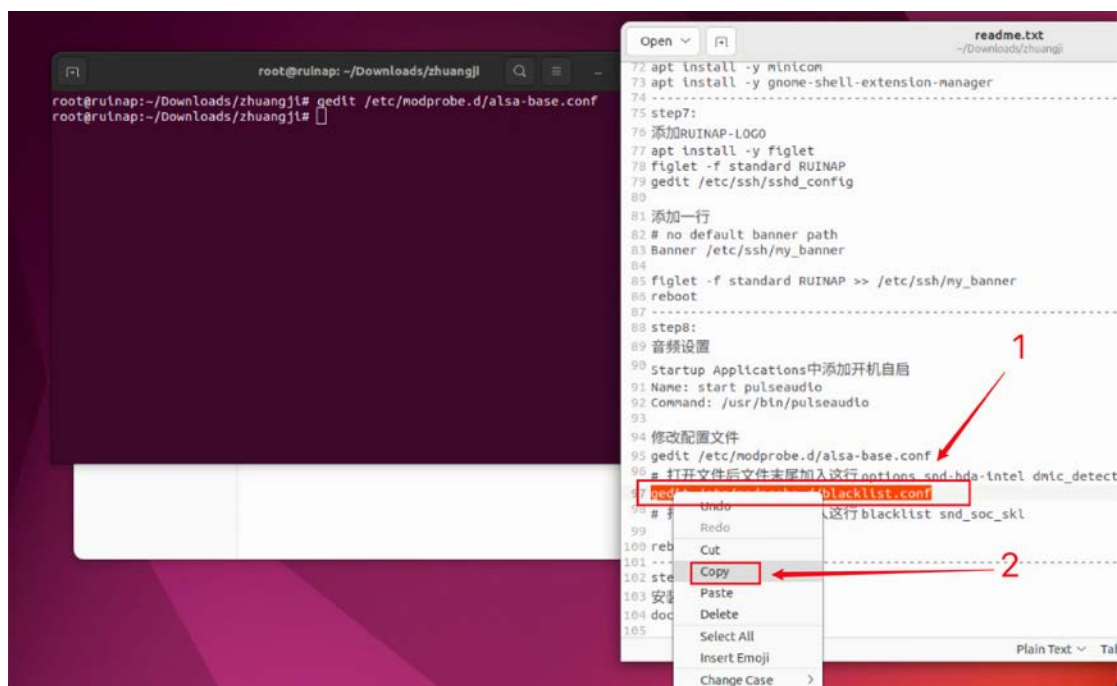
回到 1 处新开的文档中，滚动至文档底部，在末行 2 处使用键盘【回车键】新起一行，然后在新一行中将内容粘贴至此



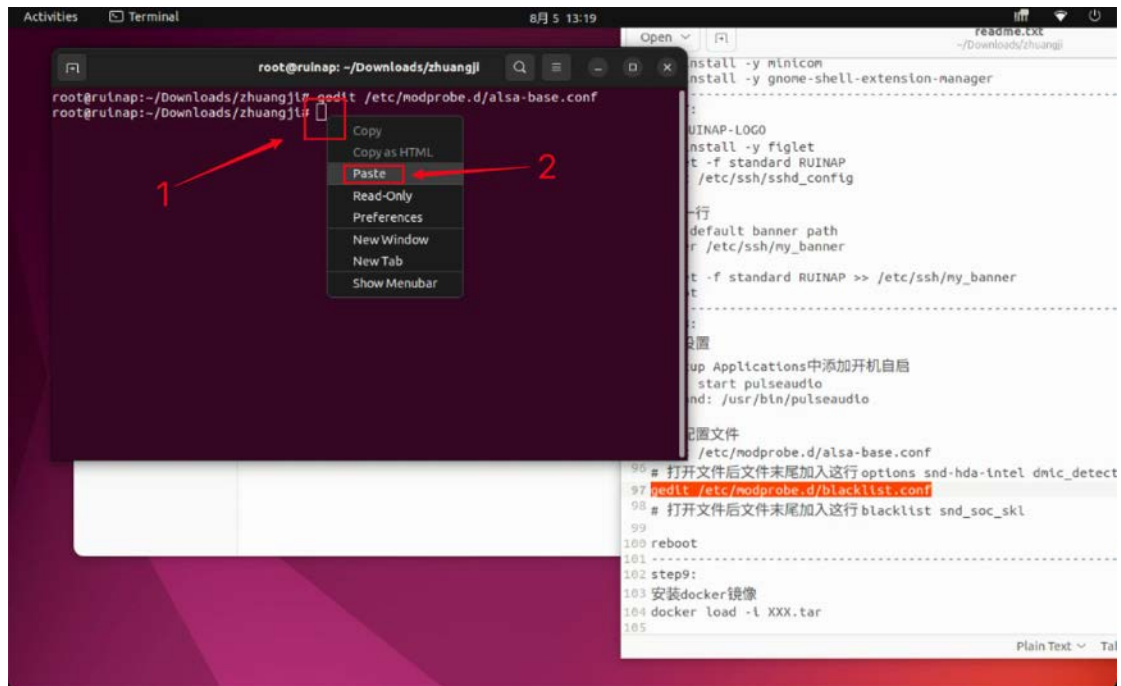
粘贴后，鼠标单击 2 处的【Save】进行保存，然后单击 3 处关闭文档



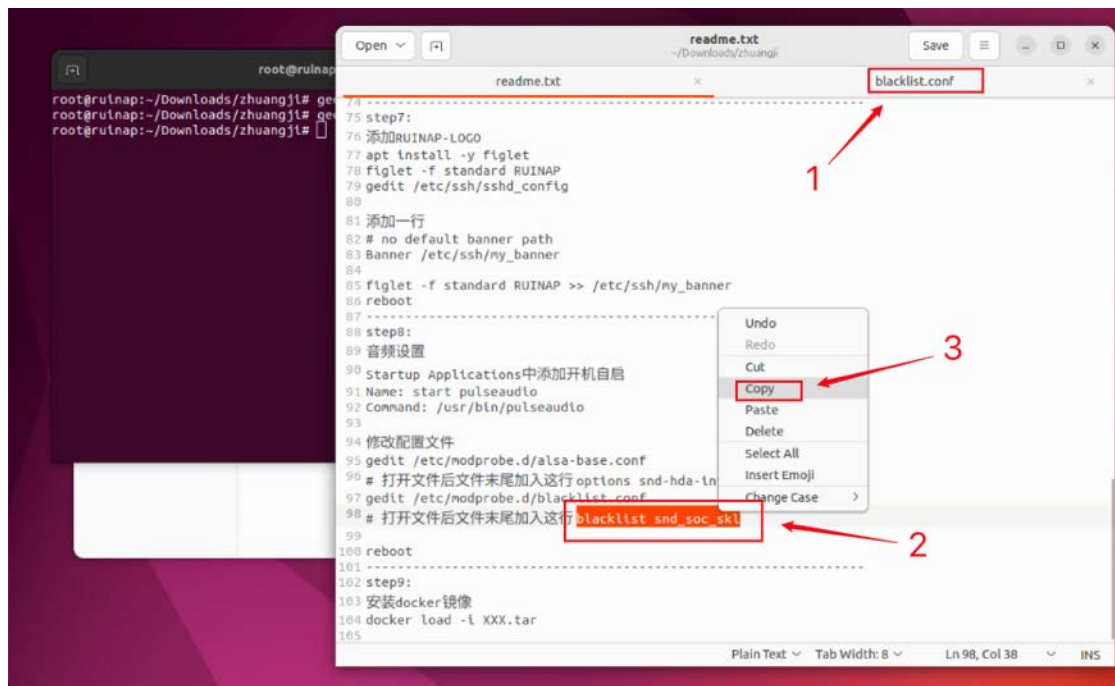
关闭后，回到 readme 中，复制 1 处的内容



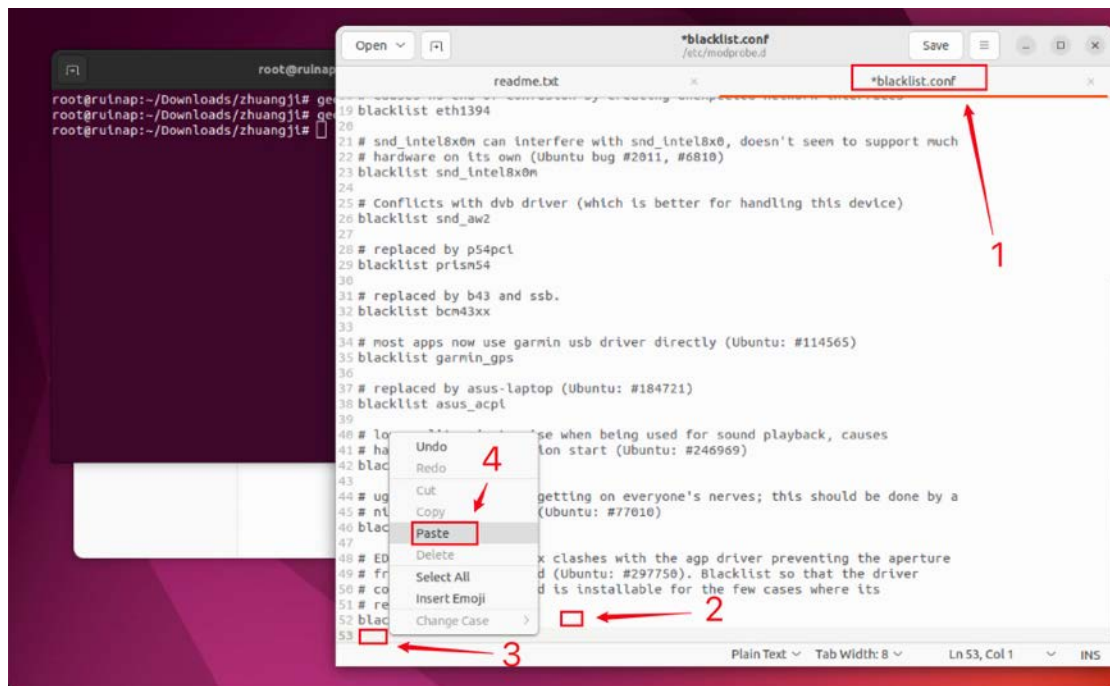
回到终端，将内容粘贴，并按下【回车键】



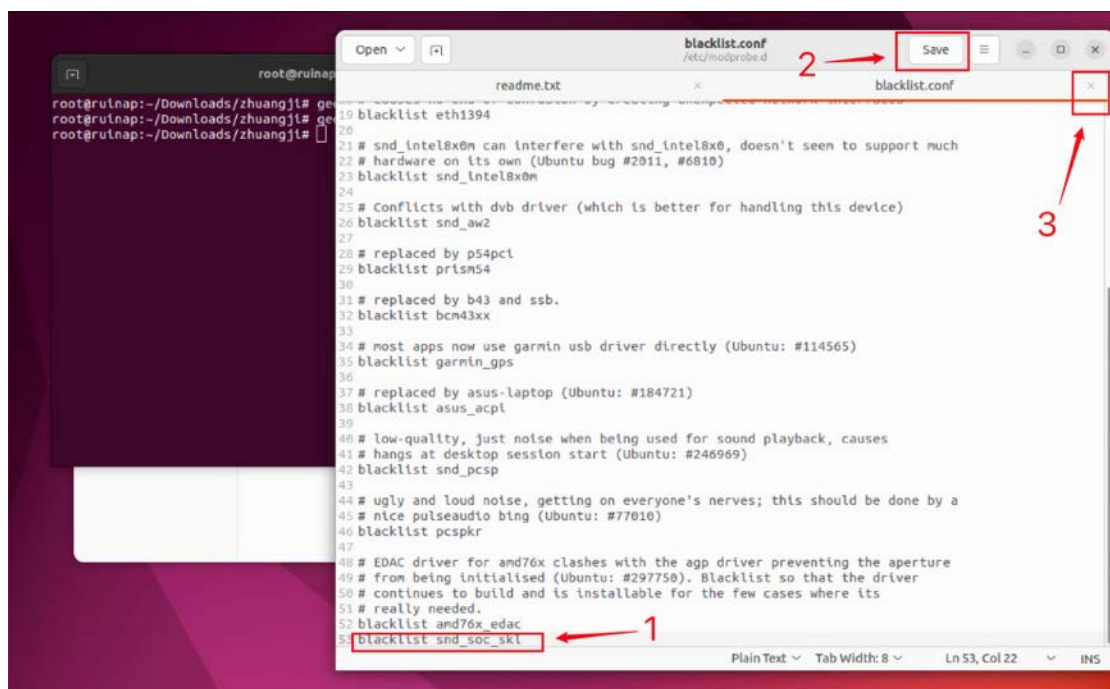
此时会新开一个如下图 1 处的文档，先回到 readme 中复制如下 2 处的内容



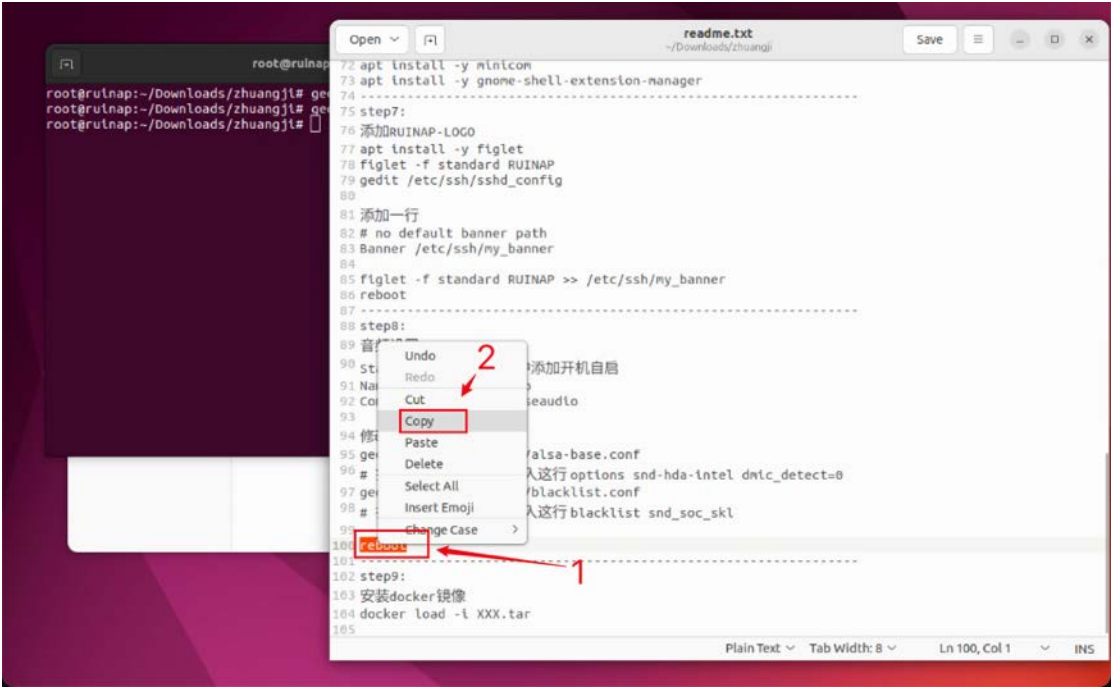
回到新开的 1 处文档中，滚动到文档底部，在末行 2 处按下键盘的【回车键】新起一行，在新一行中将内容粘贴



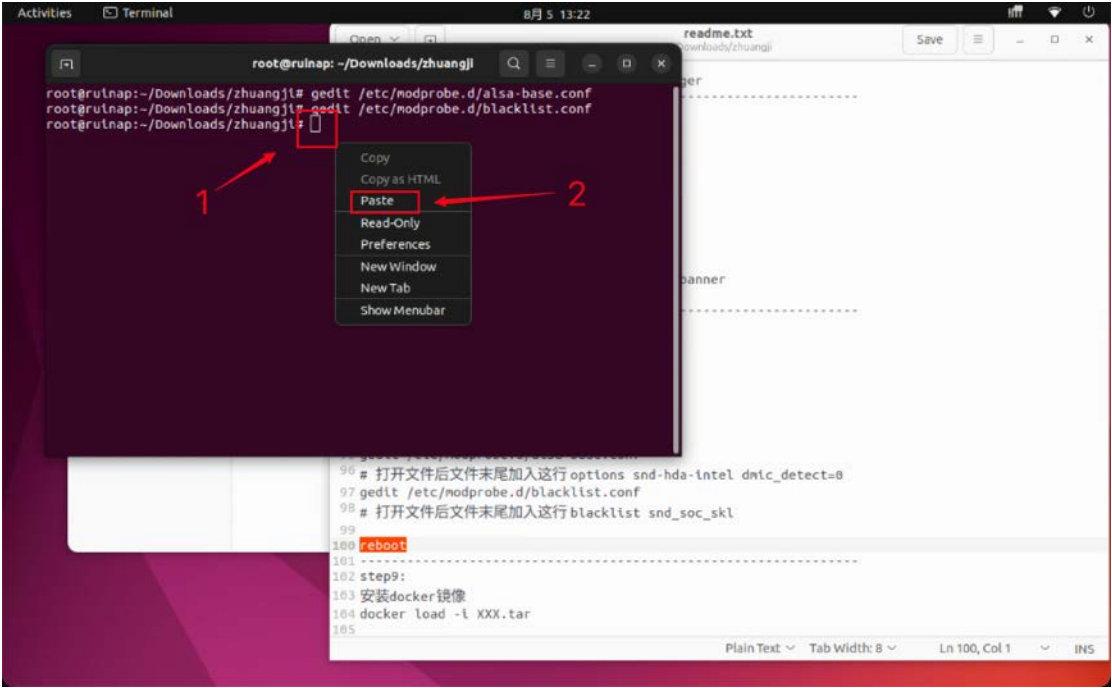
粘贴内容后，鼠标单击【Save】进行保存，并单击 3 处关闭文档



关闭后，回到 readme 中，复制如下内容



回到终端粘贴内容，并按下键盘【回车键】后，工控机会进行重启



至此，设置音频完毕

如果身边有 3.5mm 接口的有线耳机，可以在工控机重启后进行测试。插入耳机后，鼠标拖动如下 2 处音量至最大处并松开时，应该能够听到音效。

